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650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the set of protocols and APIs for 5G Media Streaming (5GMS) services based on the 5G Media Streaming Architecture (5GMSA). 5GMS supports services including MNO and third-party Downlink Media Streaming Services, and MNO and third-party Uplink Media Streaming Services.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 26.501: "5G Media Streaming (5GMS); General description and architecture".

[3] DASH Industry Forum, "Specification of Live Media Ingest", https://dashif-documents.azurewebsites.net/Ingest/master/DASH-IF-Ingest.pdf

[4] 3GPP TS 26.247: "Transparent end-to-end Packet-switched Streaming Service (PSS); Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH)".

[5] Standard ECMA-262, 5.1 Edition: "ECMAScript Language Specification", June 2011.

[6] IETF RFC 6234: "US Secure Hash Algorithms (SHA and SHA-based HMAC and HKDF)".

[7] 3GPP TS 23.003: "Numbering, addressing and identification".

[8] ITU-T Recommendation X.509 (2005) | ISO/IEC 9594-8:2005: "Information Technology – Open Systems Interconnection – The Directory: Public-key and attribute certificate frameworks".

[9] Void

[10] IETF RFC 4648: "The Base16, Base32, and Base64 Data Encodings".

[11] IEEE Standard 1003.1™, Issue 7: "The Open Group Base Specifications", 2018. https://pubs.opengroup.org/onlinepubs/9699919799/

[12] 3GPP TS 29.571: "Common Data Types for Service Based Interfaces; Stage 3".

[13] 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification".

[14] 3GPP TS 36.321: "Evolved Universal Terrestrial Radio Access (E-UTRA); Medium Access Control (MAC) protocol specification".

[15] 3GPP TS 27.007: "AT Command set for User Equipment (UE)".

[16] Void

[17] IETF RFC 7468: "Textual Encodings of PKIX, PKCS, and CMS Structures", April 2015.

[18] ISO 3166-1: "Codes for the representation of names of countries and their subdivisions — Part 1: Country codes".

[19] ISO 3166-2: "Codes for the representation of names of countries and their subdivisions — Part 2: Country subdivision code".

[20] IETF RFC 5280: "Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile", May 2008.

[21] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[22] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[23] OpenAPI: "OpenAPI 3.0.0 Specification", https://github.com/OAI/OpenAPI-Specification/blob/master/versions/3.0.0.md.

[24] IETF RFC 9112: "HTTP/1.1", June 2022.

[25] IETF RFC 9110: "HTTP Semantics", June 2022.

[26] Void

[27] Void

[28] IETF RFC 9111: "HTTP Caching", June 2022.

[29] Void

[30] IETF RFC 8446: "The Transport Layer Security (TLS) Protocol Version 1.3", August 2018.

[31] IETF RFC 9113: "HTTP/2", June 2022.

[32] ISO/IEC 23009-1: "Information technology; Dynamic adaptive streaming over HTTP (DASH) — Part 1: Media presentation description and segment formats".

[33] 3GPP TS 23.503: "Policy and charging control framework for the 5G System (5GS); Stage 2".

[34] 3GPP TS 29.514: "5G System; Policy Authorization Service; Stage 3".

[35] 3GPP TS 26.511: "5G Media Streaming (5GMS); Profiles, codecs and formats".

[36] Void.

[37] 3GPP TS 26.244: "Transparent end-to-end packet switched streaming service (PSS); 3GPP file format (3GP)".

[38] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format", December 2017.

[39] ISO/IEC 14496-12: "Information technology – Coding of audio-visual objects – Part 12: ISO base media file format".

[40] ISO/IEC 23000-19: "Information technology – Multimedia application format (MPEG-A) – Part 19: Common media application format (CMAF) for segmented media".

[41] IETF RFC 3986: "URI Generic Syntax".

[42] 3GPP TS 26.118: "Virtual Reality (VR) profiles for streaming applications".

[43] 3GPP TS 24.558: "Enabling Edge Applications; Protocol specification".

[44] 3GPP TS 29.558: "Enabling Edge Applications; Application Programming Interface (API) specification; Stage 3".

[45] 3GPP TS 23.502: "Procedures for the 5G System (5GS); Stage 2".

[46] 3GPP TS 29.517: "5G System; Application Function Event Exposure Service; Stage 3".

[47] 3GPP TS 23.288: "Architecture enhancements for 5G System (5GS) to support network data analytics services".

[48] 3GPP TS 26.531: "Data Collection and Reporting; General Description and Architecture".

[49] 3GPP TS 26.532: "Data Collection and Reporting; Protocols and Formats".

[50] 3GPP TS 29.522: "5G System. Network Exposure Function Northbound APIs; Stage 3".

[51] 3GPP TS 26.346: "Multimedia Broadcast/Multicast Service (MBMS); Protocols and codecs".

[52] 3GPP TS 26.347: "Multimedia Broadcast/Multicast Service (MBMS); Application Programming Interface and URL".

[53] IETF draft-bhutton-json-schema-validation: "JSON Schema Validation: A Vocabulary for Structural Validation of JSON", June 2022.

[54] IETF RFC 3339: "Date and Time on the Internet: Timestamps", July 2002.

[55] 3GPP 29.591: "Network Exposure Function Southbound Services; Stage 3".

[56] 3GPP TS 26.510: "Media delivery; interactions and APIs for provisioning and media session handling".

[57] IETF RFC 2045: "Multipurpose Internet Mail Extensions (MIME) Part One: Format of Internet Message Bodies".

[58] IETF RFC 9000: "QUIC: A UDP-Based Multiplexed and Secure Transport", May 2021.

[59] IETF RFC 9001: "Using TLS to Secure QUIC", May 2021.

[60] IETF RFC 9114: "HTTP/3", June 2022.

[61] IETF RFC 8673: "HTTP Random Access and Live Content", November 2019.

[62] Consumer Technology Association CTA-5005-A: "Web Application Video Ecosystem – DASH-HLS Interoperability Specification".

[63] DASH-IF Guidelines: "Low-latency Modes for DASH", available here: https://dash-industry-forum.github.io/docs/CR-Low-Latency-Live-r8.pdf

[64] 3GPP TS 26.517: "5G Multicast-Broadcast User Services; Protocols and Formats".

[65] Consumer Technology Association CTA-5004: "Web Application Video Ecosystem – Common Media Client Data", September 2020, https://cdn.cta.tech/cta/media/media/resources/standards/pdfs/cta-5004-final.pdf.

[66] IETF RFC 4122: "A Universally Unique IDentifier (UUID) URN Namespace", July 2005.

[67] IETF RFC 9330:"Low Latency, Low Loss, Scalable Throughput (L4S) Internet Service: Architecture", January 2023.

[68] IETF RFC 9331: "Explicit Congestion Notification (ECN) Protocol for Very Low Queuing Delay (L4S)", January 2023.

[69] IETF RFC 9332: "Dual-Queue Coupled Active Queue Management (AQM) for Low Latency, Low Loss, and Scalable Throughput (L4S)", January 2023.

[70] CTA-5004: "Web Application Video Ecosystem: Common Media Client Data (CMCD)", September 2020.

[71] IETF RFC 9293: "Transmission Control Protocol (TCP)", August 2022.

[72] IETF RFC 8684: "TCP Extensions for Multipath Operation with Multiple Addresses", March 2020.

[73] IETF Draft: "Multipath Extension for QUIC", draft-ietf-quic-multipath-15, July 2025.

[74] DASH Interoperability Forum: "DASH-IF Interoperability Points; Part 6: Content protection and security".

[75] ETSI TS 103 799: "Content Protection Information Exchange Format (CPIX)".

[76] ETSI TS 103 998: "Content Steering for DASH".

[77] ETSI TS 103 973: "Coded Multisource Media Format (CMMF) for Content Distribution and Delivery".

[78] IETF RFC 5053: "Raptor Forward Error Correction Scheme for Object Delivery", October 2007.

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1], TS 26.501 [2] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1] or TS 26.501 [2].

media resource: A media object that is rendered by the 5GMSd Media Player or captured by the 5GMSu Media Streamer.

transport resource: A uniquely identifiable object constructed for the purposes of transmission from a service location at reference points M2, M4 or M10 that contains a variant or representation of a media resource.3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5GMS	5G Media Streaming
5GMSd	5GMS downlink
5GMSu	5GMS uplink
5GMSA	5GMS Architecture
5GMS EAS	Edge-enabled 5GMS Application Server
BMFF	(ISO) Base Media File Format
ABR	Adaptive Bit Rate
ACR	Application Context Relocation
AF	Application Function
ANBR	Access Network Bit rate Recommendation
AS	Application Server
BM-SC	Broadcast Multicast Switching Centre
CBCS	Cipher Block Chaining
CDN	Content Delivery Network / Content Distribution Network
CENC	Common Encryption
CGI	Cell Global Identifier
CMAF	Common Media Application Format
CMMF	Coded Multisource Media Format
CRUD	Create, Read, Update, Delete
CNAME	Canonical Name
CORS	Cross-Origin Resource Sharing
CPIX	Content Protection Information Exchange
CRL	Certificate Revocation List
DANE	DASH-Aware Network Element
DASH	Dynamic Adaptive Streaming over HTTP
DER	Distinguished Encoding Rule
DNN	Domain Name News
DNS	Domain Name Server
EAS	Edge Application Server
ECGI	E-UTRAN Cell Global Identifier
ECMA	European Computer Manufacturers Association
ECN	Explicit Congestion Notification

EES	Edge Enabler Server
EFDT	Extended File Delivery Table
eMBMS	evolved MBMS
FQDN	Fully Qualified Domain Name
GPSI	Generic Public Subscription Identifier
HLS	HTTP Live Streaming
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
L4S	Low Latency, Low Loss and Scalable Throughput
LCID	Logical Channel IDentifier
MBMS	Multimedia Broadcast Multicast Service
MBS	Multicast-Broadcast Service
MBSTF	MBS Transport Function
MFBR	Maximum Flow Bit Rate
MIME	Multipurpose Internet Mail Extensions
MNO	Mobile Network Operator
MPD	Media Presentation Description
MSISDN	Mobile Subscriber ISDN number
NCGI	NR Cell Global Identifier
NEF	Network Exposure Function
OAM	Operations, Administration and Maintenance
PCC	Policy Control and Charging
PCF	Policy Control Function
PEM	Privacy-Enhanced Mail
PFD	Packet Flow Description
PFDF	Packet Flow Description Function
QoE	Quality of Experience
QoS	Quality of Service
SDF	Service Data Flow
SHA	Secure Hash Algorithm
TLS	Transport Layer Security
TOI	Transport Object Identifier
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
UTC	Coordinated Universal Time

4 Procedures for Downlink Media Streaming

4.1 General

This clause defines all procedures for Downlink Media Streaming using the different 5G Media Streaming Reference Points.

NOTE: The descriptions of certain M1 interface procedures in clause 4.3, and of certain M5 interface procedures in clause 4.7, indicate applicability of those procedures to both downlink and uplink media streaming. This avoids redundant duplication of normative text in clause 5, regarding M1 and M5 procedures for uplink media streaming.

4.2 APIs relevant to downlink media streaming

Table 4.2-1 summarises the APIs used to provision and use the various downlink media streaming features specified in TS 26.501 [2].

Table 4.2-1: Summary of APIs relevant to downlink media streaming features

5GMSd feature	Abstract	Relevant APIs		
		Interface	API name	Clause
Content protocols discovery	Used by the 5GMSd Application Provider to interrogate which content ingest protocols are supported by 5GMSd AS(s).	M1d	Content Protocols Discovery API	7.5
Content hosting	Content is ingested, hosted and distributed by the 5GMSd AS according to a Content Hosting Configuration associated with a Provisioning Session.	M1d	Provisioning Sessions API	7.2
			Server Certificates Provisioning API	7.3
			Content Preparation Templates Provisioning API	7.4
			Content Hosting Provisioning API	7.6
		M2d	HTTP pull-based content ingest protocol	8.2
			DASH-IF push-based content ingest protocol	8.3
			HTTP low-latency pull-based content ingest protocol	8.4
		M3d	Server Certificates configuration API	9.2
			Content Preparation Templates configuration API	9.3
			Content Hosting configuration API	9.4
		M4d	MPEG-DASH [4] or 3GP-DASH [37] or DASH-IF push-based content distribution	10.2
			HTTP low-latency content distribution	10.3
		M5d	Service Access Information API	11.2
		M10d	HTTP pull-based content ingest protocol	8.2
			DASH-IF push-based content ingest protocol	8.3
			HTTP low-latency pull-based content ingest protocol	8.4
Metrics reporting	The 5GMSd Client uploads metrics reports to the 5GMSd AF according to a provisioned Metrics Reporting Configuration it obtains from the Service Access Information for its Provisioning Session.	M1d	Provisioning Sessions API	7.2
			Metrics Reporting Provisioning API	7.8
		M3d	Service Access Information API	11.2
			Metrics Reporting API	11.4
		M5d	Service Access Information API	11.2
			Metrics Reporting API	11.4
Consumption reporting	The 5GMSd Client or 5GMSd AS provides feedback reports on currently consumed content according to a provisioned Consumption Reporting Configuration it obtains from the Service Access Information for its Provisioning Session.	M1d	Provisioning Sessions API	7.2
			Consumption Reporting Provisioning API	7.7
		M5d	Service Access Information API	11.2
			Consumption Reporting API	11.3
Dynamic Policy invocation	The 5GMSd Client activates different traffic treatment policies selected from a set of Policy Templates configured in its Provisioning Session.	M1d	Provisioning Sessions API	7.2
			Policy Templates Provisioning API	7.9
		M5d	Service Access Information API	11.2
			Dynamic Policies API	11.5
		M5d	Service Access Information API	11.2

5GMSd feature	Abstract	Relevant APIs		
		Interface	API name	Clause
Network Assistance	The 5GMSd Client requests bit rate recommendations and delivery boosts from the 5GMSd AF.		Network Assistance API	11.6
Edge content processing	Edge resources are provisioned for processing content in 5GMS downlink media streaming sessions.	M1d	Provisioning Sessions API	7.2
			Edge Resources Provisioning API	7.10
		M5d	Service Access Information API	11.2
5GMS via eMBMS	The 5GMSd AF provisions the delivery of content via eMBMS and MBMS User Services.	M1d	Provisioning Sessions API	7.2
		M5d	Service Access Information API	11.2
		M4d	MPEG-DASH [4] or 3GP-DASH [37] or HLS	10
5GMS via MBS	The 5GMSd AF provisions the delivery of content via MBS User Services.	M1d	Provisioning Sessions API	7.2
		M5d	Service Access Information API	11.2
		M4d	MPEG-DASH [4] or 3GP-DASH [37] or HLS	10
5GMS via eMBMS	The 5GMSd AF provisions the delivery of content via eMBMS.	M1d	Provisioning Sessions API	7.2
		M5d	Service Access Information API	11.2
		M4d	MPEG-DASH [4] or 3GP-DASH [37] or HLS content distribution	10
UE data collection, reporting and exposure	UE data related to downlink 5G Media Streaming is reported to the Data Collection AF instantiated in the 5GMSd AF for exposure to Event consumers.	M1d	Event Data Processing Provisioning API	7.11
		R4	*Ndcaf_DataReporting* service	17
		R5, R6	*Naf_EventExposure* service	18

4.3 Procedures of the M1 (5GMS Provisioning) interface

4.3.1 General

A 5GMS Application Provider may use the procedures in this clause to provision the network for media streaming sessions that are operated by that 5GMS Application Provider. For downlink media streaming, these sessions may be DASH streaming sessions, progressive download sessions, or any other type of media streaming or distribution (e.g. HLS) sessions. For uplink media streaming, the content format and delivery protocol are defined by the 5GMSu Application Provider and may be either non-fully standardized or employ standardized HTTP-based streaming of ISO BMFF content fragments as profiled by CMAF [39].

Reference point M1 offers three different sets of procedures:

- For downlink media streaming, configuration of content ingest at reference point M2d or M10d for onward distribution by the 5GMSd AS over reference point M4d or M10d, or via other distribution systems such as eMBMS or MBS. The API at this reference point is designed to offer equivalent functionality as that exposed by a public CDN. For uplink media streaming, configuration of content egest at reference point M2u or M10u for the media content received by the 5GMSu AS from the 5GMSu Client over reference point M4u or M10u. The resource types involved in content hosting configuration are provisioning session (see clause 4.3.2), content hosting procedures (see clause 4.3.3), ingest protocols (see clause 4.3.4), content preparation template (see clause 4.3.5), and server certificates (see clause 4.3.6).

- Configuration of dynamic policies: allows the configuration of Policy Templates at M5 that can be applied to M4 downlink/uplink media streaming sessions.

- Configuration of reporting: permits the MNO to collect, at M5, QoE metrics and consumption reports about M4 downlink sessions, as well as permits the MNO to collect, at M5, QoE metrics reports about M4 uplink sessions.

A 5GMS Application Provider may use any of these procedures, in any combination, to support its media streaming sessions.

4.3.2 Provisioning Session procedures

4.3.2.1 General

Prior to configuring content hosting, dynamic policies, or reporting, the 5GMS Application Provider shall create a new Provisioning Session in the 5GMS AF. The 5GMS Application Provider shall use the operations specified in clause 5.2.2 of TS 26.510 [56] at reference point M1 to create and subsequently manipulate Provisioning Sessions in the 5GMS AF.

4.3.2.2 Void

4.3.2.3 Void

4.3.2.4 Void

4.3.2.5 Void

4.3.3 Content Hosting provisioning procedures

4.3.3.1 General

The 5GMSd Application Provider shall use the operations specified in clause 5.2.8 of TS 26.510 [56] at reference point M1d when it wants to create and subsequently manipulate Content Hosting Configurations in the 5GMSd AF in order to provision the content hosting feature for downlink media streaming.

The *canonicalDomainName* property nominated by the 5GMS AF for each distribution configuration in the Content Hosting Configuration shall comply with clause 6.0.2.2.

In the case where the *entryPoint* of a distribution configuration describes a Media Entry Point resource (e.g. MPEG-DASH MPD), the 5GMS AF shall not prefix additional path elements to the *relativePath* when it nominates the *baseURL* property.

4.3.3.2 Void

4.3.3.3 Void

4.3.3.4 Void

4.3.3.5 Void

4.3.3.6 Void

4.3.3A Content Publishing provisioning procedures

4.3.3A.1 General

The 5GMSu Application Provider shall use the operations specified in clause 5.2.9 of TS 26.510 [56] at reference point M1u when it wants to create and subsequently manipulate Content Publishing Configurations in the 5GMSu AF in order to provision the content publishing feature for uplink media streaming.

The *canonicalDomainName* property nominated by the 5GMS AF for each contribution configuration in the Content Publishing Configuration shall comply with clause 6.0.2.2.

In the case where the *entryPoint* of a contribution configuration describes a Media Entry Point resource (e.g. MPEG-DASH MPD), the 5GMS AF shall not prefix additional path elements to the *relativePath* when it nominates the *baseURL* property.

4.3.4 Content Protocols Discovery procedures

4.3.4.1 General

The 5GMS Application Provider shall use the operations specified in clause 5.2.3 of TS 26.510 [56] at reference point M1 when it wants to discover the set of downlink content ingest or uplink content egest protocols supported by the 5GMS AS at reference point M2 and M10.

4.3.4.2 Void

4.3.4.3 Void

4.3.4.4 Void

4.3.4.5 Void

4.3.5 Content Preparation Template provisioning procedures

4.3.5.1 General

For downlink media streaming, the 5GMSd AS may be required to process content ingested at reference point M2d or M10d before serving it from reference point M4d service locations. For uplink media streaming, the 5GMSu AS may be required to process content it receives from the 5GMSu Client at reference point M4u before passing it to the 5GMSu Application Provider on the egest interface at reference point M2u or to another 5GMSu AS at reference point M10u.

The 5GMS Application Provider shall use the operations specified in clause 5.2.5 of TS 26.510 [56] at reference point M1 when it wants to create and subsequently manipulate Content Preparation Templates in the 5GMS AF.

4.3.5.2 Void

4.3.5.3 Void

4.3.5.4 Void

4.3.5.5 Void

4.3.6 Server Certificate provisioning procedures

4.3.6.1 General

Each X.509 server certificate [8] presented by the 5GMSd AS at reference point M4d service locations or at reference point xMB-U is represented by a Server Certificate resource at M1d. The 5GMS Application Provider shall use the operations specified in clause 5.2.4 of TS 26.510 [56] at reference point M1 when it wants to create and subsequently manipulate Server Certificates in the 5GMS AF. These enable a Server Certificate resource to be created within the scope of a Provisioning Session, and subsequently referenced by a Content Hosting Configuration created in the scope of the same Provisioning Session.

NOTE: As a consumer of media from the 5GMSd AS in a combined architecture using 5GMS and eMBMS, the BM-SC needs to be able to trust the content it is receiving comes from a *bona fide* source. This issue is left to implementation. Likewise, in the case of a combined architecture using 5GMS and MBS, the MBSTF needs to be able to trust the content it ingests.

4.3.6.2 Void

4.3.6.3 Void

4.3.6.4 Void

4.3.6.5 Void

4.3.6.6 Void

4.3.6.7 Void

4.3.7 Dynamic Policy provisioning procedures

4.3.7.1 General

The 5GMS Application Provider shall use the operations specified in clause 5.2.7 of TS 26.510 [56] at reference point M1 when it wants to create and subsequently manipulate Policy Templates available for the use of downlink or uplink media streaming sessions of a particular Provisioning Session in the 5GMS AF.

As specified in clause 5.2.7.1 of TS 26.510 [56], the 5GMS AF is responsible for ensuring that 5GMS AS supports L4S according to RFC 9330 [67], RFC 9331 [68] and RFC 9332 [69] at reference point M4 before accepting a Policy Template that declares a preference for ECN marking for L4S functionality.

4.3.7.2 Void

4.3.7.3 Void

4.3.7.4 Void

4.3.7.5 Void

4.3.8 Consumption Reporting Configuration provisioning procedures

4.3.8.1 General

The 5GMSd Application Provider shall use the interactions specified in clause 5.2.12 of TS 26.510 [56] at reference point M1 when it wants to activate and configure consumption reporting for a Provisioning Session in the 5GMSd AF.

4.3.8.2 Void

4.3.8.3 Void

4.3.8.4 Void

4.3.8.5 Void

4.3.9 Metrics Reporting provisioning procedures

4.3.9.1 General

The 5GMS Application Provider shall use the operations specified in clause 5.2.11 of TS 26.510 [56] at reference point M1 when it wants to configure QoE metrics reporting functionality associated with a downlink or uplink media streaming Provisioning Session in the 5GMS AF.

4.3.9.2 Void

4.3.9.3 Void

4.3.9.4 Void

4.3.9.5 Void

4.3.10 Edge Resources provisioning procedures

4.3.10.1 General

The 5GMS Application Provider shall use the operations specified in clause 5.2.6 of TS 26.510 [56] at reference point M1 when it wants to provision edge resources for downlink or uplink media streaming associated with a Provisioning Session in the 5GMS AF.

4.3.10.2 Void

4.3.10.3 Void

4.3.10.4 Void

4.3.10.5 Void

4.3.11 Event Data Processing provisioning procedures

4.3.11.1 General

The 5GMS Application Provider shall use the operations specified in clause 5.2.13 of TS 26.510 [56] at reference point M1 to configure the collection and processing of UE data related to 5G Media Streaming and to restrict its exposure over reference points R5 and R6 by configuring the Data Collection AF instantiated in the 5GMS AF (as defined in TS 26.531 [46] and clause 4.7 of TS 26.501 [2]) with one or more Event Data Processing Configurations and Data Access Profiles for a particular Event ID.

Each instance of a Data Access Profile specifies a set of data processing operations to be performed by the Data Collection AF on its collected UE data in order to synthesize the event data to be exposed to a specific Event service consumer entity. In this release, eligible Event service consumer entities of 5GMS event data are the NWDAF, the Event Consumer AF and the NEF.

4.3.11.2 Void

4.3.11.3 Void

4.3.11.3A Void

4.3.11.4 Void

4.4 Procedures of the M2d (5GMS content ingest) interface

The following 5GMS AS content ingest protocols are specified by the present document at reference point M2d to support downlink media streaming:

- An *HTTP pull-based content ingest protocol* is specified in clause 8.2, including specific handling for HTTP redirects issued to the 5GMS AS by the 5GMS Application Provider's origin server.
- A *DASH-IF push-based content ingest protocol* is specified in clause 8.3.

4.5 Procedures of the M3d interface

4.5.1 General

The procedures at reference point M3 defined in the following clauses should be used by the 5GMS AF to configure Server Certificate and Content Preparation Template resources in the 5GMS AS as well as Content Hosting and/or Content Publishing configuration resources that refer to them. The corresponding APIs specified in clause 9 should be used to realise these procedures.

4.5.2 Server Certificate configuration procedures

4.5.2.1 General

The 5GMS AF configures Server Certificate resources in the 5GMS AS using the procedures defined in this clause such that they remain synchronised with the Server Certificates provisioned at reference point M1 using the procedures defined in clause 4.3.6.

Each Server Certificate resource at reference point M3 is represented by a PEM-formatted X.509 certificate bundle.

4.5.2.2 Enumerate Server Certificates collection

This procedure is used by the 5GMS AF to determine the configuration state of the Server Certificates collection in a running 5GMS AS instance. The HTTP `GET` method shall be used for this purpose. The request URL shall be that of the Server Certificates collection on the 5GMS AS instance.

If the operation is successful, the 5GMS AS shall return an HTTP *200 (OK)* response. The resource body shall be a JSON array of Server Certificate resource identifiers. The array shall be empty if no Server Certificate resources currently exist in the collection.

4.5.2.3 Create Server Certificate

This procedure is used by the 5GMS AF to create a new Server Certificate resource in the target 5GMS AS instance. The HTTP `POST` method shall be used for this purpose. The 5GMS AF shall nominate the resource identifier to be used to identify the new resource in the Server Certificates collection as part of the request URL. A representation of the resource, in the form of a PEM-formatted X.509 certificate bundle that includes the private key, shall be provided as the request body.

If the operation is successful, the 5GMS AS shall return an HTTP *201 (Created)* response with an empty response body. The content of the `Location` header may differ from the request URL if the request was satisfied by a different 5GMS AS instance.

If a Server Certificate resource already exists at the request URL an HTTP *409 (Conflict)* error response shall be returned by the 5GMS AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

If a Server Certificate resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.2.4 Retrieve Server Certificate

Retrieval of individual Server Certificate resources from the 5GMS AS is not permitted at reference point M3.

4.5.2.5 Update Server Certificate

This procedure is used by the 5GMS AF to replace a Server Certificate resource in the target 5GMS AS instance. The HTTP `PUT` method shall be used for this purpose. The target resource shall be indicated in the request URL. The replacement resource representation, in the form of a PEM-formatted X.509 certificate bundle that includes the private key, shall be provided as the request body.

If the operation is successful, the 5GMS AS shall return an HTTP *200 (OK)* response with an empty response body.

If the replacement PEM bundle is identical to the current resource representation the 5GMS AS shall return an HTTP *204 (No Content)* response with an empty response body to indicate a "no-op".

If a Server Certificate resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMS AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.2.6 Destroy Server Certificate

This procedure is used by the 5GMS AF to destroy a Server Certificate resource in the target 5GMS AS instance. The HTTP `DELETE` method shall be used for this purpose. The target resource shall be indicated in the request URL. The request body shall be empty.

If the operation is successful, the 5GMS AS shall return an HTTP *204 (No Content)* response with an empty response body. Any subsequent attempt to recreate a Server Certificate resource with the same resource identifier shall fail as specified in clause 4.5.2.3.

If the target Server Certificate resource is still referenced by a Content Hosting Configuration or Content Publishing Configuration, an HTTP *409 (Conflict)* error response shall be returned by the 5GMS AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12]

If a Server Certificate resource previously existed at the request URL and was destroyed using this procedure an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. If no server certificate resource has ever existed at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMS AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.3 Content Preparation Template configuration procedures

4.5.3.1 General

The 5GMS AF shall configure Content Preparation Template resources in the 5GMS AS using the procedures defined in this clause such that they remain synchronised with the Content Preparation Templates provisioned at reference point M1 using the procedures defined in clause 4.3.5.

The format of the Content Preparation Template resource representation is not specified in the present document.

4.5.3.2 Enumerate Content Preparation Templates collection

This procedure is used by the 5GMS AF to determine the configuration state of the Content Preparation Templates collection in a running 5GMS AS instance. The HTTP `GET` method shall be used for this purpose. The request URL shall be that of the Content Preparation Templates collection on the 5GMS AS instance.

If the operation is successful, the 5GMS AS shall return an HTTP *200 (OK)* response. The resource body shall be a JSON array of Content Preparation Template resource identifiers. The array shall be empty if no Content Preparation Template resources currently exist in the collection.

4.5.3.3 Create Content Preparation Template

This procedure is used by the 5GMS AF to create a new Content Preparation Template resource in the target 5GMS AS instance. The HTTP `POST` method shall be used for this purpose. The 5GMS AF shall nominate the resource identifier to be used to identify the new resource in the Content Preparation Templates collection as part of the request URL. A representation of the resource shall be provided as the request body.

If the operation is successful, the 5GMS AS shall return an HTTP *201 (Created)* response with an empty response body. The content of the `Location` header may differ from the request URL if the request was satisfied by a different 5GMS AS instance.

If a Content Preparation Template resource already exists at the request URL an HTTP *409 (Conflict)* error response shall be returned by the 5GMS AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

If a Content Preparation Template resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.3.4 Retrieve Content Preparation Template

Retrieval of individual Content Preparation Template resources is not permitted at reference point M3.

4.5.3.5 Update Content Preparation Template

This procedure is used by the 5GMS AF to replace a Content Preparation Template resource in the target 5GMS AS instance. The HTTP `PUT` method shall be used for this purpose. The target resource shall be indicated in the request URL. The replacement resource representation shall be provided as the request body.

If the operation is successful, the 5GMS AS shall return an HTTP *200 (OK)* response with an empty response body.

If the replacement Content Preparation Template is identical to the current resource representation the 5GMS AS shall return an HTTP *204 (No Content)* response with an empty response body to indicate a "no-op".

If the Media AS does not support modification of the Content Preparation Template, the update operation shall fail with a *403 (Forbidden)* HTTP response that includes a *ProblemDetails* error message body as specified in clause 5.2.4.1 of TS 29.571 [12].

If a Content Preparation Template resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMS AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.3.6 Destroy Content Preparation Template

This procedure is used by the 5GMS AF to destroy a Content Preparation Template resource in the target 5GMS AS instance. The HTTP `DELETE` method shall be used for this purpose. The target resource shall be indicated in the request URL. The request body shall be empty.

If the operation is successful, the 5GMS AS shall return an HTTP *204 (No Content)* response with an empty response body. Any subsequent attempt to recreate a Content Preparation Template resource with the same resource identifier shall fail as specified in clause 4.5.3.3.

If the target Content Preparation Template resource is still referenced by a Content Hosting Configuration or Content Publishing Configuration, an HTTP *409 (Conflict)* error response shall be returned by the 5GMS AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12]

If a Content Preparation Template resource previously existed at the request URL and was destroyed using this procedure an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMS AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.4 Content Hosting Configuration procedures

4.5.4.1 General

The 5GMSd AF shall configure Content Hosting Configuration resources of type *ApplicationServerContentHosting Configuration* in the 5GMSd AS using the procedures defined in this clause such that they remain synchronised with the Content Hosting Configurations provisioned at reference point M1 using the procedures defined in clause 4.3.3.

The format of the Content Hosting Configuration resource representation at reference point M3d is as specified in clause 8.8.3.1 of TS 26.510 [56]. The 5GMSd AF shall populate the *externalServiceIdentifier* property with the external service identifier from the parent Provisioning Session to enable the 5GMSd AS to retrieve Service Access Information

from the 5GMSd AF at reference point M3d (see clause 4.5.6). The 5GMSd AF shall populate the *canonicalDomainName* and *baseURL* properties of each distribution configuration with the values it has chosen.

NOTE: The 5GMSd AS implementation converts received Content Hosting Configurations into a format suitable for configuring the HTTP server that realises the content hosting feature.

4.5.4.2 Enumerate Content Hosting Configurations collection

This procedure is used by the 5GMSd AF to determine the configuration state of the Content Hosting Configurations collection in a running 5GMSd AS instance. The HTTP `GET` method shall be used for this purpose. The request URL shall be that of the Content Hosting Configuration collection on the 5GMSd AS instance.

If the operation is successful, the 5GMSd AS shall return an HTTP *200 (OK)* response. The resource body shall be a JSON array of Content Hosting Configuration resource identifiers. The array shall be empty if no Content Hosting Configuration resources currently exist in the collection.

4.5.4.3 Create Content Hosting Configuration

This procedure is used by the 5GMSd AF to create a new Content Hosting Configuration resource in the target 5GMSd AS instance. The HTTP `POST` method shall be used for this purpose. The 5GMSd AF shall nominate the resource identifier to be used to identify the new resource in the Content Hosting Configurations collection as part of the request URL. A representation of the resource shall be provided as the request body.

If the operation is successful, the 5GMSd AS shall return an HTTP *201 (Created)* response with an empty response body. The content of the `Location` header may differ from the request URL if the request was satisfied by a different 5GMSd AS instance.

If a Content Hosting Configuration resource already exists at the request URL an HTTP *409 (Conflict)* error response shall be returned by the 5GMSd AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

If any resources referenced by the supplied Content Hosting Configuration resource representation are invalid, the create operation shall fail with an HTTP response status code of *400 (Bad Request).* The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12]. In this case, the Content Hosting Configuration resource shall remain in an uncreated state in the target 5GMSd AS instance.

If a Content Hosting Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSd AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.4.4 Retrieve Content Hosting Configuration

Retrieval of individual Content Hosting Configuration resources is not permitted at reference point M3.

4.5.4.5 Update Content Hosting Configuration

This procedure is used by the 5GMSd AF to replace a Content Hosting Configuration resource in the target 5GMSd AS instance. The HTTP `PUT` method shall be used for this purpose. The target resource shall be indicated in the request URL. The replacement resource representation shall be provided as the request body.

If the operation is successful, the 5GMSd AS shall return an HTTP *200 (OK)* response with an empty response body.

If the replacement Content Hosting Configuration is identical to the current resource representation the 5GMSd AS shall return an HTTP *204 (No Content)* response with an empty response body to indicate a "no-op".

If any resources referenced by the replacement Content Hosting Configuration resource representation are invalid, the update operation shall fail with an HTTP response status code of *400 (Bad Request).* The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12]. In this case, the target Content Hosting Configuration resource shall remain in the state immediately prior to the update operation.

If a Content Hosting Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSd AS. Otherwise, if no resource exists at the request URL an

HTTP *404 (Not Found)* error response shall be returned by the 5GMSd AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.4.6 Destroy Content Hosting Configuration

This procedure is used by the 5GMSd AF to destroy a Content Hosting Configuration resource in the target 5GMSd AS instance. The HTTP `DELETE` method shall be used for this purpose. The target resource shall be indicated in the request URL. The request body shall be empty.

If the operation is successful, the 5GMSd AS shall return an HTTP *204 (No Content)* response with an empty response body. Any subsequent attempt to recreate a Content Hosting Configuration resource with the same resource identifier shall fail as specified in clause 4.5.4.3.

If a Content Hosting Configuration resource previously existed at the request URL and was destroyed using this procedure an HTTP *410 (Gone)* error response should be returned by the 5GMSd AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMSd AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.4.7 (De)activate Content Hosting Configuration

This procedure is used by the 5GMSd AF to activate or deactivate a Content Hosting Configuration resource in the target 5GMSd AS instance. The HTTP `POST` method shall be used for this purpose. The target resource shall be identified in the request URL. The request body shall be a single JSON *boolean* indicating the desired state of the target resource: *true* for active and *false* for inactive.

If the Content Hosting Configuration can be (de)activated immediately by the 5GMSd AS, or if it is already in the desired state indicated in the request, the HTTP *204 (No Content)* response shall be returned by the 5GMSd AS with an empty response body. If the activation request is otherwise acceptable to the 5GMSd AS, but the Content Hosting Configuration cannot be (de)activated immediately, the HTTP *202 (Accepted)* response shall be returned by the 5GMSd AS with an empty response body to indicate that (de)activation is in progress. The state of the Content Hosting Configuration may then be polled using the procedure specified in clause 4.5.4.8.

If a Content Hosting Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSd AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMSd AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.4.8 Interrogate Content Hosting Configuration state

This procedure is used by the 5GMSd AF to interrogate the current state of a Content Hosting Configuration resource in the target 5GMSd AS instance. The HTTP `GET` method shall be used for this purpose. The target resource shall be indicated in the request URL. The request body shall be empty.

If the Content Hosting Configuration is known to the 5GMSd AS, the HTTP *200 (OK)* response shall be returned with an indication of its current state provided in the response body as a JSON object of type *boolean* with the value *true* for active and *false* for inactive.

If a Content Hosting Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMS AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.4.9 Purge Content Hosting cache

This procedure is used by the 5GMSd AF to purge one or more content items from the 5GMSd AS cache associated with a Content Hosting Configuration resource in the target 5GMSd AS instance. The HTTP `POST` method shall be used for this purpose. A Content Hosting Configuration resource shall be indicated in the request URL. The request body shall be a single key–value pair encoded per the *application/x-www-form-urlencoded* MIME content type. The key shall

be the string *pattern* and the value shall be a regular expression [5] identifying the media resource URL(s) to be purged from the associated content cache.

If the operation is successful, the 5GMSd AS shall return an HTTP *200 (OK)* response. The resource body should indicate the total number of cache entries purged encoded as a positive JSON integer. If the purge request is otherwise acceptable to the 5GMSd AS, but no cache entries were purged because none matched the supplied regular expression, the HTTP *204 (No Content)* response shall instead be returned by the 5GMSd AS with an empty response body.

If the request message body – or the regular expression contained in it – are found by the 5GMSd AS to be syntactically malformed the HTTP *400 (Bad Request)* error response shall be returned. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

If a Content Hosting Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSd AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMSd AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.5 Content Publishing Configuration procedures

4.5.5.1 General

The 5GMSu AF shall configure Content Publishing Configuration resources of type *ApplicationServerContentPublishingConfiguration* in the 5GMSu AS using the procedures defined in this clause such that they remain synchronised with the Content Publishing Configurations provisioned at reference point M1 using the procedures defined in clause 4.3.3.

The format of the Content Publishing Configuration resource representation at reference point M3 is as specified in clause 8.9.3.1 of TS 26.510 [56]. The 5GMSu AF shall populate the *externalServiceIdentifier* property with the external service identifier from the parent Provisioning Session to enable the 5GMSu AS to retrieve Service Access Information from the 5GMSu AF at reference point M3u (see clause 4.5.6). The 5GMSu AF shall populate the *canonicalDomainName* and *baseURL* properties of each contribution configuration with the values it has chosen.

NOTE: The 5GMSu AS implementation converts received Content Publishing Configurations into a format suitable for configuring the HTTP server that realises the content publishing feature.

4.5.5.2 Enumerate Content Publishing Configurations collection

This procedure is used by the 5GMSu AF to determine the configuration state of a running 5GMSu AS instance. The HTTP `GET` method shall be used for this purpose. The request URL shall be that of the Content Publishing Configuration collection on the 5GMSu AS instance.

If the operation is successful, the 5GMSu AS shall return an HTTP *200 (OK)* response. The resource body shall be a JSON array of Content Publishing Configuration resource identifiers. The array shall be empty if no Content Publishing Configuration resources currently exist in the collection.

4.5.5.3 Create Content Publishing Configuration

This procedure is used by the 5GMSu AF to create a new Content Publishing Configuration resource in the target 5GMSu AS instance. The HTTP `POST` method shall be used for this purpose. The 5GMSu AF shall nominate the resource identifier to be used to identify the new resource in the Content Publishing Configurations collection as part of the request URL. A representation of the resource shall be provided as the request body.

If the operation is successful, the 5GMSu AS shall return an HTTP *201 (Created)* response with an empty response body. The content of the `Location` header may differ from the request URL if the request was satisfied by a different 5GMSu AS instance.

If a Content Publishing Configuration resource already exists at the request URL an HTTP *409 (Conflict)* error response shall be returned by the 5GMSu AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

If any resources referenced by the supplied Content Publishing Configuration resource representation are invalid, the create operation shall fail with an HTTP response status code of *400 (Bad Request)*. The response body shall be a

ProblemDetails document as specified in clause 5.2.4.1 of TS 29.571 [12]. In this case, the Content Publishing Configuration resource shall remain in an uncreated state in the target 5GMSu AS instance.

If a Content Publishing Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSu AS. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.5.4 Retrieve Content Publishing Configuration

Retrieval of individual Content Publishing Configuration resources is not permitted at reference point M3.

4.5.5.5 Update Content Publishing Configuration

This procedure is used by the 5GMSu AF to replace a Content Publishing Configuration resource in the target 5GMSu AS instance. The HTTP `PUT` method shall be used for this purpose. The target resource shall be indicated in the request URL. The replacement resource representation shall be provided as the request body.

If the operation is successful, the 5GMSu AS shall return an HTTP *200 (OK)* response with an empty response body.

If the replacement Content Publishing Configuration is identical to the current resource representation the 5GMSu AS shall return an HTTP *204 (No Content)* response with an empty response body to indicate a "no-op".

If any resources referenced by the replacement Content Publishing Configuration resource representation are invalid, the update operation shall fail with an HTTP response status code of *400 (Bad Request).* The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12]. In this case, the target Content Publishing Configuration resource shall remain in the state immediately prior to the update operation.

If a Content Publishing Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSu AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMSu AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.5.6 Destroy Content Publishing Configuration

This procedure is used by the 5GMSu AF to destroy a Content Publishing Configuration resource in the target 5GMSu AS instance. The HTTP `DELETE` method shall be used for this purpose. The target resource shall be indicated in the request URL. The request body shall be empty.

If the operation is successful, the 5GMSu AS shall return an HTTP *204 (No Content)* response with an empty response body. Any subsequent attempt to recreate a Content Publishing Configuration resource with the same resource identifier shall fail as specified in clause 4.5.3.3.

If a Content Publishing Configuration resource previously existed at the request URL and was destroyed using this procedure an HTTP *410 (Gone)* error response should be returned by the 5GMSu AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMSu AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.5.7 (De)activate Content Publishing Configuration

This procedure is used by the 5GMSu AF to activate or deactivate a Content Publishing Configuration resource in the target 5GMSu AS instance. The HTTP `POST` method shall be used for this purpose. The target resource shall be identified in the request URL. The request body shall be a single JSON *boolean* indicating the desired state of the target resource: *true* for active and *false* for inactive.

If the Content Publishing Configuration can be (de)activated immediately by the 5GMSu AS, or if it is already in the desired state indicated in the request, the HTTP *204 (No Content)* response shall be returned by the 5GMSu AS with an empty response body. If the activation request is otherwise acceptable to the 5GMSu AS, but the Content Publishing Configuration cannot be (de)activated immediately, the HTTP *202 (Accepted)* response shall be returned by the

5GMSu AS with an empty response body to indicate that (de)activation is in progress. The state of the Content Publishing Configuration may then be polled using the procedure specified in clause 4.5.4.8.

If a Content Publishing Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSu AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMSu AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.5.8 Interrogate Content Publishing Configuration state

This procedure is used by the 5GMSu AF to interrogate the current state of a Content Publishing Configuration resource in the target 5GMSu AS instance. The HTTP `GET` method shall be used for this purpose. The target resource shall be indicated in the request URL. The request body shall be empty.

If the Content Publishing Configuration is known to the 5GMSu AS, the HTTP *200 (OK)* response shall be returned with an indication of its current state provided in the response body as a JSON object of type *boolean* with the value *true* for active and *false* for inactive.

If a Content Publishing Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMS AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMS AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.5.9 Purge Content Publishing cache

This procedure is used by the 5GMSu AF to purge one or more content items from the 5GMSu AS cache associated with a Content Publishing Configuration resource in the target 5GMSu AS instance. The HTTP `POST` method shall be used for this purpose. A Content Publishing Configuration resource shall be indicated in the request URL. The request body shall be a single key–value pair encoded per the *application/x-www-form-urlencoded* MIME content type. The key shall be the string *pattern* and the value shall be a regular expression [5] identifying the media resource URL(s) to be purged from the associated content cache.

If the operation is successful, the 5GMSu AS shall return an HTTP *200 (OK)* response. The resource body should indicate the total number of cache entries purged encoded as a positive JSON integer. If the purge request is otherwise acceptable to the 5GMSu AS, but no cache entries were purged because none matched the supplied regular expression, the HTTP *204 (No Content)* response shall instead be returned by the 5GMSu AS with an empty response body.

If the request message body – or the regular expression contained in it – are found by the 5GMSu AS to be syntactically malformed the HTTP *400 (Bad Request)* error response shall be returned. The response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

If a Content Publishing Configuration resource previously existed at the request URL and was destroyed an HTTP *410 (Gone)* error response should be returned by the 5GMSu AS. Otherwise, if no resource exists at the request URL an HTTP *404 (Not Found)* error response shall be returned by the 5GMSu AS. In either case the response body shall be a *ProblemDetails* document as specified in clause 5.2.4.1 of TS 29.571 [12].

4.5.6 Service Access Information procedures

Service Access Information includes configuration information to allow the 5GMSd AS to invoke at reference point M3d procedures for dynamic policy (see clause 4.5.7), consumption reporting (clause 4.5.8), metrics reporting (clause 4.5.9) and network assistance (clause 4.5.10). For any of these purposes, the 5GMSd AS shall obtain Service Access Information from the 5GMSd AF using the operations specified in clause 5.3.2 of TS 26.510 [56] at reference point M3d, citing an external service identifier and the Service Access Information is derived by the 5GMSd AF from the Provisioning Session established at reference point M1d (see clause 4.3.2) that is tagged with the same external service identifier.

In this version of the present document, the properties of the Service Access Information resource shall be populated as follows at reference point M3 based on the properties and sub-resources of the corresponding Provisioning Session:

- The *provisioningSessionId* and *provisioningSessionType* properties are present, as required by TS 26.510 [56].
- The *locationReporting* property is present, as required by TS 26.510 [56], and shall be set to *false*.

- The *streamingAccess* object shall be present if provisioned in the Content Hosting Configuration or Content Publishing Configuration resource.

- If any of the metrics reporting configurations listed in table 7.8.1-1 are provisioned per clause 4.3.9, the *clientMetricsReportingConfigurations* array shall be present and populated with an object corresponding to each metrics reporting configuration. Metrics Reporting Configurations for any other metrics schemes shall not be included in the *clientMetricsReportingConfigurations* array.

All other properties of the Service Access Information resource shall be omitted when served at reference point M3d.

4.5.7 Dynamic policy invocation procedures

Dynamic policy invocation at reference point M3d is not supported by the 5GMS System in this release.

4.5.8 Consumption reporting procedures

Consumption reporting at reference point M3d is not supported by the 5GMS System in this release.

4.5.9 Metrics reporting procedures

These procedures are used by the metrics reporting function of the 5GMSd AS to submit QoE metrics reports (including CMCD reporting) to the 5GMSd AF via reference point M3d if metrics reporting is applied for a media streaming session. To determine whether and how to send metrics reports to the 5GMSd AF at reference point M3d, the 5GMSd AS shall use the procedures and operations specified in clause 5.3.5 of TS 26.510 [56].

- If any of the CMCD metrics schemes specified in table 7.8.1-1 are indicated in a client metrics reporting configuration of the Service Access Information provided to the 5GMSd AS at reference point M3d (see clause 4.5.6), the 5GMSd AS shall report CMCD information to the 5GMSd AF at reference point M3d using the API specified in clause 11.4.1 and the report format specified in clause 11.4.3.3.

4.5.10 Network Assistance procedures

Network Assistance is not supported by the 5GMS System at reference point M3 in this release.

4.6 Procedures of the M4d (Media Streaming) interface

4.6.0 General

4.6.0.1 ECN marking for L4S

When a Dynamic Policy Instance (see clause 4.7.3) enabling the ECN marking for L4S functionality is successfully activated for a downlink media streaming session, the 5GMSd AS is required to support L4S according to RFC 9330 [67], RFC 9331 [68] and RFC 9332 [69] at reference point M4d.

4.6.1 Procedures for DASH session

This procedure is used by a Media Player to establish a DASH session with the 5GMSd AS via reference point M4d. In order to establish such a session, the 5GMSd AS shall host an MPD as defined in ISO/IEC 23009-1 [32] or TS 26.247 [4] and the MPD URL is known to the 5GMSd Client typically using M8d.

To initiate a DASH session using downlink media streaming, the Media Player receives the URL of a Media Player Entry (MPD) resource from the 5GMSd-Aware Application through M7d by methods defined in clause 13. The Media Player shall send an HTTP `GET` message to the 5GMSd AS including the URL of the Media Player Entry (MPD) resource. On success, the 5GMSd AS shall respond with a *200 (OK)* message that includes the requested Media Player Entry (MPD) resource.

Additional procedures for reactions to different HTTP status codes are provided in TS 26.247 [4], clause A.7 and ISO/IEC 23009-1 [32] clause A.7.

Additional procedures for handling partial file responses are provided in TS 26.247 [4], clause A.9.

This information is provided through M7d to the application for selection. In addition, the currently used service description parameters are provided as status information through M7d in order for the Media Session Handler to make use of this information, for example for Dynamic Policy and Network Assistance.

The detailed handling of service description information is documented in clause 13.2 of the present document.

For the delivery of DASH content, the delivery protocol in use to connect to a reference point M4d service location is assumed to be HTTP/1.1 [21], HTTP/2 [34] or HTTP/3 [60]. The Media Player shall support HTTP/1.1 and may support HTTP/2 and HTTP/3.

The Media Player may use a multipath transport protocol supported by the UE as specified in clause 6.2.1.2 to connect to a reference point M4d service location on the 5GMSd AS. The different multipath transport connection subflows/paths may span one or more access networks available to the UE. The different subflows/paths may be used to increase the resilience of the transport connection to packet loss or to increase data throughput.

4.6.2 Procedures for Progressive Download Session

This procedure is used by a Media Player to establish a Progressive Download session with the 5GMSd AS at reference point M4d. In order to establish such a session, the 5GMSd AS shall host an 3GP/MP4 file as defined in TS 26.247 [4]. The 3GP/MP4 URL is known to the Media Player (in this case a progressive download player), typically by using M8d.

To initiate progressive download using downlink media streaming, the Media Player receives the URL of a Media Player Entry (3GP/MP4) resource from the 5GMSd-Aware Application through reference point M7d by methods defined in clause 13. The Media Player shall send an HTTP `GET` message to the 5GMSd AS including the URL of the Media Player Entry (3GP/MP4) resource. On success, the 5GMSd AS shall respond with a *200 (OK)* message that includes the requested Media Player Entry (3GP/MP4) resource.

Additional procedures for reactions to different HTTP status codes are provided in TS 26.247 [4].

For the progressive download of content, the delivery protocol in use to connect to a reference point M4d service location is assumed to be HTTP/1.1 [21], HTTP/2 [34] or HTTP/3 [60]. The Media Player shall support HTTP/1.1 and may support HTTP/2 and HTTP/3.

The Media Player may use a multipath transport protocol supported by the UE as specified in clause 6.2.1.2 to connect to a reference point M4d service location on the 5GMSd AS. The different multipath transport connection subflows/paths may span one or more access networks available to the UE. The different subflows/paths may be used to increase the resilience of the transport connection to packet loss or to increase data throughput.

4.6.3 Procedures for using multiple service locations

These procedures may be used to augment the procedures described in clauses 4.6.1 and 4.6.2 above to allow for media resources to be obtained from multiple service locations exposed by the 5GMSd AS at reference point M4d.

Information required by the 5GMSd Client to access media from multiple service locations exposed at M4d by the 5GMSd AS is contained within a Media Player Entry document. This information may exist, for example, within:

- An MPD as XML elements or attributes containing the required information.
- A Media Player Entry document containing a pointer (e.g., URL) to an MPD or 3GP/MP4 file.
- A document pointed to by a Media Player Entry.

A 5GMSd Client may use this information to do any or all of the following:

1. Switch between service locations exposed at reference point M4d during a downlink media streaming session, as specified in clause 10.3A.2.
2. Obtain signalling via reference point M4d from a content steering service provided by the 5GMSd AS or 5GMSd Application Provider that can be used to influence the choice of one service location over another, as specified in clause 10.3A.4.

3. Access media resources from multiple service locations simultaneously, for example using multi-source object coding, as specified in clause 10.3A.3.

4.7 Procedures of the M5 (Media Session Handling) interface

4.7.1 Introduction

The procedures at reference point M5 are used by a Media Session Handler within a 5GMS Client to invoke services relating to downlink or uplink media streaming on the 5GMS AF.

4.7.2 Procedures for Service Access Information

4.7.2.1 General

Service Access Information is the set of parameters and addresses needed by the 5GMSd Client to activate reception of a downlink media streaming session or by a 5GMSu Client to activate an uplink media streaming session for contribution. Service Access Information additionally includes configuration information to allow the Media Session Handler to invoke procedures for dynamic policy (see clause 4.7.3), consumption reporting (clause 4.7.4), metrics reporting (clause 4.7.5) and network assistance (clause 4.7.6).

The Media Session Handler may obtain Service Access Information from either the 5GMS-Aware Application (via reference point M6) or from the 5GMS AF (via reference point M5). In the former case, the Service Access Information is initially acquired by the 5GMS-Aware Application from the 5GMS Application Provider via reference point M8. In the latter case, the Media Session Handler shall use the operations specified in clause 5.3.2 of TS 26.510 [56] at reference point M5 to acquire Service Access Information from the 5GMS AF, citing an external service identifier and the Service Access Information is derived by the 5GMS AF from the Provisioning Session established at reference point M1 (see clause 4.3.2) that is tagged with the same external service identifier.

Typically, the Service Access Information for media streaming includes a set of *Media Entry Points*. Examples include:

- A URL to a DASH MPD,
- A URL to a document that provides additional details for different streaming session configurations and/or that references or includes equivalent media presentations such as a DASH MPD or
- A URL to a progressive download file that can be consumed by the Media Stream Handler (Media Player or Media Streamer).

Based on the MIME media type or protocol, as well as the conformance profiles declared in the Service Access Information, one of these Media Entry Points is selected by the Media Session Handler or by the 5GMS-Aware Application and is handed to the Media Player via reference point M11 or M7 respectively.

NOTE: The Media Session Handler and 5GMS-Aware Application are assumed to have prior knowledge of the types of Media Entry Point supported by the Media Player.

For downlink media streaming exclusively via eMBMS and for hybrid 5GMSd/eMBMS services as defined in clauses 5.10.2 and 5.10.5 respectively of TS 26.501 [2], the Service Access Information indicates that the 5GMSd Client acts as an MBMS-Aware Application.

For dynamically provisioned downlink media streaming via eMBMS as defined in clause 5.10.6 of TS 26.501 [2], the 5GMSd AS creates a presentation manifest that is regularly polled by the Media Player for a potential update. When an eMBMS User Service carrying the 5GMSd content is dynamically provisioned or removed by the 5GMSd AF, the 5GMSd AS shall update the presentation manifest with the locations where the updated manifest and the media segments are now available, for example to add or change to the media server in the MBMS Client.

For downlink media streaming exclusively via MBS and for hybrid 5GMSd/MBS services as defined in clauses 5.12.2 and 5.12.4 respectively of TS 26.501 [2], the Service Access Information indicates that the 5GMSd Client acts as an MBS-Aware Application.

For dynamically provisioned downlink media streaming via MBS as defined in clause 5.12.4 of TS 26.501 [2], the 5GMSd AS creates or hosts a presentation manifest that is regularly polled by the Media Player for a potential update. When an MBS User Service carrying the 5GMSd content is dynamically provisioned or removed by the 5GMSd AF, the 5GMSd AS shall update the presentation manifest with the resource locations where the updated manifest and the media segments are now available, for example to additionally or alternatively point to the Media Server in the MBSTF Client.

If one or more Metrics Reporting Configurations are provisioned per clause 4.3.9, a subset of these is included in the corresponding Service Access Information provided to the Media Session Handler at reference point M5d as follows:

- If the metrics scheme *urn:3GPP:ns:PSS:DASH:QM10* is indicated in the metrics reporting configuration of a Provisioning Session for downlink media streaming, a corresponding client metrics reporting configuration shall be included in the Service Access Information provided to the Media Session Handler at reference point M5d to enable QoE metrics reporting for DASH-based downlink media streaming by the 5GMSd Client at reference point M5d per clause 11.4.3.2.

- If the metrics scheme specified in clause 9.3 of TS 26.118 [42] is indicated in the metrics reporting configuration of a Provisioning Session for downlink media streaming, a corresponding client metrics reporting configuration shall be included in the Service Access Information provided to the Media Session Handler at reference point M5d to enable QoE metrics reporting for Virtual Reality media by the 5GMSd Client at reference point M5d per clause 11.4.3.2.

- If any of the CMCD metrics schemes listed in table 7.8.1-1 are indicated in the metrics reporting configuration of a Provisioning Session for downlink media streaming, a corresponding client metrics reporting configuration shall be included in the Service Access Information provided to the Media Session Handler at reference point M5d to enable metrics reporting of CMCD information by the 5GMSd Client to the 5GMSd AS at reference point M4d per [70].

If an Edge Resources Configuration with client-driven management is provisioned per clause 4.3.10, a Client Edge Resources Configuration is included in the corresponding Service Access Information.

4.7.2.2 Void

4.7.2.3 Void

4.7.2.4 Void

4.7.2.5 Void

4.7.3 Procedures for dynamic policy invocation

These procedures are used by a Media Session Handler to manage Dynamic Policy Instance resources in the 5GMS AF. To do this, the Media Session Handler shall use the operations specified in clause 5.3.3 of TS 26.510 [56] at reference point M5 to instantiate Policy Templates in the 5GMS AF that are described in the Dynamic Policy Configuration provided in the Service Access Information (see clause 4.7.2).

The choice of Policy Template by the Media Session Handler shall be influenced by the following considerations:

- When instantiating a Policy Template whose Policy Template Binding indicates *l4SEnablementPreference* set to *true*, the Media Session Handler shall set the *l4SRequired* flag to *true* in the Dynamic Policy request only if an L4S-capable media transport stack is present and in use. This is the case if the Media Session Handler detects that an L4S-capable media transport stack is present and in use by the Media Player or if the Media Player configuration and settings API (see clause 13.2.4) declares *CAPABILITY_L4S.*

- When instantiating a Policy Template whose Policy Template Binding indicates *qoSMonitoringEnablement Preference* set to *true*, the Media Session Handler shall set the *qoSMonitoringRequired* flag to *true* in the Dynamic Policy request only if the Media Player is capable of consuming QoS monitoring results.

  The Media Session Handler shall interrogate the capabilities of the Media Player in order to discover whether it is capable of consuming QoS monitoring results and shall select a Policy Template that indicates the preference for QoS monitoring functionality only if the Media Player has this capability.

4.7.4 Procedures for consumption reporting

These procedures are used by the Media Session Handler and the Consumption Reporting function of the 5GMSd Client to submit a consumption report to the 5GMSd AF via reference point M5d if consumption reporting is applied for a downlink media streaming session. This is indicated by the presence of a Client Consumption Reporting Configuration in the Service Access Information. To determine whether and how to send consumption reports to the 5GMSd AF at reference point M5, the Media Session Handler shall use the procedures and operations specified in clause 5.3.6 of TS 26.510 [56].

4.7.5 Procedures for metrics reporting

These procedures are used by the Media Session Handler and the Metrics Reporting function of the 5GMS Client to submit QoE metrics reports to the 5GMS AF via reference point M5 if metrics reporting is applied for a media streaming session. To determine whether and how to send metrics reports the 5GMS AF at reference point M5, the Media Session Handler shall use the procedures and operations specified in clause 5.3.5 of TS 26.510 [56].

When the metrics collection and reporting feature is provisioned for a media streaming Provisioning Session, one or more Client Metrics Reporting Configurations, each associated with a metrics scheme, are provided to the 5GMS Client in the Service Access Information (see clause 4.7.2).

- For progressive download and DASH streaming services, the listed metrics in a given Client Metrics Reporting Configuration are associated with the 3GPP metrics scheme and shall correspond to one or more of the metrics as specified in clauses 10.3 and 10.4, respectively, of TS 26.247 [4]. When individual metrics or groups of metrics are to be reported, these are identified using controlled terms from the vocabulary specified in clause E.2.1.

- For downlink DASH streaming services, CMCD information within the scope of one of the CMCD metrics schemes listed in table 7.8.1-1 may be listed in a Metrics Reporting Configuration. When individual metrics or groups of metrics are to be reported as in-band client data, these are identified using controlled terms from the vocabulary specified in clause E.2.5. As a result, the Media Player is configured by the Media Session Handler at reference point M11d to report the corresponding CMCD information to the 5GMSd AS at reference point M4d using either HTTP request headers or URL query parameters, as specified in [70].

- Metrics relating to Virtual Reality media, as specified in clause 9.3 of TS 26.118 [42], may be listed in a Client Metrics Reporting Configuration. When individual metrics or groups of metrics are to be reported, these are identified using controlled terms from the vocabulary specified in clause E.2.2.

- Metrics relating to eMBMS delivery, as specified in clause 9.4.6 of TS 26.346 [51], may be listed in a Client Metrics Reporting Configuration.

NOTE: Metrics reporting for MBS is not specified by TS 26.517 [64] in this release.

4.7.6 Procedures for network assistance

These procedures are used by the 5GMS Client to request Network Assistance from one of the 5GMS AF instances listed in the Network Assistance Configuration of the Service Access Information. To do this, the Media Session Handler shall use the procedures and operations specified in clause 5.3.4 of TS 26.510 [56] at reference point M5.

4.8 Procedures of the M6d (UE Media Session Handling) interface

4.8.1 General

This clause contains the procedures for the interaction between the 5GMSd-Aware application or the Media Player and the Media Session Handler through the M6d API. Details are provided in clause 12.

4.8.2 Consumption reporting procedures

When consumption reporting for this session is active for a given media delivery session, the Media Session Handler and Media Streamer (downlink Media Player or uplink Media Streamer) shall follow the procedures specified in clause 5.4.6 of TS 26.510 [56] to support consumption reporting.

4.8.3 3GPP Service URL handling procedures

4.8.3.1 Launch of 5G Media Streaming session

The Media Session Handler shall play the role of 3GPP Service Handler for Service URLs as defined in clause 6 of TS 26.510 [56] by registering itself as the URL handler for the domain name *launch.3gppservices.org* and the initial path element */ms* (e.g., by declaring an intent filter in the application manifest). Hence, the Media Session Handler is launched when a 5GMS-Aware Application requests a 5GMS Service URL (e.g., by means of an intent filter).

As a result of being invoked in this way, the Media Session Handler shall initiate a new media delivery session according to clause 5.4.2.2 of TS 26.510 [56].

4.8.3.2 Retrieval of Service Access Information from 5GMS AF

If it needs to retrieve whole Service Access Information from the 5GMS AF (because a full set of Service Access Information has not been supplied as additional parameters of the 3GPP Service URL) the Media Session Handler shall decompose the 3GPP Service URL into the prefix and suffix, and shall form the M5 request URL for Service Access Information as specified in clause 9.2.2 of TS 26.510 [56].

1. If the 3GPP Service URL carries one or more *af-host-address* query parameters, the Media Session Handler shall choose one to substitute into *{apiRoot}* in the above request URL. If the port number is omitted from any *af-host-address*, port 80 (for HTTP) or 443 (for HTTPS) shall be assumed.

NOTE 1: This corresponds to collaboration scenarios where the 5GMS AF is deployed in the External DN.

The party operating the 5GMS AF is responsible for ensuring that the hostname(s) resolve to the correct IP address(es) in the External DN.

If more than one *af-host-address* query parameter is supplied in the 3GPP Service URL, the Media Session Handler may use an alternative host endpoint address at reference point M5 if the one it is using fails to respond after some implementation-specific number of retries.

2. If the *af-host-address* query parameter is omitted from the 3GPP Service URL, the default host name *ms.af.3gppservices.org* and port number 80 (HTTP) or 443 (HTTPS) shall be used instead.

NOTE 2 This corresponds to collaboration scenarios where the 5GMS AF is deployed in the Trusted DN.

The 5G System operator is responsible for supporting resolution of this well-known host name to the correct IP address(es) in the Trusted DN, e.g., by managing appropriate DNS records.

The 5G System operator is responsible for ensuring that a resilient service is available at this host endpoint address. If the hostname resolves to multiple IP addresses, the Media Session Handler may use a different one at reference point M5 if the one it is using fails to respond after some implementation-specific number of retries.

NOTE 3 It is recognised that correct resolution of the hostname may be hampered if the end user configures an alternative DNS resolution service. Unless the Media Session Handler is able to override this and use the 5G System DNS resolution service, this is considered a failure case for 5G Media Streaming session initiation that is reportable to the end user.

4.8.3.3 Processing of 5GMS-specific parameters in 3GPP Service URL

Once the Media Session Handler is in possession of whole Service Access Information, it shall process the 5GMS-specific parameters present in the 3GPP Service URL as follows. URL query parameters not specified in clause 12.4 may be ignored by the Media Session Handler.

1. If the 3GPP Service URL includes the optional *media-entry-point* query parameter, the Media Session Handler shall invoke the Media Stream Handler with the URL of the Media Entry Point conveyed in the value of this parameter.

2. Otherwise, if the 3GPP Service URL omits the optional *media-entry-point* query parameter, the Media Session Handler shall instead select one of the Media Entry Points listed in the *streamingAccess.entryPoints* array of the Service Access Information (see clause 9.2.3.1 of TS 26.510 [56]). In making its selection, the Media Session Handler can take into account the capabilities of available Media Stream Handler(s) as well as the requesting application's preferences for particular content type(s) and/or media profile(s) expressed respectively in (possibly multiple instances of) the *content-type* and *profile* query parameters if they are present in the 3GPP Service URL.

3. If the 3GPP Service URL includes the optional *service-operation-point* query parameter, the Media Session Handler shall, before instantiating a Media Stream Handler, attempt to instantiate a Dynamic Policy for the Policy Template binding listed in the whole Service Access Information that carries an *externalReference* with the same value as this query parameter (see *policyTemplateBindings* property in clause 9.2.3.1 of TS 26.510 [56]) using the procedures specified in clause 4.7.3. If no matching Policy Template binding can be identified by the Media Session Handler no Media Stream Handler shall be instantiated.

3. If the 3GPP Service URL includes the optional *estimated-volume* query parameter, the *service-operation-point* query parameter shall also be present. The Media Session Handler shall, before instantiating a Media Stream Handler, attempt to instantiate a Dynamic Policy for the Policy Template binding listed in the whole Service Access Information that carries an *externalReference* with the same value as the *service-operation-point* query parameter (see *policyTemplateBindings* property in clause 9.2.3.1 of TS 26.510 [56]) using the procedures specified in clause 4.7.3. If no matching Policy Template binding can be identified by the Media Session Handler, or if the matching Policy Template binding does not advertise any upcoming Background Data Transfer windows in its *bdtWindows* array no Media Stream Handler shall be instantiated.

4.8.3.4 Responding to the requesting application

Depending on the success (or otherwise) of retrieving the Service Access Information (and optionally chaining the Media Stream Handler with the Media Entry Point), the Media Session Handler should return an appropriate HTTP status code to the application that requested the 3GPP Service URL.

4.9 Procedures of the M7d/M11d (UE Media Player) interface

4.9.1 General

This clause contains the procedures for the interaction between the 5GMSd-Aware Application or the Media Session Handler and the Media Player through the M7d API. Details are provided in clause 13.

4.9.1A Procedures for configuring downlink content delivery

The 5GMSd-Aware Application and the Media Session Handler may configure content delivery for downlink media streaming between the 5GMSd AS and the Media Player using the client API at reference points M7d and M11d respectively by populating the *desiredContentDeliveryConfiguration* object in the Media Player configuration as specified in clause 13.2.4.

- A list of content delivery protocols (see table 13.2.4-3) to be used at reference point M4d may be provided in order of preference.
- A preference for using a multipath transport protocol at reference point M4d may be expressed.

The Media Player should attempt to satisfy the desired content delivery configuration within the limits of its capabilities and those of the underlying UE.

In the absence of a desired content delivery configuration, the Media Player may select any content delivery protocol that meets the requirement of the provided Media Player Entry, and may select a multipath transport protocol compatible with its choice of content delivery protocol.

The Media Player should inform the 5GMSd-Aware Application and the Media Session Handler using reference point M7d and M11d notifications respectively about changes to the status of reference point M4d transport connections.

4.9.2 Metrics reporting procedures

These procedures shall be used by the Media Session Handler function to control metrics reporting when such reporting is configured via metadata sent in-band via the media manifest.

When a streaming session is started, the Media Session Handler shall check if the manifest contains any client metrics configuration, as specified in TS 26.247 [4] clauses 10.4 and 10.5, or TS 26.118 [42] clause 9.3. If such a configuration is found, the Media Session Handler shall use it for the current streaming session.

The Media Session Handler shall first determine whether metrics from this session shall be reported. The determination shall be based on the *samplePercentage* attribute specified in the metrics configuration, according to TS 26.247 [4] clause 10.5.

If metrics are reported for the session, the Media Session Handler shall request the Media Player to create a metrics collection job. The Media Player shall return a reference to the created job, which the Media Session Handler shall use in all subsequent actions related to this job.

The Media Session Handler shall configure the metrics collector job with the set of metrics which shall be collected during the session. The format of the configuration shall be according to TS 26.247 [4] clause L.2, but note that only the *metrics* attribute in the configuration shall be used for this purpose.

If one of the CMCD metrics schemes listed in table 7.8.1-1 is indicated in the client metrics configuration, the Media Session Handler shall set a transmission mode in the metrics collector job to enable and configure in-band reporting of CMCD information [70] at reference point M4d. The default transmission mode in this version of the present document is to report using HTTP request headers. The 5GMS-Aware Application may override this configuration.

The Media Session Handler shall regularly request the collected metrics from the Media Player according to the *reportingInterval* specified in the metrics configuration. The metrics returned by the Media Player shall use the format as described in TS 26.247 [4] clause 10.6, and (for virtual reality media) in TS 26.118 [42] clause 9.4 and the Media Session Handler shall forward these to the server address(es) specified in the metrics configuration using the specified *DNN* according to the procedures described in TS 26.247 [4] clause 10.6.

When the session is finished the Media Session Handler shall delete the metrics collection job.

4.9.3 Dynamic Policy procedures

These procedures are used between a Media Player and a Media Session Handler at reference point M11.

If the Media Player is capable of consuming QoS monitoring results, it shall subscribe to receive QoS monitoring results notifications from the Media Session Handler at reference point M11 as specified in clause 5.4.3 of TS 26.510 [56].

4.10 Procedures of the M8d interface

This clause defines basic procedures for M8d.

No specific procedures are defined but it is expected that the 5GMSd Application Provider can provide media session entry points to a 5GMSd-Aware Application through M8d. The 5GMSd-Aware Application would then initiate the media session by providing such an entry point to the 5GMSd Client through M7d. Multiple alternative entry points of the same media streaming service may be provided.

4.10A Procedures of the M10d interface

The procedures at reference point M10d are used by a 5GMSd AS to ingest content from a 5GMSd AS in another 5GMS System. The procedures at this reference point are referred to as *service chaining*.

The following 5GMS AS service chaining protocols specified by the present document may be used at reference point M10d to support downlink media streaming:

- An *HTTP pull-based content ingest protocol* is specified in clause 8.2, including specific handling for HTTP redirects issued to the 5GMS AS by the 5GMS Application Provider's origin server.
- A *DASH-IF push-based content ingest protocol* is specified in clause 8.3.
- A *HTTP low-latency pull-based content ingest protocol* is specified in clause 8.4.

4.10B Procedures of the M13d interface

No specific procedures are defined at reference point M13d, but it is expected that the Media Player follows similar procedures when interacting with the 5GMSd Application Provider as those defined between the Media Player and the 5GMSd AS at reference point M4d, as outlined in clause 4.6.

4.11 Data collection and reporting procedures at reference point R4

4.11.1 General

These procedures are used by the 5GMS AS, as a type of data collection client, to acquire its data collection and reporting client configuration from, and subsequently report media streaming access activity to, the Data Collection AF instantiated in the 5GMS AF. It does so by invoking the *Ndcaf_DataReporting* service offered by the Data Collection AF at reference point R4, as specified in clause 7 of TS 26.532 [49].

4.11.2 5GMS AS data collection and reporting client configuration

The 5GMS AS shall use the service operations and procedures specified in TS 26.532 [49] to obtain its data collection and reporting client configuration from the Data Collection AF instantiated in the 5GMS AF at reference point R4:

- In the case where both the 5GMS AS and the 5GMS AF reside in the trusted domain, the 5GMS AS shall obtain its data collection client configuration directly from the Data Collection AF by invoking appropriate *Ndcaf_DataReporting* service operations as specified in clause 7.2 of [49] according to the procedures specified in clauses 4.2.5 of [49].

- Should the 5GMS AS and the 5GMS AF reside in different trust domains, the 5GMS AS shall instead obtain its configuration from the Data Collection AF indirectly via the NEF by invoking the equivalent operations on the *Nnef_DataReporting* service as defined in TS 26.531 [48].

In both cases, the 5GMS AS shall declare that it supports the *MS_ACCESS_ACTIVITY* data reporting domain in *DataReportingSession.supportedDomains* (see clauses 7.2.3.2.1 and 7.2.3.3.1 of TS 26.532 [49]) and the Data Collection AF instantiated in the 5GMS AF shall request reporting for this domain by including the same value as a key to the *DataReportingSession.reportingConditions* dictionary. The value of the *ReportingCondition.type* property at this key shall be *INTERVAL*, *THRESHOLD* or *OFF*.

No data packaging strategies are specified in this release for the data reporting domains associated with the 5GMS AS at reference point R4.

Default data packaging strategies for the data reporting domains associated with the 5GMS AS at reference point R4 are not specified in this release.

The 5GMS AS shall refresh its data collection and reporting client configuration according to the procedures specified in clause 4.2.5.3 of TS 26.532 [49].

4.11.3 5GMS AS data reporting

The 5GMS AS shall use the service operations and procedures specified in TS 26.532 [49] to report media streaming access activity to the Data Collection AF instantiated in the 5GMS AF at reference point R4:

- In the case where both the 5GMS AS and the 5GMS AF reside in the trusted domain, the 5GMS AS shall submit media streaming access activity reports directly to the Data Collection AF by invoking appropriate *Ndcaf_DataReporting* service operations as specified in clause 7.3 of [49] according to the procedures specified in clauses 4.2.7 of [49].

- Should the 5GMS AS and the 5GMS AF reside in different trust domains, the 5GMS AS shall instead submit media streaming access activity reports indirectly to the 5GMS AF via the NEF by invoking the equivalent operations on the *Nnef_DataReporting* service as defined in TS 26.531 [48].

In both cases, the conditions for sending a downlink media streaming access activity report by the 5GMSd AS and the parameters of such report are signalled in the data collection and reporting client configuration obtained using the procedure specified in clause 4.11.2.

In order to submit a media streaming access activity report, the 5GMS AS shall send an HTTP `POST` message to the Data Collection AF instantiated in the 5GMS AF. The request body shall be a JSON document of type *DataReport* (as defined in clause 7.3.3.2.1 of TS 26.532 [49]) containing one or more *MediaStreamingAccessRecord* structures, as specified in clauses 17.2 and C.5.1 of the present document.

The Data Collection AF shall respond with a *200 (OK)* message to acknowledge successful processing of the media streaming access activity report.

4.11A Data collection and reporting procedures at reference point R2

4.11A.1 General

These procedures are used by the Direct Data Collection Client instantiated in the Media Session Handler to acquire its data collection and reporting client configuration from, and subsequently report media streaming access activity to, the Data Collection AF instantiated in the 5GMS AF. It does so by invoking the *Ndcaf_DataReporting* service offered by the Data Collection AF at reference point R2, as specified in clause 7 of TS 26.532 [49].

4.11A.2 Data collection and reporting client configuration

The Direct Data Collection Client instantiated in the Media Session Handler shall use the service operations and procedures specified in TS 26.532 [49] to obtain its data collection and reporting client configuration from the Data

Collection AF instantiated in the 5GMS AF at reference point R2 by invoking appropriate *Ndcaf_DataReporting* service operations as specified in clause 7.2 of [49] according to the procedures specified in clauses 4.3.2 of [49].

The Direct Data Collection Client instantiated in the Media Session Handler shall declare that it supports the *MS_ANBR_NETWORK_ASSISTANCE* data reporting domain in *DataReportingSession.supportedDomains* (see clauses 7.3.2.1 and 7.3.3.1 of TS 26.532 [49]) and the Data Collection AF instantiated in the 5GMS AF shall request reporting for this domain by including the same value as a key to the *DataReportingSession.reportingConditions* dictionary. The value of the *ReportingCondition.type* property at this key shall be *EVENT* or *OFF*.

No data packaging strategies are specified in this release for the data reporting domains associated with the Direct Data Collection Client instantiated in the Media Session Handler at reference point R2.

Default data packaging strategies for the data reporting domains associated with the Direct Data Collection Client instantiated in the Media Session Handler at reference point R2 are not specified in this release.

The Direct Data Collection Client instantiated in the Media Session Handler shall refresh its data collection and reporting client configuration according to the procedures specified in clause 4.3.2.3 of TS 26.532 [49].

4.11A.3 ANBR-based Network Assistance invocation reporting

The Direct Data Collection Client instantiated in the Media Session Handler shall use the service operations and procedures specified in TS 26.532 [49] to report ANBR-based Network Assistance invocations to the Data Collection AF instantiated in the 5GMS AF at reference point R2 by invoking appropriate *Ndcaf_DataReporting* service operations as specified in clause 7.3 of [49] according to the procedures specified in clauses 4.3.3 of [49].

The conditions for sending an ANBR-based Network Assistance invocation report and the parameters of such report are signalled in the data collection and reporting client configuration obtained using the procedure specified in clause 4.11A.2.

In order to submit an ANBR-based Network Assistance invocation report, the Direct Data Collection Client instantiated in the Media Session Handler shall send an HTTP `POST` message to the Data Collection AF instantiated in the 5GMS AF. The request body shall be a JSON document of type *DataReport* (as defined in clause 7.3 of TS 26.532 [49]) containing a *ANBRNetworkAssistanceInvocationRecord* structure, as specified in clauses 17A.2 and C.5.2 of the present document.

The Data Collection AF shall respond with a *200 (OK)* message to acknowledge successful processing of the ANBR-based Network Assistance invocation report.

4.12 Event Exposure procedures at reference points R5 and R6

4.12.1 General

As specified in clauses 4.7.1 and 4.7.4 of TS 26.501 [2], Events relating to 5G Media Streaming are exposed to Event consumers at reference points R5 and R6 by the Data Collection AF instantiated in the 5GMS AF (playing the role of Event service provider). Procedures for event exposure are specified in clauses 5.11.3 (for downlink media streaming) and 6.8.3 (for uplink media streaming) respectively of [2].

The following *Naf_EventExposure* service operations as defined in TS 23.502 [45] apply for such interactions between the Data Collection AF and consumer entities:

- *Naf_EventExposure_Subscribe*,
- *Naf_EventExposure_Unsubscribe*, and
- *Naf_EventExposure* Notify.

In this release, eligible Event service consumers of 5GMS Event services are the NWDAF and NEF as NFs, and the Event Consumer AF of the 5GMS Application Provider.

4.12.2 Event Exposure subscription procedure

The definition of input and output parameters of the *Naf_EventExposure_Subscribe* service operation is identical to that specified in clause 5.2.19.2.2 of TS 23.502 [45], except that "AF" is replaced by "Data Collection AF" and "consumer NF" (or "NF consumer") is replaced by "Event service consumer".

4.12.3 Event Exposure unsubscription procedure

The definition of input and output parameters of the *Naf_EventExposure_Unsubscribe* service operation is identical to that specified in clause 5.2.19.2.3 of TS 23.502 [45], except that "AF" is replaced by "Data Collection AF" and "consumer NF" (or "NF consumer") is replaced by "Event service consumer".

4.12.4 Event Exposure notification procedure

The definition of input and output parameters of the *Naf_EventExposure_Notify* service operation is identical to that specified in clause 5.2.19.2.4 of TS 23.502 [45], except that "AF" is replaced by "Data Collection AF" and "consumer NF" (or "NF consumer") is replaced by "Event service consumer".

4.13 Procedures for downlink media streaming via eMBMS

This procedure is used by a 5GMSd Client to establish a 5GMSd session either completely, or at least partially, through eMBMS.

- For downlink media streaming exclusively via eMBMS and for hybrid 5GMSd/eMBMS services, as defined in clauses 5.10.2 and 5.10.5 respectively of TS 26.501 [2]:
  - The 5GMSd Application Provider shall provision a supplementary distribution network of type *DISTRIBUTION_NETWORK_EMBMS* in the Content Hosting configuration at reference point M1d, as specified in clause 7.6.3.1, with either *MODE_EXCLUSIVE* or *MODE_HYBRID* (as appropriate).
  - The 5GMSd Application Provider may additionally provision access reporting in the Consumption Reporting Configuration at M1d, as specified in clause 7.7.3.1.
  - The MBMS Client shall host an MPD as defined in ISO/IEC 23009-1 [32] or in TS 26.247 [4], or any other presentation manifest as the 5GMSd Media Entry Point such as an HLS Master Playlist.
  - The URL of this presentation manifest shall be signalled to the 5GMSd Client through the 5GMSd session establishment procedure.
  - The MBMS Client shall be invoked by the Media Session Handler via reference point MBMS-API-C using the procedures defined in TS 26.347 [52].
- For dynamically provisioned downlink media streaming via eMBMS as defined in clause 5.10.6 TS 26.501 [2]:
  - The 5GMSd Application Provider shall provision a supplementary distribution network of type *DISTRIBUTION_NETWORK_EMBMS* in the Content Hosting configuration at reference point M1d, as specified in clause 7.6.3.1, with *MODE_DYNAMIC*.
  - The 5GMSd Application Provider shall additionally provision access reporting in the Consumption Reporting Configuration at M1d, as specified in clause 7.7.3.1.
  - The 5GMSd AS shall host an MPD as defined in ISO/IEC 23009-1 [32] or in TS 26.247 [4], or any other presentation manifest as the 5GMSd Media Entry Point.
  - The URL of this presentation manifest shall be signalled to the 5GMSd Client through the 5GMSd session establishment procedure. If the 5GMSd service is currently available as an MBMS User Service, the 5GMSd Client forwards the manifest request to the MBMS Client; otherwise, it forwards the request to the 5GMSd AS via reference point M4d.

NOTE: The detailed execution of dynamically handling this decision is left to implementation.

- The MBMS Client shall be invoked dynamically, paused or destroyed by the Media Session Handler via reference point MBMS-API-C using the procedures defined in TS 26.347 [52].

Additional procedures for reactions to different HTTP status codes are provided in clause A.7 of TS 26.247 [4] and clause A.7 of ISO/IEC 23009-1 [32].

Additional procedures for handling partial file responses are provided in clause A.9 of TS 26.247 [4].

4.14 Procedures for downlink media streaming via MBS

This procedure is used by a 5GMSd Client to establish a downlink media streaming session either completely, or at least partially, through MBS.

- For downlink media streaming exclusively via MBS and for hybrid 5GMSd/MBS services, as defined in clauses 5.12.2 and 5.12.4 respectively of TS 26.501 [2]:
  - The 5GMSd Application Provider shall provision a supplementary distribution network of type *DISTRIBUTION_NETWORK_MBS* in the Content Hosting Configuration at reference point M1d, as specified in clause 8.8.3.1 of TS 26.510 [56], with either *MODE_EXCLUSIVE* or *MODE_HYBRID* (as appropriate).
  - The 5GMSd Application Provider may additionally provision access reporting in the Consumption Reporting Configuration at M1d, as specified in clause 8.12.3.1 of TS 26.510 [56].
  - The MBSTF Client shall host an MPD as defined in ISO/IEC 23009-1 [32] or in TS 26.247 [4], or any other presentation manifest such as an HLS Variant Playlist, as the 5GMSd Media Entry Point.
  - The URL of this presentation manifest shall be signalled to the 5GMSd Client through the 5GMSd session establishment procedure.
  - The MBSTF Client shall be invoked by the Media Session Handler via reference point MBS-7 using the procedures defined in TS 26.517 [64].
- For dynamically provisioned downlink media streaming via MBS as defined in clause 5.12.5 TS 26.501 [2]:
  - The 5GMSd Application Provider shall provision a supplementary distribution network of type *DISTRIBUTION_NETWORK_MBS* in the Content Hosting configuration at reference point M1d, as specified in clause 8.8.3.1 of TS 26.510 [56], with *MODE_DYNAMIC*.
  - The 5GMSd Application Provider shall additionally provision access reporting in the Consumption Reporting Configuration at M1d, as specified in clause 8.12.3.1 of TS 26.510 [56].
  - The 5GMSd AS shall host an MPD as defined in ISO/IEC 23009-1 [32] or in TS 26.247 [4], or any other presentation manifest such as an HLS Variant Playlist as the 5GMSd Media Entry Point.
  - The URL of this presentation manifest shall be signalled to the 5GMSd Client through the 5GMSd session establishment procedure. If the 5GMSd service is currently available as an MBS User Service, the 5GMSd Client forwards the manifest request to the Media Server in the MBSTF Client via reference point MBS-7; otherwise, it forwards the request to the 5GMSd AS via reference point M4d.

NOTE: The detailed execution of dynamically handling this decision is left to implementation.

- The MBS Client shall be invoked dynamically, paused or destroyed by the Media Session Handler via reference point MBS-7 using the procedures defined in TS 26.517 [64].

Additional procedures for reactions to different HTTP status codes are provided in clause A.7 of TS 26.247 [4] and clause A.7 of ISO/IEC 23009-1 [32].

Additional procedures for handling partial file responses are provided in clause A.9 of TS 26.247 [4].

5 Procedures for Uplink Media streaming

5.1 General

Uplink media streaming functional entities in the 5GMS System include the 5GMSu Application Provider, 5GMSu AF, 5GMSu AS and the UE. To make use of these other entities, the UE includes a 5GMSu-Aware Application that is provided by the 5GMSu Application Provider and a 5GMSu Client comprising the Media Session Handler and the Media Streamer.

The M1 Provisioning API enables the 5GMSu Application Provider to establish and manage the uplink media session handling and streaming options of the 5GMSu System.

The content egest interface enables uplink media streaming content that has been sent by the 5GMSu Client to the 5GMSu AS at reference point M4u to be subsequently delivered to the 5GMSu Application Provider via reference point M2u or to another 5GMSu AS via reference point M10u. Uplink media streaming media transfer from the 5GMSu AS to the 5GMSu Application Provider at reference point M2u or from one 5GMSu AS to another at reference point M10u may be either pull-based and initiated by the 5GMSu Application Provider using the HTTP `GET` method, or push-based and initiated by the 5GMSu AS using the HTTP `PUT` method. The 5GMSu Application Provider's target endpoint for push-based streaming content delivery at reference point M2u is provided to the 5GMSu AF as part of the M1 Provisioning Session and this is passed to the 5GMSu AS as part of the M3u configuration procedures.

The 5GMSu AF, having been successfully provisioned at reference point M1u, sets up corresponding resources at a reference point M5u endpoint from which Service Access Information for uplink media streaming session management, metrics reporting, network assistance and request for policy and/or charging treatment may be retrieved using its provisioned external application identifier. Certain types of configuration and policy information accessed over reference point M5u by the Media Session Handler, such as uplink metrics reporting, QoS policy, or support for AF-based network assistance are further passed to the Media Streamer via the M7u API.

The 5GMSu-Aware Application initiates a new uplink media streaming session by launching the Media Stream Handler at reference point M6u using a 3GPP Service URL for 5GMS (see clause 4.8.3). The 3GPP Service URL indicates the external application identifier. This may be used to retrieve Service Access Information from the 5GMSu AF at reference point M5. Alternatively, if the 5GMSu-Aware Application has already acquired all necessary Service Access Information via private means at reference point M8, this may be supplied directly to the Media Session Handler at reference point M6u as additional parameters.

Based on a request from the 5GMSu-Aware Application or from the Media Streamer received over the M6u API, and based on the Service Access Information acquired from the 5GMSu AF via reference point M5u, the Media Session Handler sets up an uplink media streaming session with a unique media delivery session identifier. Upon successful session establishment, the Media Session Handler triggers the Media Streamer to begin uplink media streaming of media content to the 5GMSu AS over reference point M4u.

Subscription to status and other event notification services is offered by the Media Session Handler to the 5GMSu-Aware Application and to the Media Streamer via the UE media session handling APIs exposed by the Media Session Handler at reference point M6u.

Subscription to status and other event notification services is also offered by the Media Streamer to the 5GMSu-Aware Application and to the Media Session Handler via the UE media stream handling APIs exposed by the Media Player at reference point M11u.

5.2 APIs relevant to Uplink Media Streaming

Table 5.2-1 summarises the APIs used to provision and use the various uplink media streaming features specified in TS 26.501 [2].

Table 5.2-1: Summary of APIs relevant to uplink media streaming features

5GMSu feature	Abstract	Relevant APIs		
		Interface	API name	Clause
Content protocols discovery	Used by the 5GMSu Application Provider to query which content egest protocols are supported by 5GMSu AS(s).	M1u	Content Protocols Discovery API	7.5
Content publishing	Content is contributed to the 5GMSu AS and published to 5GMSu Application Providers according to a Content Publishing Configuration associated with a Provisioning Session.	M1u	Provisioning Sessions API	7.2
			Server Certificates Provisioning API	7.3
			Content Preparation Templates Provisioning API	7.4
			Content Publication Provisioning API	7.6A
		M2u	HTTP pull-based content egest protocol	8.5
			DASH-IF push-based content egest protocol	8.6
			HTTP low-latency pull-based content egest protocol	8.7
		M3u	Server Certificates configuration API	9.2
			Content Preparation Templates configuration API	9.3
			Content Publication configuration API	9.5
		M4u	DASH-IF push-based contribution protocol	10.4.2
		M5u	Service Access Information API	11.2
		M10u	HTTP pull-based content egest protocol	8.5
			DASH-IF push-based content egest protocol	8.6
			HTTP low-latency pull-based content egest protocol	8.7
Metrics reporting	The 5GMSu Client uploads metrics reports to the 5GMSu AF according to a provisioned Metrics Reporting Configuration it obtains from the Service Access Information for its Provisioning Session.	M1u	Provisioning Sessions API	7.2
			Metrics Reporting Provisioning API	7.8
		M5u	Service Access Information API	11.2
			Metrics Reporting API	11.4
Dynamic Policy invocation	The 5GMSu Client activates different traffic treatment policies selected from a set of Policy Templates configured in its Provisioning Session.	M1u	Provisioning Sessions API	7.2
			Policy Templates Provisioning API	7.9
		M5u	Service Access Information API	11.2
			Dynamic Policies API	11.5
Network Assistance	The 5GMSu Client requests bit rate recommendations and delivery boosts from the 5GMSu AF.	M5u	Service Access Information API	11.2
			Network Assistance API	11.6
Edge content processing	Edge resources are provisioned for processing content in 5GMS uplink media streaming sessions.	M1u	Provisioning Sessions API	7.2
			Edge Resources Provisioning API	7.10
		M5u	Service Access Information API	11.2
UE data collection, reporting and exposure	UE data related to uplink 5G Media Streaming is reported to the Data Collection AF instantiated in the 5GMSu AF for exposure to Event consumers.	M1u	Event Data processing Provisioning API	7.11
		R4	*Ndcaf_DataReporting* service	17
		R5, R6	*Naf_EventExposure* service	18

5.3 Procedures of the M1u (5GMSu Provisioning) interface

Procedures at this reference point are for further study.

5.4 Procedures of the M2u (5GMSu content egest) interface

Procedures at this reference point are for further study.

5.5 Procedures of the M3u interface

Procedures at this reference point are for further study.

5.6 Procedures of the M4u (Media Streaming) interface

5.6.1 General

This procedure is used by a Media Streamer to establish an uplink media streaming session with the 5GMSu AS at reference point M4u.

To initiate uplink media streaming, the Media Streamer receives the URL of a Media Streamer Entry from the 5GMSu-Aware Application through reference point M7u. The Media Streamer shall send an HTTP `GET` message to the 5GMSu AS including the URL of the Media Streamer Entry resource. On success, the 5GMSu AS shall respond with a *200 (OK)* message that includes the requested Media Streamer Entry resource.

The Media Streamer may use a multipath transport protocol supported by the UE as specified in clause 6.2.1.2 to connect to a reference point M4u service location on the 5GMSu AS. The different multipath transport connection subflows/paths may span one or more access networks available to the UE. According to Media Streamer configuration (see clause 5.9.2) the different subflows/paths may be used to increase the resilience of the transport connection to packet loss or to increase data throughput.

Additional details of procedures at this reference point are for further study.

5.6.2 ECN marking for L4S

When a Dynamic Policy Instance (see clause 4.7.3) enabling the ECN marking for L4S functionality is successfully activated for an uplink media streaming session, the 5GMSu AS is required to support L4S according to RFC 9330 [67], RFC 9331 [68] and RFC 9332 [69] at reference point M4u.

5.7 Procedures of the M5u (Media Session Handling) interface

Procedures at this reference point are for further study.

5.8 Procedures of the M6u (UE Media Session Handling) interface

Procedures at this reference point are for further study.

5.9 Procedures of the M7u/M11u (UE Media Streamer) interface

5.9.1 General

This clause specifies procedures for the interaction between the 5GMSu-Aware Application or the Media Session Handler and the Media Streamer using the client API at reference points M7u and M11u respectively.

5.9.2 Procedures for configuring uplink content delivery

In the absence of a desired content delivery configuration, the Media Streamer may select any content delivery protocol that meets the requirements of the provided Media Streamer Entry, and may select a multipath transport protocol compatible with its choice of content delivery protocol.

The Media Streamer should inform the 5GMSu-Aware Application and the Media Session Handler using reference point M7u and M11u notifications respectively about changes to the status of reference point M4u transport connections.

Additional details of procedures at this reference point are for further study.

5.10 Procedures of the M8u interface

This clause defines basic procedures at reference point M8u.

No specific procedures are defined but it is expected that the 5GMSu Application Provider can provide media session entry points to a 5GMSu-Aware Application through reference point M8u. The 5GMSu-Aware Application would then initiate the uplink media session by providing such an entry point to the 5GMSu Client through reference point M7u. Multiple alternative entry points of the same uplink media streaming service may be provided.

6 General aspects of APIs for 5G Media Streaming

6.0 HTTP endpoint addresses

6.0.1 Default 5GMS AF endpoint addresses

6.0.1.1 Default 5GMS AF endpoint address at reference point M1

This version of the present document does not specify a default host name for the 5GMS AF at reference point M1.

The 5GMS AF shall expose the Provisioning network APIs specified in clause 7 via the default listening port number(s) for the version(s) of HTTP specified for reference point M1 in clause 6.2.1.1.

6.0.1.2 Default 5GMS AF endpoint address at reference point M3

This version of the present document does not specify a default host name for the 5GMS AF at reference point M3.

The 5GMS AF shall expose the Provisioning network APIs specified in clause 7 via the default listening port number(s) for the version(s) of HTTP specified for reference point M3 in clause 6.2.1.1.

6.0.1.3 Default 5GMS AF endpoint address at reference point M5

Except where the host name of an externally deployed 5GMS AF is explicitly signalled as a parameter of a 3GPP Service URL passed to the Media Session Handler at reference point M6, the following default 5GMS AF host name shall be inserted into *{apiRoot}* (see clause 6.1) for the Media Session Handler's first request at reference point M5 to the 5GMS AF for each media streaming session (see clause 11.2):

```
ms.af.3gppservices.org
```

The DNS service provided by the 5G System shall resolve this host name to the IP address(es) of deployed 5GMS AF instance(s), if any.

The 5GMS AF shall expose the Media Session Handling network APIs specified in clause 11 via the default listening port number(s) for the version(s) of HTTP specified for reference point M5 in clause 6.2.1.1.

6.0.2 Default 5GMS AS endpoint addresses

6.0.2.1 Default 5GMS AS endpoint address at reference point M3

This version of the present document does not specify a default host name for the 5GMS AS at reference point M3.

The 5GMS AS shall expose the Application Server Configuration APIs specified in clause 9 via the default listening port number(s) for the version(s) of HTTP specified for reference point M3 in clause 6.2.1.2.

6.0.2.2 Canonical 5GMS AS authority at reference point M4

Media Entry Points provisioned in distribution configurations of a Content Hosting Configuration or in contribution configurations of Content Publishing Configuration shall be exposed by the 5GMS AS at reference point M4 from service locations with the following canonical domain name respectively:

{modifiedDistributionId}.{modifiedExternalServiceId}`.ms.as.3gppservices.org`

or

{modifiedContributionId}.{modifiedExternalServiceId}`.ms.as.3gppservices.org`

where:

- *{modifiedDistributionId}* is a modified form of the *DistributionConfiguration.distributionId* property assigned by the 5GMS Application Provider upon provisioning of a Content Hosting Configuration (see clause 5.2.8.2 of TS 26.510 [56]).
- *{modifiedContributionId}* is a modified form of the *ContributionConfiguration.contributionId* property assigned by the 5GMS Application Provider upon provisioning of a Content Publishing Configuration (see clause 5.2.9.2 of TS 26.510 [56]).
- *{modifiedExternalServiceId}* is a modified form of the external service identifier indicated by the 5GMS Application Provider in the parent Provisioning Session resource at reference point M1 (see clause 5.2.2.1 of TS 26.510 [56]).

In all of the above identifiers, every period character ('`.`') is replaced with a single hyphen character ('`-`').

For example, the canonical 5GMS AS domain name for a Content Hosting Configuration with a distribution configuration assigned a *distributionId* property value of *distributionA.service* or Content Publishing Configuration with a contribution configuration assigned a *contributionId* property value of *contributionA.service* created under the Provisioning Session with external service identifier `com.provider.service` is respectively:

`distributionA-service.com-provider-service.ms.as.3gppservices.org`

or

`contributionA-service.com-provider-service.ms.as.3gppservices.org`

The DNS service provided by the 5G System shall resolve each such canonical domain name to the IP address(es) of deployed 5GMS AS service location(s) providing content hosting or content publishing endpoint(s) at reference point M4 on behalf of the parent Provisioning Session in question.

NOTE: Access to the 5GMS AS using domain name aliases at this reference point is not precluded.

The 5GMS AS shall expose all service locations at reference point M4 via the default listening port number(s) for the version(s) of HTTP specified in clause 6.2.1.2 for use at this reference point.

6.1 HTTP resource URIs

6.1.1 5GMS AF resource URIs

The resource URI used in each HTTP request to the *Maf_Provisioning* API provider at reference point M1 shall have the structure defined in clause 7.1.3 of TS 29.510 [56].

The resource URI used in each HTTP request to the *Maf_SessionHandling* API provider at reference points M3 or M5 shall have the structure defined in clause 7.1.3 of TS 29.510 [56].

6.1.2 5GMS AS resource URIs

The resource URI used in each HTTP request to the API provider shall have the structure defined in clause 4.4.1 of TS 29.501 [22], i.e.:

{apiRoot}/{apiName}/{apiVersion}/{apiSpecificResourceUriPart}

with the following components:

- *{apiRoot}* shall be set as described in TS 29.501 [22].
- *{apiName}* shall be set as defined by the following clauses.
- *{apiVersion}* shall be set to "v1" in this release.
- *{apiSpecificResourceUriPart}* shall be set as specified in clause 9.

6.2 Usage of HTTP

6.2.1 HTTP protocol version

6.2.1.1 5GMS AF

Implementations of the 5GMS AF shall comply with clause 7.1.1 of TS 26.510 [56].

6.2.1.2 5GMS AS

Implementations of the 5GMS AS shall expose HTTP/1.1 [24] endpoints at reference points M2, M4 and M10; and implementations may additionally expose HTTP/2 [31] or HTTP/3 [60] endpoints at these reference points. For each protocol version, TLS [16] shall be supported and HTTPS interactions should be used in preference to cleartext HTTP. Multipath TCP per RFC 8684 [72] may be used as the underlying transport protocol at these two reference points.

For pull-based content ingest into the 5GMSd AS:

- The 5GMSd Application Provider shall expose an HTTP/1.1-based origin endpoint to the 5GMSd AS at reference point M2d and may additionally expose HTTP/2- and/or HTTP/3-based origin endpoints at this reference point.
- The 5GMSd AS shall expose an HTTP/1.1-based origin endpoint at reference point M10d and may additionally expose HTTP/2- and/or HTTP/3-based origin endpoints at this reference point.

For push-based content ingest into the 5GMSd AS:

- The 5GMSd Application Provider may use any supported HTTP protocol version to push content at reference point M2d.
- The 5GMSd AS may use any supported HTTP protocol version to push content at reference point M10d.

For pull-based content egest from the 5GMSu AS:

- The 5GMSu AS shall expose an HTTP/1.1-based origin endpoint to the 5GMSu Application Provider at reference point M2u and may additionally expose HTTP/2- and/or HTTP/3-based origin endpoints at this reference point.
- The 5GMSu AS shall expose an HTTP/1.1-based origin endpoint at reference point M10u and may additionally expose HTTP/2- and/or HTTP/3-based origin endpoints at this reference point.

For push-based content egest from the 5GMSu AS:

- The 5GMSu AS may use any supported HTTP protocol version to push content to the 5GMSu Application Provider at reference point M2u.
- The 5GMSu AS may use any supported HTTP protocol version to push content at reference point M10u.

Implementations of the 5GMS AS should expose HTTP/3 [60] endpoints at reference point M4. In HTTP/3, the QUIC protocol [58] or its multipath variant [73] may be used for transport, and in both cases TLS [59] is used for the initial handshake and key exchange.

The 5GMS AF may use any supported HTTP protocol version at reference point M3.

The Media Stream Handler (Media Player or Media Streamer) may use any supported HTTP protocol version at reference point M4. The Media Stream Handler may use multipath TCP per RFC 8684 [72] as the underlying transport protocol for HTTP/1.1 [24] or HTTP/2 [31] at this reference point. The Media Stream Handler may use multipath QUIC [73] to support HTTP/3 [60] connections at this reference point.

6.2.2 HTTP message bodies for API resources

The OpenAPI [23] specification of HTTP messages and their content bodies is contained in annex C of the present document and in annex A of TS 26.510 [56].

6.2.3 Usage of HTTP headers

6.2.3.1 General

Standard HTTP headers shall be used in accordance with clause 5.2.2 of TS 29.500 [21] for all versions of HTTP/1.1 [24], HTTP/2 [31] and HTTP/3 [60] messages.

6.2.3.2 User Agent identification

6.2.3.2.1 Media Stream Handler identification

The Media Stream Handler in the 5GMS Client shall identify itself to the 5GMS AS at reference point M4 using a `User-Agent` request header (see section 5.5.3 of RFC 7231 [25]) that should include the *product* token `5GMSMedia StreamHandler`. If this product identifier is supplied, the optional *product-version* suffix shall be present. This should indicate the version number of the present document (without the leading "V") with which the Media Stream Handler implementation complies and shall, at minimum, indicate the 3GPP release number with which the implementation complies.

The Media Stream Handler may additionally supply a *comment* element in the `User-Agent` request header containing vendor-specific information.

EXAMPLE 1: `5GMSMediaStreamHandler/17.5.0 (build2634) ExoPlayerLib/2.17.1`

EXAMPLE 2: `5GMSMediaStreamHandler/17`

6.2.3.2.2 Media Session Handler identification

The Media Session Handler in the 5GMS Client shall identify itself to the 5GMS AF at reference point M5 using a User-Agent request header (see section 5.5.3 of RFC 7231 [25]) in which the first element shall be a *product* identified

by the token `5GMSMediaSessionHandler`. The optional *product-version* suffix shall be present. This should indicate the version number of the present document (without the leading "V") with which the Media Session Handler implementation complies and shall, at minimum, indicate the 3GPP release number with which the implementation complies.

The Media Session Handler may supply additional vendor-specific product identifiers in the `User-Agent` request header and may additionally supply a *comment* element containing vendor-specific information.

EXAMPLE 1: `5GMSMediaSessionHandler/17.5.0 (build1536) lib5gmsclient/0.3.1`

EXAMPLE 2: `5GMSMediaSessionHandler/17`

6.2.3.3 Server identification

6.2.3.3.1 5GMS AF identification

The 5GMS AF shall identify itself at reference points M1 and M5 using a `Server` response header (see section 7.4.2 of RFC 7231 [25]) that includes a product string of the following form:

`5GMSAF-`*{FQDN}*`/`*{complianceInformation}*

where *{FQDN}* shall be the Fully-Qualified Domain Name of the 5GMS AF exposed to the requesting client, and *{complianceInformation}* should indicate the version number of the present document (without the leading "V") with which the 5GMS AF implementation complies and shall, at minimum, indicate the 3GPP release number with which the implementation complies.

The `Server` response header may also include comments strings and vendor-specific subproduct strings compliant with the syntax and guidance provided in section 7.4.2 of [25].

EXAMPLE 1: `5GMSAF-vm10664.mno.net/17.4.0 (api=2.1.0) libsbi/2.1 libnf/1.2 libaf/1.1`

EXAMPLE 2: `5GMSAF-vm10664.mno.net/17 (api=2.1.0) libsbi/2.1 libnf/1.2 libaf/1.1`

6.2.3.4 Support for conditional HTTP GET requests

All responses from the 5GMS AF that carry a resource message body shall comply with clause 7.1.4.2 of TS 26.510 [56].

6.2.3.5 Support for conditional HTTP POST, PUT, PATCH and DELETE requests

All API endpoints on the 5GMS AF that expose the HTTP `POST`, `PUT`, `PATCH` or `DELETE` methods shall comply with clause 7.1.4.3 of TS 26.510 [56].

6.2.3.6 Media delivery session identifier

At applicable reference points, the media delivery session identifier specified in clause 7.3.2 of TS 26.510 [56] shall be conveyed in the HTTP header `CMCD-Session:sid` as specified in table 1 of CTA-5004 [65]. As such, the media delivery session identifier for 5G Media Streaming shall be a UUID [66].

EXAMPLE: `CMCD-Session:sid: 8bf9f090-82fd-4686-aa4a-39e6a9381b76`

6.3 HTTP response codes

Guidelines for error responses to the invocation of APIs of NF services are specified in clause 4.8 of TS 29.501 [22]. API-specific error responses are specified in the respective technical specifications.

6.4 Common API data types

6.4.1 General

The data types defined in this clause are intended to be used by more than one of the 5GMS APIs.

6.4.2 Simple data types

Table 6.4.2-1 below specifies common simple data types used within the 5GMS APIs, including a short description of each. In cases where types from other specifications are reused, a reference is provided.

Table 6.4.2-1: Simple data types

Type name	Type definition	Description	Reference
ResourceId	`string`	String chosen by the 5GMS AF to serve as an identifier in a resource URL.	TS 26.510 [56] table 7.3.2-1
Uri	`string`	Uniform Resource Identifier conforming with the URI Generic Syntax.	TS 29.571 [12] table 5.2.2-1
Url	`string`	Uniform Resource Locator, conforming with the URI Generic Syntax.	TS 26.510 [56] table 7.3.2-1. IETF RFC 3986 [41]
RelativeUrl	`string`	Relative Uniform Resource Locator, conforming with the *relative-ref* production of the URI Generic Syntax. Both *query* and *fragment* suffixes are permitted.	TS 26.510 [56] table 7.3.2-1. IETF RFC 3986 [41].
AbsoluteUrl	`string`	Absolute Uniform Resource Locator, conforming with the *absolute-URI* production of the URI Generic Syntax in which the scheme part is *http* or *https*. The *query* suffix is permitted but the *fragment* suffix is not.	TS 26.510 [56] table 7.3.2-1. IETF RFC 3986 [41]
Percentage	`number`	A percentage expressed as a floating-point value between 0.0 and 100.0 (inclusive).	TS 26.510 [56] table 7.2.2-1
Duration	`string`	A period of time expressed as a string compliant with the *duration* format specified in section 7.3.1 of the JSON Schema specification [52].	TS 26.510 [56] table 7.3.2-1. IETF RFC 3339 [53] appendix A.
DurationSec	`integer`	An unsigned integer identifying a period of time expressed in units of seconds.	TS 29.571 [12] table 5.2.2-1
DateTime	`string`	An absolute date and time expressed using the OpenAPI *date-time* string format.	TS 29.571 [12] table 5.2.2-1
Ipv4Addr	`string`	Ipv4 address formatted in “dotted decimal” notation	TS 29.571 [12] table 5.2.2-1
Ipv6Addr	`string`	Ipv6 address formatted in colon-separated hexadecimal quartet notation.	TS 29.571 [12] table 5.2.2-1
Uinteger	`integer`	Unsigned integer.	TS 29.571 [12] table 5.2.2-1
Dnn	`string`	Data Network Name.	TS 29.571 [12] table 5.2.2-1
BitRate	`string`	A bit rate expressed as a string-encoded decimal value and unit.	TS 29.571 [12] table 5.2.2-1
MediaDeliverySessionId	`string`	A unique identifier for a media delivery session. This should not contain any user-identifiable data.	TS 26.510 [56] table 7.3.2-1

6.4.3 Structured data types

6.4.3.1 Void

6.4.3.2 Void

6.4.3.3 Void

6.4.3.4 Void

6.4.3.5 Void

6.4.3.6 Void

6.4.3.7 Void

6.4.3.8 Void

6.4.3.9 Void

6.4.3.9A MediaStreamingBaseReportingRecord type

This data type is intended to be used as a building block in other data types.

Table 6.4.3.9A-1: Definition of MediaStreamingBaseReportingRecord type

Property name	Data Type	Cardinality	Description
sliceInfo	`Snssai`	0..1	Identifying the target slice to which the reporting record pertains.
dataNetworkName	`Dnn`	0..1	The name of the target Data Network to which the reporting record pertains.
location	`TypedLocation`	0..1	The location of the UE to which the reporting record pertains.

6.4.3.10 MediaStreamingSessionIdentification type

This data type is intended to be used as a building block in other data types.

Table 6.4.3.10-1: Definition of MediaStreamingSessionIdentification type

Property name	Data Type	Cardinality	Description
sessionId	`MediaDelivery SessionId`	1..1	An identifier for a media streaming session. This should not contain any user-identifiable data.

6.4.3.11 MediaStreamingAccess type

This data type is intended to be used as a building block in other data types.

Table 6.4.3.11-1: Definition of MediaStreamingAccess type

Property name	Data Type	Cardinality	Description
mediaStreamHandler EndpointAddress	`Endpoint Address`	1..1	The endpoint address of the Media Stream Handler accessing the 5GMS AS. (See clause 6.4.3.8.)
applicationServer EndpointAddress	`Endpoint Address`	1..1	The service endpoint on the 5GMS AS to which the Media Stream Handler is connected. (See clause 6.4.3.8.)

Property name	Data Type	Cardinality	Description
requestMessage	`object`	1..1	Details of the HTTP request message submitted to the 5GMS AS by the Media Stream Handler for this media access.
method	`string`	1..1	The request method.
url	`AbsoluteUrl`	1..1	The request URL. (See table 6.4.2-1.)
protocolVersion	`string`	1..1	The HTTP protocol version, e.g., "HTTP/1.1".
range	`string`	0..1	The value of the Range request header, if present.
size	`Uinteger`	1..1	The total number of bytes in the request message.
bodySize	`Uinteger`	1..1	The number of bytes supplied by the Media Stream Handler in the HTTP request message body. Zero if there is no request body.
contentType	`string`	0..1	The MIME content type of the request message, if any.
userAgent	`string`	0..1	A string describing the requesting Media Stream Handler, if it supplies a `User-Agent` request header.
userIdentity	`string`	0..1	A string identifying the user that made the access, if supplied.
referer	`AbsoluteUrl`	0..1	The URL that the Media Player reports being referred from, if the `Referer` request header is supplied. (See table 6.4.2-1.)
cacheStatus	`CacheStatus`	0..1	An indication of whether the 5GMS AS is able to serve an object corresponding to *requestMessage,url* from cache (*HIT*) or whether there is a stale object cached (*EXPIRED*) or the requested object is not present in cache (*MISS*). (See table 6.4.4.4.) For non-caching implementations of the 5GMS AS, the property shall be omitted.
responseMessage	`object`	1..1	Details of the HTTP response message returned by the 5GMS AS to the Media Stream Handler for this media access.
responseCode	`Uinteger`	1..1	The HTTP response code.
size	`Uinteger`	1..1	The total number of bytes in the response message.
bodySize	`Uinteger`	1..1	The number of bytes in the HTTP response message body.
contentType	`string`	0..1	The MIME content type of response message, if any.
processingLatency	`Float`	1..1	The time, expressed in milliseconds, taken by the 5GMS AS to respond to the Media Stream Handler request, measured from the first byte of the HTTP request being processed by the 5GMS AS to the last byte of the response being sent.
connectionMetrics	`object`	0..1	Metrics about the performance of the transport connection underlying the HTTP session serving this media access.
meanNetwork RoundTripTime	`Float`	1..1	A rolling mean average, expressed in milliseconds, of the network round-trip time for the HTTP session.
networkRoundTrip TimeVariation	`Float`	1..1	The variation in *meanNetworkRoundTripTime*, expressed in milliseconds, during the averaging period.
congestion WindowSize	`Uinteger`	1..1	The current size (in bytes) of the congestion window for the transport connection underlying the HTTP session.

6.4.3.12 NetworkAssistanceInvocation type

This data type is intended to be used as a building block in other data types.

Table 6.4.3.11-1: Definition of NetworkAssistanceInvocation type

Property name	Data Type	Cardinality	Description
policyTemplateId	`ResourceId`	0..1	Identifying the Policy Template (if any) referenced by the Media Session Handler in the parent Network Assistance Session.
applicationFlow Descriptions	`array( ApplicationFlow Description)`	0..1	If present, a set of one or more Application Flow Descriptions (see clause 7.3.3.2 of TS 26.510 [56]) to which the Network Assistance session has been applied. Present only for individual data samples and, in exposed events, only when exposure is permitted by the data exposure restrictions in force.
requestedQoS	`Unidirectional QoSSpecification`	0..1	The network QoS parameters (if any) requested by the Media Session Handler from the 5GMS AF in the parent Network Assistance Session or from the RAN (see clause 6.4.3.13).
recommendedQoS	`object`	0..1	The network QoS parameters (if any) recommended to the Media Session Handler by the 5GMS AF in the parent Network Assistance Session or by the RAN.
maximumBitRate	`BitRate`	1..1	The maximum recommended bit rate.
minimumBitRate	`BitRate`	1..1	The minimum recommended bit rate.

6.4.3.13 Void

6.4.4 Enumerated data types

6.4.4.1 Void

6.4.4.2 Void

6.4.4.3 Void

6.4.4.4 Void

6.4.4.5 CacheStatus enumeration

Table 6.4.4.5-1: Definition of CacheStatus enumeration

Enumeration value	Description
HIT	The requested object is present in the 5GMS AS cache and is still valid.
MISS	The requested object is not present in the 5GMS AS cache.
EXPIRED	The requested object is present in the 5GMS AS cache but is stale.

6.5 Void

6.6 Security

6.6.1 General

The 5GMS AF shall enable secure provision of information in the 5GMS System by authenticated and authorised 5GMS Application Providers or 5GMS-Aware Applications or the 5GMS AS.

6.6.2 Authorising 5GMS Application Provider access to the 5GMS AF at reference point M1

Implementations of the 5GMS Application Provider shall comply with the authorisation provisions specified in clause 7.4.2 of TS 26.510 [56].

6.6.3 Authorising 5GMS AF access to the 5GMS AS at reference point M3

When a 5GMS AF deployed outside the Trusted DN attempts to access a 5GMS AS deployed inside the Trusted DN, the 5GMS System shall authenticate and authorise the 5GMS AF.

Access to the *Mas_Configuration* API of the 5GMS AS by the 5GMS AF at reference point M3 may be authorised by means of the OAuth 2.0 protocol specified in RFC 6749 [47]), using the *client credentials* authorization grant. Based on the content of the access token, the 5GMS AS determines whether the 5GMS AF is authorised to invoke a given configuration operation in terms of at least the targeted resource and HTTP method, and hence the underlying 5GMS feature.

NOTE: The provisioning and negotiation of the security method is not specified in this release.

When CAPIF (see TS 29.222 [48]) is used for external API exposure:

- The CAPIF core function shall play the role of authorization server, the 5GMS AS shall play the role of resource server and the 5GMS AF shall play the role of client.
- Before invoking any service operation exposed by the 5GMS AS, the 5GMS AF shall negotiate the security method (PKI, TLS-PSK or OAuth 2.0) with the CAPIF core function and shall ensure that the 5GMS AS has the required credentials to authenticate access tokens subsequently presented by the 5GMS Application Provider (see clauses 5.6.2.2 and 6.2.2.2 of TS 29.222 [48]).
- If PKI or TLS-PSK is the selected security method between the 5GMS AF and the 5GMS AS shall, upon invocation of a *Mas_Configuration* service operation by the 5GMS AF at reference point M3, retrieve the authorisation information from the CAPIF core function as described in clause 5.6.2.4 of TS 29.222 [48].
- If OAuth 2.0 [47] is the selected security method between the 5GMS AF and the 5GMS AS, the 5GMS AF shall, prior to invoking *Mas_Configuration* service operations on the 5GMS AS at reference point M3, obtain an access token from the authorization server (CAPIF core function) by invoking the *Obtain_Authorization* service operation specified in clause 5.6.2.3.2 of TS 29.222 [48].

Otherwise:

- The 5GMS AS shall play the role of both authorization server and resource server, and the 5GMS AF shall play the role of client.
- The 5GMS AF shall obtain an access token from the authorization server (5GMS AS) using the client credentials authorization grant specified in section 4.4 of RFC 6749 [47] prior to invoking *Mas_Configuration* service operations on the resource server (5GMS AS) at reference point M3.

6.6.4 Authorising Media Session Handler access to the 5GMS AF at reference point M5

Implementations of the Media Session Handler shall comply with the authorisation provisions specified in clause 7.4.3 of TS 26.510 [56].

6.6.5 Authorising 5GMS AS access to the 5GMS AF at reference point M3

Implementations of the 5GMS AS shall comply with the authorisation procedures for the Media AS specified in clause 7.4.3 of TS 26.510 [56].

7 Provisioning (M1) APIs

7.1 General

This clause defines the provisioning API used by a 5GMS Application Provider to configure downlink or uplink 5G Media Streaming services.

7.2 Provisioning Sessions API

7.2.1 Overview

The API used by the 5GMS Application Provider to instantiate and manipulate downlink or uplink media streaming Provisioning Sessions in the 5GMS AF at reference point M1 is specified in clause 8.2 of TS 26.510 [56]. Having created a Provisioning Session, the 5GMS Application Provider then goes on to provision other 5GMS features in the context of this Provisioning Session, using the APIs specified in clause 7.3 *et seq*. Certain of these features are only applicable to the type of Provisioning Session created.

7.2.2 Void

7.2.3 Void

7.3 Server Certificates Provisioning API

7.3.1 Overview

The API used by the 5GMS Application Provider at reference point M1 to instantiate and manipulate Server Certificates associated with a particular downlink or uplink media streaming Provisioning Session in the 5GMS AF is specified in clause 8.4 of TS 26.510 [56]. It is used to provision X.509 [8] server certificates that can be referenced by a Content Hosting Configuration or Content Publishing Configuration and subsequently presented to the 5GMS Client when it connects to a 5GMS AS endpoint at reference point M4 using Transport Layer Security [30]. Server Certificate resources are provisioned within the scope of an enclosing Provisioning Session.

7.3.2 Void

7.3.3 Void

7.4 Content Preparation Templates Provisioning API

7.4.1 Overview

The API used by the 5GMS Application Provider at reference point M1 to instantiate and manipulate Content Preparation Templates associated with a particular downlink or uplink media streaming Provisioning Session in the 5GMS AF is specified in clause 8.5 of TS 26.510 [56]. Content Preparation Templates are used to specify manipulations applied by a 5GMS AS to downlink media resources ingested at reference point M2d or M10d for distribution at reference point M4d, or to uplink media resources contributed at reference point M4u or M10u for egest at reference point M2u. The Content Preparation Templates Provisioning API is used to provision a Content Preparation Template within the scope of a Provisioning Session that can subsequently be referenced from a Content Hosting Configuration.

7.4.2 Void

7.4.3 Void

7.4.4 Void

7.5 Content Protocols Discovery API

7.5.1 Overview

The API used by the 5GMS Application Provider at reference point M1 to discover which content ingest or egest protocols are supported at reference point M2 by the 5GMS AS instance(s) associated with a 5GMS AF is specified in clause 8.3 of TS 26.510 [56]. The controlled vocabulary of term identifiers used in this API to advertise support for these content protocols is specified in clause 8.1 of the present document.

- One of the supported ingest protocols may subsequently be provisioned in a Content Hosting Configuration for downlink media streaming (see clause 7.6).
- One of the supported egest protocols may subsequently be provisioned in a Content Publishing Configuration for uplink media streaming (see clause 7.6A).

The Content Protocols resource shall specify the set of geo-fencing locator types supported by the 5GMS System in relation to the Content Hosting feature using the controlled vocabulary specified in clause B.1 of TS 26.510 [56]. Every 5GMS System shall support at least the ISO 3166 administrative area locator type. Refer to clause 7.6.4.6 of the present document for usage.

7.5.2 Void

7.5.3 Void

7.6 Content Hosting Provisioning API

7.6.1 Overview

The API used by the 5GMSd Application Provider at reference point M1d to create and manipulate the 5GMSd AS Content Hosting Configuration associated with a particular downlink media streaming Provisioning Session in the 5GMSd AF is specified in clause 8.8 of TS 26.510 [56].

Within a Content Hosting Configuration, one or more distribution configurations may be defined where each may specify different caching, purging and content preparation behaviours for media resources ingested at reference point M2d or M10d. The Content Hosting Configuration may further specify, through the declaration of affinity groups how reference point M4d service locations associated with each distribution configuration are deployed in the 5GMS System.

7.6.2 Void

7.6.3 Void

7.6.4 5GMSd AS functions supporting Content Hosting

7.6.4.1 Overview

This clause defines the behaviour that is expected from the 5GMSd AS when the Content Hosting Configuration has been successfully provisioned as specified in clause 5.2.8 of TS 26.510 [56]. The main operations that are performed affect content caching and purging of cached content, as well as media processing for content preparation prior to distribution from one or more service locations.

7.6.4.2 Content caching

A distribution configuration defined within the Content Hosting Configuration may specify caching rules to be applied to transport resources when they are distributed by the 5GMSd AS from reference point M4d service locations. The 5GMSd AS shall use the *DistributionConfiguration.CachingConfigurations[ ].urlPatternFilter* property of the Content Hosting Configuration resource specified in clause 8.8.3.1 of TS 26.510 [56] to determine which caching directives apply to that transport resource. In the case where a distribution configuration has multiple caching configurations and a transport resource's URL matches the pattern filter of more than one, the first match shall apply. In case no caching configuration is identified as a match, the 5GMSd AS shall apply the caching directives that were received from the upstream ingest source at reference point M2d or M10d. In the absence of these, the 5GMSd AS shall apply default caching directives as specified in clause 8.8.3.1 of TS 26.510 [56] based on the media resource type.

A caching directive shall indicate that a matching transport resource is:

- Not to be cached by the 5GMSd AS, nor by downstream M4d clients, when *noCache* is set to *true*, or
- To be cached for *maxAge* seconds by the 5GMSd AS, and potentially by downstream M4d clients, when *noCache* is set to *false*.

The *maxAge* value applies relative to the time when a transport resource was ingested by the 5GMSd AS, defined here as *t_ingest* regardless of whether or not it is further modified by a Content Preparation Template. For an HTTP-based ingest, this corresponds to the `Date` header field in the HTTP request/response that carries the transport resource at M2d or M10d. At the time *t_ingest + maxAge*, the transport resource is considered stale and should not be served from the 5GMSd AS cache. The 5GMSd AS shall compensate for any synchronization skew between the origin and its own clock. For instance, this can be done by including the `max-stale` HTTP cache directive in HTTP responses sent from reference point M4d service locations.

The *maxAge* value may be signalled by the 5GMSd AS at reference point M4d service locations using the `Expires` HTTP response header or the HTTP `Cache-Control` directives `max-age` or `s-maxage`.

When distributing a transport resource using HTTP, a *no-cache* request may be translated into a `no-cache` and `no-store` HTTP `Cache-Control` directive and/or a `max-age=0` HTTP `Cache-Control` directive.

By default, all origin HTTP header fields shall be assumed as not forwarded by the 5GMSd AS, unless specified otherwise by setting the flag *originCacheHeaders* to *true*.

7.6.4.3 Cache purging

The 5GMSd Application Provider shall use the procedures and operations specified in clause 5.2.8.6 of TS 26.510 [56] to invalidate some or all cached transport resources of a particular Content Hosting Configuration. As a consequence, the 5GMSd AF shall invoke an operation on the 5GMSd AS at reference point M3d to remove those transport resources (e.g., in the case the underlying media resource has been modified by a Content Preparation Template – see clause 7.6.4.4) from the 5GMSd AS cache across all distribution configurations associated with that Content Hosting Configuration, as specified in clause 9.

7.6.4.4 Content processing

The 5GMSd AS may be required to perform various content processing tasks (such as repackaging, encryption, ABR transcoding, multi-source object coding, etc.) on media resources ingested at reference point M2d or M10d prior to distributing them from reference point M4d service locations. These processing tasks shall be specified in a Content Preparation Template resource referenced from a distribution configuration within the Content Hosting Configuration.

7.6.4.5 URL signing

The URL signing procedure allows the 5GMSd Application Provider to prevent deep linking and unauthorized access to M4d transport resources. It works by cryptographically signing some elements of the M4d request URL and then appending this authentication token to the URL as an additional query parameter. The token is generated by the 5GMSd Application Provider and supplied to the player, for example as part of an initial URL. When it receives a request that requires URL signing, the 5GMSd AS verifies the presence and validity of the token in the M4d request URL before allowing access to the requested transport resource. The 5GMSd AS instance(s) and the origin share a secret that is encoded as part of the query parameter hash, but not shared with the 5GMSd Media Player.

The validity of the authentication token can also be limited to a single UE. If *useIPAddress* is set to True, then the public IP address of the UE as viewed by the 5GMSd AS, *ue_public_ip_address*, shall be incorporated into the token calculation. The parameter name shall be indicated by *ipAddressName*.

The shared secret shall be provided in the *urlSignature.passphrase* property of the Content Hosting Configuration resource. The parameter name for the passphrase to be used in the authentication token shall be provided by *passphraseName*.

The expiry time of the signed URL, *tokenExpiry*, shall be included as an additional query parameter in the URL exposed at M4d with the name indicated in *tokenExpiryName*. The expiry time shall be the string representation of the number of seconds from 1970-01-01T00:00:00Z UTC until the desired expiry UTC date/time, ignoring leap seconds, as defined in section 4.16 of POSIX.1 [11].

Given the above, the authentication token shall be calculated as:

token := SHA512(*url*&*UrlSignature.tokenExpiryName*=*token_expiry*&*UrlSignature.ipAddressName*=*ue_public_ip_address*&*UrlSignature.passphraseName*=*passphrase*)

where the SHA512 function shall be the SHA-512 hash [6] of the enclosed string. The *url* parameter shall be the original M4d transport resource request URL, including the scheme, authority and path components but excluding any query and fragment components.

The resulting token value shall be “base64url” encoded, as specified in section 5 of RFC 4648 [10], prior to inclusion in the M4d URL.

The query part of the signed URL presented by the 5GMSd Media Player at M4d as proof of authenticity shall be composed as follows:

query := *urlSignature.tokenExpiryName*=*token_expiry*&*urlSignature.tokenName*=base64url(*token*)

For all transport resources requested at reference point M4d that match the regular expression specified in *urlSignature.urlPattern*, whether modified by the 5GMSd AS or not, the 5GMSd AS shall validate the *query* presented in the request URL according to the following steps:

1) If the parameter indicated by *urlSignature.tokenName* is absent from *query*, or if the supplied *token* value is malformed, the 5GMSd AS shall respond with a *403 (Forbidden)* error response message and terminate further processing of the M4d request.

2) If the parameter indicated by *urlSignature.tokenExpiryName* is absent from *query*, or if the supplied *token_expiry* value has expired, or if the supplied *token_expiry* is malformed, the 5GMSd AS shall respond with a *403 (Forbidden)* error response message and terminate further processing of the M4d request.

3) The 5GMSd AS shall compute the authentication token according to the *token* production specified above using the requesting UE's public IP address as the value of *ue_public_ip_address* if required by *urlSignature.useIPAddress* being set to *true*. After applying "base64url" encoding, the 5GMSd AS shall compare this with the value supplied in the URL *query* parameter whose name is *urlSignature.tokenName*. If the two values differ, the 5GMSd AS shall respond with a *403 (Forbidden)* error response message and terminate further processing of the M4d request.

4) Otherwise, the presented authentication token is valid. The 5GMSd AS shall either return the requested transport resource in a *200 (OK)* response message (if it is able to serve that transport resource), or else return an appropriate error response, such as *404 (Not Found)* or *503 (Service Unavailable)*.

7.6.4.6 Geofencing

The 5GMSd Application Provider may wish to limit access to the media content it makes available at reference point M2d to UEs located in certain geographical zones. Geofencing is used to configure the zone from which content is accessible.

The *geoFencing.locatorType* shall be set to one of the controlled term identifiers in the first column of table B.1-1 of TS 26.510 [56] and each member of the *geoFencing.locators* array in the distribution configuration shall then be set as specified in the third column of that table.

7.6.4.7 Service chaining

The 5GMSd Application Provider may chain content hosting services by provisioning two or more Content Hosting Configurations as described in clause 5.2.8.2 of TS 26.510 [56] where:

- At least one Content Hosting Configuration ingests media content from the 5GMSd Application Provider at reference point M2d.

- Additional Content Hosting Configurations ingest media content from an upstream 5GMSd AS at reference point M10d where the *IngestConfiguration.baseURL* property of each Content Hosting Configuration corresponds to the *DistributionConfiguration.baseURL* property of an already provisioned Content Hosting Configuration in the upstream 5GMSd AS.

7.6.4.8 Service location deployment

Reference point M4d service locations associated with distribution configurations within the Content Hosting Configuration are deployed within the 5GMS System at the discretion of the 5GMSd AF, taking into account any deployment affinity requirements as specified below.

The *DistributionConfiguration.affinityGroup* property within the Content Hosting Configuration (see clause 5.2.8.2 of TS 26.510 [56]) may be defined by the 5GMSd Application Provider to guide deployment of service locations within the 5GMSd AS according to the following:

- The scope of the *affinityGroup* property applies only to those distribution configurations defined within a single Content Hosting Configuration. Service locations associated with distribution configurations with the same *affinityGroup* value but belonging to different Content Hosting Configurations may or may not be deployed together within the 5GMSd AS at the discretion of the 5GMSd AF.

- When any two distribution configurations have the same *affinityGroup* value or when this property is not defined anywhere within a Content Hosting Configuration, the deployment of service locations within the 5GMSd AS is at the discretion of the 5GMSd AF.

- When any two distribution configurations defined within a single Content Hosting Configuration have different *affinityGroup* values (including those that are not defined), a service location associated with one distribution

configuration should not be deployed together with (e.g., at the same physical location) the service location associated with the other distribution configuration.

7.6A Content Publishing Provisioning API

7.6A.1 Overview

The API used by the 5GMSu Application Provider at reference point M1u to create and manipulate the 5GMSu AS Content Publishing Configuration associated with a particular uplink media streaming Provisioning Session in the 5GMSu AF is specified in clause 8.9 of TS 26.510 [56].

Within a Content Publishing Configuration, one or more contribution configurations may be defined where each may specify different caching, purging and content preparation behaviours for media resources contributed at reference point M4u or M10u. The Content Publishing Configuration may further specify, through the declaration of affinity groups, how reference point M4u or M10u service locations associated with each contribution configuration are deployed in the 5GMS System.

7.7 Consumption Reporting Provisioning API

7.7.1 Overview

The API used by the 5GMSd Application Provider at reference point M1d to configure the Consumption Reporting Procedure associated with a particular downlink media streaming Provisioning Session in the 5GMSd AF is specified in clause 8.12 of TS 26.510 [56].

7.7.2 Void

7.7.3 Void

7.8 Metrics Reporting Provisioning API

7.8.1 Overview

The API used by the 5GMS Application Provider at reference point M1 to instantiate and manipulate Metrics Reporting Configurations associated with a particular downlink or uplink media streaming Provisioning Session in the 5GMS AF is specified in clause 8.11 of TS 26.510 [56].

In the case of downlink media streaming, the metrics scheme indicated in the *scheme* property of the Metrics Reporting Configuration shall be *urn:3GPP:ns:PSS:DASH:QM10* and the QoE metrics (if any) listed in the *metrics* property shall be term identifiers from the vocabulary specified in clause E.2.1.

In the case of downlink DASH media streaming, any of the CMCD metrics schemes listed in table 7.8.1-1 below may be provisioned, each one in the *scheme* property of a separate Metrics Reporting Configuration. The metrics (if any) listed in the *metrics* property shall be term identifiers from the vocabulary specified in clause E.2.5.

Table 7.8.1-1: CMCD metrics scheme identifiers

CMCD information class	Applicability	Metrics scheme identifier
CMCD per-session information	Downlink	*urn:3gpp:5gms:metrics:common-media-client-data:session*
CMCD per-object information	Downlink	*urn:3gpp:5gms:metrics:common-media-client-data:object*
CMCD per-request information	Downlink	*urn:3gpp:5gms:metrics:common-media-client-data:request*
CMCD status information	Downlink	*urn:3gpp:5gms:metrics:common-media-client-data:status*

Metrics related to Virtual Reality media, as specified in TS 26.118 [42] clause 9.3, may be listed in the *metrics* property of a metrics configuration when the *scheme* property indicates that metrics scheme. These shall be term identifiers from the vocabulary specified in clause E.2.2.

No standardised metrics schemes are defined in the present document for uplink media streaming.

7.8.2 Void

7.8.3 Void

7.9 Policy Templates Provisioning API

7.9.1 Overview

The API used by the 5GMS Application Provider at reference point M1 to instantiate and manipulate Policy Templates associated with a particular downlink or uplink media streaming Provisioning Session in the 5GMS AF is specified in clause 8.7 of TS 26.510 [56]. These Policy Template may subsequently be applied to downlink or uplink media streaming sessions belonging to that Application Provider using the Dynamic Policies API specified in clause 11.5. A Policy Template is used to specify the traffic shaping and charging policies to be applied to these media streaming sessions.

7.9.2 Void

7.9.3 Void

7.10 Edge Resources Provisioning API

7.10.1 General

The API used by the 5GMS Application Provider at reference point M1 to provision edge resource usage for media streaming sessions associated with a particular downlink or uplink media streaming Provisioning Session in the

5GMS AF is specified in clause 8.6 of TS 26.510 [56]. The information serves as a template to select or instantiate an appropriate 5GMS AS EAS instance that will support the media streaming session at reference point M4.

7.10.2 Void

7.10.3 Void

7.11 Event Data Processing Provisioning API

7.11.1 General

The API used by the 5GMS Application Provider at reference point M1 to instantiate and manipulate Event Data Processing Configurations associated with a particular downlink or uplink media streaming Provisioning Session in the Data Collection AF instantiated in the 5GMS AF is specified in clause 8.13 of TS 26.510 [56]. Each such configuration comprises processing rules and parameters expressed by Data Access Profiles each of which defines a level of access by Event consumers to the UE data collected by the Data Collection AF.

7.11.2 Void

7.11.3 Void

8 Media ingest and publish (M2 and M10) protocols

8.1 General

The set of content protocols supported by the 5GMS AS is listed in table 8.1-1 below:

Table 8.1-1: Supported content protocols

Description	Term identifier	Clause
Content ingest protocols at reference point M2d or M10d		
HTTP pull-based content ingest protocol	*urn:3gpp:5gms:content-protocol:http-pull* or *urn:3gpp:5gms:content-protocol:http-pull-ingest* (see NOTE)	8.2
DASH-IF push-based content ingest protocol	*http://dashif.org/ingest/v1.2/interface-1* or *http://dashif.org/ingest/v1.2/interface-2* or *urn:3gpp:5gms:content-protocol:dash-if-ingest* (see NOTE)	8.3
HTTP low-latency pull-based content ingest protocol	*urn:3gpp:5gms:content-protocol:http-ll-pull*	8.4
Content egest protocols at reference point M2u or M10u		
HTTP pull-based content egest protocol	*urn:3gpp:5gms:content-protocol:http-pull*	8.5
DASH-IF push-based content egest protocol	*http://dashif.org/ingest/v1.2/interface-1* or *http://dashif.org/ingest/v1.2/interface-2*	8.6
HTTP low-latency pull-based content egest protocol	*urn:3gpp:5gms:content-protocol:http-ll-pull*	8.7
NOTE: Term identifier deprecated in this version of the present document.		

CMCD information conveyed to the 5GMSd AS at reference point M4d (see clause 10.5) shall not be included in requests to the 5GMSd Application Provider at reference point M2d.

8.2 HTTP pull-based content ingest protocol

The following provisions shall apply if *IngestConfiguration.protocol* is set to *urn:3gpp:5gms:content-protocol:http-pull* or to the deprecated value *urn:3gpp:5gms:content-protocol:http-pull* in the Content Hosting Configuration:

- Media resources shall be ingested by the 5GMSd AS from the 5GMSd Application Provider or from another 5GMSd AS using HTTP [25].

NOTE 0: Any supported HTTP protocol version may be used for HTTP pull-based content ingest at reference point M2d or M10d.

- The *IngestConfiguration.mode* property shall be set to *PULL*, indicating that a pull-based protocol is used.
- The *IngestConfiguration.baseURL* property shall point at the 5GMSd Application Provider's origin server or the *DistributionConfiguration.baseURL* property of another Content Hosting Configuration, as specified in table 8.8.3.1-1 of TS 26.510 [56], and may indicate the use of HTTPS [30].

When the 5GMSd AS receives a request for a media resource at reference point M4d service location that cannot be satisfied from its content cache, the request shall be transformed into a corresponding HTTP `GET` request directed to the 5GMSd Application Provider's origin server via interface M2d or to another 5GMSd AS via reference point M10d as follows:

1. The prefix of the request URL indicated in the *DistributionConfiguration.baseURL* of the applicable Content Hosting Configuration is replaced with that of the corresponding *IngestConfiguration.baseURL*.

NOTE 1: It is the responsibility of the 5GMSd AF to assign unique M4d and M10d base URLs to each provisioned Content Hosting Configuration so as to ensure that this substitution is unambiguous.

2. The path rewrite rules (if provisioned in *DistributionConfiguration.pathRewriteRules*) are applied in strict order to the remainder of the request URL (i.e., the path segments following *DistributionConfiguration.baseURL*). The *requestPathPattern* of the first matching path rewrite rule is replaced with the corresponding *mappedPath*.

In the case where the 5GMSd Application Provider's origin server issues an HTTP *3xx* redirect at reference point M2d pointing to another location, or an upstream 5GMSd AS issues such a redirect at reference point M10d, the 5GMSd AS shall issue an equivalent HTTP redirect to the Media Player via reference point M4d whose location is a dynamically generated M4d endpoint. Requests to this location shall be rewritten by the 5GMSd AS to the target location of the M2d or M10d redirection, as appropriate.

NOTE 2: This explicit handling of HTTP redirects received by the 5GMSd AS at reference point M2d or M10d ensures that it is not bypassed by the Media Player. The general concept underlying this is commonly referred to as a "reverse mapping rule" by HTTP reverse proxies.

8.3 DASH-IF push-based content ingest protocol

The following provisions shall apply if *IngestConfiguration.protocol* is set to *http://dashif.org/ingest/v1.2/interface-1* or *http://dashif.org/ingest/v1.2/interface-2* or to the deprecated value *urn:3gpp:5gms:content-protocol:dash-if-ingest* in the Content Hosting Configuration:

- Media resources shall be published by the 5GMSd Application Provider to the 5GMSd AS as specified by the DASH-IF Live Media Ingest specification [3].

NOTE: The protocol in [3] is specified for use with HTTP/1.1 [24] only.

- The *IngestConfiguration.mode* property shall be set to *PUSH*, indicating that a push-based protocol is used.
- The *IngestConfiguration.baseURL* property shall be set by the 5GMSd AF to the base URL that is to be used by the 5GMSd Application Provider or by an upstream 5GMSd AS to upload the DASH segments and MPD(s) to the 5GMSd AS at reference point M2d or M10d respectively.

8.4 HTTP low-latency pull-based content ingest protocol

The provisions specified in clause 8.2 shall apply if *IngestConfiguration.protocol* is set to *urn:3gpp:5gms:content-protocol:http-ll-pull*.

In addition, if HTTP/1.1 [24] is used by at reference point M2d or M10d:

- The requesting 5GMSd AS shall make partially received media segments available immediately for retrieval by 5GMS Clients at reference point M4d or M10d instead of waiting until the full segment is received.
- The 5GMSd Application Provider should use HTTP chunked transfer coding as defined in section 7.1 of [24]. In this case, the requesting 5GMSd AS shall accept chunked HTTP/1.1 response messages and shall make partially received media segments (i.e., HTTP Chunks) available immediately for retrieval by 5GMS Clients at reference point M4d or M10d instead of waiting until the full segment is received.
- If the DASH-IF Low Latency mode as defined in [63] is used, then the content is packaged as a series of CMAF Segments [40]. Further, each CMAF Segment is typically subdivided into one or more multiple CMAF Chunks to support low-latency content generation. According to the DASH-IF Live Media Ingest specification [3], each HTTP Chunk should contain at most one CMAF Chunk in order to minimise the latency.

NOTE: Usage of HTTP/2.0 and HTTP/3 at reference points M2d and M10d is for future study.

8.5 HTTP pull-based content egest protocol

If *EgestConfiguration.protocol* is set to *urn:3gpp:5gms:content-protocol:http-pull-egest* in the Content Publishing Configuration, media resources shall be retrieved by the 5GMSu Application Provider from the 5GMSu AS at reference point M2u or by an upstream 5GMSu AS at reference point M10u using HTTP [25]. Media segments contributed to the 5GMSu AS by the 5GMSu Client shall be processed according to the Content Preparation Template(s) specified in the corresponding Content Publishing Configuration (if any) prior to making them available at reference point M2u or M10u.

In this case:

- The *EgestConfiguration.mode* property shall be set to *PULL*, indicating that a pull-based protocol is used.
- The *EgestConfiguration.baseURL* property shall be set by the 5GMSu AF to the base URL on the 5GMSu AS where it will publish media segments, presentation manifests and metadata for retrieval by the 5GMSu Application Provider at reference point M2u or by an upstream 5GMSu AS at reference point M10u.
- The *EgestConfiguration.entryPoint.relativePath* property shall point at a Media Entry Point document below this base URL, as specified in table 8.9.3.1-1 of TS 26.510 [56], and may indicate the use of HTTPS [30]. This document describes the location of media content and associated metadata exposed by the 5GMSu AS at reference point M2u or M10u which are expected to be pulled by the 5GMSu Application Provider or by an upstream 5GMSu AS respectively.

In the absence of content preparation, the 5GMSu AS shall publish media resources by replacing the prefix *Contribution Configuration.baseURL* of its URL at M4u with that of the corresponding *EgestConfiguration.baseURL*.

8.6 DASH-IF push-based content egest protocol

If *EgestConfiguration.protocol* is set to *http://dashif.org/ingest/v1.2/interface-1* or *http://dashif.org/ingest/v1.2/interface-2* in the Content Publishing Configuration, media resources shall be published by the 5GMSu AS to the 5GMSu Application Provider at reference point M2u or to an upstream 5GMSu AS at reference point M10u as specified in the DASH-IF Live Media Ingest specification [3]. Media segments contributed to the 5GMSu AS by the 5GMSu Client shall be processed according to the Content Preparation Template(s) specified in the corresponding Content Publishing Configuration (if any) prior to publishing them at reference point M2u or M10u.

NOTE 1: The protocol in [3] is specified for use with HTTP/1.1 [24] only.

NOTE 2: A 5GMSu AS implementation that advertises support for the egest of content at reference point M2u or M10u using interface 2 of the DASH-IF Live Media Ingest specification [3] is required to produce a suitable DASH presentation manifest.

In this case:

- The *EgestConfiguration.mode* property shall be set to *PUSH*, indicating that a push-based protocol is used.
- The *EgestConfiguration.baseURL* property shall be set by the 5GMSu Application Provider to the base URL that is to be used by the 5GMSu AS to upload media segments and presentation manifests to the 5GMSu Application Provider at reference point M2u or to an upstream 5GMSu AS at reference point M10u.

If the 5GMSu Application Provider has provisioned an egest Media Entry Point, and if such document has been contributed to or produced by 5GMSu AS, the 5GMSu AS shall publish this document to the URL formed by the concatenation of *EgestConfiguration.baseURL* with *EgestConfiguration.entryPoint.relativePath*, as specified in table 8.9.3.1-1 of TS 26.510 [56]. This URL may indicate the use of HTTPS [30].

In the absence of any content preparation, each media resource uploaded at reference point M4u shall be published to the 5GMSu Application Provider at the URL formed by replacing the prefix *ContributionConfiguration.baseURL* of its URL at M4u with that of the corresponding *EgestConfiguration.baseURL*.

8.7 HTTP low-latency pull-based content egest protocol

The following provisions shall apply if *EgestConfiguration.protocol* is set to *urn:3gpp:5gms:content-protocol:http-ll-pull* the following provisions shall apply.

The content shall be packaged as a series of CMAF Segments [40]. Each CMAF Segment shall be subdivided into multiple one or more CMAF Chunks.

In addition:

- If HTTP/1.1 [24] is used at reference point M2u or M10u, partially available media segments may be accessed by the 5GMSu Application Provider using an HTTP byte range request, as specified in section 14 of RFC 9110 [25]. If the 5GMS Application Provider makes a byte-range request for a partially available media segment (the first media segment it retrieves) and the `first-pos` of that range is non-zero and the 5GMS Application Provider is expecting an aggregating response, then the 5GMS Application Provider should signal that expectation following the convention of IETF RFC 8673 [61]. Specifically, it should use a `last-pos` value of 9007199254740991. In this case, the 5GMSu AS is required to respond with a *206 (Partial Content)* HTTP response without a `Content-length` response header instead of waiting for the end of the segment and responding with a *200 (OK)* HTTP response code.

8.8 Key exchange for encrypted content

If the 5GMSd AS supports content encoding and packaging, but the DRM server (including License and Key Servers) is deployed externally in the 5GMSd Application Provider domain, then the 5GMSd AS needs to communicate with the License Server for content encoding and packaging.

In order to support exchange of public keys between the Encryptor/Packager on the 5GMSd AS and the external License Server and Authorization Server via reference point M2d, the 5GMSd AS shall support the Content Protection Information Exchange Format (CPIX) as specified in ETSI TS 103 799 [75] at reference point M2d.

9 Application Server Configuration (M3) APIs

9.1 General

This clause defines the *Mas_Configuration* API invoked by a 5GMS AF on a 5GMS AS instance at reference point M3 to configure downlink or uplink 5G Media Streaming services in that 5GMS AS instance. The OpenAPI definitions for this API are specified in clause C.3A.

9.2 Server Certificates Configuration API

9.2.1 Overview

The Server Certificates Configuration API is used to configure X.509 [8] server certificates in a 5GMS AS instance that may then be referenced by a Content Hosting Configuration or by a Content Publishing Configuration in that 5GMS AS instance and subsequently presented by the 5GMS AS instance to 5GMS Clients connecting to it at reference point M4 using Transport Layer Security [30]. All Server Certificate resources are held by the 5GMS AS instance in a flat collection. The resource identifier for each Server Certificate resource is nominated by the 5GMS AF and need not be the same as that used at reference point M1.

9.2.2 Resource structure

The Server Certificates Configuration API is exposed by the 5GMS AS through the following URL base path:

{apiRoot}`/3gpp-mas-configuration/`*{apiVersion}*`/certificates/`

Table 9.2.2-1 specifies the operations and the corresponding HTTP methods that are supported by this API. In each case, the sub-resource path (if any) specified in the second column shall be appended to the URL base path.

Table 9.2.2-1: Operations supported by the Server Certificates Configuration API

Operation	Sub-resource path	Allowed HTTP method(s)	Description
Enumerate Server Certificates		`GET`	Used by the 5GMS AF to retrieve a list of Server Certificate resources currently configured in a 5GMS AS instance. The request message body shall be empty.
Create Server Certificate	*{afResourceId}*	`POST`	Used by the 5GMS AF to create a new Server Certificate resource in a 5GMS AS instance. The *{afResourceId}* is nominated by the 5GMS AF in this operation and shall be unique in the collection of Server Certificates in a given 5GMS AS instance. The request message body shall be a PEM-encoded X.509 certificate [8] bundle that includes the private key.
Update Server Certificate		`PUT`	Used by the 5GMS AF to replace an existing Server Certificate resource *{afResourceId}* in a 5GMS AS instance. The request message body shall be a replacement PEM-encoded X.509 certificate [8] bundle that includes the private key.
Destroy Server Certificate		`DELETE`	Removes the Server Certificate resource *{afResourceId}* from the collection. Attempting to destroy a Server Certificate resource that is referenced by a Content Hosting Configuration or Content Publishing Configuration is an error.
NOTE: The Server Certificate resource identifier *{certificateId}* differs from the serial number of the X.509 certificate.			

9.2.3 Data model

Server Certificate resource representations shall be PEM-encoded X.509 certificate [8] bundles including their private key.

9.3 Content Preparation Templates Configuration API

9.3.1 Overview

The Content Preparation Templates Configuration API is used to configure Content Preparation Template resources in a 5GMS AS instance that may then be referenced by a Content Hosting Configuration or Content Publishing Configuration in that 5GMS AS instance. All Content Preparation Template resources are held by the 5GMS AS instance in a flat collection. The resource identifier for each Content Preparation Template resource is nominated by the 5GMS AF and need not be the same as that used at reference point M1.

9.3.2 Resource structure

The Content Preparation Templates Configuration API is exposed by the 5GMS AS through the following URL base path:

{apiRoot}`/3gpp-mas-configuration/`*{apiVersion}*`/content-preparation-templates/`

Table 9.3.2-1 specifies the operations and the corresponding HTTP methods that are supported by this API. In each case, the sub-resource path (if any) specified in the second column shall be appended to the URL base path.

Table 9.3.2-1: Operations supported by the Content Preparation Templates Configuration API

Operation	Sub-resource path	Allowed HTTP method(s)	Description
Enumerate Content Preparation Templates		`GET`	Used by the 5GMS AF to retrieve a list of Content Preparation Template resources currently configured in a 5GMS AS instance. The request message body shall be empty.
Create Content Preparation Template	*{afResourceId}*	`POST`	Used by the 5GMS AF to create a new Content Preparation Template resource in a 5GMS AS instance. The *{afResourceId}* is nominated by the 5GMS AF in this operation and shall be unique in the collection of Content Preparation Templates in a given 5GMS AS instance. The request message body shall be a Content Preparation Template document.
Update Content Preparation Template		`PUT`	Used by the 5GMS AF to replace an existing Content Preparation Template resource *{afResourceId}* in a 5GMS AS instance. The request message body shall be a replacement Content Preparation Template document.
Destroy Content Preparation Template		`DELETE`	Removes the Content Preparation Template resource *{afResourceId}* from the collection. The request message body shall be empty. Attempting to destroy a Content Preparation Template resource that is referenced by a Content Hosting Configuration or Content Publishing Configuration is an error.

9.3.3 Data model

The representation of Content Preparation Template resources shall be the same as that specified in clause 8.5.3 of TS 26.510 [56].

9.4 Content Hosting Configuration API

9.4.1 Overview

The Content Hosting Configuration API is used to configure Content Hosting Configuration resources in a 5GMS AS instance. All Content Hosting Configuration resources are held by the 5GMS AS instance in a flat collection. The resource identifier for each Content Hosting Configuration resource is nominated by the 5GMS AF and need not be the same as that used at reference point M1.

9.4.2 Resource structure

The Content Hosting Configuration API is exposed by the 5GMS AS through the following URL base path:

{apiRoot}/3gpp-mas-configuration/*{apiVersion}*/content-hosting-configurations/

Table 9.4.2-1 specifies the operations and the corresponding HTTP methods that are supported by this API. In each case, the sub-resource path (if any) specified in the second column shall be appended to the URL base path.

Table 9.4.2-1: Operations supported by the Content Hosting Configuration API

Operation	Sub-resource path	Allowed HTTP method(s)	Description
Enumerate Content Hosting Configurations		GET	Used by the 5GMSd AF to retrieve a list of Content Hosting Configuration resources currently configured in a 5GMSd AS instance. The request message body shall be empty.
Create Content Hosting Configuration	*{afResourceId}*	POST	Used by the 5GMSd AF to create a new Content Hosting Configuration resource in a 5GMSd AS instance. The *{afResourceId}* is nominated by the 5GMSd AF in this operation and shall be unique in the collection of Content Hosting Configurations in a given 5GMSd AS instance. The request message body shall be a Content Hosting Configuration document.
Update Content Hosting Configuration		PUT	Used by the 5GMSd AF to replace an existing Content Hosting Configuration resource *{afResourceId}* in a 5GMSd AS instance. The request message body shall be a replacement Content Hosting Configuration document.
Destroy Content Hosting Configuration		DELETE	Used by the 5GMS AF to remove the Content Hosting Configuration resource *{afResourceId}* from the collection. The request message body shall be empty. Attempting to destroy a Content Hosting Configuration resource that is referenced by a Content Hosting Configuration or Content Publishing Configuration is an error.
(De)activate Content Hosting Configuration	*{afResourceId}*/active	POST	Used by the 5GMSd AF to (de)activate Content Hosting Configuration *{afResourceId}* in the 5GMSd AS instance.
Interrogate Content Hosting Configuration state		GET	Used by the 5GMSd AF to interrogate the state of Content Hosting Configuration *{afResourceId}* in the 5GMSd AS instance.
Purge Content Hosting cache	*{afResourceId}*/purge	POST	Used by the 5GMSd AF to purge one or more media resources associated with Content Hosting Configuration *{afResourceId}* from the 5GMSd AS content cache. The request message body shall specify the set of media resource URLs to be purged.

9.4.3 Data model

The representation of the Content Hosting Configuration resource used by the 5GMSd AF to configure the 5GMSd AS at reference point M3d shall be the same as that specified in clause 8.8.3.1 of TS 26.510 [56] except that:

- *externalServiceId* is an additional read/write property of *ApplicationServerContentHostingConfiguration* populated by the 5GMSd AF from the parent Provisioning Session.
- *canonicalDomainName* is a read/write property of *ApplicationServerDistributionConfiguration* set by the 5GMS AF.

9.5 Content Publishing Configuration API

9.5.1 Overview

The Content Publishing Configuration API is used to configure Content Publishing Configuration resources in a 5GMS AS instance. All Content Publishing Configuration resources are held by the 5GMS AS instance in a flat collection. The resource identifier for each Content Publishing Configuration resource is nominated by the 5GMS AF and need not be the same as that used at reference point M1.

9.5.2 Resource structure

The Content Publishing Configuration API is exposed by the 5GMS AS through the following URL base path:

{apiRoot}/3gpp-mas-configuration/*{apiVersion}*/content-publishing-configurations/

Table 9.5.2-1 specifies the operations and the corresponding HTTP methods that are supported by this API. In each case, the sub-resource path (if any) specified in the second column shall be appended to the URL base path.

Table 9.5.2-1: Operations supported by the Content Publishing Configuration API

Operation	Sub-resource path	Allowed HTTP method(s)	Description
Enumerate Content Publishing Configurations		GET	Used by the 5GMSu AF to retrieve a list of Content Publishing Configuration resources currently configured in a 5GMSu AS instance. The request message body shall be empty.
Create Content Publishing Configuration	*{afResourceId}*	POST	Used by the 5GMSu AF to create a new Content Publishing Configuration resource in a 5GMSu AS instance. The *{afResourceId}* is nominated by the 5GMSu AF in this operation and shall be unique in the collection of Content Publishing Configurations in a given 5GMSu AS instance. The request message body shall be a Content Publishing Configuration document.
Update Content Publishing Configuration		PUT	Used by the 5GMSu AF to replace an existing Content Publishing Configuration resource *{afResourceId}* in a 5GMSu AS instance. The request message body shall be a replacement Content Publishing Configuration document.
Destroy Content Publishing Configuration		DELETE	Used by the 5GMSu AF to remove the Content Publishing Configuration resource *{afResourceId}* from the collection. The request message body shall be empty. Attempting to destroy a Content Publishing Configuration resource that is referenced by a Content Publishing Configuration or Content Publishing Configuration is an error.
(De)activate Content Publishing Configuration	*{afResourceId}*/active	POST	Used by the 5GMSd AF to (de)activate Content Publishing Configuration *{afResourceId}* in the 5GMSd AS instance.
Interrogate Content Publishing Configuration state		GET	Used by the 5GMSd AF to interrogate the state of Content Publishing Configuration *{afResourceId}* in the 5GMSd AS instance.
Purge Content Publishing cache	*{afResourceId}*/purge	POST	Used by the 5GMSu AF to purge one or more media resources associated with the Content Publishing Configuration *{afResourceId}* from the 5GMSu AS content cache. The request message body shall specify the set of media resource URLs to be purged.

9.5.3 Data model

The representation of the Content Publishing Configuration resource used by the 5GMSu AF to configure the 5GMSu AS at reference point M3u shall be the same as that specified in clause 8.9.3.1 of TS 26.510 [56] except that:

- *externalServiceId* is an additional read/write property of *ApplicationServerContentPublishingConfiguration* populated by the 5GMSu AF from the parent Provisioning Session.

- *canonicalDomainName* is a read/write property of *ApplicationServerContributionConfiguration* set by the 5GMS AF.

10 Media Streaming (M4) interface

10.1 General

This clause specifies the interface for downlink and uplink media streaming at reference point M4 using different distribution formats and protocols. TS 26.511 [35] defines several media codecs and distribution formats for 5G Media Streaming. It also provides requirements and recommendations for the support of these media codecs and formats in profiles specific to 5G Media Streaming. However, 5GMS is not restricted to the media profiles and distribution formats defined in TS 26.511 [35]. For example, any CMAF media profile that integrates with the APIs specified in the present document may be used for media streaming at reference point M4.

A media resource shall be made available at reference point M4 as one or more transport resources where each transport resource shall be uniquely addressable.

10.1A Media delivery session identification

All media requests addressed by the Media Stream Handler (Media Player or Media Streamer) to the 5GMS AS at reference point M4 shall cite a media delivery session identifier using the HTTP header specified in clause 6.2.3.6. The value of this identifier shall be different for every media streaming session. When a media delivery session uses multiple service locations, the value of this identifier shall be the same regardless of the service location used.

10.2 DASH distribution

In the case of DASH distribution, M4d is relevant for the distribution as shown in figure 10.2-1.

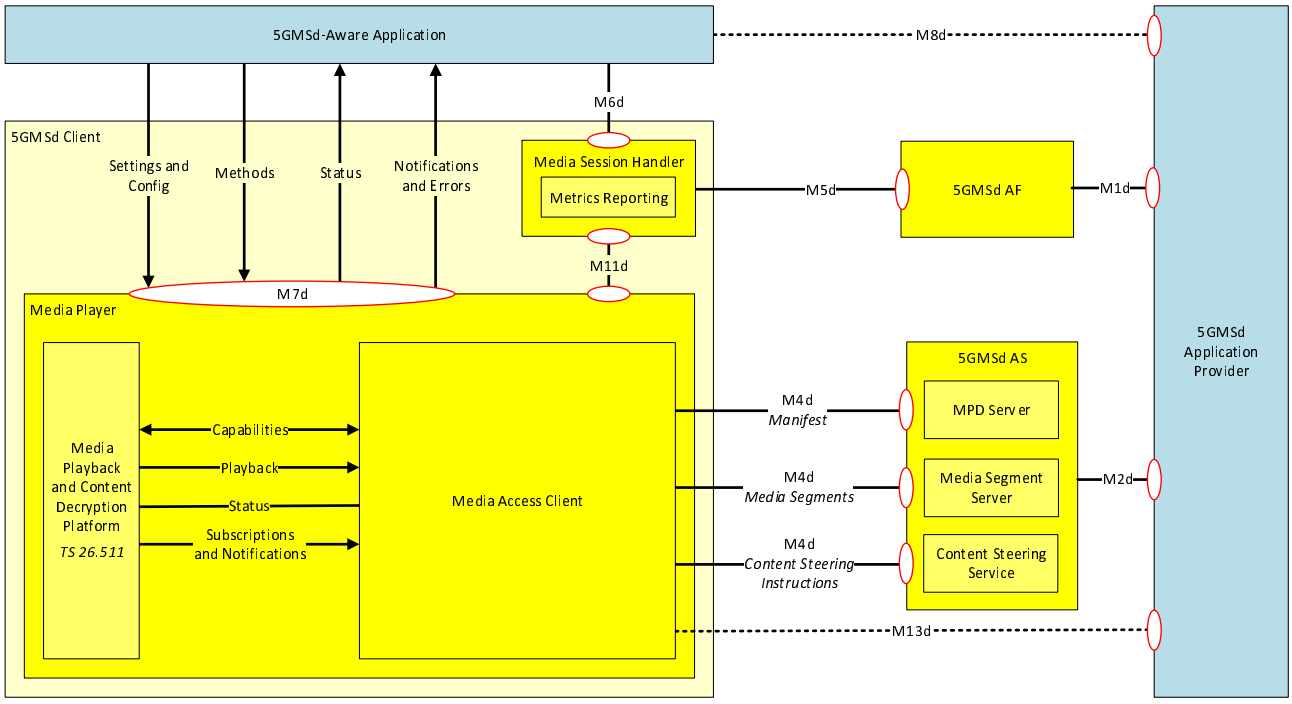


Figure 10.2-1: M4d usage for DASH distribution

For DASH-based distribution according to TS 26.247 [4] and ISO/IEC 23009-1 [32], two main formats are of relevance:

1) The Media Presentation Description (MPD) that is processed in the DASH Access Client.

2) The Segment formats that are passed through the DASH Access Client and processed in the Media Playback and Content Decryption Platform. Note that the DASH Access Client may parse Segments to extract, for example, In-band Events or producer reference times.

Other resources may be referenced in the MPD. Examples include:

- Service locations in the form of **`BaseURL`** elements from which Segments can be downloaded.
- Content steering instructions provided by a content steering service.
- DRM-related information.

The Segment formats for DASH Streaming in the context of 5G Media Streaming are defined in TS 26.511 [35] based on the CMAF encapsulation. The DASH Access Client downloads the Segments from the 5GMSd AS based on the instructions in the MPD and the instructions from the 5GMSd-Aware Application through M7d (see clause 13 for details).

The interface between the DASH Access Client and the Media Playback and Content Decryption Platform as well as the 5GMSd Client requirements for media codecs are documented in TS 26.511 [35].

The following requirements apply at reference point M4d:

1) The Media Presentation Description (MPD) and Segments shall conform to an MPD according to ISO/IEC 23009-1 [32] or TS 26.247 [4].
2) The Segment formats should conform to CMAF addressable resources as well as to the requirements in TS 26.511 [35].
3) The Media Presentation should conform to the 5G Media Streaming DASH Interoperability Point as defined in clause 7.3.11 of TS 26.247 [4].

A 5GMSd Client shall support the 5G Media Streaming DASH Interoperability Point as defined in TS 26.247 [4], clause 7.3.11. A 5GMSd Client may support additional DASH profiles and interoperability points.

The MPD may contain a one or several **`ServiceDescription`** elements that include operational parameters. The MPD may also include multiple configurations for the media (different codecs, different content protection, different resolutions, etc.), for example for playback under different operating policies. For more details refer to annex G.

The handling of this information within the 5GMSd Client is documented in clause 13.2.

If the media segment formats conform to CMAF addressable resources as defined ISO/IEC 23000-19 [27], the same CMAF content may then be provided for DASH and HLS. In order to support common deployment, the media segment content should conform to CTA-5005-A [62].

10.3 HTTP low-latency content distribution

When low-latency distribution of media content at reference point M4d is provisioned, then the following provisions shall apply:

- The 5GMSd AS shall make partially received media segments available immediately for retrieval by 5GMS Clients at reference point M4d instead of waiting until the full segment is received.
- the 5GMSd AS should use HTTP chunked transfer coding as defined in section 7.1 of [24]. In this case, the 5GMSd client shall accept chunked HTTP/1.1 response messages.
- If the DASH-IF Low Latency mode as defined in [63] is used as identified in the MPD by the profile indicator *http://www.dashif.org/guidelines/low-latency-live-v5*, then the content is packaged as a series of CMAF Segments [40]. Further, each CMAF Segment is typically subdivided into one or more multiple CMAF Chunks to support low-latency content generation. According to the DASH-IF Live Media Ingest specification [3], each HTTP Chunk should contain at most one CMAF Chunk in order to minimise the latency.
- At reference point M4d, the Media Player may access partially available media segments using an HTTP byte range request, as specified in section 14 of RFC 9110 [25]. (For details see for example [63] on Resynchronization Points.) If the Media Player makes a byte-range request for a partially available media segment (the first media segment it retrieves) and the `first-pos` of that range is non-zero and the Media Player is expecting an aggregating response, then the Media Player should signal that expectation following the convention of IETF RFC 8673 [61]. Specifically, it should use a `last-pos` value of 9007199254740991. In this case, the 5GMSd AS is required to respond with a *206 (Partial Content)* HTTP response without a `Content-`

`length` response header instead of waiting for the end of the segment and responding with a *200 (OK)* HTTP response code.

10.3A Content distribution from multiple service locations

10.3A.1 General

This clause extends clauses 10.2 and 10.3 to provide Media Access Client capabilities that allow for content distribution using multiple service locations exposed by the 5GMSd AS at reference point M4d. These capabilities may be used independently or in combination.

Necessary information required by the Media Access Client to use multiple service locations exposed by the 5GMSd AS at reference point M4d during a media streaming session may be provided within a Media Player Entry (or a document pointed to by a Media Player Entry) and/or by a content steering service. Service locations provided in the Media Player Entry and/or by a content steering service may be distinguishable and identifiable via their base URLs.

A Media Access Client that supports content distributed from multiple service locations should support the following:

- Identification of the base URL(s) of the service location(s) to be used and the construction of the absolute URL(s) for the transport resources that apply to the media resource(s) to be acquired.
- Acquiring transport resources associated with a media resource from one or more service locations.
- Recovering or reconstituting media resources from the acquired transport resources and making them available to the Media Playback and Content Decryption Platform (as specified in TS 26.511 [35]) for immediate or delayed consumption.

Additional Media Access Client capabilities, depending on the method(s) used to acquire media resources from one or more service locations, are provided in the following clauses.

A Media Access Client accessing multiple service locations should take into account that the throughput estimation for one service location may not accurately predict the throughput of another service location.

10.3A.2 Media Access Client capabilities to support switching between multiple service locations during content distribution

The Media Access Client of the 5GMSd Client may have the capability to switch between service locations during a media streaming session.

When switching between multiple service locations, the Media Access Client of the 5GMSd Client should have the ability to select one or more service locations to use when acquiring a transport resource. For example, the service location(s) selected have the highest priority based on the contents of the Media Player Entry or signalling from a content steering service; or the Media Access Client selects the service location(s) using internal logic.

In case a previous request using a particular service location fails or the Media Access Client decides to switch to a new service location, another service location may be selected to acquire media resources.

10.3A.3 Media Access Client capabilities to support concurrent use of multiple service locations for content distribution

The Media Access Client of the 5GMSd Client may have the ability to acquire media resources from more than one service location in parallel (i.e., through simultaneous use of multiple service locations). When using multiple service locations concurrently, the Media Access Client of the 5GMSd Client should have the additional capabilities:

- The ability to select more than one service location to be used to acquire transport resources. For example, the service locations selected have the highest priority for usage based on the contents of the Media Player Entry or signalling from a content steering service; the Media Access Client selects the service locations using internal logic; or all the identified service locations are used. In another example, the video transport resources and the audio transport resources may be offered from different service locations, and the Media Player needs to access both service locations in parallel to provide a full media experience.

- The ability to acquire more than one transport resource simultaneously from the identified service locations, where each may be the original or a transformed (e.g., encoded) representation of the media resource. This may include downloading (either partially or in full) transport resources from more than one reference point M4d service location simultaneously.

10.3A.4 Media Access Client capabilities to support content steering for content distribution

In the case when content steering is used, the Media Access Client of the 5GMSd Client may communicate via reference point M4d with a content steering service provisioned within the 5GMSd AS to obtain information about which service location(s) to use to acquire transport resources at reference point M4d. Content steering information may be provided within the Media Player Entry (e.g., a DASH MPD) or in a document pointed to by the Media Player Entry.

When obtaining content steering information from the content steering service, the Media Access Client shall use a protocol appropriate to that service.

10.4 Contribution protocols

10.4.1 General

The contribution protocols supported by the 5GMSu AS at reference point M4u and M10u are listed in table 10.4.1-1 below:

Table 10.4.1-1: Supported contribution protocols at reference point M4u and M10u

Description	Term identifier	Clause
DASH-IF push-based content ingest protocol	*http://dashif.org/ingest/v1.2/interface-1*	10.4.2

10.4.2 DASH-IF push-based contribution protocol

If *streamingAccess.entryPoints.protocol* is set to *http://dashif.org/ingest/v1.2/interface-1* in the Service Access Information, media resources shall be streamed to the 5GMSu AS as specified by the DASH-IF Live Media Ingest specification Interface-1 [3]. The content shall conform to at least one of the conformance profiles listed in *streamingAccess.profiles*, if any.

The content uploaded to the 5GMSu AS using this protocol is processed according to the Content Preparation Template(s) specified in the corresponding Content Publishing Configuration (if any), and the result is made available to the 5GMSu Application Service Provider at reference point M2u or to another 5GMSu AS at reference point M10u using the egest protocol indicated in *EgestConfiguration* as specified in clause 8.

10.4.3 Concurrent use of service locations for content contribution

The Media Streamer Entry (or a document pointed to by the Media Streamer Entry) shall be used to identify multiple service locations exposed by the 5GMSu AS at reference point M4u and the method used to contribute transport resources to the 5GMSu through simultaneous use of those multiple service locations. The following applies:

- Service locations provided in the Media Streamer Entry shall be distinguishable and identifiable via their base URLs.

- Any necessary configuration information required by the Access Client of the 5GMSu Client to contribute transport resources to multiple service locations concurrently shall be provided within the Media Streamer Entry.

When using multiple service locations concurrently, it is the responsibility of the Media Access Client of the 5GMSu Client to:

- Identify the different base URLs in the Media Streamer Entry that apply to the media resource to be contributed.

- Select one or more service locations to be used to upload the media resource. For example, the service locations selected have the highest priority for usage based on the contents of the Media Streamer Entry, the Access Client selects the service locations using internal logic, or all the identified service locations are used.

- If applicable, identify the absolute URLs or combine the base URLs of the selected service locations with the relative path of the transport resource identified in the Media Streamer Entry. In the case where uniquely identifiable transport resources are used to contribute partial representations of the media resource to each of the selected service locations, the Media Access Client shall either identify or construct the absolute URLs of the transport resource used to transmit the original media resource to each service location.

- Prepare and upload the transport resources containing complete or partial representations of the media resource to the identified service locations. This may include uploading (either partially or in full) the media resource and/or the transport resources to one or more reference point M4u service locations concurrently.

10.5 In-band client data reporting

10.5.1 CMCD-based client data reporting

If one or more of the CMCD metrics schemes specified in table 7.8.1-1 is indicated in the client metrics reporting configurations provided to the Media Session Handler, the corresponding class(es) of CMCD information shall be collected by the Media Player and reported to the 5GMSd AS at reference point M4d per [70]. In addition:

- The currently effective Media Player Entry may restrict the subset of Service Locations and/or media object types for which CMCD information is reported. These are honoured by the Media Player.

- The Media Player shall include in every CMCD report the current media delivery session identifier in the *sessionId* property of the *MetricsSession* (see clause 9.5.3.2 of TS 26.510 [56]). If the currently effective Media Player Entry declares a session identifier of its own for the purpose of client data reporting, this is typically used to populate the CMCD *sid* key (e.g., as required by clauses K.3.7 and K.4.2.7 of ISO/IEC 23009-1 [32]); otherwise, this key shall be populated with the current media delivery session identifier.

- If available in its configuration (i.e., if provided via the configuration and settings API specified in clause 13.2.4 or via the relevant query parameter of the 3GPP Service URL specified in clause 12.4, as applicable), the Media Player shall include in every CMCD report the content identifier of the media streaming asset being consumed in the *contentId* property of the *MetricsSession* (see clause 9.5.3.2 of TS 26.510 [56]). If the currently effective Media Player Entry declares a content identifier of its own for the purpose of client data reporting, this is typically used to populate the CMCD *cid* key (e.g., as required by clauses K.3.7 and K.4.2.7 of ISO/IEC 23009-1 [32]); otherwise, this key shall be populated with the current content identifier in the Media Player's configuration, if available.

- The Media Player Entry may explicitly specify a set of CMCD keys to be reported (e.g., as required by clauses K.3.7 and K.4.2.7 of ISO/IEC 23009-1 [32]). These are typically reported by the Media Player in addition to those configured in the currently effective client metrics reporting configuration.

    - A Media Player supporting CMCD metrics may implement the collection and reporting of all metrics defined in CMCD or only a subset. In the context of the present document, a Media Player shall support at least the *sid* key.

    - If both the *metricsConfiguration[ ]* and the *cmcdReporting* are present in the configuration and settings API specified in clause 13.2.4 and they configure different metrics, the Media Player shall report the union of the two sets.

- The use by the Media Player of HTTP request headers or URL query parameters to report CMCD information is governed by the configuration and settings API specified in clause 13.2.4 or by the relevant query parameter of the 3GPP Service URL specified in clause 12.4 (as applicable). This value shall take precedence over any client data reporting mode indicated in the currently effective Media Player Entry.

NOTE: Section 3.1 of the CMCD specification [70] recommends that HTTP request headers and URL query parameters are sent in alphabetical order of key name when transmitted. This reduces the fingerprinting surface exposed by the Media Player.

Clause G.5 specifies how to instantiate CMCD-based client data reporting in DASH.

11 Media Session Handling (M5/M3) APIs

11.1 General

This clause defines the Media Session Handling APIs used by the Media Session Handler at reference point M5 and by the 5GMS AS at reference point M3 to access resources exposed by the 5GMS AF.

NOTE: While the entirety of the Media Session Handling APIs apply to downlink media streaming, only a subset is applicable to uplink media streaming. Specifically, the Consumption Reporting API is not applicable to uplink media streaming.

11.2 Service Access Information API

11.2.1 General

The API used by the Media Session Handler at reference point M5 and by the 5GMS AS at reference point M3 to acquire Service Access Information from the 5GMS AF is specified in clause 9.2 of TS 26.510 [56]. The Service Access Information enables the Media Session Handler and/or 5GMS AS (as applicable) to go on to use the other Media Session Handling APIs specified in clause 11.3 *et seq.*

11.2.2 Void

11.2.3 Void

11.3 Consumption Reporting API

11.3.1 General

The API used by the Media Session Handler to submit consumption reports to the 5GMS AF at reference point M5 is specified in clause 9.6 of TS 26.510 [56].

11.3.2 Void

11.3.3 Report format

11.3.3.1 ConsumptionReport data type

The *ConsumptionReport* data type is specified in clause 9.6.3.1 of TS 26.510 [56].

In the case of downlink media streaming with DASH [32]:

- The *mediaPlayerEntry* property shall be populated with the URL of the Media Player Entry (e.g., a MPD resource or a document pointing to the MPD resource) that was retrieved at reference point M4d after following any HTTP redirects.
- A separate Consumption Reporting Unit shall be reported in the *consumptionReportingUnits* array for each DASH Adaptation Set currently selected for presentation by the Media Player.

For other types of media streaming, the content of these properties is undefined.

11.3.3.2 ConsumptionReportingUnit type

The *ConsumptionReport* data type is specified in clause 9.6.3.2 of TS 26.510 [56].

In the case of downlink media streaming with DASH [32]:

- The *mediaConsumed* property shall indicate the content identifier currently being consumed (if known to the 5GMS Client) and which DASH Representation is currently selected by the Media Player for the DASH Adaptation Set in question by citing the value of its **`Representation`**`@id` from the DASH MPD. If both the content identifier and the DASH Representation identifier are available, they shall be provided in that order with a single pipe character "|" delimiting them, and this character shall not be present in the value of either identifier.

For other types of media streaming, the content of this property is undefined.

11.3.3.2 ConsumptionReportingUnit type

The *ConsumptionReport* data type is specified in clause 9.6.3.2 of TS 26.510 [56].

In the case of downlink media streaming with DASH [32]:

- The *mediaConsumed* property shall indicate which DASH Representation is currently selected by the Media Player for the DASH Adaptation Set in question by citing the value of its **`Representation`**`@id` from the DASH MPD.

For other types of media streaming, the content of this property is undefined.

11.4 Metrics Reporting API

11.4.1 General

The API used by the Media Session Handler at reference point M5 and by the 5GMS AF at reference point M3 to submit metrics reports to the 5GMS AF is specified in clause 9.5 of TS 26.510 [56]. Metrics reports shall be submitted according to the metrics scheme indicated in each client metrics reporting configuration described by the Service Access Information resource (see clause 11.2) at the applicable reference point.

NOTE: Multiple metrics configurations may be active at the same time in the Service Access Information.

- If the metrics scheme *urn:3GPP:ns:PSS:DASH:QM10* is indicated in the client metrics reporting configuration, QoE metrics for DASH-based downlink media streaming shall be reported by the Media Session Handler to the 5GMSd AF at reference point M5d per clause 11.4.3.2.

- If the metrics scheme specified in clause 9.3 of TS 26.118 [42] is indicated in the client metrics reporting configuration, QoE metrics related to Virtual Reality media shall be reported by the Media Session Handler to the 5GMSd AF at reference point M5d per clause 11.4.3.2.

- If one or more of the metrics schemes specified in table 7.8.1-1 is indicated in the client metrics reporting configurations provided to the 5GMSd AS, QoE metrics reports containing the corresponding class(es) of CMCD information shall be reported by the 5GMSd AS to the 5GMSd AF at reference point M3d per clause 11.4.3.3.

11.4.2 Void

11.4.3 Report format

11.4.3.1 General

Metrics reports shall be submitted to the 5GMS AF in a format specified by the metrics reporting scheme in question. The `Content-Type` HTTP request header shall be set in accordance with the relevant metrics reporting scheme specification.

11.4.3.2 QoE metrics reporting for DASH-based downlink media streaming

For DASH-based downlink media streaming sessions, the following applies when QoE metrics reporting at reference point M5d has been provisioned:

- When the metrics scheme 3GPP *urn:3GPP:ns:PSS:DASH:QM10* is indicated in the client metrics reporting configuration of the Service Access Information at reference point M5d, metrics shall be reported by the Media Session Handler according to the quality reporting scheme specified in clause 10.5 of TS 26.247 [7]. Clauses 10.6.1 and 10.6.2 respectively of TS 26.247 [7] specify the required MIME media type and document format for submitting metrics reports to the 5GMSd AF at reference point M5d.

- When the metrics scheme for Virtual Reality media is indicated in the client metrics reporting configuration of the Service Access Information at reference point M5d, the format of metrics reports submitted to the 5GMSd AF at reference point M5d is further extended as defined in clause 9.4 of TS 26.118 [42].

In both cases, the **`ReceptionReport`**`@clientID` attribute should be populated with a GPSI value as defined by TS 23.003 [7], if present and available to the Media Session Handler. Otherwise, this attribute should be represented by a stable and globally unique string.

In both cases, the **`QoEReport`**`@recordingSessionId` attribute shall be populated with the media delivery session identifier.

11.4.3.3 Client data reporting

11.4.3.3.1 General

The syntax and MIME media type of metrics reports conveying client data to the 5GMSd AF shall comply with the JSON-based metrics reporting envelope specified in clause 9.5.3.2 of TS 26.510 [56]. The *mediaStreamingClientData* property (of type *MediaStreamingClientData*, as specified in table 11.4.3.3.1-1 below) shall be present in every *MetricsSample* object contained in this metrics reporting envelope.

Table 11.4.3.3.1-1: Definition of MediaStreamingClientData data type

Property name	Type	Cardinality	Description
requestUrl	`AbsoluteUrl`	0..1	The client request URL to which this client data sample pertains (see NOTE).
cmcdSessionInfo	`CmcdSessionInfo`	0..1	CMCD per-session information (see table 11.4.3.3-2).
cmcdObjectInfo	`CmcdObjectInfo`	0..1	CMCD per-request information (see table 11.4.3.3-3).
cmcdRequestInfo	`CmcdRequestInfo`	0..1	CMCD per-object information (see table 11.4.3.3-4).
cmcdStatusInfo	`CmcdStatusInfo`	0..1	CMCD status information (see table 11.4.3.3-5).
NOTE: Data type *AbsoluteUrl* is specified in TS 26.510 [56].			

The OpenAPI syntax of the above data type is specified annex C.7.

11.4.3.3.2 Client data reporting for DASH-based downlink media streaming

For DASH-based downlink media streaming sessions, the following applies for the reporting of client data by the 5GMSd AS to the 5GMSd AF at reference point M3d:

- When one or more of the CMCD metrics schemes listed in table 7.8.1-1 is indicated in client metrics reporting configurations of the Service Access Information provided to the 5GMSd AS at reference point M3d, the syntax and MIME media type of metrics reports conveying CMCD information [70] to the 5GMSd AF at reference point M3d shall comply with clause 11.4.3.3.1 above, and at least one of the following properties shall be populated in this *ClientData* object: *cmcdSessionInfo*, *cmcdObjectInfo*, *cmcdRequestInfo*, *cmcdStatusInfo*.

NOTE: The syntax of these objects aligns with the *payload definition for JSON transmission* specified in section 3.1 of the CMCD specification [70].

Aligned with section 3.1 of the CMCD specification [70], key–value pairs conveyed by these objects should appear in alphabetical order of key name when transmitted. (This reduces the fingerprinting surface exposed by different sources of client data.)

The OpenAPI syntax of the following data types is specified annex C.7.

Table 11.4.3.3.2-1: Definition of CmcdSessionInfo data type

Property name	Type	Cardinality	Description
v	`integer`	0..1	CMCD specification version. If omitted, version 1 is implied.
sid	`string`	1..1	Media delivery session identifier.
cid	`string`	0..1	Content identifier.
st	`CmcdStreamType`	0..1	Stream type (see table 11.4.3.3.2-5).
sf	`CmcdStramingFormat`	0..1	Streaming format (see table 11.4.3.3.2-6).
pr	`number`	0..1	Playback rate.

Table 11.4.3.3.2-2: Definition of CmcdObjectInfo data type

Property name	Type	Cardinality	Description
ot	`CmcdObjectType`	0..1	Object type (see table 11.4.3.3.2-7).
d	`integer`	0..1	Object duration expressed in milliseconds.
br	`integer`	0..1	Encoded bit rate expressed in kilobits per second.
tb	`integer`	0..1	Top bit rate expressed in kilobits per second.

Table 11.4.3.3.2-3: Definition of CmcdRequestInfo data type

Property name	Type	Cardinality	Description
su	`boolean`	0..1	Start-up.
mtp	`integer`	0..1	Measured throughput expressed in kilobits per second.
dl	`integer`	0..1	Deadline expressed in milliseconds.
bl	`integer`	0..1	Buffer length expressed in milliseconds.
nor	`RelativeUri`	0..1	Next object request (see NOTE).
nrr	`string`	0..1	Next range request.
NOTE: Data type *RelativeUri* is specified in TS 26.510 [56].			

Table 11.4.3.3.2-4: Definition of CmcdStatusInfo data type

Property name	Type	Cardinality	Description
rtp	`integer`	0..1	Requested throughput expressed in kilobits per second.
bs	`boolean`	0..1	Buffer starvation.

Table 11.4.3.3.2-5: Definition of CmcdStreamType enumeration

Enumeration value	Description
v	All media segments are available, e.g. Video-on-Demand.
l	Media segments become available over time, e.g. live.

Table 11.4.3.3.2-6: Definition of CmcdStreamingFormat enumeration

Enumeration value	Description
d	MPEG-DASH [32].
h	HTTP Live Streaming (HLS).
s	Smooth Streaming.
o	Other.

Table 11.4.3.3.2-7: Definition of CmcdObjectType enumeration

Enumeration value	Description
m	Text, e.g. manifest or playlist.
a	Audio only.
v	Video only.
av	Multiplexed audio and video.
i	Initialisation segment.
c	Caption or subtitle.
tt	ISO BMFF Timed Text track.
k	Cryptographic key, licence or certificate.
o	Other.

11.5 Dynamic Policies API

11.5.1 Overview

The API used by the Media Session Handler at reference point M5 to request a specific policy and charging treatment to be applied to a particular application data flow of a downlink or uplink media streaming session by the 5GMS AF is specified in clause 9 of TS 26.510 [56].

11.5.2 Void

11.5.3 Void

11.5.4 Void

11.6 Network Assistance API

11.6.1 Overview

The API used by the Media Session Handler at reference point M5 to obtain bit rate recommendations from the 5GMS AF and to issue delivery boost requests to the 5GMS AF during an ongoing media streaming session is specified in clause 9.4 of TS 26.510 [56].

11.6.2 Void

11.6.3 Void

11.6.4 Void

12 UE Media Session Handling (M6/M11) APIs for uplink and downlink

12.1 General

This clause defines the client APIs for Media Session Handling to be used by other 5G System components such as a Media Player in a 5GMSd Client or the Media Streamer in a 5GMSu Client.

NOTE: Client-driven management of edge processing resources via reference point M6 or M11 is not specified in this release.

12.2 Media session handling for downlink media streaming – APIs and functions

12.2.1 Overview

In the following, it is assumed that the Media Session Handler for downlink media streaming adheres to a basic set of functionalities as shown in figure 12.2.1-1.

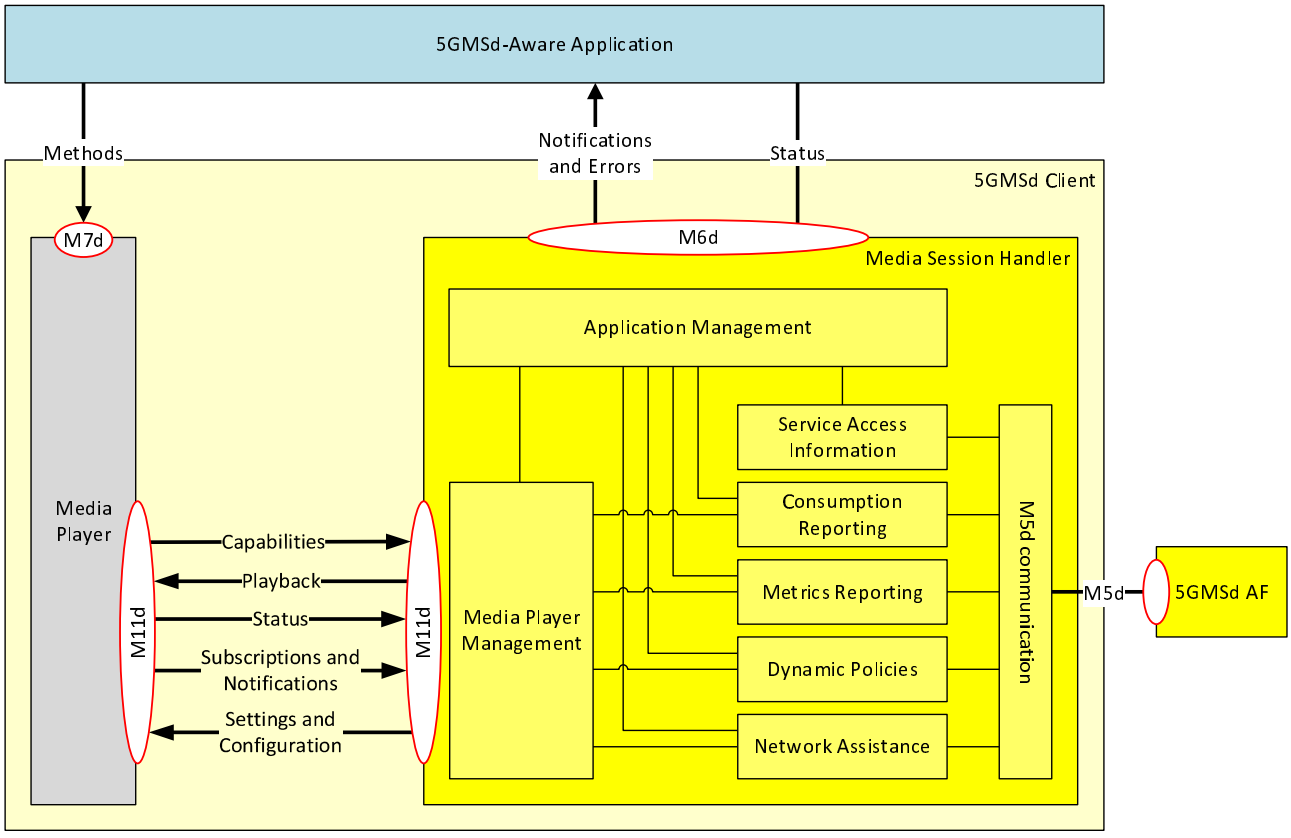


Figure 12.2.1-1: Usage of M6d in Media Downlink Streaming

The Media Session Handler is considered to run as a service in the background and is invoked for a media session once a media player in the 5GMSd streaming client is activated with an MPD URL of media MIME type `"application/dash+xml"`. Based on the MPD URL, the Media Session Handler may initiate communication with the 5GMSd AF through reference point M5d.

NOTE: The initiation of the Media Session Handler for other media types than DASH is for further study.

For an ongoing 5G Media Streaming session, the Media Session Handler is given the following authorizations:

1) The ability to query the status of the Media Player at reference point M11d. For details see clause 13.
2) The ability to process notifications and errors received from the Media Player at reference point M11d. For details see clause 13.
3) The ability to configure certain parameters on the Media Player using methods exposed at reference point M11d. For details see clause 13.

In addition, the Media Session Handler provides information to the 5GMS-Aware Application at reference point M6d, possibly delegated to Media Player at reference point M11d for each of the Media Session Handler functionalities, namely providing:

1) Notification and Error Events;
2) Status Information.

The client API used for downlink media session handling at reference point M6d by the 5GMSd-Aware Application and at reference point M11d by the Media Player in a 5GMSd Client is specified in clause 11 of TS 26.510 [56].

12.2.2 Void

12.2.3 Void

12.2.4 Void

12.2.5 Void

12.2.6 Void

12.2.7 Void

12.3 Media session handling for uplink Streaming – APIs and functions

The client API used for uplink media session handling at reference point M6u by the 5GMSu-Aware Application and at reference point M11u by the Media Streamer in a 5GMSu Client is specified in clause 11 of TS 26.510 [56].

12.4 3GPP Service URL for 5G Media Streaming

The 3GPP Service URL for 5G Media Streaming is based on the generic 3GPP Service URL defined in clause 6 of TS 26.510 [56].

If the service type discriminator *service* in the URL indicates `ms`, then the target service is a 5G Media Streaming service.

The parameters of the 3GPP Service URL for 5G Media Streaming are defined in table 12.4-1.

Table 12.4-1: 3GPP Service URL parameters for 5G Media Streaming

Path element	Cardinality	Description
service_id	1	An External Service Identifier that resolves to a Provisioning Session in the 5GMS System.
Query parameter	**Cardinality**	**Description**
af-host-address	0..*	The Fully Qualified Domain Name and optional port number of a 5GMS AF endpoint to be used by the Media Session Handler at reference point M5 with the format *hostname[:port]*. More than one occurrence of this parameter may be present in the Service URL to indicate alternative host endpoint addresses. Any of these may be used by the Media Session Handler at reference point M5. Supplied by the invoking 5GMS-Aware Application when the 5GMS AF is deployed in an External DN. The endpoint address(es) may, for example, have been passed to the 5GMS-Aware Application via reference point M8. If omitted, the Media Session Handler assumes the default 5GMS AF host endpoint address *ms.af.3gppservices.org:443* is to be used at reference point M5.
access-token	0..1	A token that is presented by the Media Session Handler to the 5GMS AF at reference point M5 that asserts its right to invoke the media session handling operations exposed by the 5GMS AF.
service-operation-point	0..1	A reference to the target Service Operation Point of the 5G Media Streaming session. Overrides any default Service Operation Point signalled in the resource pointed to by *media-entry-point*.
content-id	0..1	Identifies the media consumed. If provided, this value shall be used by the Media Session Handler to populate the *mediaConsumed* property of *ConsumptionReportingUnit* (see clause 11.3.3.2). If provided, this value shall be used to populate the content identifier field for in-band client data reporting, such as the *cid* key defined by CMCD [70].
estimated-volume	0..1	An estimate of the volume of media data to be transferred between the 5GMS Client and the 5GMS AS at reference point M4 during the 5G Media Streaming session. Required if *service-operation-point* or *media-entry-point* refers to a Policy Template with an associated Background Data Transfer policy.
media-entry-point	0..1	A Media Entry Point reference expressed as a fully qualified URL per RFC 3986 [41], suitable for presentation to a 5GMS AS at reference point M4. If supplied, used by the Media Session Handler to launch the Media Stream Handler (Media Player or Media Streamer) after successfully initiating media session handling.
delivery-protocol	0..*	A content delivery protocol suggested to be used by the Media Stream Handler (Media Player or Media Streamer) for media delivery at reference point M4. - The value *HTTP_1_1* indicates the use of the HTTP/1.1 protocol specified in RFC 9112 [24]. - The value *HTTP_2* indicates the use of the HTTP/2 protocol specified in RFC 9113 [31]. - The value *HTTP_3* indicates the use of the HTTP/3 protocol specified in RFC 9114 [60]. More than one occurrence of this parameter may be present in the Service URL indicating multiple choices in decreasing order of preference.

multipath-mode	0..1	Indicates a preference for using multipath connections at reference point M4. When present, the Media Stream Handler (Media Player or the Media Streamer) may establish multipath connections over one or more access networks active in the UE (see clause 4.9.1A). If omitted, the Media Stream Handler may independently choose whether or not to enable multipath delivery at reference point M4 depending on the selected content delivery protocol (see *delivery-protocol* parameter above) and its capability to use multipath protocols (see table 13.2.4-2).
client-data-reporting-mode	0..1	A string from the following enumeration that controls the mode of in-band client data reporting by the Media Stream Handler (Media Player or Media Streamer) to the 5GMS AS at reference point M4: - *cmcd-headers* indicates that CMCD key–value pairs are required to be reported in HTTP request headers per sections 2.1 and 3.2 of [70]. - *cmcd-query* indicates that CMCD key–value pairs are required to be reported as query parameters of the request URL per sections 2.2 and 3.2 of [70]. It is an error to request more than one of the above CMCD reporting modes in the same 3GPP Service URL. This value shall take precedence over any client data reporting mode indicated in the selected Media Entry Point. If omitted, no CMCD-based in-band client data reporting is required by the invoker, but a directive in the selected Media Entry Point (e.g., the *mode* parameter specified in clauses K.3.7 and K.4.2.7.2 of ISO/IEC 23009-1 [32]) is honoured by the Media Stream Handler as required.
content-type	0..*	A MIME content type string conforming to section 5 of RFC 2045 [57] identifying a type of Media Entry Point that is acceptable to the Media Stream Handler (Media Player or Media Streamer). More than one occurrence of this parameter may be present in the Service URL to indicate that more than one type of Media Entry Point is acceptable. Used by the Media Session Handler to eliminate unacceptable Media Entry Points from those listed in the Service Access Information. It is an error to supply this parameter if an explicit Media Entry Point is specified using *media-entry-point*.
profile	0..*	A fully-qualified term identifier from a controlled vocabulary specified outside the scope of the present document identifying a profile of Media Entry Point that is acceptable to the Media Stream Handler (Media Player or Media Streamer). The term identifier shall be formatted as a URI according to RFC 3986 [41]. More than one occurrence of this parameter may be present in the Service URL to indicate that more than one type of Media Entry Point is acceptable. Used by the Media Session Handler to eliminate unacceptable Media Entry Points from those listed in the Service Access Information. It is an error to supply this parameter if an explicit Media Entry Point is specified using *media-entry-point*.

The *service_id* path element, and the *af-host-address* and *access-token* query parameters correspond to the baseline Service Access Information for downlink media streaming specified in clause 4.2.3 of TS 26.501 [2] and the baseline parameters of the 3GPP Service URL for 5G Media Streaming defined in clause 4.10.2 of [2]. Together, they enable a full set of Service Access Information to be retrieved by the Media Session Handler from the 5GMS AF using the Service Access Information API at reference point M5 specified in clause 11.2 of the present document.

The *content-id* parameter is used to populate consumption reports (see clauses 4.7.4 and 11.3.3) and/or metrics reports conveying client data (see clauses 4.7.5, 10.5 and 11.4.3).

The *service-operation-point* parameter is used to support the procedure where the desired Service Operation Point is known *a priori* to the invoker and/or is not encoded in the Media Entry Point.

The *estimated-volume* parameter is used to support the procedure where the invoker intends the launched 5G Media Streaming session to be used for the purpose of Background Data Transfer.

The *media-entry-point* query parameter is used to support the procedure where the Media Session Handler launches media playback in the Media Stream Handler (Media Player or Media Streamer) after successfully retrieving a full set of Service Access Information via reference point M5 (if needed) and after successfully initiating media session handling.

The *delivery-protocol* query parameter is used to suggest to the Media Stream Handler (Media Player or Media Streamer) the use of a specific content delivery protocol for media delivery at reference point M4. If multiple instances of this parameter are present in the URL, they indicate a decreasing order of preference, and the Media Stream Handler (Media Player or Media Streamer) should select the most preferred content delivery protocol that it supports, taking into account any preference to use a multipath transport protocol (see *multipath-mode* parameter below).

The *multipath-mode* parameter is used to indicate a preference to use multipath transport connections for media delivery at reference point M4. When this parameter is included, the Media Stream Handler (Media Player or Media Streamer) may establish multipath transport connections with the 5GMS AS at reference point M4 as specified in clauses 4.6.1, 4.6.2, and 4.9.1A of the present document. In the absence of this parameter, the Media Stream Handler (Media Player or Media Streamer) is not precluded from establishing multipath connections for media delivery at reference point M4 according to its capabilities

The *client-data-reporting-mode* query parameter is used to select between different reporting modes for in-band client data provided by the 5GMS Client at reference point M4.

The remaining query parameters are used for client-side filtering of Media Entry Point information provided in the Service Access Information and selection of one Media Entry Point by the Media Session Handler. (They are mutually exclusive with the *media-entry-point* parameter.) In this case, media playback by the Media Stream Handler (Media Player or Media Streamer) is launched by the Media Session Handler with its chosen Media Entry Point.

If the 5GMS-Aware Application prefers to launch media streaming itself (rather than have the Media Session Handler launch media streaming on its behalf), the *media-entry-point* query parameter and all client-side filtering parameters shall be omitted from the 3GPP Service URL. In this case, the Media Session Handler initiates only media session handling for the 5GMS Provisioning Session identified by the External Service Identifier.

13 UE Media Stream Handler (M7/M11) APIs

13.1 General

This clause defines a set of APIs and methods that permit a 5GMS-Aware Application at reference point M7 or a Media Session Handler at reference point M11 to communicate with a Media Stream Handler (Media Player or Media Streamer). The main focus of this clause is to formalize and harmonize commonly available proprietary APIs in order to support the usage of a Media Player or a Media Streamer in a 5G Media Streaming context.

The APIs specified in this clause are language- and runtime-independent. Implementations are expected to provide language bindings appropriate to the UE runtime environment.

13.2 DASH Media Player APIs and functions

13.2.1 Overview

In the following, it is assumed that the Media Player (in this case a DASH client) adheres to a basic set of functionalities as shown in figure 13.2-1. The DASH client downloads, processes and presents a DASH Media Presentation under the control of a 5GMSd-Aware Application via reference point M7d or of the Media Session Handler via reference point M11d.

The 5GMSd-Aware Application may, in addition, configure the presentation of the media, receive notifications on events, or query the internal status of the DASH Player, also supported through reference point M7d. Different functions of the DASH Access Client that are typically necessary to process a DASH Media Presentation, are shown in figure 13.2-1. Additional functions may be available as well.

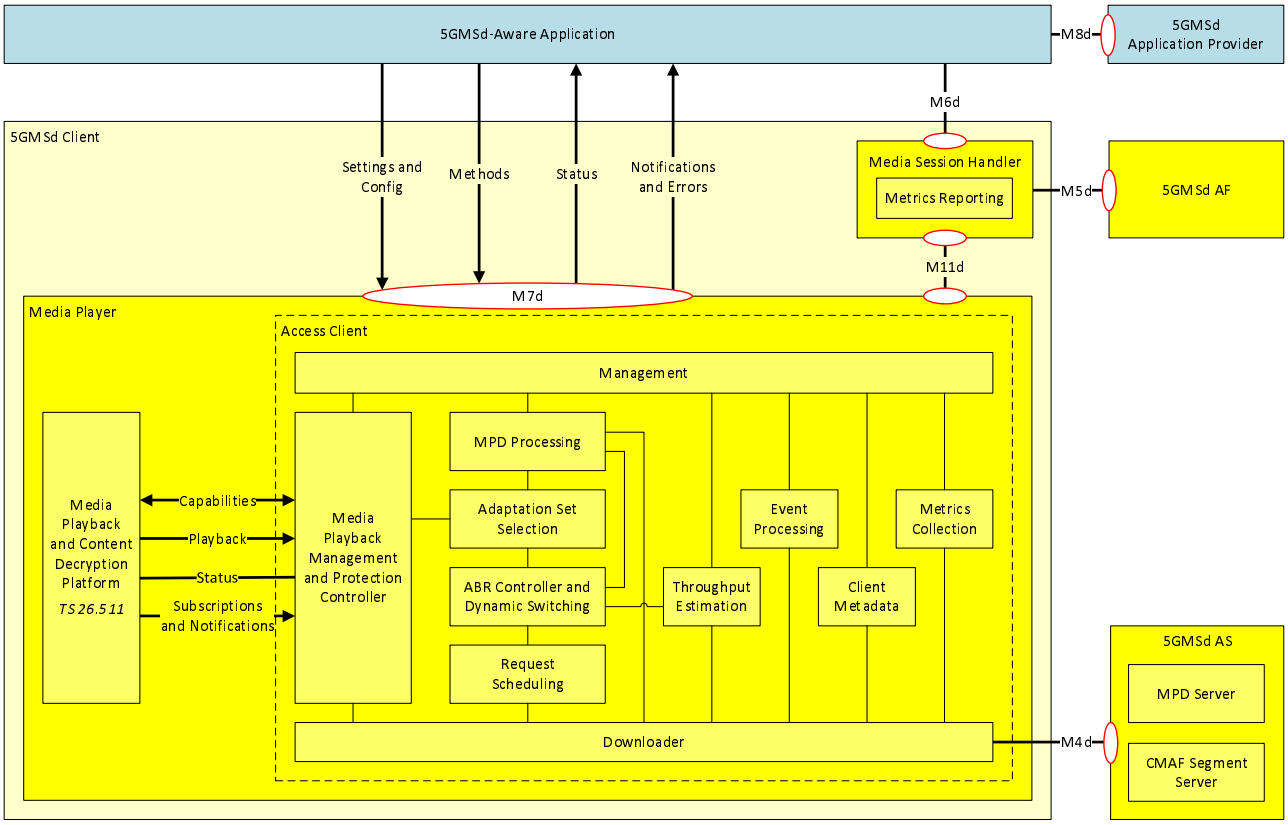


Figure 13.2.1-1: Architecture of DASH-based 5GMSd Client

The key functionalities of each of the functions as shown in figure 13.2-1 are summarized in the following:

- *5GMSd-Aware Application:* Application that makes use of the DASH-based Media Player to play back a DASH Media Presentation using the APIs defined in this clause.
- *Media Player:* A complete player for the playback of a Media Presentation, including the Media Playback and Content Decryption Platform as defined in TS 26.511 [35].
- *Access Client:* A part of the DASH Player that accesses and downloads of the resources and provides the downloaded resources to the Media Playback Platform and Content Decryption for the playback of DASH content.
- *Management:* Controls all internal processes and the communication with the 5GMSd-aware application. In particular this includes the handling of service descriptions and operation points.
- *MPD Processing:* parses and processes the MPD and extracts the relevant information.
- *Adaptation Set Selection:* selects the Adaptation Set based on user, application and/or device capability information. Information provided through M7d may be used.
- *ABR Controller and Dynamic Switching:* runs adaptive bit rate logic and triggers adaptive switching of Representations. Information provided to the Media Player via reference point M7d may be used.
- *Throughput Estimation:* estimates the network status, i.e. throughput, congestion information on the transmission link between the Media Player and the 5GMSd Application Server. Information provided to the Media Player via reference point M11d (i.e., QoS monitoring results) may be considered as input to the throughput estimation.

  When ECN marking for L4S according to RFC 9330 [67], RFC 9331 [68] and RFC 9332 [69] is activated (as notified by the Media Session Handler using the *L4S_ENABLED* notification at reference point M11d) ECN marking of congestion in downlink packets may additionally be considered as input to throughput estimation, subject to the availability of a suitable low-level API.
- *Metrics Collection:* logs relevant low-level metrics and provides those to the metrics aggregation and reporting functions in the Media Session Handler or to the Downloader for in-band client data reporting.

- *Media Playback Management and Protection Controller:* manages the media playback by moving downloaded information into the media playback platform and also addresses handling of protection and DRM-related information.
- *Media Playback and Content Decryption Platform:* plays back CMAF-based media content according to the playback requirements in TS 26.511 [35]. It also provides status information as well as events that may be provided through M7d.
- *Event Processing:* Processes DASH events and provides information to the 5GMSd-Aware Application as defined in TS 26.247 [4].
- *Downloader:* Retrieves resources from one or more reference point M4d service locations. It may optionally support switching between service locations as described in clause 10.3A.2, communication with a content steering service as described in clause 10.3A.4, and the download and decoding of coded objects as described in clause 10.3A.3.

This clause focuses on interactions with the Media Player through reference point M7d. In particular, the following aspects of the API are defined:

1) Methods to interact with the Media Player at this reference point are defined in clause 13.2.3.
2) Notification and Error Events raised by the Media Player at this reference point are defined in clause 13.2.4.
3) Configuration and Settings of the Media Player at this reference point are defined in clause 13.2.5.
4) Status Information exposed by the Media Player at this reference point is defined in clause 13.2.6.

Communication between the Access Client and the media playback platform of the Media Player is defined in TS 26.511 [35].

A 5GMSd Client for DASH distribution shall support the APIs defined in this clause 13.

NOTE: The initial APIs have largely been designed based on the dash.js APIs documented here: http://cdn.dashjs.org/latest/jsdoc.

13.2.2 Media Player model

Figure 13.2.2-1 provides an informative client state model in order to appropriately describe the messages on the Media streaming service API. The different states are defined in table 13.2.2-1.

State changes may happen based on:

- Calls from application.
- Information provided in the Media Presentation Description (MPD).

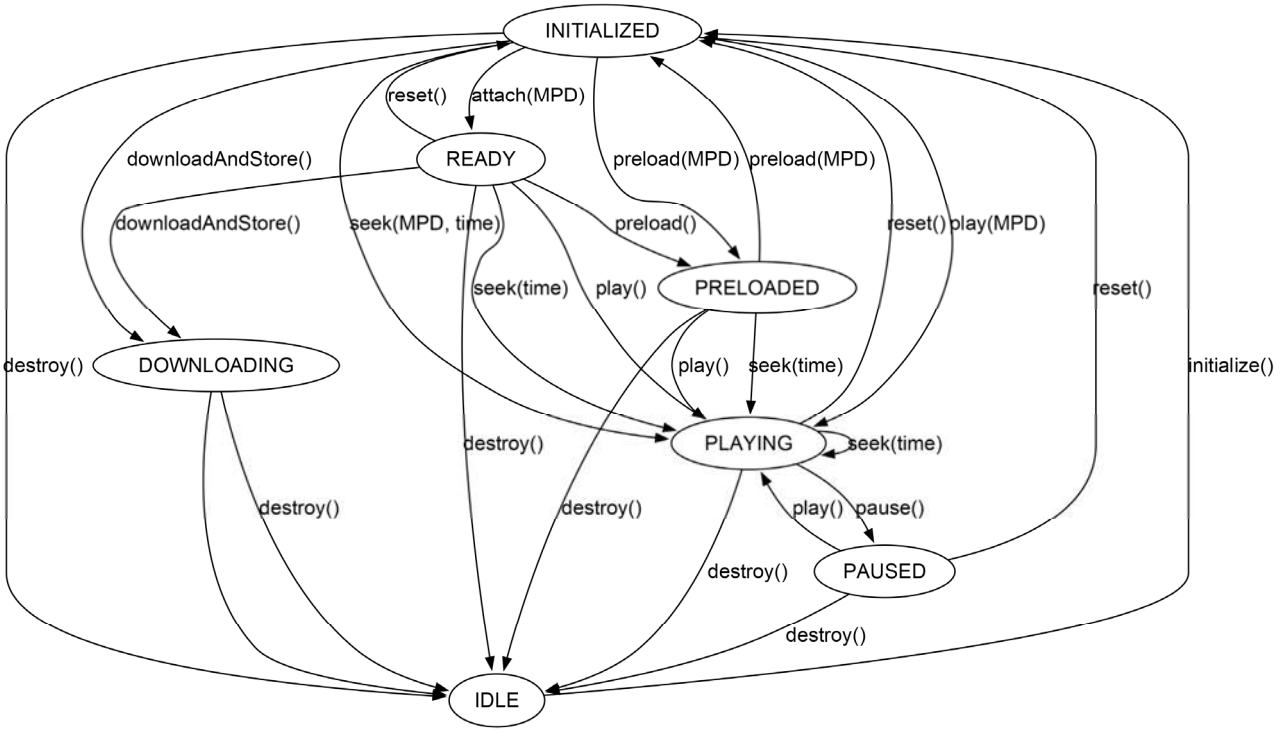


Figure 13.2.2-1: State diagram for Media Player

Table 13.2.2-1 defines states for the Media Player. Detailed descriptions are provided in the following clauses.

Table 13.2.2-1: States of Media Player

States	Definition
IDLE	The Media Player is not associated with any application.
INITIALIZED	The Media Player is associated with an application and the M7d API communication is established.
READY	The Media Player has loaded an MPD and is able to playback the media in this Media Presentation. It also updates the MPD according to the MPD update mechanism.
PRELOADED	The Media Player has pre-loaded all media information in order to start playback instantaneously. It also updates the MPD according to the MPD update mechanism.
PLAYING	The Media Player is playing the Media Presentation. It also updates the MPD according to the MPD update mechanism.
DOWNLOADING	The Media Player is downloading the Media Presentation and storing it in the Media Player for later playback.
PAUSED	The playback of the Media Presentation is paused. It also updates the MPD according to the MPD update mechanism.

It is assumed that the DASH Access Client manages the playback of at most one CMAF track for each media type, namely one for video, one for audio and one for subtitles as defined in TS 26.511 [35]. Playback of multiple CMAF tracks of the same media type is not excluded by the present document, but details are for further study.

13.2.3 Media Player methods

13.2.3.1 General

Based on the state model in clause 13.2.2, this clause introduces relevant procedures and API calls exposed by the Media Player to the 5GMSd-Aware Application at reference point M7d and to the Media Session Handler at reference point M11d.

Table 13.2.3.1-1 provides an overview over the methods defined for the DASH-based streaming API. Note that in implementations, additional methods may be supported.

Table 13.2.3.1-1: Methods defined for DASH Streaming API

Method	State after success	Brief description	Clause
`initialize()`	*INITIALIZED*	The Media Player is created.	13.2.3.2
`attach()`	*READY*	Sets a source URL to an MPD file or a previously downloaded and parsed MPD.	13.2.3.3
`preload()`	*PRELOADED*	Streaming the media is initiated.	13.2.3.4
`play()`	*PLAYING*	Playback of the media is initiated.	13.2.3.5
`pause()`	*PAUSED*	Playback of the media is paused.	13.2.3.6
`seek()`	*PLAYING*	The playback time of the media is altered.	13.2.3.7
`downloadAndStore()`	*DOWNLOADING*	Download a media presentation and store it locally for later playback.	13.2.3.7A
`reset()`	*INITIALIZED*	All media related information is reset.	13.2.3.8
`destroy()`	*IDLE*	All media player related information is reset and API communication is stopped.	13.2.3.9

13.2.3.2 Initialize

This clause defines the `initialize()` method.

No pre-conditions apply.

This method is invoked to create a new Media Player instance. A media delivery session identifier may be assigned to the media delivery session by the Media Player and returned if a value is not supplied by the invoker of the method.

The input parameters of this method are specified in table 13.2.3.2-1.

Table 13.2.3.2-1: Input parameters of `initialize()` method

Name	Type	O	Description
serviceId	`string`	M	An external service identifier, as specified in clause 5.4.2.1 of TS 26.510 [56].
sessionId	`string`	C	A media delivery session identifier nominated by the Media Session Handler, as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document. This parameter shall be included when the method is invoked by the Media Session Handler at reference point M6d; it shall be omitted when the method is invoked by a 5GMSd-Aware Application at reference point M7d.

The following functions are initialized:

- Media Playback Management in order to enable API-based communication through M7d and/or M11d. In particular, the *Notifications and Errors API* (see clause 13.2.5) and the *Status Information* (see clause 13.2.6) are established.

If the *sessionId* parameter is provided by the Media Session Handler at reference point M11d, the newly created Media Player instance shall use this value in its further interactions, in particular the *Notifications and Errors API* (see clause 13.2.5) and the *Status Information* (see clause 13.2.6).

If the *sessionId* parameter is omitted when the method is invoked by the 5GMSd-Aware Application at reference point M6d, the newly created Media Player instance shall rely on the Media Session Handler assigning a media delivery session identifier as a side-effect of invoking the method specified in clause 11.2.2.1 of TS 26.510 [56] at reference

point M11d. The Media Player shall then use this value in its further interactions, in particular the *Notifications and Errors API* (see clause 13.2.5) and the *Status Information* (see clause 13.2.6).

The return value of the method is specified in table 13.2.3.2-2.

Table 13.2.3.2-2: Return value of `initialize()` method

Type	Description
`string`	A media delivery session identifier as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document.

The remaining Media Player methods specified below require the media delivery session identifier to be cited as an input parameter.

13.2.3.3 Attach

This clause defines the `attach()` method.

The following pre-conditions apply:

- The Media Player transitions to the *INITIALIZED* state.

The `attachMPD()` method is invoked on a Media Player instance to set a source URL to an MPD file or a previously downloaded and parsed MPD.

The input parameters of the method are specified in table 13.2.3.3-1.

Table 13.2.3.3-1: Input parameters of `attachMPD()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.
urlOrMPD	`string | Object`	A URL to a valid MPD or a valid MPD as defined in ISO/IEC 23009-1 [32] or TS 26.247 [4]. The URL may be augmented by MPD Anchors as defined in clause C.4 of ISO/IEC 23009-1 [32].

The following Media Player Actions are expected:

- The *Request Scheduling* and *Download* functions are established.
- If the input is a URL, the Media Player requests the MPD at the corresponding URL through reference point M4d.
- If the MPD is not found after multiple retries, an error *ERROR_MPD_NOT_FOUND* is returned and the process is terminated.
- The *MPD Processing* function is established and the MPD parsed.
- If the MPD is not valid, an error *ERROR_MPD_NOT_VALID* is returned and the process is terminated.
- If the DASH Player does not support the profiles as indicated in the MPD, an error *ERROR_PROFILE_NOT_SUPPORTED* is returned and the process is terminated.
- Depending on the type of the MPD, possibly present anchors as well as the wall-clock time, the Media Player selects the Period in the content that is expected to be played next.
- The *Media Playback Management and Protection Controller* is established.
- The MPD is parsed for available Service Descriptions (including Media Subsets and Adaptation Sets). By using capability mechanisms defined in TS 26.511 [35] as well as using other information (language settings, output capabilities, accessibility settings), the Media Player identifies a set of permissible Service Descriptions

including Media Subsets and Adaptation Sets. If no Adaptation Sets are capable to be played, an error *ERROR_MEDIA_NOT_SUPPORTED* is returned and the process is terminated.

- The available Service Descriptions including included Adaptation Sets are provided to the 5GMS-Aware Application through reference point M7d.
- The application may select a Service Description instance as well as Adaptation Sets. Additional Service Descriptions parameters may be configured by the 5GMSd-Aware Application through reference point M7d.
- Based on the service description parameters and selected Adaptation Sets:
  - the Operation Point parameters are set.
  - the *Media Playback Platform and Content Decryption* is established using the methods defined in TS 26.511 [35].
  - The selected Adaptation Sets are initialized by downloading the relevant Initialization Segments/CMAF Headers through M4d in the Media Playback Platform as in TS 26.511 [35] establishing a track buffer for each selected media type.
- Depending on the MPD information and/or M7d configuration, one or more of the following functions may be established:
  - Metrics Logging and Collection
  - Event Processing and Notification
  - Client Metadata handling
- The Media Player transitions to the *READY* state.

An application may use this method to load an MPD and in order to prepare playback. In case of errors notifications, it is up to the application to initiate appropriate actions.

13.2.3.4 Pre-load

This clause defines the `preload()` method.

The following pre-conditions apply:

- The Media Player is in *INITIALIZED* or *READY* state.

The `preload()` method is invoked on a Media Player instance to cause it to begin streaming the media as set by the `attach()` method (see clause 13.2.3.3) in preparation for playing.

The input parameters of the method are specified in table 13.2.3.4-1.

Table 13.2.3.4-1: Input parameters of `preload()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.
urlOrMPD	`string | Object`	A URL to a valid MPD or a valid MPD as defined in ISO/IEC 23009-1 [32] or TS 26.247 [4]. The URL may be augmented by MPD Anchors as defined in clause C.4 of ISO/IEC 23009-1 [32].

The following Media Player Actions are expected:

- If in *INITIALIZED* state, the `attach()` method is invoked with the value of the *urlOrMPD* input parameter.
- Depending on the type of the MPD, possibly present anchors as well as the wall-clock time, and other MPD information, the earliest media time span for pre-loading is identified.

- The Access Client schedules and generates requests for the relevant media segments based on the ABR Controller information, as well as the throughput estimation and downloads this media.
- The segments are downloaded from the corresponding URLs through reference point M4d no earlier than the segment availability start time of the Segments.
- The downloaded segments are appropriately appended to the track buffers as established according to *Media Playback Platform and Content Decryption* APIs, following the description in TS 26.511 [35] for playback requirements.
- Configuration and service description parameters are taken into account, for example the content is continuously loaded to remain at the live edge following the latency requirements provided in the service description setting. Content not at the live edge is removed. For static services, the content is loaded from the beginning up to a suitable buffer duration, possibly as configured, and then downloading is stopped.
- The Media Player transitions to the *PRELOADED* state.
- Appropriate notifications and error messages are generated. For details, refer to clause 13.2.5.
- Appropriate Status Information is generated. For details, refer to clause 13.2.6.

An application may use this method to preload media into the player in order minimize the start-up time.

13.2.3.5 Play

This clause defines the `play()` method.

The following pre-conditions apply:

- The Media Player is in *INITIALIZED* or *READY* or *PRELOADED* or *PAUSED* state.

The `play()` method is invoked on a Media Player instance to cause it to begin playback of the media as set by the `attach()` method (see clause 13.2.3.3).

The input parameters of the method are defined in table 13.2.3.5-1.

Table 13.2.3.5-1: Input parameters of `play()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.
urlOrMPD	`string | Object`	A URL to a valid MPD or a valid MPD as defined in ISO/IEC 23009-1 [32] or TS 26.247 [4]. The URL may be augmented by MPD Anchors as defined in clause C.4 of ISO/IEC 23009-1 [32].

The following Media Player Actions are expected:

- If in *INITIALIZED* state, the `attach()` method (see clause 13.2.3.3) is invoked internally, passing the value of the *urlOrMPD* input parameter.
- If in *PAUSED* state, the earliest media time is *MEDIA_TIME* (for details see clause 13.2.3.6), else, depending on the type of the MPD, possibly present anchors as well as the wall-clock time, and other MPD information, the earliest media time for start-up is identified.
- The Access Client checks the available buffer state of media in the Media Playback Platform. Based on this, the Access Client schedules and generates requests for the relevant media segments based on the ABR Controller information, as well as the throughput estimation and downloads this media.
- Media segments are acquired from their corresponding URLs (through M4d or, if already downloaded and stored, from local storage) earliest at the segment availability start times.

- The media is appropriately appended to the *Media Playback Platform and Content Decryption* APIs, following the description in TS 26.511 [35] for playback requirements.
- Once a threshold for sufficient buffering is reached, the Media Playback platform is initiated to be started, i.e. a playback is initiated, following the description in TS 26.511 [35] for playback requirements.
- The content is continuously streamed, downloaded and played back.
- The Media Player transitions to the *PLAYING* state.
- Appropriate notifications and error messages are generated. For details, refer to clause 13.2.5.
- Appropriate Status Information is generated. For details, refer to clause 13.2.6.

An application may use this method to initiate playback of media.

13.2.3.6 Pause

This clause defines `pause()` method.

The following pre-conditions apply:

- The Media Player is in *PLAYING* state.

The `pause()` method is invoked on a Media Player instance to cause the Media Playback Platform to pause playback.

The input parameters of the method are specified in table 13.2.3.6-1.

Table 13.2.3.6-1: Input parameters of `pause()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.

The following Media Player Actions are expected:

- The playback on the playback platform is paused and the media time is maintained as *MEDIA_TIME*.
- The Access Client checks the available buffer state of media in the Media Playback Platform. Based on this, the Access Client schedules and generates requests for the relevant media segments based on the ABR Controller information, as well as the throughput estimation and downloads this media.
- The media is downloaded from the corresponding URL through reference point M4d earliest at the segment availability start time of the media.
- The media is appropriately appended to the *Media Playback Platform and Content Decryption* APIs, following the description in TS 26.511 [35] for playback requirements.
- Once the buffers are sufficiently filled, the client stops downloading.
- The Media Player transitions to the *PAUSED* state.
- Appropriate notifications and error messages are generated. For details, refer to clause 13.2.5.
- Appropriate Status Information is generated. For details, refer to clause 13.2.6.

An application may use this method to play back media.

13.2.3.7 Seek

This clause defines `seek()` method.

The following pre-conditions apply:

- The Media Player is in *INITIALIZED*, *READY*, *PRELOADED* or *PAUSED* state.

The `seek()` method is invoked on a Media Player instance to cause it to go to a specific media time.

The input parameters of the method are specified in table 13.2.3.7-1.

Table 13.2.3.7-1: Input parameters of `seek()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.
urlOrMPD	`string | Object`	A URL to a valid MPD or a valid MPD. The URL may be augmented by MPD Anchors as defined in clause C.4 of ISO/IEC 23009-1 [32].
mediaTime	`Unsigned integer`	The media time in milliseconds for playback.

The following Media Player actions are expected:

- If in *INITIALIZED* state, the `attach()` method (see clause 13.2.3.3) is invoked internally, passing the value of the *urlOrMPD* input parameter.
- If the `mediaTime` is not accessible, an *ERROR_MEDIA_TIME_NOT_ACCESSIBLE* event is issued (see clause 13.2.5) and the following actions are not carried out.
- The earliest media time is set to the value of the `mediaTime` input parameter.
- The Media Player transitions to the *PAUSED* state.
- The `play()` method (see clause 13.2.3.5) is invoked internally.

An application may use this method to initiate playback of media.

13.2.3.7A Download and store

This clause defines the `downloadAndStore()` method.

The following pre-conditions apply:

- The Media Player is in *INITIALIZED* or *READY* state.

The `downloadAndStore()` is invoked on a Media Player instance to cause it to download and store the media as set by the `attach()` method (see clause 13.2.3.3).

The input parameters of the method are specified in table 13.2.3.5A-1.

Table 13.2.3.7A-1: Input parameters of `downloadAndStore()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.
url	`string`	The URL of a Media Entry Point which may be a valid MPD as defined in ISO/IEC 23009-1 [32] or TS 26.247 [4].
serviceComponents	`array(string)`	A list of MPEG-DASH Adaptation Set and Representation identifiers to be downloaded. If omitted or empty, all Representations are downloaded.
background	`boolean`	If true, the content is required to be acquired using a Background Data Transfer.
deadline	`dateTime`	An optional date–time by which content acquisition is required to be complete. If this deadline is missed, any resources downloaded already are removed from the Media Player.

expiry	`dateTime`	A date–time by which any download resources are required to be removed from the 5GMS Client. The value of this parameter shall be after *deadline*, if present.

The anonymous output parameter of the method is specified in table 13.2.3.7A-2.

Table 13.2.3.7A-2: Output parameter of `downloadAndStore()` method

Name	Type	Description
—	`string`	The URL of a Media Entry Point in local storage which may be a valid MPD as defined in ISO/IEC 23009-1 [32] or TS 26.247 [4]. The downloaded media asset can be played back by passing this URL to the methods specified in clauses 13.2.3.4 and/or 13.2.3.5.

The following Media Player Actions are expected:

- If in *INITIALIZED* state, the `attach()` method (see clause 13.2.3.3) is invoked internally, passing the value of the *urlOrMPD* input parameter.
- The downloaded Media Entry Point is stored locally in the 5GMS Client for later playback. In the case where the Media Entry Point is a presentation manifest, such as an MPD, Base URLs may be modified to suit offline playback of the media resources it references.
- Depending on the type of the Media Entry Point, possibly present anchors as well as the wall-clock time, and other information, the earliest media time span for pre-loading is identified.
- The Access Client schedules and generates requests for the relevant media resources referenced by the Media Entry Point.
- The media resources are downloaded through reference point M4d at the earliest available opportunity. In the case where the Media Entry Point is a presentation manifest, such as an MPD, they are downloaded no earlier than the availability start time indicated in the presentation manifest.
- The downloaded media resources are stored locally in the 5GMS Client for later playback.
- The Media Player transitions to the *DOWNLOADING* state.
- Appropriate notifications and error messages are generated. For details, refer to clause 13.2.5.
- Appropriate Status Information is generated. For details, refer to clause 13.2.6.
- When the download completes (either successfully or unsuccessfully) the Media Player transitions to the *IDLE* state.

An application may use this method to preload media into the player in order minimize the start-up time.

13.2.3.8 Reset

This clause defines the `reset()` method.

The following pre-conditions apply:

- The Media Player may be in any state.

The `reset()` method is invoked on a Media Player instance to reset all information related to the media. As a side-effect, the Media Presentation described by the MPD is terminated.

The input parameters of the method are specified in table 13.2.3.8-1.

Table 13.2.3.8-1: Input parameters of `reset()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.

The following Media Player Actions are expected:

- The playback on the playback platform terminated.
- All open requests are cancelled.
- All scheduled requests are deleted.
- The current MPD is removed.
- Any media downloaded and stored in the 5GMSd Client by this Media Player instance is discarded.
- The Media Player transitions to the *INITIALIZED* state.

An application may use this method to terminate the playback of any media.

13.2.3.9 Destroy

This clause defines `destroy()` method.

The following pre-conditions apply:

- The Media Player may be in any state.

The `destroy()` method is invoked on a Media Player instance to reset all information related to the media and the network.

The input parameters of the method are specified in table 13.2.3.9-1.

Table 13.2.3.9-1: Input parameters of `destroy()` method

Name	Type	Description
sessionId	`string`	The media delivery session identifier (as specified in clause 7.3.2 of TS 26.510 [56] and in clause 10.1A of the present document) of an initialised downlink media streaming session.

The following Media Player actions are expected:

- The playback on the playback platform terminated.
- All open download requests are cancelled.
- All scheduled download requests are deleted.
- All network information is history is cleared.
- Any unexpired media downloaded and stored in the 5GMSd Client by this Media Player instance is retained.
- The Media Player transitions to the *IDLE* state.

An application may use this method to terminate the playback of any media clear and download related information.

If successful, the Media Player instance no longer exists once control has passed back from this method to the invoker.

13.2.4 Configurations and settings API

DASH streaming for a particular downlink media delivery session may be configured by the 5GMSd-Aware Application at reference point M7d or by the Media Session Handler at reference point M11d with the parameters provided in table 13.2.4-1. Note that these parameters may be set and they may also be observed.

Table 13.2.4-1: Media Player Configuration API

Status	Type	Definition
sessionId	`string`	A media delivery session identifier for the downlink media streaming session that has been initialised using the method specified in clause 13.2.3.2.
capabilities	`array(Enumeration)`	A read-only list of Media Player capabilities. See table 13.2.4-2.
source	`Object`	Provides the MPD and all contained information.
consumptionMode	`Enum`	Defines two modes: *live*: in this case the target latency is maintained, if specified in the service description, according to the parameters *vod*: in this case the latency is set by the application and the latency settings are ignored.
maxBufferTime	`Integer`	Maximum buffer time in milliseconds for the service.
serviceDescriptionId	`id`	Selects a service description by selecting an identifier.
serviceDescriptions[]	`Service description parameters`	Configures a service description as defined in annex K of ISO/IEC 23009-1 [32]. This allows the application to define additional service descriptions beyond those defined in the MPD, including a content identifier.
id	`id`	Sets a service description identifier different from the ones available in the service descriptions in the MPD or modifies existing service descriptions.
serviceLatency	`Object`	Sets service description parameters for the service latency, as defined in table K.3.1-1 of ISO/IEC 23009-1 [32].
playBackRate	`Object`	Sets service description parameters for the playback rate, as defined in table K.3.2-1 of ISO/IEC 23009-1 [32] when the service is consumed in live mode.
operatingQuality	`Object`	Sets service description parameters for the operating quality, as defined in table K.3.4-1 of ISO/IEC 23009-1 [32].
operatingBandwidth	`Object`	Sets service description parameters for the operating bandwidth, as defined in table K.3.5-1 of ISO/IEC 23009-1 [32].
clientDataReporting	`Object`	Sets service description parameters for client data reporting as defined in table K.3.7-1 of ISO/IEC 23009-1 [32]. These parameter values shall take precedence over the equivalent value specified in the selected Media Entry Point.
cmcdReporting	`Object`	Sets service description parameters for the CMCD reporting scheme as defined in table K.3.7-2 of ISO/IEC 23009-1 [32]. If the *sid* CMCD key is present in the service description parameters, it shall have the same value as *sessionId* above.
mediaSettings[]	Media type `audio`, `video`, `subtitle`	Sets the selected Adaptation Set based on the available Adaptation Sets for each media type.
metricsConfiguration[ ]	`Object`	Zero or more sets of settings for collecting and/or reporting metrics in relation to the downlink media streaming session. See table 13.2.4-3. The key of the array is a metrics reporting scheme identifier URI.
desiredContentDelivery Configuration	`Object`	Desired configuration of content delivery at reference point M4d.
deliveryProtocols	`array(Enumeration)`	Enumerated values from table 13.2.4-4, listed in decreasing order of preference, to be used by the Media Player for media delivery at reference point M4d.

	multipathMode	`Enumeration`	An enumerated value from table 13.2.4-5 indicating a preference on the use of multipath transport connections at reference point M4. When this is preferred, a multipath-capable Media Player should negotiate the establishment of multipath transport connections according to RFC 8484 [72] or [73] as appropriate for the selected delivery protocol.

Table 13.2.4-2: Media Player capabilities enumeration

Value	Definition
CAPABILITY_L4S	The Media Player has a protocol stack capable of handling ECN marking for L4S according to RFC 9330 [67], RFC 9331 [68] and RFC 9332 [69] (see NOTE).
CAPABILITY_QOS	The Media Player is capable of reacting to QoS monitoring results (see NOTE).
NOTE: Monitoring of progress on industry support for exposure of ECN marking for L4S and QoS monitoring is expected.	

Table 13.2.4-3: Metrics configuration object

Status	Type	Definition
reportingStartOffset	`DurationSec`	The time offset (expressed in seconds) from the start of a media delivery session when the Media Client is required to begin submitting metrics reports. The value shall not be negative. If omitted, the value of this parameter is assumed to be zero, i.e., directing the Media Player to start collecting and reporting metrics from the start of the media delivery session.
reportingDuration	`DurationSec`	The period of time (expressed in seconds) measured relative to the reporting start point, after which the Media Client is required to stop reporting metrics. The value shall not be negative. If set to zero, a single report shall be sent at *reportingStartOffset*. If omitted, reporting is required to continue until the end of the media delivery session.
reportingInterval	`DurationSec`	The time interval between successive metrics reports to be sent by the Media Session Handler. The value shall be greater than zero. If not specified, a single final report shall be sent after the media delivery session has ended.
samplePercentage	`Percentage`	The proportion of media delivery sessions for which QoE metrics shall be reported, expressed as a floating-point value between 0.0 and 100.0. If not specified, reports shall be sent for all media delivery sessions.
urlFilters	`array(string)`	A list of Media Entry Point URL patterns for which QoE metrics shall be reported. If not specified, reporting shall be done for all media delivery sessions initiated within the scope of the parent Provisioning Session.
samplingPeriod	`DurationSec`	The time interval the Media Player should wait between sampling the QoE metrics specified by this metrics configuration object. The value shall be greater than zero.
positiveCrossing Thresholds	`map(Uri -> array(Float)`	A map of QoE metrics to their respective threshold values. - The index of the associative array shall be an unqualified term identifier from the controlled vocabulary described by this metrics configuration object. - The value of each associative array member shall be an array of floating-point threshold values. A metric in this associative array shall be reported once when its value exceeds one of the associated threshold values, and shall not be reported again until it falls below that threshold and subsequently exceeds it.
negativeCrossing Thresholds	`map(Uri -> array(Float)`	A map of QoE metrics and their respective threshold values. - The index of the associative array shall be an unqualified term identifier from the controlled vocabulary described by this metrics configuration object.

		- The value of each associative array member shall be an array of floating-point threshold values. A metric in this associative array shall be reported once when its value falls below one of the associated threshold values, and shall not be reported again until it exceeds that threshold and subsequently falls below it.
locationFilter	`array(LocationArea5G)`	A list of one or more locations where QoE metrics collection is required. When present, a Media Player shall collect metrics only when it is located in these locations, and shall report them according to the other properties of the metrics configuration object. If omitted, QoE metrics are to be collected and reported regardless of the UE location.
metrics	`array(Uri)`	A list of QoE metrics, each indicated using an unqualified term identifier from the controlled vocabulary described by this metrics configuration object, which are required to be collected and reported by the Media Player. If omitted, the complete (or default, as applicable) set of metrics associated with the specified metrics scheme shall be collected and reported.

Table 13.2.4-4: Content Delivery Protocols enumeration

Value	Definition
DELIVERY_PROTOCOL_HTTP_1_1	HTTP/1.1 protocol as specified in RFC 9112 [24].
DELIVERY_PROTOCOL_HTTP_2	HTTP/2 protocol as specified in RFC 9113 [31]
DELIVERY_PROTOCOL_HTTP_3	HTTP/3 protocol as specified in RFC 9114 [60]

Table 13.2.4-5: Multipath connection preference enumeration

Value	Definition
MULTIPATH_PREFERRED	Preference for using multipath transport connections.
MULTIPATH_NOT_PREFERRED	Preference for not using multipath transport connections.

13.2.5 Notifications and error events

Table 13.2.5-1 provides a list of notification events that are provided by the Media Player to 5GMSd-Aware Applications at reference point M7d and to the Media Session Handler at reference point M11d. Every notification and error event is disambiguated by a media delivery session identifier.

Table 13.2.5-1: Media Player Notification events

Status	Definition	Payload
AST_IN_FUTURE	Triggered when playback will not start yet as the MPD's *availabilityStartTime* is in the future.	Media delivery session identifier, Time before playback will start.
AVAILABLE_MEDIA_CHANGED	The list of available media has changed.	Media delivery session identifier, Media type: - video - audio - subtitle - all
BUFFER_EMPTY	Triggered when the media playback platform's buffer state changes to stalled.	Media delivery session identifier, Media Type
BUFFER_LOADED	Triggered when the media playback platform's buffer state changes to loaded.	Media delivery session identifier, Media Type
CAN_PLAY	Sent when enough data is available that the media can be played.	Media delivery session identifier

CMCD_DATA_GENERATED	Triggered when CMCD data was generated for a HTTP request	Media delivery session identifier
MANIFEST_LOADED	Triggered when the manifest load is complete	Media delivery session identifier
METRIC_ADDED	Triggered every time a new metric is added.	Media delivery session identifier
METRIC_CHANGED	Triggered every time a metric value changes.	Media delivery session identifier, Metric identifier
METRIC_UPDATED	Triggered when the configuration of a metric is updated.	Media delivery session identifier, Metric identifier
METRICS_CHANGED	Triggered whenever there is a change to the overall metrics.	Media delivery session identifier
OPERATION_POINT_CHANGED	Triggered whenever there is a change of a Service Operation Point parameter.	Media delivery session identifier, External reference identifier of currently selected Service Operation Point.
PLAYBACK_ENDED	Sent when media playback completes normally.	Media delivery session identifier
PLAYBACK_ERROR	Sent when an error occurs during media playback. The element's error attribute contains more information.	Media delivery session identifier, Error reason (see table 13.2.5-2).
PLAYBACK_PAUSED	Sent when media playback is paused.	Media delivery session identifier
PLAYBACK_PLAYING	Sent when the media begins to play (either for the first time, after having been paused, or after ending and then restarting).	Media delivery session identifier
PLAYBACK_SEEKED	Sent when a media playback seek operation completes.	Media delivery session identifier
PLAYBACK_SEEKING	Sent when a media playback seek operation begins.	Media delivery session identifier
PLAYBACK_STALLED	Sent when the media playback platform reports stalled.	Media delivery session identifier
PLAYBACK_STARTED	Sent when playback of the media starts after having been paused; that is, when playback is resumed after a prior pause event.	Media delivery session identifier
PLAYBACK_WAITING	Sent when the media playback has stopped because of a temporary lack of data.	Media delivery session identifier
SERVICE_DESCRIPTION_SELECTED	sent when the DASH client has selected a service description.	Media delivery session identifier
SERVICE_DESCRIPTION_CHANGED	Sent when the DASH client has changed a service description.	Media delivery session identifier
SERVICE_DESCRIPTION_VIOLATED	Provides notification that the service description parameters are currently not met.	Media delivery session identifier, Parameters of service description that are not met
SOURCE_INITIALIZED	Triggered when the source is set up and ready.	Media delivery session identifier
DOWNLOAD_STARTED	Sent when a non-real-time content download begins.	Media delivery session identifier
DOWNLOAD_COMPLETED	Sent when a non-real-time content download is complete.	Media delivery session identifier
DOWNLOAD_ERROR	Send when an error occurs during non-real-time content download	Media delivery session identifier, Error reason (see table 13.2.5-2).
TRANSPORT_CONNECTION_ STATUS_CHANGED	Triggered when the status of a media transport connection changes, including its initial establishment.	Media delivery session identifier, Transport connection status information (see table

Table 13.2.5-2 provides a list of error reasons that are indicated for notifications of type *PLAYBACK_ERROR* and *DOWNLOAD_ERROR*.

Table 13.2.5-2: Media Player Error reasons

Error reason	Definition
ERROR_MEDIA_ENTRY_NOT_FOUND	The Media Entry Point resource requested by the Media Player could not be located.
ERROR_CONTENT_NOT_FOUND	Other content requested by the Media Player could not be located.
ERROR_MEDIA_PLAYBACK	There is an error from the media playback platform buffer.
ERROR_INVALID_MEDIA_ENTRY	The Media Entry Point resource supplied is not syntactically valid.
ERROR_INACCESSIBLE_MEDIA_TIME	The media time requested in a seek operation is not accessible in the current media presentation.
ERROR_UNSUPPORTED_PROFILE	The profile of the media presentation described by the Media Entry Point resource is not supported by the media playback platform.
ERROR_DOWNLOAD_DEADLINE_MISSED	The download of content was not completed before the requested deadline and the incomplete download has been discarded.
ERROR_UNSUPPORTED_ METRICS_SCHEME	One of the configured metrics schemes (see table 13.4.2-1) is not supported by the Media Player.

13.2.6 Dynamic Status Information

Table 13.2.6-1 provides a list of dynamically changing status information that can be obtained from the Media Player via reference point M7d or M11d. A separate set of Dynamic Status Information is provided for each active downlink media streaming session, indexed by its media delivery session identifier initialised per clause 13.2.3.2.

Table 13.2.6-1: Media Player Dynamic Status information

Status	Type	Parameter	Definition
state	Enumeration		An enumerated value from table 13.2.2-1 indicating the current state of the Media Player.
averageThroughput	`float`	none	Current average throughput computed in the ABR logic in bit/s.
bufferLength	`float`	`MediaType` "video", "audio" and "subtitle"	Current length of the buffer for a given media type, in seconds. If no type is passed in, then the minimum of video, audio and subtitle buffer length is returned. NaN is returned if an invalid type is requested, the presentation does not contain that type, or if no arguments are passed and the presentation does not include any adaption sets of valid media type.
liveLatency	`float`	none	Current live stream latency in seconds based on the latency measurement.
mediaSetting[ ]	`MPDAdaptationSet`	`MediaType` "video", "audio" and "subtitle"	Current media settings for each media type based on the CMAF Header and the MPD information based on the selected Adaptation Set for this media type.
mediaTime	`float`	None	Current media playback time from media playback platform. The media time is in seconds and is relative to the start of the playback and provides the media that is actually rendered.
playbackRate	`float`	None	The current rate of playback. For a video that is playing twice as fast as the default playback, the *playbackRate* value should be 2.00.

availableServiceDescriptions[ ]	Array of service descriptions		Provides the list of available selectable service descriptions with an id to select from. Those are either configured ones or the ones in the MPD.
availableMediaOptions[ ]	List of Adaptation Set or Preselection ids	`MediaType` "video", "audio" "subtitle" "all"	Provides the list of available media options that can be selected by the application based on the capability discovery and the subset information.
serviceOperationPoints	`array(Service OperationPoint)`		The set of Service Operation Points declared in the presentation manifest (e.g. DASH MPD) of the current media presentation. See table 13.2.6-2.
operativeServiceOperation Point	`integer`		A zero-based index into the *service OperationPoints* array indicating the Service Operation Point currently operative in the playback session. Set to -1 if the array is empty.
pathmetrics[ ][ ]	`Metrics`		A data blob of metrics for each configured metrics collecting scheme.
transportConnectionStatuses[ ]	`array(Transport Connection Status)`		Status information about the current set of reference point M4d transport connection(s) for media delivery. See table 13.2.6-3.

Table 13.2.6-2 provides a list of configured operation point information that can be obtained from the client. Any change to a parameter below shall be announced with a notification *OPERATION_POINT_CHANGED* as specified in table 13.2.5-1.

Table 13.2.6-2: Media Player Service Operation Point Information

Parameter	Type	Definition
ServiceOperationPoint	`Object`	The currently configured Service Operation Point parameters according to which the DASH client is operating.
externalIdentifier	`String`	The external identifier uniquely identifying this Service Operation Point in the presentation manifest (e.g. DASH MPD).
mode	`Enum`	The following operation modes are defined: *live*: The DASH client operates to maintain configured target latencies using playback rate adjustments and possibly resynchronisation. *vod*: The DASH client operates without latency requirements and rebuffering may result in additional latencies
maxBufferTime	`Integer`	maximum buffer time in milliseconds for the service.
switchBufferTime	`Integer`	buffer time threshold below which the DASH clients attempt to switch Representations.
latency	`Object`	Defines the latency parameters used by the DASH client when operating in live mode.
target	`Integer`	The target latency for the service in milliseconds.
max	`Integer`	The maximum latency for the service in milliseconds.
min	`Integer`	The maximum latency for the service in milliseconds.
playbackRate	`MediaType` *audio*, *video*, *all*	Defines the playback rate parameters used by the DASH client for catchup mode and deceleration to avoid buffer underruns and maintaining target latencies.
max	`Real`	The maximum playback rate for the purposes of automatically adjusting playback latency and buffer occupancy during normal playback, where 1.0 is normal playback speed.
min	`Real`	The minimum playback rate for the purposes of automatically adjusting playback latency and buffer occupancy during normal playback, where 1.0 is normal playback speed.
bitRate		Defines the operating bit rate parameters used by the DASH client used for a specific media type or aggregated. The values are on IP level.
target	`Integer`	The target bit rate for the service in bit/s that the client is configured to consume.
max	`Integer`	The maximum bit rate for the service in bit/s that the client is configured to consume.
min	`Integer`	The minimum bit rate for the service in bit/s that the client is configured to consume.
playerSpecificParameters		Player-specific parameters may be provided, for example about the used algorithm, etc.

Table 13.2.6-3 specifies the transport connection status information object. Any change to a parameter below shall be announced with the *TRANSPORT_CONNECTION_STATUS_CHANGED* notification as specified in table 13.2.5-1.

Table 13.2.6-3: Transport connection status information

Parameter	Type	Definition
TransportConnectionStatus	`Object`	Status information of a transport connection used for media delivery at reference point M4d.
componentIds	`array(String)`	Identifies which media presentation components are conveyed by this transport connection. Depending on implementation this could, for example, be a set of MPEG-DASH Representation identifiers or a set of Service Location labels.
transportProtocol	`Enumeration`	An enumerated value from table 13.2.6-4 indicating the transport protocol used by this transport connection.
numberOfActivePaths	`Integer`	The current number of active subflows/paths in this transport connection.

Table 13.2.6-4: Transport Protocols enumeration

Value	Definition
TRANSPORT_PROTOCOL_TCP	TCP as specified in RFC 9293 [71].
TRANSPORT_PROTOCOL_QUIC	QUIC as specified in RFC 9000 [58].
TRANSPORT_PROTOCOL_MPTCP	Multipath TCP as specified in RFC 8684 [72].
TRANSPORT_PROTOCOL_MPQUIC	Multipath QUIC as specified in [73].

13.2.7 Usage of Media Player information by Media Session Handler

The Media Session Handler may use the notifications, errors and status information provided by the Media Player at reference point M11 to execute relevant tasks, such as compiling QoE metrics reports.

14 Application (M8) APIs for uplink and downlink

APIs of this reference point are not specified within this release.

15 Miscellaneous UE-internal APIs

15.1 General

The use of UE modem APIs by the Media Session Handler to support ANBR-based Network Assistance and ANBR-based metrics reporting is specified in clause 5.6 of TS 26.510 [56].

15.2 Void

15.3 Void

16 Usage of 5GC interfaces and APIs

16.1 General

The use of 5GC APIs by the 5GMS AF is specified in clause 5.5 of TS 26.510 [56].

16.2 Void

16.3 Void

17 Media Streaming data reporting at R4

17.1 General

The following record types shall be used with the *Ndcaf_DataReporting_Report* service operation specified in clause 7.2.3.4.1 of TS 26.532 [47]. In each case, one or more records are included in a *DataReport*, as specified in clause 7.3.3.2.1 of [47].

The OpenAPI definitions of the record types are found in clause C.5.1 of the present document.

17.2 MediaStreamingAccessRecord type

As specified in clause 4.11.3, the *MediaStreamingAccessRecord* type shall be used by the 5GMS AS to report media streaming access. This data type is defined in clause C.5.1 as the combination of:

- *BaseRecord*, as defined in clause B.4 of TS 26.532 [49], which provides a record timestamp for the data report and which also cites the set of reporting context identifiers that gave rise to the data report, as specified in clause 4.1 of [49].
- *MediaStreamingBaseReportingRecord*, as defined in clause 6.4.3.9A of the present document, which enables the network slice and/or Data Network of the media streaming session at the time of the media streaming access to be reported by the 5GMS AS, as well as the location of the UE when the media streaming access was made.
- *MediaStreamingSessionIdentification*, as defined in clause 6.4.3.10 of the present document.

  If a media streaming session identifier is not explicitly provided by the 5GMS Client at reference point M4, the 5GMS AS should instead synthesise a value for the *sessionId* property, for example a one-way hash of the transport connection identifier (if available from the underlying transport protocol) or a one-way hash of the 5-tuple formed from the Media Stream Handler and 5GMSd AS endpoint addresses and a transport protocol identifier, combined with a randomly chosen salt to prevent reverse engineering of the original values.
- *MediaStreamingAccess*, as defined in clause 6.4.3.11 of the present document. These properties shall be populated by the 5GMS AS.

17A Data reporting at R2

17A.1 General

The following record types shall be used with the *Ndcaf_DataReporting_Report* service operation specified in clause 7.2.3.4.1 of TS 26.532 [49]. In each case, one or more records are included in a *DataReport*, as specified in clause 7.3.2.3 of [49].

The OpenAPI definitions of the record types are found in clause C.5.2 of the present document.

17A.2 ANBRNetworkAssistanceInvocationRecord type

As specified in clause 7.3.2.3 of TS 26.532 [49], the *ANBRNetworkAssistanceInvocationRecords* type shall be used by the Media Session Handler to report ANBR-based Network Assistance invocations at reference point R2.

The *ANBRNetworkAssistanceInvocationRecords* type structure is defined in clause C.5.2 as the combination of:

- *BaseRecord*, as defined in clause B.4 of [49], which provides a record timestamp for the data report and which also cites the set of reporting context identifiers that gave rise to the data report, as specified in clause 4.1 of [49].
- *MediaStreamingBaseReportingRecord*, as defined in clause 6.4.3.9A of the present document, which enables the network slice and/or Data Network of the media streaming session on which ANBR-based Network Assistance

was invoked to be reported by the Media Session Handler, as well as the location of the UE when the ANBR-based Network Assistance was invoked.

- *MediaStreamingSessionIdentification*, as defined in clause 6.4.3.10 of the present document.
- *NetworkAssistanceInvocation*, as defined in clause 6.4.3.12 of the present document.

18 Event exposure at R5 and R6

18.1 General

The *Naf_EventExposure* service specified in TS 29.517 [46] shall be used by event consumers to subscribe to the following types of 5G Media Streaming event notifications, identified by their respective Event IDs, from the Data Collection AF instantiated in the 5GMS AF, and subsequently to receive such notifications in the form of *AfEventNotification* objects as specified in table 5.6.2.6-1 of [46] and as further specified in this clause:

1. Media Streaming QoE Event notifications, as specified in clause 18.3, comprising
   a) 3GPP-defined QoE metrics information, and
   b) non-3GPP-defined QoE metrics information.
2. Media Streaming Consumption Event notifications, as specified in clause 18.4.
3. Media Streaming Network Assistance Invocation Event notifications, as specified in clause 18.5.
4. Media Streaming Dynamic Policy Invocation Event notifications, as specified in clause 18.6.
5. Media Streaming Access Event notifications, as specified in clause 18.7.

OpenAPI data types for these event notifications are found in clause C.6.

In this release, eligible event consumer subscribers to the *Naf_EventExposure* service as specified in [46] are the following:

- The NWDAF defined in TS 23.288 [47].
- The Event Consumer AF defined in TS 26.531 [46] when it is deployed in the Trusted DN.
- The NEF defined in TS 23.501 [2] when it is used to expose the *Naf_EventExposure* service to functions outside the Trusted DN via the *Nnef_EventExposure* service defined in TS 23.502 [45] and specified in TS 29.591 [55].

Implementations of the Data Collection AF instantiated the 5GMS AF shall support negotiation of the optional features relating to 5G Media Streaming specified in table 5.8-1 of TS 29.517 [46]. Feature negotiation by event consumers is achieved as specified in clause 5.8 of [46].

18.2 Common data types for event exposure

18.2.1 BaseEventCollection data type

BaseEventCollection is an abstract base data type for describing a collection of media streaming event records. It is combined with other collection data types to form concrete collections. As such, this data type shall not be exposed by the Data Collection AF instantiated in the 5GMS AF.

Table 18.2.1-1: BaseEventCollection data type

Property name	Data Type	Cardinality	Description
collectionTimestamp	`DateTime`	1..1	The date–time at which this collection was exposed by the Data Collection AF as an event to its subscribed event consumers.
startTimestamp	`DateTime`	1..1	Date–time of earliest data sample included in or summarised by this collection.
endTimestamp	`DateTime`	1..1	Date–time of latest data sample included in or summarised by this collection.
sampleCount	`integer`	1..1	The number of data samples included in or summarised by this collection.
streamingDirection	`Provisioning SessionType`	1..1	Indicates whether the collection concerns a downlink media streaming session or an uplink media streaming session.
summarisations	`array(Data Aggregation Function Type)`	1..1	A set of one or more data aggregation functions (see TS 26.532 [49]) that have been applied to the UE data to produce summary records present in this collection.
records	`array()`	1..1	A set of zero or more records, the type of which is specified in subsequent clauses. Empty when the *summarisations* property indicates that this collection describes only a count of UE data samples.

18.2.2 BaseEventRecord data type

BaseEventRecord data type is an abstract base data type for describing a single UE data record or summarising a summarising a set of UE data records. It is combined with other collection data types to form concrete collections. As such, this data type shall not be exposed by the Data Collection AF instantiated in the 5GMS AF.

Table 18.2.2-1: BaseEventRecord data type

Property name	Data Type	Cardinality	Description
recordType	`EventRecordType`	1..1	The type of this record. See table 18.2.2-2.
recordTimestamp	`DateTime`	1..1	Date–time at which the UE data carried in this record was sampled or summarised.
appId	`ApplicationId`	1..1	Identifying the application (see table 5.4.2-1 of TS 29.571 [12]) to which the UE data carried in this record pertains.
provisioningSessionId	`ResourceId`	0..1	The identifier of the Provisioning Session to which this record pertains. Present only for individual data sample *recordType*.
sessionId	`MediaDelivery SessionId`	0..1	A value that uniquely identifies the media streaming session to which this record pertains. Present only for individual data sample *recordType*.
ueIdentification	`string`	0..1	GPSI of the requesting UE or a stable globally unique string identifying the requesting Media Session Handler. Present only for individual data sample *recordType* and only when exposure is permitted by the data exposure restrictions in force for the event type in question.
dataNetworkName	`Dnn`	0..1	Identifying the Data Network of the M4 media streaming session. Present only for individual data sample *recordType*.
sliceId	`Snssai`	0..1	The S-NSSAI identifying the Network Slice of the M4 media streaming session.

Property name	Data Type	Cardinality	Description
			Present only for individual data sample *recordType*.
ueLocations	`array(LocationArea5G)`	0..1	The location(s) of the UE when the data described by this record was sampled. Present only for individual data sample *recordType* and only when exposure is permitted by the data exposure restrictions in force for the event in question.

Table 18.2.2-2: EventRecordType enumeration

Enumeration value	Description
INDIVIDUAL_SAMPLE	The event record is an individual UE data sample.
SUMMARY_MEAN	The event record summarises the mean average value over the period indicated by the parent collection.
SUMMARY_MINIMUM	The event record summarises the minimum value over the period indicated by the parent collection.
SUMMARY_MAXIMUM	The event record summarises the maximum value over the period indicated by the parent collection.
SUMMARY_SUM	The event record summarises the sum over the period indicated by the parent collection.

18.3 QoE metrics event notifications

18.3.1 QoEMetricsCollection data type

QoEMetricsCollection is a concrete data type describing a collection of QoE Metrics Event records.

Table 18.3.1-1: QoEMetricsCollection data type

Property name	Data Type	Cardinality	Description
collectionTimestamp	`DateTime`	1..1	The date–time at which this collection was exposed by the Data Collection AF as an event to its subscribed event consumers.
startTimestamp	`DateTime`	1..1	Date–time of earliest QoE metrics report included in or summarised by this collection.
endTimestamp	`DateTime`	1..1	Date–time of latest QoE metrics report included in or summarised by this collection.
sampleCount	`integer`	1..1	The number of QoE metrics reports included in or summarised by this collection. (Where summary records are included in the collection, the number of records in the collection differs from this number.)
streamingDirection	`Provisioning SessionType`	1..1	Collections of QoE metrics records only apply to downlink media streaming.
summarisations	`array(Data Aggregation Function Type)`	1..1	One or more of the following data aggregation functions (see clause 6.3.3.2 of TS 26.532 [49]) that have been applied to the UE data to produce summary records present in this collection with the semantics indicated in table 4.7.4.4-1 of TS 26.501 [2]: - *NULL* - *COUNT* - *MEAN* - *MINIMMUM* - *MAXIMUM* - *SUM*
records	`array(QoE Metrics Event)`	1..1	A set of zero or more QoE Metrics Event records, each one describing a QoE metrics report or summarising a set of QoE metrics reports. Empty when the *summarisations* property indicates that this collection describes only a count of UE data samples.

18.3.2 QoEMetricsEvent data type

QoEMetricsEvent is a concrete data type describing a set of, or summaries of, QoE metric samples of the same type.

Table 18.3.2-1: QoEMetricsEvent data type

Property name	Data Type	Cardinality	Description
recordType	`EventRecordType`	1..1	One of the following: - *INDIVIDUAL_SAMPLE* - *SUMMARY_MEAN* - *SUMMARY_MINIMMUM* - *SUMMARY_MAXIMUM* - *SUMMARY_SUM*
recordTimestamp	`DateTime`	1..1	For individual records, the date–time at which the parent QoE metrics report was generated by the Media Session Handler for submission at reference point M5 or generated by the 5GMS AS for submission at reference point M3. Otherwise, the date–time at which the summary record was generated by the Data Collection AF instantiated in the 5GMS AF.
appId	`ApplicationId`	1..1	Identifying the application (see table 5.4.2-1 of TS 29.571 [12]) to which the UE data carried in this record pertains.
provisioningSessionId	`ResourceId`	0..1	The identifier of the Provisioning Session to which this record pertains. Present only for individual data sample *recordType*.
sessionId	`MediaDelivery SessionId`	0..1	A value that uniquely identifies the media streaming session to which this record pertains. Present only for individual data sample *recordType*.
ueIdentification	`string`	0..1	GPSI of the requesting UE or a stable globally unique string identifying the requesting Media Session Handler. Present only for individual data sample *recordType* and only when exposure is permitted by the data exposure restrictions in force.
dataNetworkName	`Dnn`	0..1	Identifying the Data Network of the M4 media streaming session. Present only for individual data sample *recordType*.
sliceId	`Snssai`	0..1	The S-NSSAI identifying the Network Slice of the M4 media streaming session. Present only for individual data sample *recordType*.
ueLocations	`array(Location Area5G)`	0..1	The location(s) of the UE when the data described by this record was sampled. Present only for individual data sample *recordType* and only when exposure is permitted by the data exposure restrictions in force.

Property name	Data Type	Cardinality	Description
metricType	`Uri`	1..1	A fully-qualified term identifier that uniquely identifies the QoE metrics reporting scheme and the type of QoE metric included this record, as specified in clause E.2, up to but excluding the first hierarchical separator. For example, *urn:3GPP:ns:PSS:DASH:QM10#AvgThroughput* or *urn:3gpp:5gms:event-exposure:common-media-client-data#session*.
samples	`array(object)`	1..1	An ordered list of one or more samples of type *metricType* derived from a single QoE metrics report.
sampleTimestamp	`DateTime`	0..1	Where applicable for the metric indicated by *metricType*, the moment in time at which this QoE metric was sampled.
sampleDuration	`Duration`	0..1	Where applicable for the metric indicated by *metricType*, the time duration over which this QoE metric was sampled.
mediaTimestamp	`Duration`	0..1	Where applicable for the metric indicated by *metricType*, the time point (expressed relative to the start of the media streaming presentation) at which this QoE metric was sampled.
metrics	`array(object)`	1..1	A set of key–value pairs for the sampled metrics associated with this QoE metric sample.
key	`string`	1..1	A token that uniquely identifies metric subtype within the scope of the QoE metric type. For example: *numbytes*. There shall be at most one instance of this property's value in the parent array.
value	`{}`	0..1	A value (of any type) associated with the metric indicated by *key*.

18.3.3 Exposure of client data

18.3.3.1 Exposure of CMCD version 1 information

CMCD version 1 information [70] submitted to the 5GMSd AF and passed to the Data Collection AF instantiated in the 5GMS AF shall be packed into *QoEMetricsEvent* records (see clause 18.3.2) as follows:

- *recordTimestamp* shall be populated with the timestamp of an HTTP request received by the 5GMS AS at reference point M4d.
- *appId* shall be populated with the external service identifier of the Provisioning Session.
- *sessionId* shall be populated with the CMCD session identifier, which is typically the same value as the media delivery session identifier (see clause 6.2.3.6).
- *metricType* shall be populated with one of the fully-qualified term identifier URIs listed in table E.2.5-1, indicating the class of CMCD information conveyed in this record.

In the case of individual records (*recordType* set to *INDIVIDUAL_SAMPLE*) the *samples* array shall contain a single entry, corresponding to CMCD information of the class indicated by *metricType* from a single HTTP request at reference point M4d.

- *sampleTimestamp*, *sampleDuration* and *mediaTimestamp* shall be omitted; the timestamp of all metrics in the record is indicated by *recordTimestamp*, as specified above.
- The *metrics* array shall be populated with the set of CMCD keys and values from a single HTTP request at reference point M4d belonging to the class of CMCD information indicated in *metricType*.
  - *key* shall be one the key names reserved in section 3.3 of the CMCD specification [70] for the class of CMCD information indicated in *metricType*.

Aligned with section 3.1 of the CMCD specification [70], key–value pairs should appear in alphabetical order of key name when transmitted. (This reduces the fingerprinting surface exposed by different sources of client data.)

- *value* shall be the value of the key indicated by *key*. In the case of Boolean keys, *value* may be omitted, implying the value *true*.

18.4 Consumption reporting event notifications

18.4.1 ConsumptionReportingUnitsCollection data type

ConsumptionReportingUnitsCollection is a concrete data type describing a collection of consumption reporting event records.

Table 18.4.1-1: ConsumptionReportingUnitsCollection data type

Property name	Data Type	Cardinality	Description
collectionTimestamp	DateTime	1..1	The date–time at which this collection was exposed by the Data Collection AF as an event to its subscribed event consumers.
startTimestamp	DateTime	1..1	Date–time of earliest consumption reporting unit included in or summarised by this collection.
endTimestamp	DateTime	1..1	Date–time of latest consumption reporting unit included in or summarised by this collection.
sampleCount	integer	1..1	The number of consumption reporting units included in or summarised by this collection. (Where summary records are included in the collection, the number of records in the collection differs from this number.)
streamingDirection	ProvisioningSessionType	1..1	Collections of consumption reporting event records only apply to downlink media streaming.
summarisations	array(DataAggregationFunctionType)	1..1	One or more of the following data aggregation functions (see clause 6.3.3.2 of TS 26.532 [49]) that have been applied to the UE data to produce summary records present in this collection with the semantics indicated in table 4.7.4.5-1 of TS 26.501 [2]: - *NULL* - *COUNT*
records	array(ConsumptionReportingEvent)	1..1	A set of zero or more consumption reporting event records.

18.4.2 ConsumptionReportingEvent data type

ConsumptionReportingEvent is a concrete data type corresponding to a single consumption reporting unit.

Table 18.4.2-1: ConsumptionReportingEvent data type

Property name	Data Type	Cardinality	Description
recordType	EventRecordType	1..1	One of the following: - *INDIVIDUAL_SAMPLE*
recordTimestamp	DateTime	1..1	The date–time at which this consumption reporting unit began.
appId	ApplicationId	1..1	Identifying the application (see table 5.4.2-1 of TS 29.571 [12]) to which the UE data carried in this record pertains.
provisioningSessionId	ResourceId	1..1	The identifier of the Provisioning Session to which this record pertains.
sessionId	MediaDeliverySessionId	1..1	A value that uniquely identifies the media streaming session to which this record pertains.

Property name	Data Type	Cardinality	Description
ueIdentification	`string`	0..1	GPSI of the requesting UE or a stable globally unique string identifying the requesting Media Session Handler. Present only when exposure is permitted by the data exposure restrictions in force.
dataNetworkName	`Dnn`	1..1	Identifying the Data Network of the M4 media streaming session.
sliceId	`Snssai`	1..1	The S-NSSAI identifying the Network Slice of the M4 media streaming session.
ueLocations	`array(LocationArea5G)`	0..1	The location(s) of the UE when the data described by this record was sampled. Present only when exposure is permitted by the data exposure restrictions in force.
unitDuration	`Duration`	1..1	The duration of this consumption reporting unit.
clientEndpointAddress	`EndpointAddress`	0..1	The endpoint address of the UE that consumed media. Present only if access reporting is enabled in the Consumption Reporting Configuration.
serverEndpointAddress	`EndpointAddress`	0..1	The endpoint address of the 5GMS AS from which media was consumed. Present only if access reporting is enabled in the Consumption Reporting Configuration.
mediaPlayerEntryUrl	`AbsoluteUrl`	1..1	The Media Player Entry URL to which this consumption reporting unit pertains.
mediaComponentIdentifier	`string`	1..1	A token (e.g., an MPEG-DASH representation identifier) identifying the media component of the Media Player Entry that was consumed during this consumption reporting unit.

18.5 Network Assistance invocation event notifications

18.5.1 NetworkAssistanceInvocationsCollection data type

NetworkAssistanceInvocationsCollection is a concrete data type describing a collection of Network Assistance Invocation Event records.

Table 18.5.1-1: NetworkAssistanceInvocationsCollection data type

Property name	Data Type	Cardinality	Description
collectionTimestamp	DateTime	1..1	The date–time at which this collection was exposed by the Data Collection AF as an event to its subscribed event consumers.
startTimestamp	DateTime	1..1	Date–time of earliest Network Assistance invocation included in or summarised by this collection.
endTimestamp	DateTime	1..1	Date–time of latest Network Assistance invocation included in or summarised by this collection.
sampleCount	integer	1..1	The number of Network Assistance invocations included in or summarised by this collection. (Where summary records are included in the collection, the number of records in the collection differs from this number.)
streamingDirection	ProvisioningSessionType	1..1	Indicating whether this collection of Network assistance invocation records relates to downlink media streaming sessions or to uplink media streaming sessions.
summarisations	array(DataAggregationFunctionType)	1..1	One or more of the following data aggregation functions (see clause 6.3.3.2 of TS 26.532 [49]) that have been applied to the UE data to produce summary records present in this collection with the semantics indicated in table 4.7.4.7-1 of TS 26.501 [2]: - *NULL* - *COUNT* - *MEAN* - *MINIMMUM* - *MAXIMUM*
records	array(NetworkAssistanceInvocationEvent)	1..1	A set of zero or more Network Assistance invocation records.

18.5.2 NetworkAssistanceInvocationEvent data type

NetworkAssistanceInvocationEvent is a concrete data type describing a single invocation of the Network Assistance API (see clause 11.6).

Table 18.5.2-1: NetworkAssistanceInvocationEvent data type

Property name	Data Type	Cardinality	Description
recordType	EventRecordType	1..1	One of the following: - *INDIVIDUAL_SAMPLE* - *SUMMARY_MEAN* - *SUMMARY_MINIMMUM* - *SUMMARY_MAXIMUM*
recordTimestamp	DateTime	1..1	For individual records, the date–time at which Network Assistance was invoked by the Media Session Handler. Otherwise, the date–time at which the summary record was generated by the Data Collection AF instantiated in the 5GMS AF.

Property name	Data Type	Cardinality	Description
appId	ApplicationId	1..1	Identifying the application (see table 5.4.2-1 of TS 29.571 [12]) to which the UE data carried in this record pertains.
provisioningSessionId	ResourceId	0..1	The identifier of the Provisioning Session to which this record pertains. Present only for individual data sample *recordType*.
sessionId	MediaDelivery SessionId	0..1	A value that uniquely identifies the media streaming session to which this record pertains. Present only for individual data sample *recordType*.
ueIdentification	string	0..1	GPSI of the requesting UE or a stable globally unique string identifying the requesting Media Session Handler. Present only for individual data sample *recordType* and only when exposure is permitted by the data exposure restrictions in force.
dataNetworkName	Dnn	0..1	Identifying the Data Network of the M4 media streaming session. Present only for individual data sample *recordType*.
sliceId	Snssai	0..1	The S-NSSAI identifying the Network Slice of the M4 media streaming session. Present only for individual data sample *recordType*.
ueLocations	array(Location Area5G)	0..1	The location of the UE when Network Assistance was sought. If present, the array shall contain exactly one location for the Network Assistance invocation. Present only for individual data sample *recordType* and only when exposure is permitted by the data exposure restrictions in force.
networkAssistance Type	Network AssistanceType	1..1	The type of Network Assistance solicited by the Media Session Handler (see table 18.5.2-2).
policyTemplateId	ResourceId	0..1	Identifying the Policy Template (if any) referenced by the Media Session Handler in the parent Network Assistance Session.
serviceDataFlow Descriptions	array(Service DataFlow Description)	0..1	If present, a set of one or more Service Data Flow Descriptions (see clause 6.4.3.1) to which the Network Assistance session has been applied. Present only for individual data sample *recordType* and only when exposure is permitted by the data exposure restrictions in force.
requestedQoS	Unidirectional QoSSpecification	0..1	The network QoS parameters (if any) requested by the Media Session Handler from the 5GMS AF in the parent Network Assistance Session or from the RAN (see clause 6.4.3.13).
recommendedQoS	object	0..1	The network QoS parameters (if any) recommended to the Media Session Handler by the 5GMS AF in the parent Network Assistance Session or by the RAN.
maximumBitRate	BitRate	1..1	The maximum recommended bit rate.
minimumBitRate	BitRate	1..1	The minimum recommended bit rate.

Table 18.5.2-2: NetworkAssistanceType enumeration

Enumeration value	Description
AF_THROUGHPUT_ESTIMATION	Throughput estimation (bit rate recommendation) sought from the 5GMS AF.
AF_DELIVERY_BOOST	Delivery boost sought from the 5GMS AF.
ANBR_THROUGHPUT_ESTIMATION	Throughput estimation (bit rate recommendation) sought from the RAN.
ANBR_DELIVERY_BOOST	Delivery boost sought from the RAN.

18.6 Dynamic policy invocation event notifications

18.6.1 DynamicPolicyInvocationsCollection data type

DynamicPolicyInvocationsCollection is a concrete data type describing a collection of Dynamic Policy Invocation records.

Table 18.6.1-1: DynamicPolicyInvocationsCollection data type

Property name	Data Type	Cardinality	Description
collectionTimestamp	DateTime	1..1	The date–time at which this collection was exposed by the Data Collection AF as an event to its subscribed event consumers.
startTimestamp	DateTime	1..1	Date–time of earliest dynamic policy invocation included in or summarised by this collection.
endTimestamp	DateTime	1..1	Date–time of latest dynamic policy invocation included in or summarised by this collection.
sampleCount	integer	1..1	The number of dynamic policy invocations included in or summarised by this collection. (Where summary records are included in the collection, the number of records in the collection differs from this number.)
streamingDirection	ProvisioningSessionType	1..1	Indicating whether this collection of dynamic policy invocation records relates to downlink media streaming sessions or to uplink media streaming sessions.
summarisations	array(DataAggregationFunctionType)	1..1	One or more of the following data aggregation functions (see clause 6.3.3.2 of TS 26.532 [49]) that have been applied to the UE data to produce summary records present in this collection with the semantics indicated in table 4.7.4.6-1 of TS 26.501 [2]: - *NULL* - *COUNT*
records	array(DynamicPolicyInvocationEvent)	1..1	A set of zero or more dynamic policy invocation event records.

18.6.2 DynamicPolicyInvocationEvent data type

DynamicPolicyInvocationEvent is a concrete data type describing a single invocation of the Dynamic Policies API (see clause 11.5).

Table 18.6.2-1: DynamicPolicyInvocationEvent data type

Property name	Data Type	Cardinality	Description
recordType	EventRecordType	1..1	One of the following: - *INDIVIDUAL_SAMPLE*
recordTimestamp	DateTime	1..1	The date–time at which the dynamic policy was invoked by the Media Session Handler.
appId	ApplicationId	1..1	Identifying the application (see table 5.4.2-1 of TS 29.571 [12]) to which the UE data carried in this record pertains.
provisioningSessionId	ResourceId	1..1	The identifier of the Provisioning Session to which this record pertains.
sessionId	MediaDeliverySessionId	1..1	A value that uniquely identifies the media streaming session to which this record pertains.
ueIdentification	string	0..1	GPSI of the requesting UE or a stable globally unique string identifying the requesting Media Session Handler. Present only when exposure is permitted by the data exposure restrictions in force.

Property name	Data Type	Cardinality	Description
dataNetworkName	Dnn	1..1	Identifying the Data Network of the M4 media streaming session.
sliceId	Snssai	1..1	The S-NSSAI identifying the Network Slice of the M4 media streaming session.
ueLocations	array(Location Area5G)	0..1	The location of the UE when the dynamic policy was invoked. If present, the array shall contain exactly one location for the dynamic policy invocation. Present only when exposure is permitted by the data exposure restrictions in force.
policyTemplateId	ResourceId	1..1	Identifying the Policy Template instantiated by the Media Session Handler.
applicationFlow Descriptions	array(ApplicationFlow Description)	0..1	If present, a set of one or more Application Flow Descriptions (see clause 7.3.3.2 of TS 26.510 [56]) to which the dynamic policy has been applied. Present only when exposure is permitted by the data exposure restrictions in force.
requestedQoS	Unidirectional QoSSpecification	0..1	The network QoS parameters (if any) requested by the Media Session Handler when instantiating the Policy Template (see clause 6.4.3.13).
enforcementMethod	string	1..1	The policy enforcement method indicated by the 5GMS AF.
enforcementBitRate	BitRate	1..1	The enforcement bit rate indicated by the 5GMS AF.

18.7 Media streaming access event notifications

18.7.1 MediaStreamingAccessesCollection data type

MediaStreamingAccessesCollection is a concrete data type describing a collection of Media Streaming Access Event records.

Table 18.7.1-1: MediaStreamingAccessesCollection data type

Property name	Data Type	Cardinality	Description
collectionTimestamp	DateTime	1..1	The date–time at which this collection was exposed by the Data Collection AF as an event to its subscribed event consumers.
startTimestamp	DateTime	1..1	Date–time of earliest media streaming access event record included in or summarised by this collection.
endTimestamp	DateTime	1..1	Date–time of latest media streaming access event record included in or summarised by this collection.
sampleCount	integer	1..1	The number of media streaming access event records included in or summarised by this collection.
streamingDirection	Provisioning SessionType	1..1	Indicating whether this collection of media streaming access event records relates to downlink media streaming sessions or to uplink media streaming sessions.
summarisations	array(Data Aggregation Function Type)	1..1	One or more of the following data aggregation functions (see clause 6.3.3.2 of TS 26.532 [49]) that have been applied to the UE data to produce summary records present in this collection with the semantics indicated in table 4.7.4.8-1 of TS 26.501 [2]: - *NULL* - *COUNT*
records	array(Media Streaming AccessEvent)	1..1	A set of zero or more media streaming access event records.

18.7.2 MediaStreamingAccessEvent data type

MediaStreamingAccessEvent is a concrete data type describing a single media access by a Media Stream Handler.

Table 18.7.2-1: MediaStreamingAccessEvent data type

Property name	Data Type	Cardinality	Description
recordType	`EventRecordType`	1..1	One of the following: - *INDIVIDUAL_SAMPLE*
recordTimestamp	`DateTime`	1..1	The date–time at which media was accessed by the Media Stream Handler (Media Player or Media Streamer).
appId	`ApplicationId`	1..1	Identifying the application (see table 5.4.2-1 of TS 29.571 [12]) to which the UE data carried in this record pertains.
provisioningSessionId	`ResourceId`	1..1	The identifier of the Provisioning Session to which this record pertains.
sessionId	`MediaDelivery SessionId`	1..1	A value that uniquely identifies the media streaming session to which this record pertains.
ueIdentification	`string`	0..1	GPSI of the requesting UE or a stable globally unique string identifying the requesting Media Session Handler. Present only when exposure is permitted by the data exposure restrictions in force.
dataNetworkName	`Dnn`	1..1	Identifying the Data Network of the M4 media streaming session.
sliceId	`Snssai`	1..1	The S-NSSAI identifying the Network Slice of the M4 media streaming session.
ueLocations	`array(Location Area5G)`	0..1	The location of the UE when the media was accessed. If present, the array shall contain exactly one location for the media streaming access. Present only when exposure is permitted by the data exposure restrictions in force.
mediaStreamHandler EndpointAddress	`EndpointAddress`	1..1	The endpoint address of the Media Stream Handler accessing the 5GMS AS. (See clause 6.4.3.8.)
applicationServer EndpointAddress	`EndpointAddress`	1..1	The service endpoint on the 5GMS AS to which the Media Stream Handler is connected. (See clause 6.4.3.8.)
requestMessage	`Object`	1..1	Details of the HTTP request message submitted to the 5GMS AS by the Media Stream Handler for this media access.
method	`string`	1..1	The request method.
url	`AbsoluteUrl`	1..1	The request URL. (See table 6.4.2-1.)
protocolVersion	`string`	1..1	The HTTP protocol version, e.g., "HTTP/1.1".
range	`string`	0..1	The value of the `Range` request header, if present.
size	`Uinteger`	1..1	The total number of bytes in the request message.
bodySize	`Uinteger`	1..1	The number of bytes supplied by the Media Stream Handler in the HTTP request message body. Zero if there is no request body.
contentType	`string`	0..1	The MIME content type of the request message, if any.
userAgent	`string`	0..1	A string describing the requesting Media Stream Handler, if it supplies a `User-Agent` request header.
userIdentity	`String`	0..1	A string identifying the user that made the access, if supplied.
referer	`AbsoluteUrl`	0..1	The URL that the Media Player reports being referred from, if the `Referer` request header is supplied. (See table 6.4.2-1.)

Property name	Data Type	Cardinality	Description
cacheStatus	CacheStatus	0..1	An indication of whether the 5GMS AS is able to serve an object corresponding to *requestMessage,url* from cache (*HIT*) or whether there is a stale object cached (*EXPIRED*) or the requested object is not present in cache (*MISS*). (See table 6.4.4.4.) For non-caching implementations of the 5GMS AS, the property shall be omitted.
responseMessage	Object	1..1	Details of the HTTP response message returned by the 5GMS AS to the Media Stream Handler for this media access.
responseCode	Uinteger	1..1	The HTTP response code.
size	Uinteger	1..1	The total number of bytes in the response message.
bodySize	Uinteger	1..1	The number of bytes in the HTTP response message body.
contentType	string	0..1	The MIME content type of response message, if any.
processingLatency	Float	1..1	The time, expressed in milliseconds, taken by the 5GMS AS to respond to the Media Stream Handler request, measured from the first byte of the HTTP request being processed by the 5GMS AS to the last byte of the response being sent.
connectionMetrics	Object	0..1	Metrics about the performance of the transport connection underlying the HTTP session serving this media access.
meanNetwork RoundTripTime	Float	1..1	A rolling mean average, expressed in milliseconds, of the network round-trip time for the HTTP session.
networkRoundTrip TimeVariation	Float	1..1	The variation in *meanNetworkRoundTripTime*, expressed in milliseconds, during the averaging period.
congestion WindowSize	Uinteger	1..1	The current size (in bytes) of the congestion window for the transport connection underlying the HTTP session.

Annex A (informative): 5GMS Parameter propagation for DASH Streaming

A.1 End-to-end model

Figure A.1-1 below depicts an end-to-end model for the 5GMS parameter propagation for DASH streaming with dynamic policy. The arrows indicate the main information flow. The interfaces specified in TS 26.501 [2] are used throughout. However, there are additional interfaces (i.e. P1 or U1), which are not in the 5GMS Architecture.

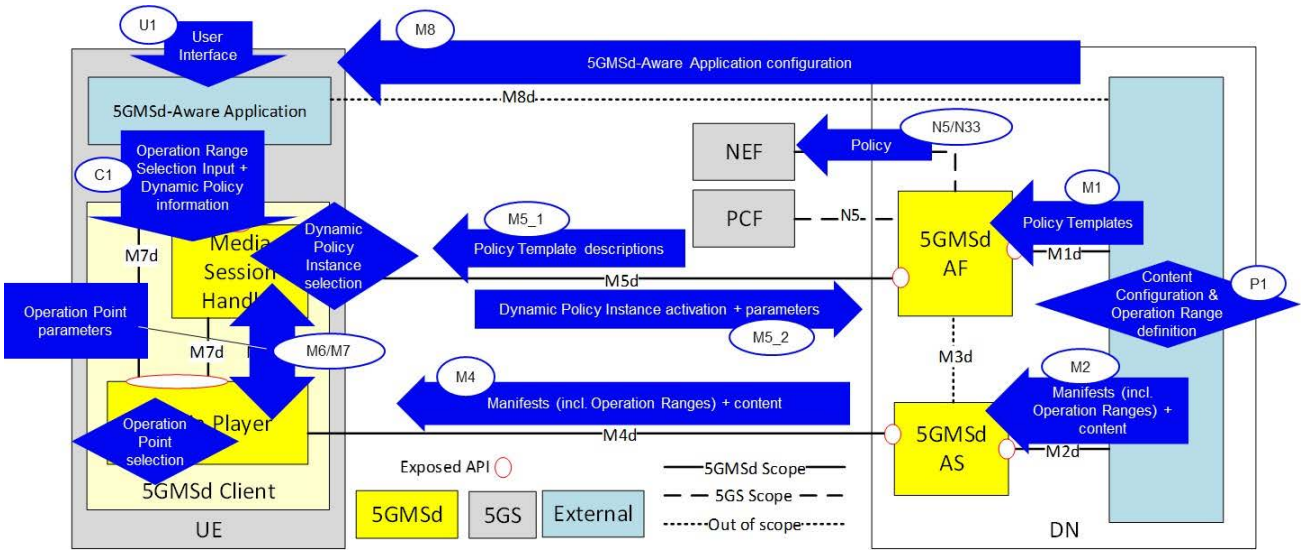


Figure A.1-1: End-to-end model for dynamic policy parameter propagation

The interfaces involved and their roles in this feature are as follows:

- M1: Provisioning interface between the 5GMS Application Provider and the 5GMS AF.
- P1: The 5GMS Application Provider provisions the DASH MPD generator, e.g. by annotating the MPD with Service Descriptions.
- U1: User Interface to the 5GMS-Aware Application.

NOTE: The 5GMS Application Provider controls the application, i.e. controls the GUI choices.

- M8: Non-standardized input from the 5GMS Application Provider to the 5GMS-Aware Application, such as country-specific application behaviours (languages, on-demand catalogue, etc).
  - Input on subscriptions (e.g. 4K subscription versus SD subscription).
  - Device-specific content selection rules (e.g. Smartphone versus Smart TV).
  - Additional service offering features (e.g. background download possible).
- C1 (one of M6 or M7): Information from the 5GMS-Aware Application to the 5GMS Client, e.g. user content selections.
- M6: Information flow from the DASH Player to the Media Session Handler.
- M7: Information flow from the Media Session Handler to the DASH Player.
- M5_1: Information flow into the Media Session Handler for parameter provisioning (Policy Descriptions, which originate from 5GMS AF and 5GMS Application Provider). The Policy Descriptions contain or reference the detailed Service Access Information, i.e. URLs to activate a certain policy.

- M5_2: Information flow from the Media Session Handler to the 5GMS AF. This includes:
  - input to create the Service Data Flow Templates (see TS 23.503 [33]) for identifying the application data flows within a PDU Session,
  - an identifier for the Dynamic Policy instance (e.g. QoS, Conditional Zero-rating, charging, etc) and
  - optionally, Network Assistance information, e.g. bit rate recommendations.

In its Annex K, the DASH standard [32] specifies so-called "Service Descriptions". The purpose of Service Descriptions is to provide additional information to a DASH player to influence its "Selection Logic", e.g. a DASH player should prefer a certain set of representations within an adaptation set. It is assumed in the following that the DASH MPD can be annotated using Service Descriptions to give hints for subscription models and different device types.

The 5G System specifies a number of different means to detect application flows. When activating a Dynamic Policy, the Media Session Handler provides a Service Data Flow Template to the 5GMS System, which identifies the application flow(s) of interest. It is assumed here that multiple applications are executing simultaneously on a given UE and that each application may independently access the network. Therefore, the Media Session Handler needs to provide (and update) these Service Data Flow Templates in order that the application traffic can be treated according to the corresponding Dynamic Policy.

In the following clauses, the parameter propagation for a number of different use cases is described.

A.2 Premium QoS dynamic policy

A.2.1 General

To realise a Premium QoS service offering, the 5GMS Client should activate a QoS Flow with characteristics matching the service needs. It is assumed that the DASH content is prepared for different subscription levels, e.g. 4K, HDR or SD, and for different target device types, e.g. SmartPhone or SmartTV. When commencing playback of a DASH presentation according to a particular subscription level (e.g. 4K), the 5GMS Client needs to activate a QoS Flow with a matching bit rate setting.

NOTE: The 5GMS Client may choose to activate a QoS Flow with a lower bit rate than the maximum supported by the 5G System, e.g. a small screen Smartphone may select different QoS settings from a large screen device.

The per-title quality and the subscription levels of an example on-demand catalogue are illustrated in the figure below. The subscription levels in this example are 4K, FullHD, HD, SD and 480p. Only devices entitled to activate a 4K quality should actually select the according representations from the MPDs. In this example, all titles are available in SD and HD quality. Often, not all titles are available in 4K quality. Thus, a device with a 4K subscription can only activate reception of the HD or SD representations.

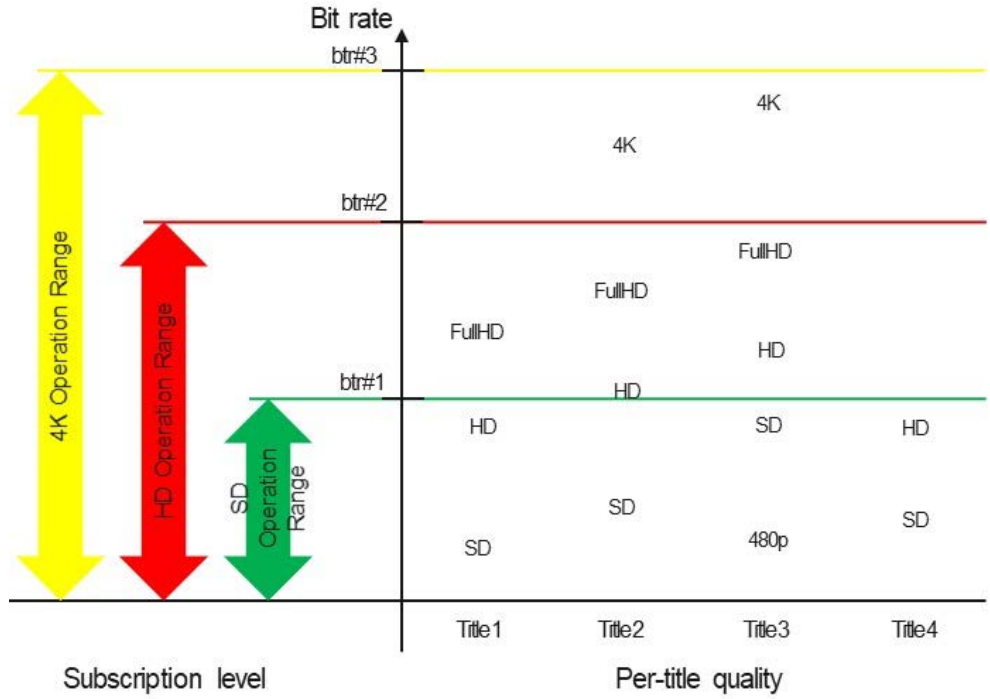


Figure A.2.1-1: Subscription Levels for Premium QoS

The bit rate required to sustain a certain quality varies from title to title. In the figure, the bit rate needed for *Title4* in HD is in the same range as SD quality of *Title3*.

The various consumer-facing Network Subscription Levels define a set of bounded Operation Ranges, as illustrated on the right side of the figure. Each such Operation Range is conveniently modelled in the 5GMS architecture as a Policy Template. The Policy Template for SD subscription level (*SD Operation Range*) is authorized to activate a maximal bit rate of *btr#1*. The Policy Template for 4K subscription level is authorized to activate between any low bit rate and a maximal bit rate of *btr#3*.

When activating a Dynamic Policy instance, the 5GMSd Client provides a desired bit rate for the selected title. The desired bit rate can be smaller than the maximal bit rate allowed by the Policy Template. The 5GMSd Client always activates a Dynamic Policy instance from its assigned Network Subscription Level, even when the desired bit rate justifies a different Policy Template.

When activating a QoS Flow for a certain subscription level and title, the 5GMSd Client should preferably select a desired bit rate matching the quality needed. For example, a device with an *HD Operation Range* subscription needs a higher desired bit rate when consuming *Title3* in HD quality and a lower desired bit rate when consuming *Title4* in HD quality.

In some cases, the system rejects a requested QoS Flow or drops an established QoS Flow due to insufficient available network resource. The 5GMSd Client can then try to activate a different QoS Flow with a lower desired bit rate.

A.2.2 Procedure

The procedure for activating a Premium QoS dynamic policy is illustrated in figure A.2.2-1 below.

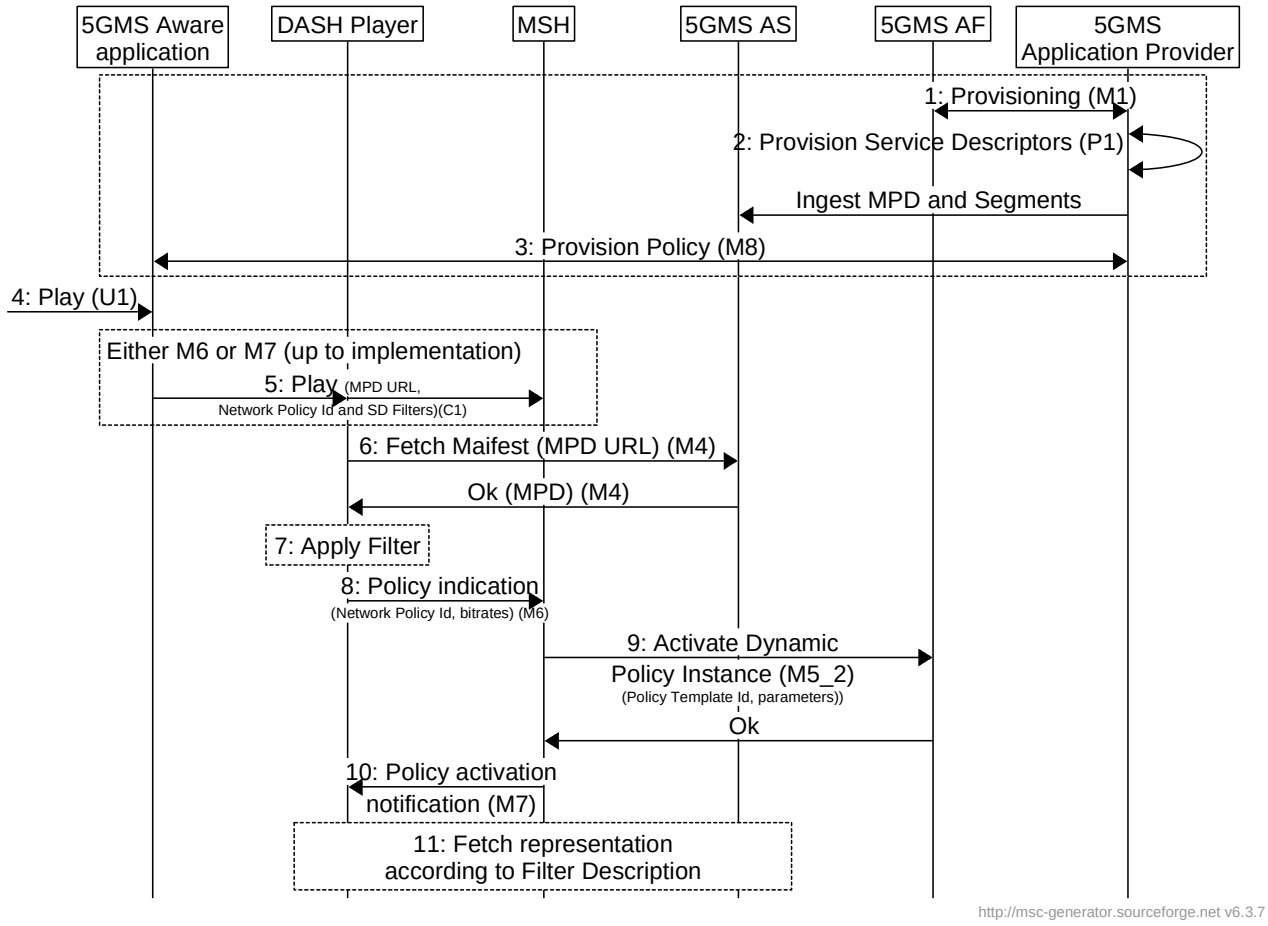


Figure A.2.2-1: Procedure for activating Premium QoS dynamic policy

Steps:

1. The 5GMS Application Provider interacts with the 5GMS AF to set up one or more Policy Templates (using M1). Each Policy Template is identified by a Policy Template identifier and contains information about how to activate the corresponding policy within the 5G System (e.g. N5 URLs and parameters).

2. The 5GMS Application Provider interacts with its DASH content generation function (e.g. an MPD provider) to annotate the DASH MPD with Service Descriptions (using P1). The Service Descriptions define the Operational Ranges within the Media Player should operate. The DASH MPD and the DASH Media Segments are then ingested by the 5GMS AS.

3. The 5GMS-Aware Application is configured via M8 (step 3) with information about the available content catalogue (e.g. resolving MPD URLs), the available subscription identifiers (e.g. the user has a 4K subscription or the user has an SD subscription), device type identifiers and network policy identifiers.

   The subscription identifiers and the device type identifiers are collectively referred to as Service Description Filters in the following.

NOTE 1: It is for further study whether network policy identifiers are embedded in the MPD Service Descriptions or derived from the Service Descriptions.

NOTE 2: The network policy identifier can be equal to a Policy Template identifier when the 5GMS-Aware Application is aware about its usage (e.g. for QoS streaming or background download). It is assumed here, that a unique Network Policy identifier is assigned to each subscription level.

4. When the user selects an item via the User Interface (U1), the 5GMS-Aware Application translates the input to the needed 5GMSd API calls.

5. The 5GMS-Aware Application provides input (via C1) on the selected presentation entry (i.e. MPD URL) together with a Network Policy Identifier (the value indicates here a "HD Premium QoS" policy (alternative Network Policy Identifiers can refer to e.g. 4K quality), i.e. make the Media Session Handler request a QoS Flow) and Service Description Filters. The Service Description Filter is used by the Media Player to identify the usable Service Descriptions from the MPD. The Network Policy Identifier is used by the Media Session Handler to find the according Policy Description containing information on the Dynamic Policy instantiation method (i.e. procedure and parameters such as Policy Template identifier).

6. The DASH player fetches the MPD.

7. The Media Player selects the Service Description and applies the Service Description Filter.

8. The DASH player indicates to the Media Session Handler (M6) that a "HD Premium QoS" network service should be activated (value of the Network Policy Identifier). The DASH player provides input on bit rate ranges (which may depend on the device type and the title quality). The Media Session Handler has received one or more Policy Descriptions together with matching Service Access Information (via M5_1). When the Media Session Handler has received the policy indication, the Media Session Handler uses the Network Policy Identifier to find the procedure and the parameters to activate the Dynamic Policy instance (i.e. find the matching Policy Description). The Media Session Handler activates a Dynamic Policy instance in the 5GMS AF, providing Service Data Flow Templates identifying the DASH media flows (audio, video, etc) and to provide the desired bit rate of the video.

9. The Media Session Handler activates a Dynamic Policy instance with the 5GMS AF. The 5GMS AF uses the Policy Template identifier to look up the matching Policy Template in order to create the PCF or NEF API invocation. As result, the Media Session Handler receives the enforcement bit rate in the 5GMS AF response. The 5GMS Client should not exceed this bit rate threshold.

   The Service Access Information (via M5_1) includes a list of recommended traffic detection methods. The Media Session Handler selects a Service Data Flow description method (e.g. 5-Tuples). When the Media Session Handler selects:

   - 5-Tuples: For each new TCP connection, the Media Session Handler updates the Dynamic Policy instances and adds a new 5-Tuple. For each closed TCP connection, the Media Session Handler updates the Dynamic Policy instances and removes the 5-Tuple of the closed TCP connection.

   - TOS or Traffic Class: The Media Session Handler sets the TOS or Traffic Class for each new TCP connection.

   - Domain name: The Media Session Handler provides the domain name with the Dynamic Policy Instance.

A.2.3 Example parameters

Table A.2.3-1: M5_1 parameters for Policy Descriptions (used by the Media Session Handler)

Parameter	Type	Purpose	Example Values
Policy Description	Object		
Network Policy Identifier	String	Identifies the Policy Description.	"4K Premium QoS", "HD Premium QoS".
Service Access Information URL	URL	References the associated Service Access Information.	

Table A.2.3-2: M5_1 parameters for Service Access Information

Parameter	Type	Purpose	
Service Access Information	Object		
Policy Template identifier	String	Identifies the Policy Template.	"HD QoS".
5GMS AF URL	URL	Used to invoke the 5GMS AF.	
Mandatory Request M5 information	List	Desired bit rate, which should be provided by the network for the application.	Policy Template identifier, Desired Bit Rate, Packet Detection Filters.
M5 Response information	List	Information to the Media Session Handler on the response parameters.	OK (requested bit rate is accepted), Proposed Lower Bit rate (requested bit rate cannot be provided).
sdfMethod	[String]	Indicates which Service Data Flow Description methods are recommended to be used by the Media Session Handler.	"5-Tuple", "domainName", "TOS=xx", etc.

A.3 (Conditional) Zero Rating dynamic policy

A.3.1 General

In the case of (Conditional) Zero Rating, the quality of a video streaming service should not exceed a certain bit rate threshold (called the policy threshold). This can be realized by deploying a traffic shaper in the network (e.g. a policing function in the UPF) or by instructing the DASH Player not to exceed a certain policy threshold bit rate. The policy threshold may be network-specific, i.e. depending on the 5G System. The following realization assumes the latter, i.e. the DASH Player is not exceeding the bit rate policy and the UPF is just monitoring the compliance of the application flows (one or more TCP and/or UDP flows). The MPD is annotated using DASH Service Descriptions in such a way that the DASH Player can identify which maximal representation bit rates still comply with the policy threshold.

Figure A.3.1-1 below illustrates the per-title quality and the policy threshold. For *Title1* and *Title2*, the 5GMSd Client can activate the SD and HD representations. For *Title3*, the 5GMSd Client can activate the 480p and the SD representations. For *Title4*, the 5GMSd Client can activate all available representations (i.e. SD and HD).

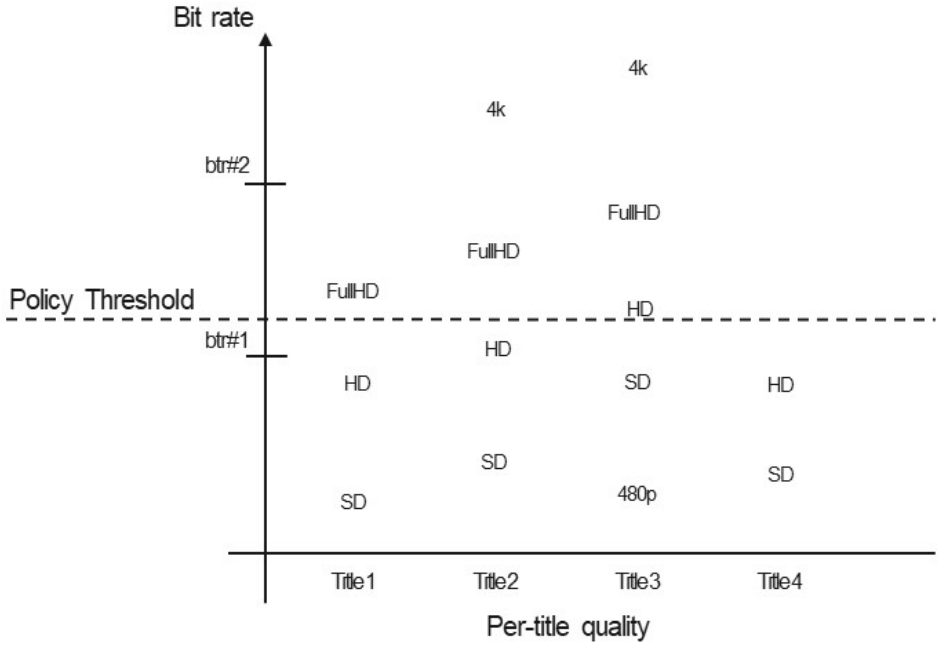


Figure A.3.1-1: Policy threshold versus quality

When the 5GMSd Client receives the bit rate of the policy threshold from the network, the 5GMSd Client filters the MPD for policy-compliant representations (i.e. those that lie at or below the policy threshold).

A.3.2 Procedure

The procedure for activating a (Conditional) Zero Rating dynamic policy is illustrated in figure A.3.2-1 below.

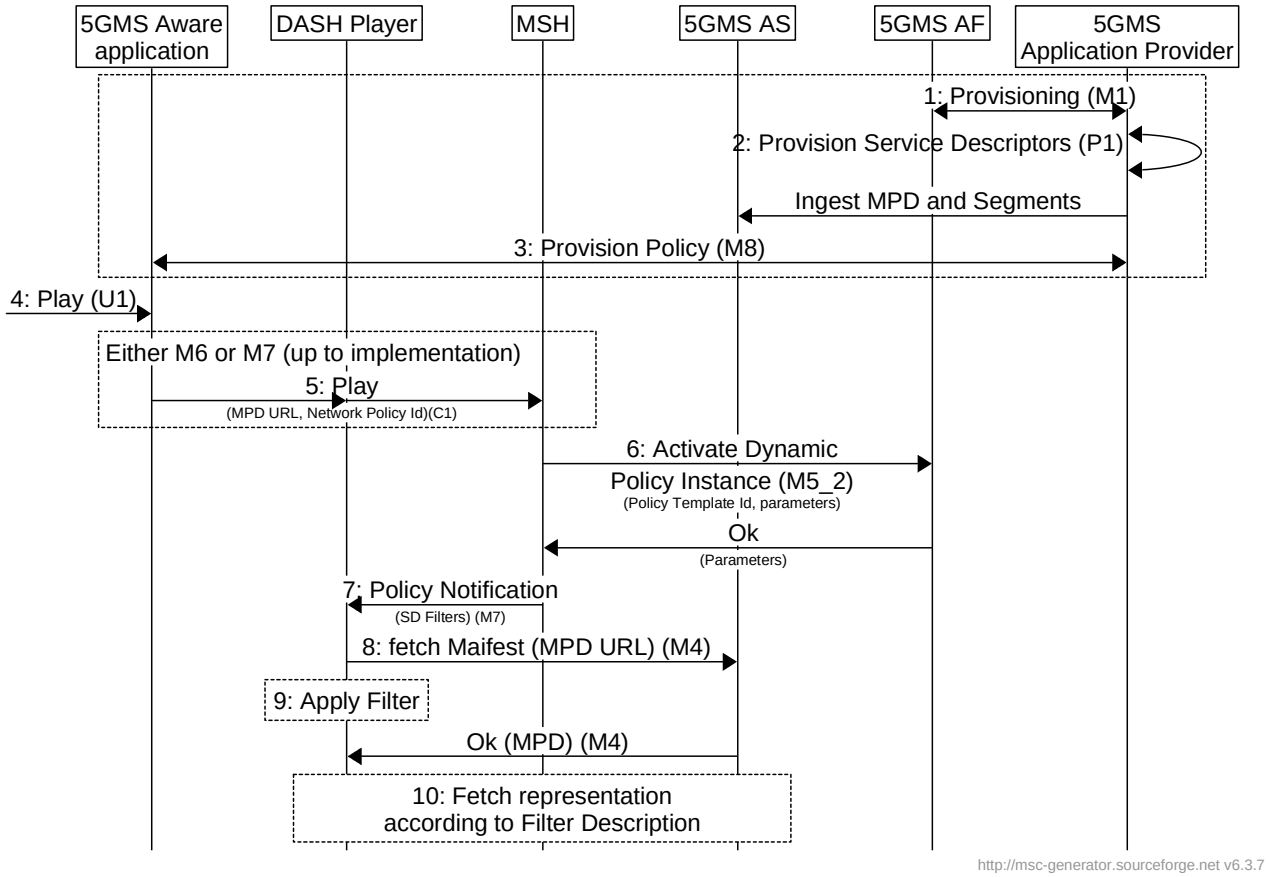


Figure A.3.2-1: Procedure for activating (Conditional) Zero Rating dynamic policy

Steps:

1. The 5GMS Application Provider interacts with the 5GMS AF to set up one or more Policy Templates. Each Policy Template is identified by a Policy Template identifier and contains information about how to activate the corresponding policy within the 5G System (e.g. N5 URLs and parameters).

2. The 5GMS Application Provider interacts with its DASH content generation function (e.g. an MPD provider) to annotate the DASH MPD with Service Descriptions (step 2). The intention of the Service Descriptions here is that the DASH Player can identify those representation combinations which do not exceed the bit rate requirement.

3. The 5GMSAware Application is configured via M8 with information about the available content catalogue (e.g. resolving MPD URLs), the available subscription identifiers (e.g. the user has a 4K content subscription or the user has an SD subscription), device type identifiers.

   The 5GMSd-Aware Application is configured via M8 about the available (Conditional) Zero Rating policy. This includes the Network Policy Ids.

4. When a user selects an item via the User Interface (U1), the 5GMS-Aware Application translates the input to the needed 5GMSd API calls.

5. The 5GMS Aware Application provides input (via C1) on the selected presentation entry (i.e. MPD URL) and also on the Network Policy Id (the value in this case indicates a (Conditional) Zero-Rating policy, i.e. make the Media Session Handler request the policy threshold parameter from the network).

NOTE: C1 is an abstract interface and indicates that the 5GMS-Aware Application may either first use M6 or M7 for the interactions with the 5GMS Client.

6. The Media Session Handler uses the Network Policy Identifier to find the procedure and the parameters to activate the Dynamic Policy Instance (here a (Conditional) Zero Rating policy). The Media Session Handler has received one or more Policy Descriptions together with matching Service Access Information (via M5_1). The Media Session Handler uses the Network Policy Identifier as a key to find the correct Policy Description. Here, the Network Policy Identifier indicates a (Conditional) Zero Rating policy. The Media Session Handler should activate a dynamic policy in the 5GMS AF, providing Service Data Flow Template information about the DASH media flows (audio, video, etc.) and retrieving the bit rate threshold, which cannot be exceeded to comply with the policy. The Media Session Handler receives (as result of the Dynamic Policy activation) some information on the policy enforcement (*enforcementMethod* and/or *enforcementBitrate*), so that the representation selection logic (bit rate adaptation function) in the DASH Player can consider the effects of the enforcement scheme.

7. The Media Session Handler activates the Dynamic Policy instance on M5, providing a Policy Template identifier. Upon positive response, the Media Session Handler notifies the DASH Player, providing Service Descriptor Filters. The Media Session Handler may receive these Service Descriptor Filters with the response, or it may look up the Service Descriptor Filter values by a response value. Alternatively, the Media Session Handler receives a maximum bit rate with the response and the Media Session Handler derives the Service Descriptor Filter. The Media Session Handler may also receive information about Policy Enforcement, e.g. what type of traffic shaper will throttle the bit rate.

   The Media Session Handler may need to update the Dynamic Policy instance, depending on the selected traffic detection method. For example, when the Media Session Handler uses 5-Tuples, the Media Session Handler needs to update the Dynamic Policy instance with every newly opened and every closed TCP connection.

8. The DASH Player fetches the MPD of the selected content.

9.The Service Descriptor Filter is used by the DASH Player to filter policy-compliant Service Descriptions from the MPD. The DASH Access Engine or Selection Logic (see ISO/IEC 23009-1 [32] figure K.1) selects only adaptation sets and representations according to the filter. Here, the DASH Player fetches the MPD after the notification from the Media Session Handler.

A.3.3 Example parameters

Table A.3.3-1: M5_1 parameters for Policy Descriptions (used by the Media Session Handler)

Parameter	Type	Purpose	Example Values
Policy Description	Object		
Network Policy Id	String	Identifies the Policy Description.	"(Conditional) Zero Rating".
Service Access Information URL	URL	References the associated Service Access Information.	

Table A.3.3.-2: M5_1 parameters for Service Access Information

Parameter	Type	Purpose	
Service Access Information	Object		
Policy Template Id	String	Identifies the Policy Template.	"not exceed bit rate"
5GMS AF URL	URL	Used to invoke the 5GMS AF.	
sdfMethods	[String]	Indicates which Service Data Flow Description methods are recommended for use by the Media Session Handler.	"5-Tuple", "domainName", "TOS=xx", etc.
Mandatory M5 Request information	List		Policy Template identifier, Service Data Flow Template.
M5 Response information	List	Information to the Media Session Handler on the response parameters.	Bit rate Policy Threshold (upper bit rate bound, which should not be exceeded).

A.4 Background Download

A.4.1 General

In the case of Background Download, the asset is acquired in the background, prior to viewing. Many application services offer the capability of acquiring a VoD item for later consumption. The 5GMS-Aware Application triggers the Media Session Handler to acquire the item, providing a background download network policy id.

NOTE: Here, the DASH Player is handling the acquisition, since the DASH Player contains the MPD processing and the DASH Access engine parts. Other realizations would use a separate background download agent, which is not even try to decode and render the video.

Figure A.4.1-1 below illustrates the representation marking for background download. The MPD may be annotated with Service Descriptions clearly identifying representations intended for download. Here, *Title1* should be downloaded in Full HD quality and all other titles in regular HD quality.

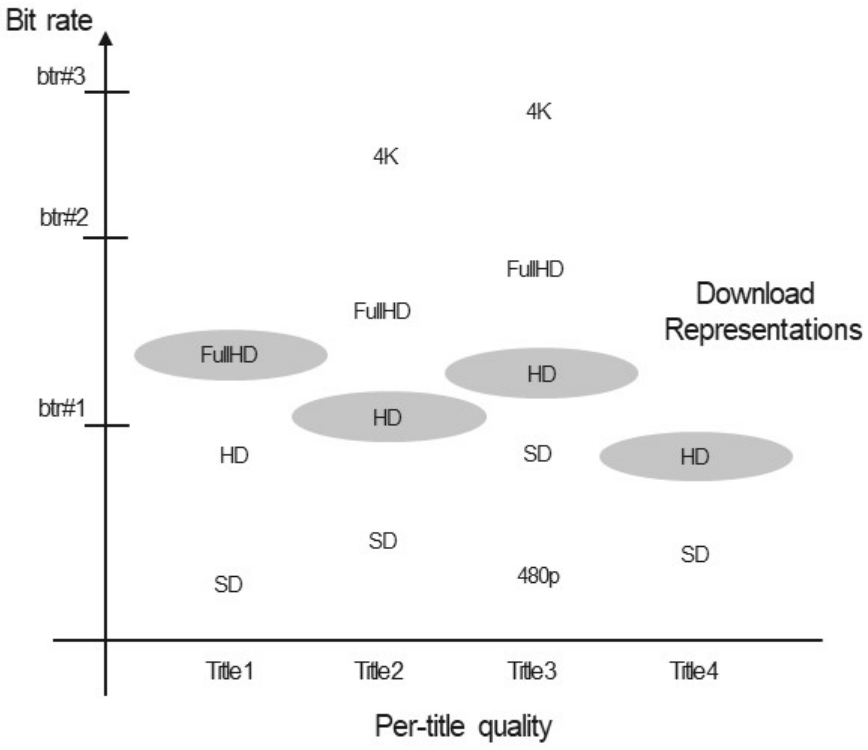


Figure A.4.1-1: Background Download Representations

A.4.2 Procedure

The procedure for activating a Background Download dynamic policy is illustrated in figure A.3.2-1 below.

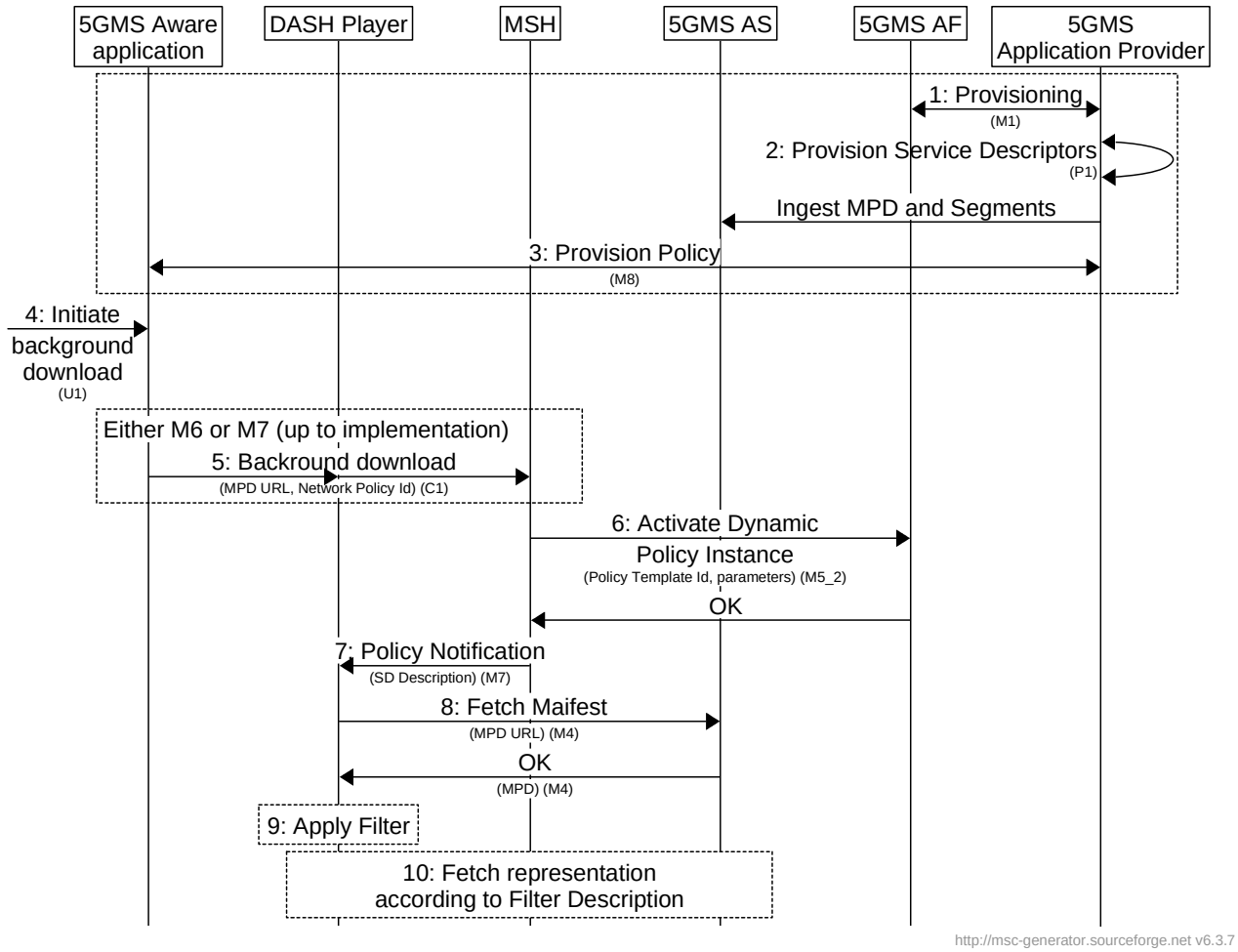


Figure A.3.2-1: Procedure for activating Background Download dynamic policy

Steps:

1. The 5GMS Application Provider interacts with the 5GMS AF to set up one or more Policy Templates (M1). Each Policy Template is identified by a Policy Template identifier and contains information about how to activate the according policy within the 5G System (e.g. N5 URLs and parameters).

2. The 5GMS Application Provider also interacts with its DASH content generation function (e.g. an MPD provider) to annotate the DASH MPD with Service Descriptions, e.g. to identify, which representation is intended for background download.

3. The 5GMS-Aware Application is configured via M8 with information about the available content catalogue (e.g. resolving MPD URLs), the available subscription identifiers (e.g. the user has a 4K subscription or the user has an SD subscription), device type identifiers.

   The 5GMSd-Aware Application is configured via M8 about the available background download policy. This includes the Network Policy Id which hints a background download policy.

4. When a user selects an item via the User Interface (U1) for Background Download the 5GMS-Aware Application translates the input to the needed 5GMSd API calls.

5. The 5GMS-Aware Application provides input (via C1) on the selected presentation entry (i.e. MPD URL) and also on the Network Policy Identifier (indicating a background download policy, i.e. make the Media Session Handler request a bearer suitable for Background Download).

NOTE: C1 is an abstract interface and indicates that the 5GMS-Aware Application may either first use M6 or M7 for the interactions with the 5GMS Client.

6. The Media Session Handler uses the Network Policy Identifier to find the procedure and the parameters to activate the Dynamic Policy Instance (here a Background Download policy). The Media Session Handler has received one or more Policy Descriptions together with matching Service Access Information (via M5_1). The Media Session Handler uses the Network Policy Identifier as a key to find the correct Policy Description. The Media Session Handler should activate a Dynamic Policy in the 5GMS AF, providing Service Data Flow Template information of the media flows (audio, video, etc). The Media Session Handler can also receive information on a bit rate policing (*enforcementMethod* and/or *enforcementBitrate*), e.g. that the bit rate is actively limited.

7. The Media Session Handler activates the Dynamic Policy instance on M5, providing the Policy Template identifier and additional parameters. Upon positive response, the Media Session handler notifies the DASH Player to start the Background Download. The notification contains a Service Descriptor Filters, which is used by the DASH Player to filter policy-compliant Service Descriptions from the MPD. The Media Session Handler may receive the Service Descriptor Filters with the response or may look up the Service Descriptor Filter values by a response value (e.g. derived from a maximum bit rate indication).

   The Media Session Handler may need to update the Dynamic Policy instance, depending on the selected traffic detection method. For example, when the Media Session Handler uses 5-Tuples, it needs to update the Dynamic Policy instance with every newly opened and every closed TCP connection.

8. The DASH Player fetches the MPD of the selected content.

9. The DASH Access Engine / Selection Logic (see ISO 23009-1 [32] figure K.1) selects only adaptation sets and representations according to the filter (i.e. suitable for Background Download). Here, the DASH Player fetches the MPD after the notification from the Media Session Handler.

A.4.3 Example parameters

Table A.4.3-1: M5_1 Parameters for Policy Descriptions (used by the Media Session Handler)

Parameter	Type	Purpose	Example Values
Policy Description	Object		
Network Policy Id	String	Identifies the Policy Description.	"Background Download".
Service Access Information URL	URL	References the associated Service Access Information.	

Table A.4.3-2: M5_1 Parameters for Service Access Information

Parameter	Type		
Service Access Information	Object		
Policy Template Id	String	Identifies the Policy Template.	"backgrounddata".
5GMS AF URL	URL	Used to invoke the 5GMS AF.	
sdfMethods	[String]	Indication, which Service Data Flow Description methods are recommended to use by the media session handler.	"5-Tuple", "domainName", "TOS=xx", etc.
Mandatory M5 Request information	List	Desired bit rate, to be provided by the network for the application.	Policy Template Id, Average Bit rate, Service Data Flow Template.
M5 Response information	List	Information to the Media Session Handler on the response parameters.	

Annex B (informative): Content Hosting Configuration examples

B.1 Pull-based content ingest example

B.1.1 Overview

1. The 5GMSd Client on the UE requests a media resource via M4d.
2. The 5GMSd AS determines that it does not have a cached copy of the requested media resource.
3. The 5GMSd AS transforms the M4d request URL into a request to the 5GMSd Application Provider's origin server via M2d.

B.1.2 Desired URL mapping

In the example shown in table B.1.2-1 below, media resources are exposed by a reference point M4d service location with a default canonical domain name *dist-loc.com-provider-service.ms.as.3gppservices.org*. This domain name is assigned by the 5GMSd AF based on a Provisioning Session with external identifier *com.provider.service* and a Content Hosting Configuration distribution configuration with distribution identifier *dist.loc*. The 5GMSd Application Provider has also assigned a custom domain name alias `5gms.provider.com` by defining *DistributionConfiguration.domainNameAlias* in the same distribution configuration. The base URL of the 5GMSd Application Provider's origin server is `https://origin.provider.com/media`.

Table B.1.2-1: Example URL mapping for pull-based ingest

M4d request from 5GMSd Client	Mapped M2d request to origin server on 5GMSd AS cache miss
https://**dist-loc.com-provider-service.ms.as.3gppservices.org**/**asset123456/video1**/segment1000.mp4	https://origin.provider.com/media/**asset123456/video1**/segment1000.mp4
https://**5gms.provider.com**/**asset123456/video1**/segment1000.mp4	
https://**dist-loc.com-provider-service.ms.as.3gppservices.org**/**asset123456/video2**/segment1000.mp4	https://origin.provider.com/media/**asset123456/video2**/segment1000.mp4
https://**5gms.provider.com**/**asset123456/video2**/segment1000.mp4	
https://**dist-loc.com-provider-service.ms.as.3gppservices.org**/**asset123456/audio1**/segment1000.mp4	https://origin.provider.com/media/**asset123456/audio1**/segment1000.mp4
https://**5gms.provider.com**/**asset123456/audio1**/segment1000.mp4	

B.1.3 Content Hosting Configuration

Table B.1.3-1 below shows the relevant Content Hosting Configuration parameters needed to achieve the example mapping described in table B.1.2-1 above.

Table B.1.3-1: Content Hosting Configuration properties relevant to pull-based ingest

Property	Example value	Set by
ProvisioningSession		
externalServiceId	com.provider.service	5GMSd Application Provider *(M1d request)*
IngestConfiguration		
protocol	urn:3gpp:5gms:content-protocol:**http-pull**	5GMSd Application Provider *(M1d request)*
mode	*PULL*	
baseURL	https://origin.provider.com/media	
DistributionConfiguration		
distributionId	dist.loc	5GMSd Application Provider *(M1d request)*
canonicalDomainName	dist.loc.com-provider-service.ms.as.3gppservices.org	5GMSd AF *(M1d response)*
domainNameAlias	5gms.provider.com	5GMSd Application Provider *(M1d request)*
baseURL	https://5gms.provider.com/	5GMSd AF *(M1d response)*

B.2 Push-based content ingest example

B.2.0 Overview

1. The 5GMSd Application Provider uploads content to the 5GMSd AS via M2d.
2. The 5GMSd AS rewrites the M2d upload URL to an M4d downlink URL that is exposed to the 5GMSd Client on the UE.

B.2.1 Desired URL mapping

In the example shown in table B.2.1-1, media resources are exposed by a reference point M4d service location with a default canonical domain name *dist-loc.com-provider-service.ms.as.3gppservices.org*. This domain name is assigned by the 5GMSd AF based on a for the Provisioning Session with external identifier *com.provider.service* and a Content Hosting Configuration distribution configuration with distribution identifier *dist-loc*. The 5GMSd Application Provider also assigns a custom domain name alias *5gms.provider.com* by defining *DistributionConfiguration.domainNameAlias* in the same distribution configuration. Media resources are pushed into the 5GMSd AS at M2d by the 5GMSd Application Provider using the ingest base URL `https://5gmsd-as.mno.net/com-provider-service` chosen by the 5GMSd AF.

Table B.2.1-1: Example URL mapping for push-based ingest

M2d ingest URL pushed to 5GMSd AS	M4d URL exposed to 5GMSd Client
https://5gmsd-as.mno.net/com-provider-service/**asset123456/video1**/segment1000.mp4	https://**dist-loc.com-provider-service.ms.as.3gppservices.org**/**asset123456/video1**/segment1000.mp4
	https://**5gms.provider.com**/**asset123456/video1**/segment1000.mp4
https://5gmsd-as.mno.net/com-provider-service/**asset123456/video2**/segment1000.mp4	https://**dist-loc.com-provider-service.ms.as.3gppservices.org**/**asset123456/video2**/segment1000.mp4
	https://**5gms.provider.com**/**asset123456/video2**/segment1000.mp4
https://5gmsd-as.mno.net/com-provider-service/**asset123456/audio1**/segment1000.mp4	https://**dist-loc.com-provider-service.ms.as.3gppservices.org**/**asset123456/audio1**/segment1000.mp4
	https://**5gms.provider.com**/**asset123456/audio1**/segment1000.mp4

B.2.2 Content Hosting Configuration

Table B.2.2-1 below shows the relevant Content Hosting Configuration parameters needed to achieve the example mapping described in table B.2.1-1 above.

Table B.2.2-1: Content Hosting Configuration properties relevant to push-based ingest

Property	Example value	Set by
ProvisioningSession		
externalServiceId	com.provider.service	5GMSd Application Provider *(M1d request)*
IngestConfiguration		
protocol	http://dashif.org/ingest/v1.2/interface-1	5GMSd Application Provider *(M1d request)*
mode	*PUSH*	
baseURL	https://5gmsd-as.mno.net/com-provider-service	5GMSd AF *(M1d response)*
DistributionConfiguration		
distributionId	dist.loc	5GMSd Application Provider *(M1d request)*
canonicalDomainName	dist-loc.com-provider-service.ms.as.3gppservices.org	5GMSd AF *(M1d response)*
domainNameAlias	5gms.provider.com	5GMSd Application Provider *(M1d request)*
baseURL	https://5gms.provider.com/	5GMSd AF *(M1d response)*

B.3 Pull-based content ingest with 5GMSd AS service chaining via M10d

B.3.1 Overview

This example shows how to provision multiple Content Hosting Configurations allowing for content hosting service chaining via reference point M10d (see clause 5.2.8.2 of TS 26.510 [56]). In this example, one 5GMSd AS (*edge.5gms.provider.com*) is acting as an edge proxy while its upstream 5GMSd AS (*shield.5gms.provider.com*) is providing an origin shield function.

1. The 5GMSd Client on the UE requests a media resource via reference point M4d.
2. The client-facing 5GMSd AS determines that it does not have a cached copy of the requested media resource.
3. The client-facing 5GMSd AS transforms the M4d request URL into a request to the origin server-facing 5GMSd AS via reference point M10d.
4. The origin server-facing 5GMSd AS transforms the M10d request URL into a request to the 5GMSd Application Provider's origin server via M2d.

B.3.2 Desired URL mapping

In the example shown in table B.3.2-1 below, the following apply:

1. Media resources for the Provisioning Session with external identifier *com.provider.service.edge* are exposed at M4d from a default canonical domain *dist-loc.com-provider-service-edge.ms.as.3gppservices.org* determined by the 5GMSd Application Provider, and a custom domain name alias *edge.5gms.provider.com* has also been configured by the 5GMSd Application Provider.
2. Media resources for the Provisioning Session with external identifier *com.provider.service.shield* are exposed at M10d (and potentially M4d) from a default canonical domain *dist-loc.com-provider-service-shield.ms.as.3gppservices.org* determined by the 5GMSd Application Provider, and a custom domain name alias *shield.5gms.provider.com* has also been configured by the 5GMSd Application Provider. The base URL of the 5GMSd Application Provider's origin server is `https://origin.provider.com/media`.

Table B.3.2-1: Example URL mapping for pull-based ingest

M4d request from 5GMSd Client to client-facing 5GMSd AS	Mapped M10d request to origin server-facing 5GMSd AS	Mapped M2d request to origin server on 5GMSd AS cache miss
https://**dist-loc.com-provider-service-edge.ms.as.3gppservices.org/asset123456/video1**/segment1000.mp4	https://**dist-loc.com-provider-service-shield.ms.as.3gppservices.org/asset123456/video1**/segment1000.mp4	https://origin.provider.com/media/**asset123456/video1**/segment1000.mp4
https://**edge.5gms.provider.com/asset123456/video1**/segment1000.mp4	https://**shield.5gms.provider.com/asset123456/video1**/segment1000.mp4	
https://**dist-loc.com-provider-service-edge.ms.as.3gppservices.org/asset123456/video2**/segment1000.mp4	https://**dist-loc.com-provider-service-shield.ms.as.3gppservices.org/asset123456/video2**/segment1000.mp4	https://origin.provider.com/media/**asset123456/video2**/segment1000.mp4
https://**edge.5gms.provider.com/asset123456/video2**/segment1000.mp4	https://**shield.5gms.provider.com/asset123456/video2**/segment1000.mp4	
https://**dist-loc.com-provider-service-edge.ms.as.3gppservices.org/asset123456/audio1**/segment1000.mp4	https://**dist-loc.com-provider-service-shield.ms.as.3gppservices.org/asset123456/audio1**/segment1000.mp4	https://origin.provider.com/media/**asset123456/audio1**/segment1000.mp4
https://**edge.5gms.provider.com/asset123456/audio1**/segment1000.mp4	https://**shield.5gms.provider.com/asset123456/audio1**/segment1000.mp4	

B.3.3 Content Hosting Configurations

Tables B.3.3-1 and B.3.3-2 below show the relevant parameters for both Content Hosting Configurations needed to achieve the example mapping described in table B.3.2-1 above.

Table B.3.3-1: Origin server-facing 5GMSd AS Content Hosting Configuration properties relevant to pull-based ingest

Property	Example value	Set by
ProvisioningSession		
externalServiceId	com.provider.service.shield	5GMSd Application Provider *(M1d request)*
IngestConfiguration		
protocol	*urn:3gpp:5gms:content-protocol:http-pull*	5GMSd Application Provider *(M1d request)*
mode	*PULL*	
baseURL	https://origin.provider.com/media	
DistributionConfiguration		
distributionId	dist.loc	5GMSd Application Provider *(M1d request)*
canonicalDomainName	dist-loc.com-provider-service-shield.ms.as.3gppservices.org	5GMSd AF *(M1d response)*
domainNameAlias	d1.5gms.provider.com	5GMSd Application Provider *(M1d request)*
baseURL	https://shield.5gms.provider.com/	5GMSd AF *(M1d response)*

Table B.3.3-2: Client-facing 5GMSd AS Content Hosting Configuration properties relevant to pull-based ingest

Property	Example value	Set by
ProvisioningSession		
externalServiceId	com.provider.service.edge	5GMSd Application Provider *(M1d request)*
IngestConfiguration		
protocol	*urn:3gpp:5gms:content-protocol:http-**pull***	5GMSd Application Provider *(M1d request)*
mode	*PULL*	
baseURL	https://dist-loc.com-provider-service-shield.ms.as.3gppservices.org	
	https://shield.5gms.provider.com/	
DistributionConfiguration		
distributionId	dist.loc	5GMSd Application Provider *(M1d request)*
canonicalDomainName	dist-loc.com-provider-service-edge.ms.as.3gppservices.org	5GMSd AF *(M1d response)*
domainNameAlias	edge.5gms.provider.com	5GMSd Application Provider (*M1d request)*
baseURL	https://edge.5gms.provider.com/	5GMSd AF *(M1d response)*

B.4 Push-based content ingest with 5GMSd AS service chaining via M10d

B.4.1 Overview

This example shows how to provision multiple Content Hosting Configurations allowing for content hosting service chaining via reference point M10d (see clause 5.2.8.2 of TS 26.510 [56]).

1. The 5GMSd Application Provider uploads content to the origin server-facing 5GMSd AS using push-based ingest via reference point M2d.
2. The origin server-facing 5GMSd AS rewrites the M2d upload URL to a M10d downlink URL that is exposed to a client-facing 5GMSd AS.
3. The client-facing 5GMSd AS rewrites the M10d upload URL to a M4d downlink URL that is exposed to the 5GMSd Client on the UE.

B.4.2 Desired URL mapping

In the example shown in table B.4.2-1 below, the following apply:

1. Media resources for the Provisioning Session with external identifier *com.provider.service.edge* are pushed into the client-facing 5GMSd AS at M10d by the origin server-facing 5GMSd AS. These media resources are exposed to the 5GMSd Client at M4d using the canonical name of the client-facing 5GMSd AS *dist-loc.com-provider-service-edge.ms.as.3gppservices.org* and an additional domain name alias *edge.5gms.provider.com* configured by the 5GMSd Application Provider. The ingest base URL populated by the 5GMSd AF is `https://5gmsd-as-edge.mno.net/com-provider-service-edge`.
2. Media resources for the Provisioning Session with external identifier *com.provider.service.shield* are pushed into the origin server-facing 5GMSd AS at M2d by the 5GMSd Application Provider and exposed to the client-facing 5GMSd AS at M10d. The canonical domain name and base URL of the distribution pushing media resources to the client-facing 5GMSd AS are determined when provisioning the client-facing 5GMSd AS. The ingest base URL populated by the 5GMSd AF is `https://5gmsd-as-shield.mno.net/com-provider-service-shield`.

Table B.4.2-1: Example URL mapping for push-based ingest

M2d ingest URL pushed to origin server-facing 5GMSd AS	M10d URL pushed to client-facing 5GMSd AS	M4d URL exposed to 5GMSd Client
https://5gmsd-as-shield.mno.net/com-provider-service-shield/**asset123456**/**video1**/segment1000.mp4	https://5gmsd-as-edge.mno.net/com-provider-service-edge/**asset123456**/**video1**/segment1000.mp4	https://**dist-loc.com-provider-service-edge.ms.as.3gppservices.org/asset123456**/**video1**/segment1000.mp4
		https://**edge**.5gms.provider.com/asset123456/video1/segment1000.mp4
https://5gmsd-as-shield.mno.net/com-provider-service-shield/**asset123456**/**video2**/segment1000.mp4	https://5gmsd-as-edge.mno.net/com-provider-service-edge/**asset123456**/**video2**/segment1000.mp4	https://**dist-loc.com-provider-service-edge.ms.as.3gppservices.org/asset123456**/**video2**/segment1000.mp4
		https://**edge**.5gms.provider.com/asset123456/video2/segment1000.mp4
https://5gmsd-as-shield.mno.net/com-provider-service-shield/**asset123456**/**audio1**/segment1000.mp4	https://5gmsd-as-edge.mno.net/com-provider-service-edge/**asset123456**/**audio1**/segment1000.mp4	https://**dist-loc.com-provider-service-edge.ms.as.3gppservices.org/asset123456**/**audio1**/segment1000.mp4
		https://**edge**.5gms.provider.com/asset123456/audio1/segment1000.mp4

B.4.3 Content Hosting Configuration

Tables B.4.3-1 and B.4.3-2 below show the relevant parameters for both Content Hosting Configurations needed to achieve the example mapping described in table B.4.2-1 above.

Table B.4.3-1: Client-facing Content Hosting Configuration properties relevant to push-based ingest

Property	Example value	Set by
ProvisioningSession		
externalServiceId	com.provider.service.edge	5GMSd Application Provider *(M1d request)*
IngestConfiguration		
protocol	*http://dashif.org/ingest/v1.2/interface-1*	5GMSd Application Provider *(M1d request)*
mode	*PUSH*	
baseURL	https://5gmsd-as-edge.mno.net/com-provider-service-edge/	5GMSd AF *(M1d response)*
DistributionConfiguration		
distributionId	dist.loc	5GMSd Application Provider *(M1d request)*
canonicalDomainName	dist-loc.com-provider-service-edge.ms.as.3gppservices.org	5GMSd AF *(M1d response)*
domainNameAlias	edge.5gms.provider.com	5GMSd Application Provider *(M1d request)*
baseURL	https://edge.5gms.provider.com/	5GMSd AF *(M1d response)*

Table B.4.3-2: Origin server-facing Content Hosting Configuration properties relevant to push-based ingest

Property	Example value	Set by
ProvisioningSession		
externalServiceId	com.provider.service.shield	5GMSd Application Provider *(M1d request)*
IngestConfiguration		
protocol	*http://dashif.org/ingest/v1.2/interface-1*	5GMSd Application Provider *(M1d request)*
mode	*PUSH*	
baseURL	https://5gmsd-as-shield.mno.net/com-provider-service-shield/	5GMSd AF *(M1d response)*
DistributionConfiguration		
mode	*PUSH*	5GMSd Application Provider *(M1d request)*
distributionId	dist.loc	
baseURL	https://5gmsd-as-edge.mno.net/com-provider-service-edge/	

Annex C (normative): OpenAPI representation of the 5GMS HTTP REST APIs

C.1 General

This annex is based on the OpenAPI 3.0.0 specification [23] and provides corresponding representations of all APIs defined in the present document.

NOTE 1: An OpenAPIs representation embeds JSON Schema representations of HTTP message bodies.

This Annex shall take precedence when being discrepant to other parts of the present document with respect to the encoding of information elements and methods within the API(s).

NOTE 2: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

C.2 Data Types applicable to several APIs

For the purpose of referencing entities defined in this clause, it shall be assumed that the OpenAPI definitions below are contained in a physical file named "TS26512_CommonData.yaml".

```yaml
openapi: 3.0.0
info:
  title: 5GMS Common Data Types
  version: 3.2.0
  description: |
    5GMS Common Data Types
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.
tags:
  - name: 5GMS Common Data Types
    description: '5G Media Streaming: Common Data Types'
externalDocs:
  description: 'TS 26.512 V19.3.0; 5G Media Streaming (5GMS); Protocols'
  url: 'https://www.3gpp.org/ftp/Specs/archive/26_series/26.512/'
paths: {}
components:
  schemas:
    ###################################
    # Clause 6.4.2: Simple data types
    ###################################

    ######################################
    # Clause 6.4.3: Structured data types
    ######################################
    MediaStreamingBaseReportingRecord:
      type: object
      properties:
        sliceInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        dataNetworkName:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        location:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/TypedLocation'

    MediaStreamingSessionIdentification:
      type: object
      required:
        - sessionId
      properties:
        sessionId:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/MediaDeliverySessionId'

    MediaStreamingAccess:
      type: object
      description: Common properties of a single media access by the Media Stream Handler.
```

```
      required:
        - mediaStreamHandlerEndpointAddress
        - applicationServerEndpointAddress
        - requestMessage
        - responseMessage
        - processingLatency
      properties:
        mediaStreamHandlerEndpointAddress:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/EndpointAddress'
        applicationServerEndpointAddress:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/EndpointAddress'
        requestMessage:
          type: object
          required:
            - method
            - url
            - protocolVersion
            - size
            - bodySize
          properties:
            method:
              type: string
            url:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/AbsoluteUrl'
            protocolVersion:
              type: string
            range:
              type: string
            size:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
            bodySize:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
            contentType:
              type: string
            userAgent:
              type: string
            userIdentity:
              type: string
            referer:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/AbsoluteUrl'
        cacheStatus:
          $ref: '#/components/schemas/CacheStatus'
        responseMessage:
          type: object
          required:
            - responseCode
            - size
            - bodySize
          properties:
            responseCode:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
            size:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
            bodySize:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
            contentType:
              type: string
        processingLatency:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
        connectionMetrics:
          type: object
          required:
            - meanNetworkRoundTripTime
            - networkRoundTripTimeVariation
            - congestionWindowSize
          properties:
            meanNetworkRoundTripTime:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
            networkRoundTripTimeVariation:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
            congestionWindowSize:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'

    NetworkAssistanceInvocation:
      description: Common properties of a single Network Assistance invocation.
      type: object
      properties:
        policyTemplateId:
```

```
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
        applicationFlowDescriptions:
          type: array
          minItems: 1
          items:
            $ref: 'TS26510_CommonData.yaml#/components/schemas/ApplicationFlowDescription'
        requestedQoS:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/UnidirectionalQoSSpecification'
        # The network QoS parameters (if any) requested by the Media Session Handler.
        recommendedQoS:
          type: object
          description: The network QoS parameters (if any) recommended to the Media Session
Handler.
          required:
            - maximumBitRate
            - minimumBitRate
          properties:
            maximumBitRate:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
            minimumBitRate:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'

    ######################################
    # Clause 6.4.4: Enumerated data types
    ######################################
    CacheStatus:
      anyOf:
        - type: string
          enum:
            - HIT
            - MISS
            - EXPIRED
        - type: string
          description: >
            This string provides forward-compatibility with future
            extensions to the enumeration but is not used to encode
            content defined in the present version of this API.
```

C.3 OpenAPI representation of the M1 APIs

C.3.0 Maf_Provisioning API

The normative code specifying the APIs defined in this clause, including JSON Schema representations of HTTP message bodies to be used with these APIs, is published on 3GPP Forge according to the OpenAPI 3.0.0 specification [23]. The YAML files corresponding to this version of the present document shall be published to the following location:

`https://forge.3gpp.org/rep/all/5G_APIs/-/tags/TSG112-Rel19`

Informative copies of these YAML files shall be distributed with the present document for convenience only. Where any discrepancy exists, the version on 3GPP Forge shall be considered definitive.

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_Maf_Provisioning.yaml".

C.3.1 Void

C.3.2 Void

C.3.3 Void

C.3.4 Void

C.3.5 Void

C.3.6 Void

C.3.7 Void

C.3.8 Void

C.3.9 Void

C.3.10 Void

C.3A OpenAPI representation of the M3 APIs

C.3A.0 Mas_Configuration API

The normative code specifying the APIs defined in this clause, including JSON Schema representations of HTTP message bodies to be used with these APIs, is published on 3GPP Forge according to the OpenAPI 3.0.0 specification [23]. The YAML files corresponding to this version of the present document shall be published to the following location:

`https://forge.3gpp.org/rep/all/5G_APIs/-/tags/TSG112-Rel19`

Informative copies of these YAML files shall be distributed with the present document for convenience only. Where any discrepancy exists, the version on 3GPP Forge shall be considered definitive.

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_Mas_Configuration.yaml".

C.3A.1 Mas_Configuration_ServerCertificates API

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_Mas_Configuration_ServerCertificates.yaml".

C.3A.2 Mas_Configuration_ContentPreparationTemplates API

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_Mas_Configuration_ContentPreparationTemplates.yaml".

C.3A.3 Mas_Configuration_ContentHosting API

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_Mas_Configuration_ContentHosting.yaml".

C.3A.4 Mas_Configuration_ContentPublishing API

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_Mas_Configuration_ContentPublishing.yaml".

C.4 OpenAPI representation of the M5 APIs

C.4.0 Maf_SessionHandling API

The normative code specifying the APIs defined in this clause, including JSON Schema representations of HTTP message bodies to be used with these APIs, is published on 3GPP Forge according to the OpenAPI 3.0.0 specification [23]. The YAML files corresponding to this version of the present document shall be published to the following location:

`https://forge.3gpp.org/rep/all/5G_APIs/-/tags/TSG112-Rel19`

Informative copies of these YAML files shall be distributed with the present document for convenience only. Where any discrepancy exists, the version on 3GPP Forge shall be considered definitive.

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_Maf_SessionHandling.yaml".

C.4.1 Void

C.4.2 Void

C.4.3 Void

C.4.4 M5_DynamicPolicies API

This file is specified for the sole purpose of maintaining backwards compatibility for Release 17 of TS 29.517 [46]. The data types it specifies are deprecated and shall not be used in the present release.

```
openapi: 3.0.0

info:
  title: M5_DynamicPolicies (deprecated)
  version: 3.1.0
  description: |
    5GMS AF M5 Dynamic Policy API
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

tags:
  - name: M5_DynamicPolicies
    description: '5G Media Streaming: Media Session Handling (M5) APIs: Dynamic Policies
(deprecated)'

externalDocs:
  description: 'TS 26.512 V19.3.0; 5G Media Streaming (5GMS); Protocols'
  url: 'https://www.3gpp.org/ftp/Specs/archive/26_series/26.512/'

paths: {}
```

```
components:
  schemas:
    DynamicPolicy:
      deprecated: true
      description: "A representation of a Dynamic Policy resource."
      type: object
      required:
        - dynamicPolicyId
        - policyTemplateId
        - serviceDataFlowDescriptions
        - provisioningSessionId
      properties:
        dynamicPolicyId:
          readOnly: true
          allOf:
            - $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
        policyTemplateId:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
        serviceDataFlowDescriptions:
          type: array
          items:
            $ref: 'TS26510_CommonData.yaml#/components/schemas/ApplicationFlowDescription'
        mediaType:
          $ref: 'TS29514_Npcf_PolicyAuthorization.yaml#/components/schemas/MediaType'
        provisioningSessionId:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
        qosSpecification:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ClientPolicySpecification'
        enforcementMethod:
          readOnly: true
          type: string
        enforcementBitRate:
          readOnly: true
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
```

C.4.5 M5_NetworkAssistance API

This file is specified for the sole purpose of maintaining backwards compatibility for Release 17 of TS 29.517 [46]. The data types it specifies are deprecated and shall not be used in the present release.

```
openapi: 3.0.0

info:
  title: M5_NetworkAssistance (deprecated)
  version: 3.1.0
  description: |
    5GMS AF M5 Network Assistance API
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

tags:
  - name: M5_NetworkAssistance
    description: '5G Media Streaming: Media Session Handling (M5) APIs: Network Assistance
(deprecated)'

externalDocs:
  description: 'TS 26.512 V19.3.0; 5G Media Streaming (5GMS); Protocols'
  url: 'https://www.3gpp.org/ftp/Specs/archive/26_series/26.512/'

paths: {}

components:
  schemas:
    NetworkAssistanceSession:
      deprecated: true
      description: "A representation of a Network Assistance Session resource."
      type: object
      required:
        - naSessionId
        - provisioningSessionId
        - serviceDataFlowDescriptions
      properties:
        naSessionId:
```

```
          readOnly: true
          allOf:
            - $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
        provisioningSessionId:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
        serviceDataFlowDescriptions:
          type: array
          items:
            $ref: 'TS26510_CommonData.yaml#/components/schemas/ApplicationFlowDescription'
          minItems: 1
        mediaType:
          $ref: 'TS29514_Npcf_PolicyAuthorization.yaml#/components/schemas/MediaType'
        policyTemplateId:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
        requestedQoS:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ClientPolicySpecification'
        recommendedQoS:
          readOnly: true
          allOf:
            - $ref: 'TS26510_CommonData.yaml#/components/schemas/ClientPolicySpecification'
        notficationURL:
          readOnly: true
          allOf:
            - $ref: 'TS26510_CommonData.yaml#/components/schemas/AbsoluteUrl'
```

C.5 OpenAPI representation of data reporting records

C.5.1 R4 data reporting records

For the purpose of referencing entities defined in this clause, it shall be assumed that the OpenAPI definitions below are contained in a physical file named "TS26512_R4_DataReporting.yaml".

```
openapi: 3.0.0

info:
  title: 5GMS R4 Data Reporting data types
  version: 2.1.0
  description: |
    5GMS R4 Data Reporting data types
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

tags:
  - name: 5GMS R4 Data Reporting data types
    description: '5G Media Streaming: R4 Data Reporting data types'

externalDocs:
  description: 'TS 26.512 V19.3.0; 5G Media Streaming (5GMS); Protocols'
  url: 'https://www.3gpp.org/ftp/Specs/archive/26_series/26.512/'

paths: {}

components:
  schemas:
    MediaStreamingAccessRecord:
      description: "An R4 data reporting record describing a single media access by the Media
Stream Handler."
      allOf:
        - $ref: 'TS26532_Ndcaf_DataReporting.yaml#/components/schemas/BaseRecord'
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/MediaStreamingBaseReportingRecord'
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/MediaStreamingSessionIdentification'
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/MediaStreamingAccess'
```

C.5.2 R2 data reporting records

For the purpose of referencing entities defined in this clause, it shall be assumed that the OpenAPI definitions below are contained in a physical file named "TS26512_R2_DataReporting.yaml".

```
openapi: 3.0.0

info:
  title: 5GMS Data Reporting data types
  version: 1.1.0
  description: |
    5GMS Data Reporting data types
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

tags:
  - name: 5GMS Data Reporting data types
    description: '5G Media Streaming: Data Reporting data types'

externalDocs:
  description: 'TS 26.512 V19.3.0; 5G Media Streaming (5GMS); Protocols'
  url: 'https://www.3gpp.org/ftp/Specs/archive/26_series/26.512/'

paths: {}

components:
  schemas:
    ANBRNetworkAssistanceInvocationRecord:
      description: "An R2 data reporting record describing a single ANBR-based Network Assistance
invocation by the Media Stream Handler."
      allOf:
        - $ref: 'TS26532_Ndcaf_DataReporting.yaml#/components/schemas/BaseRecord'
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/MediaStreamingBaseReportingRecord'
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/MediaStreamingSessionIdentification'
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/NetworkAssistanceInvocation'
```

C.6 OpenAPI representation of event exposure data types

For the purpose of referencing entities defined in this clause, it shall be assumed that the OpenAPI definitions below are contained in a physical file named "TS26512_EventExposure.yaml".

```
openapi: 3.0.0

info:
  title: 5GMS Event Exposure
  version: 1.2.0
  description: |
    5GMS Event Exposure syntax
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

tags:
  - name: 5GMS Event Exposure
    description: '5G Media Streaming: Event Exposure'

externalDocs:
  description: 'TS 26.512 V19.3.0; 5G Media Streaming (5GMS); Protocols'
  url: 'https://www.3gpp.org/ftp/Specs/archive/26_series/26.512/'

paths: {}

components:
  schemas:
    BaseEventCollection:
      type: object
      description: Abstract base data type describing a collection of event records
      required:
        - collectionTimestamp
        - startTimestamp
```

```
        - endTimestamp
        - sampleCount
        - streamingDirection
        - summarisations
        - records
      properties:
        collectionTimestamp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
          # The date-time at which this collection was exposed by the Data Collection AF as an
event to its subscribed event consumers.
        startTimestamp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
          # Date-time of earliest data sample included in or summarised by this collection.
        endTimestamp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
          # Date-time of latest data sample included in or summarised by this collection.
        sampleCount:
          type: integer
          minimum: 1
          description: The number of data samples included in or summarised by this collection.
        streamingDirection:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ProvisioningSessionType'
        summarisations:
          type: array
          minItems: 1
          items:
            $ref:
'TS26532_Ndcaf_DataReportingProvisioning.yaml#/components/schemas/DataAggregationFunctionType'
          description: One or more data aggregation functions that have been applied to the UE
data to produce summary records present in this collection.
        records:
          type: array
          minItems: 0
          items: {}
          description: Type of record is defined by concrete data type.

    BaseEventRecord:
      type: object
      description: Abstract base data type describing a single UE data record or summarising a set
of UE data records.
      required:
        - recordType
        - recordTimestamp
        - appId
      properties:
        recordType:
          $ref: '#/components/schemas/EventRecordType'
        recordTimestamp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
          # The date-time at which the UE data carried in this record was sampled or summarised.
        appId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationId'
          # Identifying the application to which the UE data carried in this record pertains.
        provisioningSessionId:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
          # The identifier of the Provisioning Session to which this record pertains. Present only
for individual data sample record type.
        sessionId:
          $ref: 'TS26510_CommonData.yaml#/components/schemas/MediaDeliverySessionId'
          # A value synthesised by the 5GMS System that uniquely identifies the media streaming
session to which this record pertains. Present only for individual data sample record type.
        ueIdentification:
          type: string
          description: GPSI of the requesting UE or a stable globally unique string identifying
the requesting Media Session Handler. Present only for individual data sample record type and only
when exposure is permitted by the data exposure restrictions in force for the event type in
question.
        dataNetworkName:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
          # Identifying the Data Network of the M4 media streaming session. Present only for
individual data sample record type.
        sliceId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
          # The S-NSSAI identifying the Network Slice of the M4 media streaming session. Present
only for individual data sample record type.
        ueLocations:
          type: array
          minItems: 0
```

```
          items:
            $ref: 'TS29122_CommonData.yaml#/components/schemas/LocationArea5G'
          description: The location of the UE when the data described by this record was sampled.
Present only for individual data sample record type and only when exposure is permitted by the
data exposure restrictions in force for the event in question.

    EventRecordType:
      description: Enumeration of event record types.
      anyOf:
        - type: string
          enum:
            - INDIVIDUAL_SAMPLE
            - SUMMARY_MEAN
            - SUMMARY_MINIMUM
            - SUMMARY_MAXIMUM
            - SUMMARY_SUM
        - type: string
          description: >
            This string provides forward-compatibility with future
            extensions to the enumeration but is not used to encode
            content defined in the present version of this API.

    QoEMetricsCollection:
      description: A collection of QoE Metrics Event records.
      allOf:
        - $ref: '#/components/schemas/BaseEventCollection'
        - type: object
          required:
            - records
          properties:
            records:
              type: array
              minItems: 0
              items:
                $ref: '#/components/schemas/QoEMetricsEvent'
              description: A set of QoE Metrics Event records, each one describing a QoE metrics
report or summarising a set of QoE metrics reports.

    QoEMetricsEvent:
      description: A QoE Metrics Event record.
      allOf:
        - $ref: '#/components/schemas/BaseEventRecord'
        - type: object
          required:
            - metricType
          properties:
            metricType:
              $ref: 'TS29122_CommonData.yaml#/components/schemas/Uri'
              # A fully-qualified term identifier that uniquely identifies the QoE metrics
reporting scheme and the type of QoE metric included in this record, as specified in clause E.2,
up to but excluding the first hierarchical separator.
            samples:
              type: array
              minItems: 1
              items:
                type: object
                required:
                  - metrics
                properties:
                  sampleTimestamp:
                    $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
                    # The moment in time at which this QoE metric was sampled.
                  sampleDuration:
                    $ref: 'TS26510_CommonData.yaml#/components/schemas/Duration'
                    # The time duration over which this QoE metric was sampled.
                  mediaTimestamp:
                    $ref: 'TS26510_CommonData.yaml#/components/schemas/Duration'
                    # The time point in the media at which this QoE metric was sampled.
                  metrics:
                    type: array
                    minItems: 1
                    description: A set of key–value pairs for the samples metrics associated with
this QoE metric sample.
                    items:
                      type: object
                      description: A name and optional value for a QoE metric within a QoE metric
sample.
                      required:
```

```
                        - key
                      properties:
                        key:
                          type: string
                          description: A token that uniquely identifies the type of metric within
the scope of the event type.
                        value: {}

    ConsumptionReportingUnitsCollection:
      description: A collection of Consumption Reporting Event records.
      allOf:
        - $ref: '#/components/schemas/BaseEventCollection'
        - type: object
          required:
            - records
          properties:
            records:
              type: array
              minItems: 0
              items:
                $ref: '#/components/schemas/ConsumptionReportingEvent'
              description: A set of records, each one describing a Consumption Reporting Unit.

    ConsumptionReportingEvent:
      description: A Consumption Reporting Event record, corresponding to a Consumption Reporting
Unit.
      allOf:
        - $ref: '#/components/schemas/BaseEventRecord'
        - type: object
          required:
            - unitDuration
            - mediaPlayerEntryUrl
            - mediaComponentIdentifier
          properties:
            unitDuration:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/Duration'
              # The duration of this consumption reporting unit.
            clientEndpointAddress:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/EndpointAddress'
              # The endpoint address of the UE that consumed media.
            serverEndpointAddress:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/EndpointAddress'
              # The endpoint address of the 5GMS AS from which media was consumed.
            mediaPlayerEntryUrl:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/AbsoluteUrl'
              # The Media Player Entry URL to which this consumption reporting unit pertains.
            mediaComponentIdentifier:
              type: string
              description: A token (e.g., an MPEG DASH representation identifier) identifying the
media component of the Media Player Entry that was consumed during this consumption reporting
unit.

    NetworkAssistanceInvocationsCollection:
      description: A collection of Network Assistance Invocation Event records.
      allOf:
        - $ref: '#/components/schemas/BaseEventCollection'
        - type: object
          required:
            - records
          properties:
            records:
              type: array
              minItems: 0
              items:
                $ref: '#/components/schemas/NetworkAssistanceInvocationEvent'
              description: A set of records, each one describing a Network Assistance invocation
or summarising a set of Network Assistance invocations.

    NetworkAssistanceInvocationEvent:
      description: A Network Assistance Invocation Event record.
      allOf:
        - $ref: '#/components/schemas/BaseEventRecord'
        - type: object
          required:
            - networkAssistanceType
          properties:
            networkAssistanceType:
              $ref: '#/components/schemas/NetworkAssistanceType'
```

```
              # The type of Network Assistance solicited by the Media Session Handler.
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/NetworkAssistanceInvocation'

    NetworkAssistanceType:
      description: The type of network assistance sought by the Media Session Handler.
      anyOf:
        - type: string
          enum:
            - AF_THROUGHPUT_ESTIMATION
            - AF_DELIVERY_BOOST
            - ANBR_THROUGHPUT_ESTIMATION
            - ANBR_DELIVERY_BOOST
        - type: string
          description: >
            This string provides forward-compatibility with future
            extensions to the enumeration but is not used to encode
            content defined in the present version of this API.

    DynamicPolicyInvocationsCollection:
      description: A collection of Dynamic Policy Invocation Event records.
      allOf:
        - $ref: '#/components/schemas/BaseEventCollection'
        - type: object
          required:
            - records
          properties:
            records:
              type: array
              minItems: 0
              items:
                $ref: '#/components/schemas/DynamicPolicyInvocationEvent'
              description: A set of records, each one describing a dynamic policy invocation.

    DynamicPolicyInvocationEvent:
      description: A Dynamic Policy Invocation Event record.
      allOf:
        - $ref: '#/components/schemas/BaseEventRecord'
        - type: object
          required:
            - policyTemplateId
          properties:
            policyTemplateId:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/ResourceId'
            applicationFlowDescriptions:
              type: array
              minItems: 1
              items:
                $ref: 'TS26510_CommonData.yaml#/components/schemas/ApplicationFlowDescription'
              description: The set of Service Data Flows to which the Media Session Handler
requested that the Policy Template be applied. At least a domain name or one packet filter shall
be present.
            requestedQoS:
              $ref: 'TS26510_CommonData.yaml#/components/schemas/UnidirectionalQoSSpecification'
              # The network QoS parameters (if any) requested by the Media Session Handler.
            enforcementMethod:
              type: string
              description: The policy enforcement method chosen by the 5GMS AF.
            enforcementBitRate:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/BitRate'
              # The enforcement bit rate (if any) selected by the 5GMS AF.

    MediaStreamingAccessesCollection:
      description: A collection of Media Streaming Access Event records.
      allOf:
        - $ref: '#/components/schemas/BaseEventCollection'
        - type: object
          required:
            - records
          properties:
            records:
              type: array
              minItems: 0
              items:
                $ref: '#/components/schemas/MediaStreamingAccessEvent'
              description: A set of records, each one describing one media streaming access.

    MediaStreamingAccessEvent:
      description: A Media Streaming Access Event record.
```

```
      allOf:
        - $ref: '#/components/schemas/BaseEventRecord'
        - $ref: 'TS26512_CommonData.yaml#/components/schemas/MediaStreamingAccess'
```

C.7 OpenAPI representation of client data

The normative code specifying the APIs defined in this clause, including JSON Schema representations of HTTP message bodies to be used with these APIs, is published on 3GPP Forge according to the OpenAPI 3.0.0 specification [23]. The YAML files corresponding to this version of the present document shall be published to the following location:

`https://forge.3gpp.org/rep/all/5G_APIs/-/tags/TSG109-Rel19`

Informative copies of these YAML files shall be distributed with the present document for convenience only. Where any discrepancy exists, the version on 3GPP Forge shall be considered definitive.

For the purpose of referencing entities specified in this clause, it shall be assumed that the OpenAPI definitions are contained in a physical file named "TS26512_ClientData.yaml".

Annex D (informative):
Void

Annex E (normative): Controlled vocabularies of 5G Media Streaming UE data parameters

E.1 General

The set of parameters used to provision processing and exposure of UE data is defined for each exposed Event by a controlled vocabulary with a unique name space identifier string in the form of a URN. Each parameter is specified as a term identifier string that is unique within the scope of its parent name space.

In the context of 5G Media Streaming, the values signalled in *DataAccessProfile.parameters* (see clause 6.3.2.3 of TS 26.532 [47]) shall be fully-qualified URIs formed from the name space identifier specified in the following clauses concatenated with a single hash character ('#') concatenated with the appropriate term identifier.

E.2 Controlled vocabularies of 5GMS QoE metrics reporting parameters

E.2.1 Reporting parameters for 3GP-DASH metrics

The name space identifier for the controlled vocabulary of DASH QoE metrics is:

urn:3GPP:ns:PSS:DASH:QM10

The term identifiers in this controlled vocabulary shall be the set of key names defined in clause 10.2 of TS 26.247 [4], using a single forward slash character ('/') as the hierarchical separator in the resulting path specifier. In the case of metrics lists, the *Entry* object shall be omitted from the term identifier path.

EXAMPLE 1: *urn:3GPP:ns:PSS:DASH:QM10#AvgThroughput/numbytes*

To select all reportable metrics below a common branch of the metrics hierarchy the relevant terminal path element(s) are pruned from the term identifier.

EXAMPLE 2: *urn:3GPP:ns:PSS:DASH:QM10#AvgThroughput*

E.2.2 Reporting parameters for Virtual Reality (VR) DASH metrics

The name space identifier for the controlled vocabulary of VR DASH metrics is:

urn:3gpp:metadata:2020:VR:metrics

The term identifiers in this controlled vocabulary shall be the set of key names defined in clause 9.3 of TS 26.118 [42], using a single forward slash character ('/') as the hierarchical separator in the resulting path specifier. In the case of metrics lists, the *Entry* object shall be omitted from the term identifier path.

EXAMPLE 1: *urn:3gpp:metadata:2020:VR:metrics#CompQualLatency/Latency*

To select all reportable metrics below a common branch of the metrics hierarchy the relevant terminal path element(s) are pruned from the term identifier.

EXAMPLE 2: *urn:3gpp:metadata:2020:VR:metrics#CompQualLatency*

E.2.3 Reporting parameters for 5GMS delivery over eMBMS

The controlled vocabularies specified in clauses E.2.1 and/or E.2.2 (as applicable) shall be used to identify metrics or groups of metrics in relation to reporting the QoE of 5GMS delivery over eMBMS.

E.2.4 Reporting parameters for 5GMS delivery over MBS

Metrics reporting for MBS User Services is not specified by TS 26.517 [64] in this release.

E.2.5 Reporting parameters for CMCD-based client data

The name space identifier for the controlled vocabulary of CMCD-based client data is:

urn:3gpp:5gms:event-exposure:common-media-client-data

The term identifiers in this controlled vocabulary are arranged in a two-level hierarchy, the two levels being delimited by a single forward slash character ('/') as the hierarchical separator in the resulting path specifier.

The first path element of the term identifier shall indicate the class of CMCD information as specified in table E.2.3-1 below.

Table E.2.3-1: Term identifiers for CMCD information exposure

CMCD information class	Term identifier	Fully-qualified term identifier
CMCD per-session information	*session*	*urn:3gpp:5gms:event-exposure:common-media-client-data#session*
CMCD per-object information	*object*	*urn:3gpp:5gms:event-exposure:common-media-client-data#object*
CMCD per-request information	*request*	*urn:3gpp:5gms:event-exposure:common-media-client-data#request*
CMCD status information	*status*	*urn:3gpp:5gms:event-exposure:common-media-client-data#status*

The second path element shall indicate one the reserved keys defined in section 3.3 of the CMCD specification [70] for the class of CMCD information indicated by the first path element.

EXAMPLE 1: *urn:3gpp:5gms:event-exposure:common-media-client-data#session/pr* (current playback rate)

EXAMPLE 2: *urn:3gpp:5gms:event-exposure:common-media-client-data#request/mtp* (measured throughput)

E.3 Controlled vocabulary of 5GMS consumption reporting parameters

The name space identifier for the controlled vocabulary of 5GMS consumption reporting parameters is:

urn:3gpp:5gms:event-exposure:consumption-reporting

The term identifiers in this controlled vocabulary are specified in table E.3-1 below.

EXAMPLE: *urn:3gpp:5gms:event-exposure:consumption-reporting#locations*

Table E.3-1: Controlled vocabulary of 5GMS consumption reporting parameters

Term identifier	Description
start-time	The date and time of the consumption reporting unit.
duration	The duration of the consumption reporting unit.
media-player-entry	The entry pointer for the media streaming session.
reporting-client-identifier	Identifier for the reporting client that consumed the media.
media-consumed	Identifies the media consumed within the context of the media player entry.
locations	The UE location(s) where the media was consumed. (Only available to trusted event consumer.)

E.4 Controlled vocabulary of 5GMS Network Assistance reporting parameters

The name space identifier for the controlled vocabulary of 5GMS Network Assistance parameters is:

urn:3gpp:5gms:event-exposure:network-assistance

The term identifiers in this controlled vocabulary are specified in table E.4-1 below.

EXAMPLE: *urn:3gpp:5gms:event-exposure:network-assistance#requested-qos*

Table E.4-1: Controlled vocabulary of 5GMS Network Assistance parameters

Term identifier	Description
timestamp	The date and time of the Network Assistance invocation by the Media Session Handler.
service-data-flow-information	Identification of the media streaming application flow for which Network Assistance was requested.
policy-template-identifier	The policy template in force for the media streaming session.
requested-qos	The network Quality of Service requested by the media streamer.
recommended-qos	The network Quality of Service recommended by the 5GMS AF.

E.5 Controlled vocabulary of 5GMS Dynamic Policy reporting parameters

The name space identifier for the controlled vocabulary of 5GMS Dynamic Policy parameters is

urn:3gpp:5gms:event-exposure:dynamic-policy

The term identifiers in this controlled vocabulary are specified in table E.5-1 below.

EXAMPLE: *urn:3gpp:5gms:event-exposure:dynamic-policy#enforcement-bit-rate*

Table E.5-1: Controlled vocabulary of 5GMS Dynamic Policy parameters

Term identifier	Description
timestamp	The date and time of the dynamic policy invocation by the Media Session Handler.
policy-template-identifier	The policy template instantiated for the media streaming session.
service-data-flow-descriptions	The set of media streaming application flows managed by the Dynamic Policy.
qos-specification	The network Quality of Service described by the policy template currently in force.
enforcement-method	The Policy Enforcement Method set by the 5GMS AF.
enforcement-bit-rate	The bit rate currently being enforced by this Dynamic Policy.

E.6 Controlled vocabulary of 5GMS media access activity parameters

The name space identifier for the controlled vocabulary of 5GMS media access activity is:

urn:3gpp:5gms:event-exposure:access-activity

The term identifiers in this controlled vocabulary are specified in table E.6-1 below.

EXAMPLE: *urn:3gpp:5gms:event-exposure:access-activity#request-message/url*

Table E.6-1: Controlled vocabulary of 5GMS media access activity parameters

Term identifier	Description
timestamp	The date and time of the media access.
media-stream-handler-endpoint-address	The endpoint address of the Media Stream Handler accessing the 5GMS AS.
application-server-endpoint-address	The service endpoint on the 5GMS AS to which the Media Stream Handler is connected.
session-identifier	An opaque identifier for the HTTP session on which the Media Stream Handler request was made.
request-message	All term identifiers below with prefix *request-message*.
request-message/method	The request method.
request-message/url	The request URL.
request-message/protocol-version	The HTTP protocol version, e.g. "HTTP/1.1".
request-message/range	The value of the Range request header.
request-message/size	The total number of bytes in the request message.
request-message/body-size	The number of bytes supplied by the Media Stream Handler in the HTTP request body.
request-message/content-type	The MIME content type of the request message, if any.
request-message/user-agent	A string describing the requesting Media Stream Handler from the `User-Agent` request header.
request-message/user-identity	A string identifying the user that made the access.
request-message/referer	The URL that the Media Player reports being referred from in the `Referer` request header.
cache-status	An indication of whether the 5GMS AS served the response object corresponding from cache.
response-message	All term identifiers below with prefix *response-message*.
response-message/response-code	The HTTP response code.
response-message/size	The total number of bytes in the response message.
response-message/body-size	The number of bytes in the HTTP response message body.
response-message/content-type	The MIME content type of response message.
processing-latency	The time, expressed in milliseconds, taken by the 5GMS AS to respond to the Media Stream Handler request.
connection-metrics	All term identifiers below with prefix *connection-metrics*.
connection-metrics/mean-rtt	Mean network round-trip time for the HTTP session, expressed in milliseconds.
connection-metrics/rtt-variation	The variation in mean network round-trip time, expressed in milliseconds.
connection-metrics/congestion-window-size	The current size (in bytes) of the congestion window for the transport connection underlying the HTTP session.

Annex F (informative): 5GMS AS Certificate provisioning and discovery

F.1 General

This annex describes 5GMS AS discovery by the 5GMS Client, including provisioning aspects that leverage the Domain Name System (DNS).

F.2 5GMS AS discovery and media streaming access with a Server Certificate created by the 5GMS System

Figure F.2-1 illustrates the initial provisioning needed to allow discovery of a 5GMS AS by the 5GMS Client as well as the eventual 5GMS AS discovery sequence using the Domain Name System (DNS). Specific focus here is on the provisioning and usage of TLS Certificates. The desired outcome is that the 5GMS Client is satisfied that it has established a TLS connection with an authorized 5GMS AS instance.

Here, the *Server Certificate Create* procedure (see clause 4.3.6.2) is used to request that the 5GMS AF creates the server certificate resource. In this case, the certificate's Common Name (*CN*) is assigned in a domain under the control of the 5GMS System operator (i.e., the 5GMS System operator is the legitimate owner of the domain name). The 5GMS System operator may use a third-party DNS service to host the domain in question.

For example, the 5GMS AF assigns the *canonicalDomainName* value `com-provider-service.ms.as.3gppservices.org` as the Common Name in the Server Certificate per clause 6.0.2.2. The 5GMS AF embeds this host name in the distribution base URL it returns to the 5GMS Application Provider.

Table F.2-1: Example Content Hosting Configuration corresponding to Create Server Certificate procedure

Content Hosting Configuration property	Example value
distributionConfigurations[n]	
canonicalDomainName	com-provider-service.ms.as.3gppservices.org
domainNameAlias	Not present.
baseURL	https://com-provider.ms.as.3gppservices.org/
certificateId	Pointing to Server Certificate resource with: - *CN* = com-provider.ms.as.3gppservices.org - *subjectAltName[0]* = com-provider.ms.as.3gppservices.org

The Media Entry Point URL advertised via reference point M5 or M8 (used by the Media Stream Handler to access the content *<relativePath>* in the 5GMS AS at reference point M4) is then:

`https://com-provider-service.ms.as.3gppservices.org/`*<relativePath>*

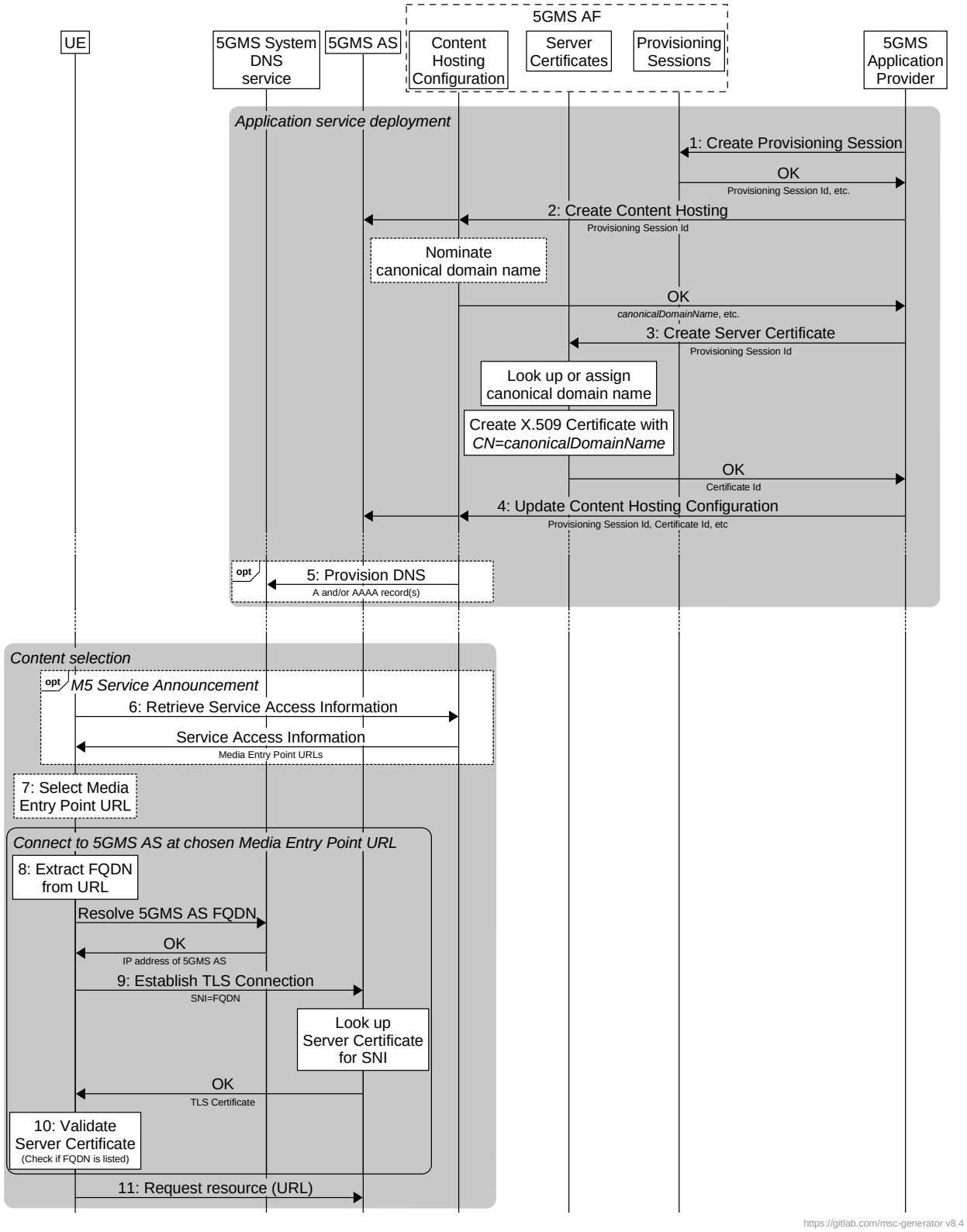


Figure F.2-1: 5GMS AS discovery using DNS using 5GMS System created server certificates

Prerequisites:

- A 5GMS-Aware Application is installed on the UE. This includes a list of API endpoint addresses for interacting with the 5GMS AF at reference point M5.

At application service deployment time:

1. The 5GMS Application Provider creates a Provisioning Session using the procedure specified in clause 4.3.2.2. The 5GMS AF provides the Provisioning Session Id in its response to the 5GMS Application Provider.

2. The 5GMS Application Provider creates a Content Hosting Configuration using the procedure specified in clause 4.3.3.2. If it has not already done so, the 5GMS AF assigns a canonical domain name for the 5GMS AS and includes this in its respose to the 5GMS Application Provider (*canonicalDomainName*).

3. The 5GMS Application Provider requests that the 5GMS AF creates a Server Certificate for this Provisioning Session using the procedure specified in clause 4.3.6.2. If it has not already done so, the 5GMS AF assigns a canonical domain name for the 5GMS AS and this is used as the Common Name and the first (and only) Subject Alternative Name of the generated server certificate. The CertificateId is provided to the 5GMS Application Provider upon success.

NOTE: The order of step 2 and step 3 may be reversed, in which case step 4 is not needed and the 5GMS AS is updated as a consequence of step 2.

4. The 5GMS Application Provider updates the Content Hosting Configuration using the procedure specified in clause 4.3.3.4, adding the Certificate Identifier of the newly created Server Certificate resource (i.e., modifying the value of *distributionConfigurations[ ].certificateId*).

   As a consequence, the 5GMS AF provisions the 5GMS AS with relevant information from the Provisioning Session, including the Server Certificate and Content Hosting Configuration.

5: If it has not already done so, the 5GMS AF provisions *A* and/or *AAAA* records in the DNS service mapping the canonical domain name of the 5GMS AS instance to its assigned IP address(es).

NOTE: The DNS service may be provided by a third-party service provider under the direction of the 5GMS System operator.

When content is selected in the 5GMS-Aware Application:

6. When Service Access Information retrieval at reference point M5 is used (see clause 4.7.2.3), the 5GMS Client in the UE retrieves the Service Access Information. This may include media entry point URLs within the *streamAccess.entryPoints* array. Otherwise, the 5GMS-Aware Application obtains this information via reference point M8.

7. The 5GMS Client in the UE selects one of the offered Media Entry Point URLs corresponding to the its media stream handling capabilities.

8. The 5GMS Client extracts the Fully-Qualified Domain Name (FQDN) from the chosen media entry point URL and uses the DNS service to resolve its IP address.

9. When establishing the TLS connection to the 5GMS AS at reference point M4, the Media Stream Handler in the UE cites the FQDN of the media entry point URL in the Server Name Indication (SNI) field of the TLS Client Hello message. The 5GMS AS uses the value of the SNI field to look up the corresponding Server Certificate and returns it to the Media Stream Handler in its Server Hello response. If the Server Certificate was provisioned with a wildcard Common Name, appropriate matching rules are followed by the 5GMS AS to identify the correct Server Certificate to present to the Media Stream Handler.

10. The 5GMS Client validates the Server Certificate, including whether the value of the Common Name (or one of its Subject Alternative Names) matches the FQDN of the chosen Media Entry Point URL.

When all server certificate validation steps are successfully passed, the following steps are executed:

11. The 5GMS Client requests the resource identified by the Media Entry Point over the TLS connection established with the 5GMS AS at reference point M4.

F.3 5GMS AS discovery and service access with a Server Certificate owned by the 5GMS Application Provider

Figure F.3-1 illustrates the initial provisioning needed to allow discovery of a 5GMS AS by the 5GMS Client as well as the eventual 5GMS AS discovery sequence using the Domain Name System (DNS). Specific focus here is on the provisioning and usage of TLS Certificates. The desired outcome is that the 5GMS Client is satisfied that it has established a TLS connection with an authorized 5GMS AS instance.

Here, the *Reserve Server Certificate* (see clause 4.3.6.3) and *Upload Server Certificate* (see clause 4.3.6.5) procedures are used for creating the Server Certificate resource and for subsequently providing it to the 5GMS AF. In this case, the certificate's Common Name (*CN*) is assigned in a domain under the control of the 5GMS Application Provider (i.e., Application Provider is the legitimate owner of the domain name). The 5GMS Application Provider may use the 5GMS System operator or any third-party DNS service to host the domain in question.

For example, the 5GMS Application Provider wishes to use the alias `5gms.provider.com` to access content through the 5GMS AS and it includes its chosen host name as a parameter to the *Reserve Server Certificate* operation. In response, the 5GMS AF generates a Certificate Signing Request (CSR) which it returns to the 5GMS Application Provider. The 5GMS AF assigns a canonical host name for the 5GMS AS in a domain under its control per clause 6.0.2.2. The 5GMS AF embeds the Fully-Qualified Domain Name of the alias in the distribution base URL of the Content Hosting Configuration it returns to the 5GMS Application Provider.

Table F.3-1: Example Content Hosting Configuration corresponding to Reserve/Upload Server Certificate procedure

Content Hosting Configuration property	Example value
distributionConfigurations[n]	
canonicalDomainName	com-provider-service.ms.as.3gppservices.org
domainNameAlias	5gms.provider.com
baseURL	*https://5gms.provider.com/*
certificateId	Pointing to Server Certificate resource with: - *CN* = 5gms.provider.com - *subjectAltName[0]* = 5gms.provider.com - *subjectAltName[1]* = …

The 5GMS Application Provider additionally configures the 5GMS AF-nominated *canonicalDomainName* as a *CNAME* record for its chosen 5GMS AS domain name alias in its preferred DNS service.

Listing F.3-1: Example DNS *CNAME* record to support distribution configuration in table F.3-1

```
5gms.provider.com.  CNAME   com-provider-service.ms.as.3gppservices.org
```

The Media Entry Point URL advertised via reference point M5 or M8 (used by the Media Stream Handler to access the content *<relativePath>* in the 5GMS AS at reference point M4) is then:

`https://5gms.provider.com/`*<relativePath>*

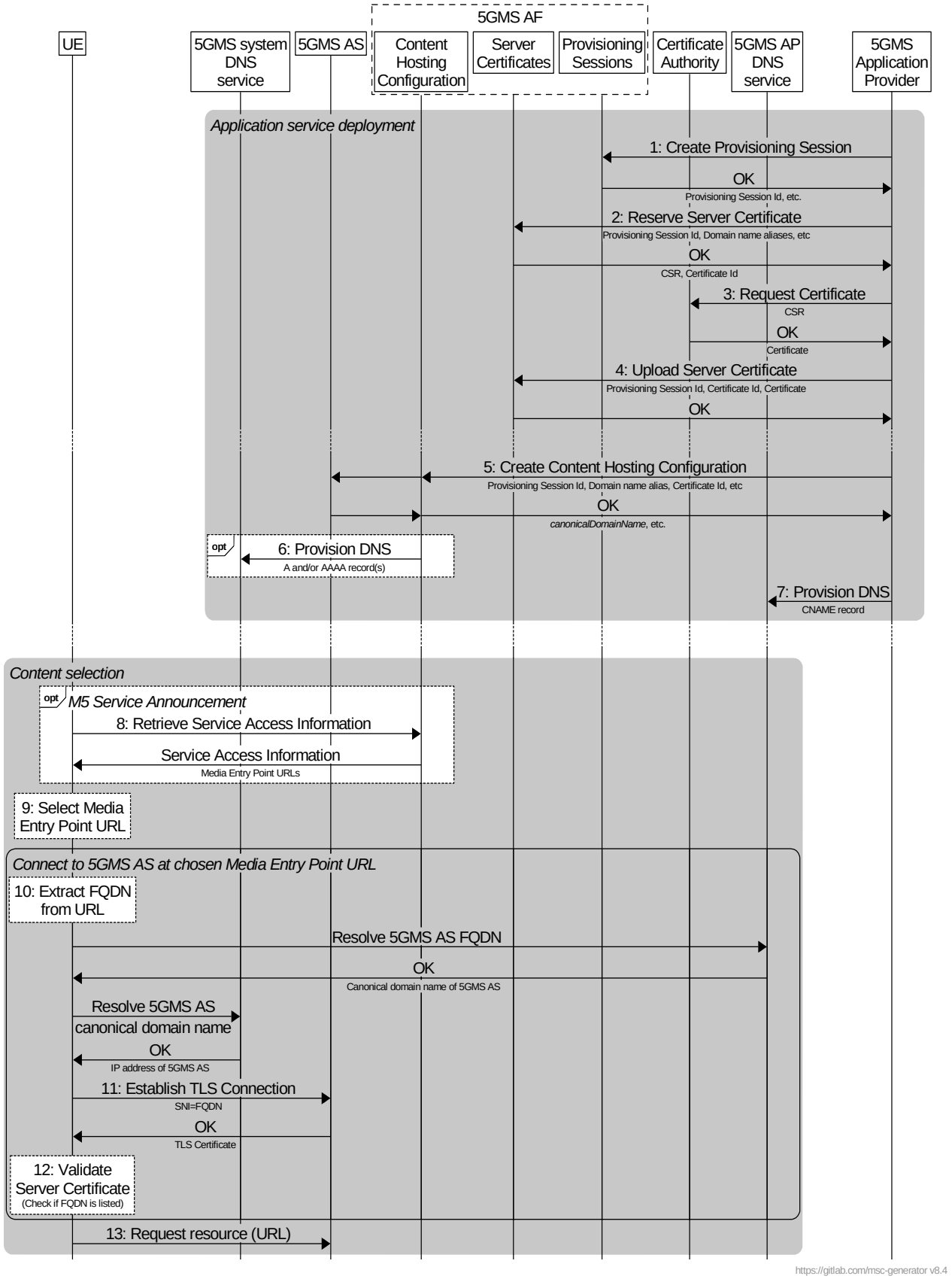


Figure F.3-1: 5GMS AS discovery using DNS using uploaded Server Certificates

Prerequisites:

- A 5GMS-Aware Application is installed on the UE. This includes a list of API endpoint addresses for interacting with the 5GMS AF at reference point M5.

At application service deployment time:

1. The 5GMS Application Provider creates a Provisioning Session using the procedure specified in clause 4.3.2.2. The 5GMS AF provides the Provisioning Session Id in its response to the 5GMS Application Provider.

2. The 5GMS Application Provider creates a CSR using the *Reserve Server Certificate* procedure of this Provisioning Session using the procedure specified in clause 4.3.6.3. The 5GMS Application Provider provides a set of 5GMS AS domain name aliases as input. The CSR and the Certificate Id are provided upon success in the response.

3. The 5GMS Application Provider uses the CSR for obtaining the Server Certificate from its prefered Certificate Authority.

4. The Application Provider uses the Upload Server Certificate procedure (see clause 4.3.6.5) to upload the received Server Certificate.

5. The 5GMS Application Provider creates the Content Hosting Configuration using the procedure specified in clause 4.3.3.2, adding the Domain Name Alias *distributionConfigurations.domainNameAlias* and CertificateId of the uploaded Server Certificate (*distributionConfigurations.certificateId*). The 5GMS AF assigns a cannonical domain name (*canonicalDomainName*) and provides its with the response.

   As a consequence, the 5GMS AF provisions the 5GMS AS with relevant information from the Provisioning Session, including the Server Certificate and Content Hosting Configuration.

6: If it has not already done so, the 5GMS AF provisions *A* and/or *AAAA* records in the 5GMS System's DNS service mapping the canonical name of the 5GMS AS instance to its assigned IP address(es).

NOTE 1: The DNS service may be provided by a third-party service provider under the direction of the 5GMS System operator.

7. The 5GMS Application Provider provisions its DNS service with the *CNAME* record mapping its chosen host name alias to the canonical name of the 5GMS AS instance.

NOTE 2: The 5GMS Application Provider may leverage the DNS of the 5GMS System.

When content is selected in the 5GMS-Aware Application:

8. When Service Access Information retrieval at reference point M5 is used (see clause 4.7.2.3), the 5GMS Client in the UE retrieves the Service Access Information. This may include media entry point URLs within the streamAccess.entryPoints array. Otherwise, the 5GMS-Aware Application obtains this information via reference point M8.

9. The 5GMS Client in the UE selects one of the offered Media Entry Point URLs corresponding to the its media stream handling capabilities.

10. The 5GMS Client extracts the Fully-Qualified Domain Name (FQDN) from the chosen Media Entry Point URL and uses the DNS service to resolve its IP address, first resolving the domain name alias to the canonical domain name using the 5GMS Application Provider's nominated DNS service, and then resolving the canonical domain name to the IP address(es) using the 5GMS System operator's nominated DNS service.

11. When establishing the TLS connection to the 5GMS AS at reference point M4, the Media Stream Handler in the UE cites the FQDN of the Media Entry Point URL in the Server Name Indication (SNI) field of the TLS Client Hello Message. The 5GMS AS uses the value of the SNI field to look up the corresponding Server Certificate and returns it to the Media Stream Handler in its Server Hello response.

NOTE 3: If the Server Certificate was provisioned with a wildcard Common Name, appropriate matching rules are followed by the 5GMS AS to identify the correct Server Certificate to present to the Media Stream Handler.

12. The 5GMS Client Validates the Server Certificate, including whether the value of the Common Name (or one of its Subject Alternative Names) matches the FQDN of the chosen Media Entry Point URL.

When all server certificate validation steps are successfully passed, the following steps are executed

13. The 5GMS Client requests the resource identified by the Media Entry Point over the TLS connection established with the 5GMS AS at reference point M4.

Annex G (normative): DASH in 5G Media Streaming

G.1 Introduction

This annex provides a binding between DASH as defined in ISO/IEC 23009-1 [32] and 3GPP TS 26.247 [4] as a media streaming protocol in the 5G Media Streaming System. In this context, it provides a mapping of DASH functionalities to generic 5G Media Streaming features. It also specifies requirements and makes recommendations on how certain features defined in 5G Media Streaming are to be used and supported when DASH is used as the media streaming protocol at reference point M4.

The annex also specifies how CMAF content as profiled in TS 26.511 [35] is required to be delivered through 5G Media Streaming using DASH.

The clause expands based on the requirements specified in clause 10.2.

G.2 General mapping of DASH to 5G Media Streaming

Based on the basic mapping of DASH in clause 10.2, the following generic aspects are documented:

- For any requirement that applies to the Media Player Entry in downlink streaming as part of the present document, for DASH this requirement applies to the Media Presentation Description (MPD) with content type `"application/dash+xml"`.
- Media resources are primarily DASH Segments, or other data referenced in the MPD. According to clause 10.2, DASH Segments may be CMAF-addressable resources per ISO/IEC 23000-19 [38], in particular CMAF media resources as profiled in TS 26.511 [35].
- The mapping of media resources into DASH MPDs, including codec parameters, is specified in TS 26.511 [35].

G.3 Content Protection and DRM

G.3.1 Overview

This clause defines the requirements and recommendations to support Content Protection and Digital Rights Management in DASH-based 5G Media Streaming according to clause 5.14 in TS 26.501 [2].

G.3.2 DASH content encoding requirements and recommendations

A DASH Media Presentation with encrypted content shall follow the requirements and recommendations in DASH-IF Interoperability Points Part 6 [74], in particular:

- DASH Segments and the corresponding CMAF-addressable resources shall follow the requirements and recommendations concerning Segments in clause 6 of [74].
- DASH MPDs shall follow the requirements and recommendations concerning the MPD in clause 7 of [74].

G.3.3 Media Player requirements and recommendations

A Media Player supporting DRM-based content protection shall implement the requirements and recommendations for DASH clients defined in clauses 6 and 7 of DASH-IF Interoperability Points Part 6 [74].

G.3.4 Content Preparation Template requirements

If content preparation is employed for DRM encryption by the 5GMS AS, the following parameters need to be supported in the Content Preparation Template:

- An indication of one of the *encryption methods* profiled in TS 26.511 [35], namely CENC or CBCS.
- The *DRM systems* that are to be supported, identified by their respective unique system identifiers.
- The URL of the *license server* from which the Media Player fetches decryption keys. The Packager subfunction of the 5GMS AS is required to embed the license acquisition URL in the Media Presentation Description (MPD).
- The *CPIX configuration* used to communicate encryption keys and DRM policies between the Packager and the Key Server.

G.3.5 Content Preparation Template syntax

NOTE: The format of an interoperable Content Preparation Template to support DRM encryption is for future study.

G.3.6 Examples (informative)

For examples of MPDs including Content Protection, see clause 7.3 of DASH-IF Interoperability Points Part 6 [74].

G.4 DASH content distribution using multiple service locations

G.4.1 Overview

This clause defines the requirements and recommendations to support content distribution from multiple service locations at reference point M4d in DASH-based downlink media streaming according to clause 5.2.6 of TS 26.501 [2]. This clause G.4 focuses on:

- Client-side switching
- Content Steering for DASH
- SAND for Multi-network access mode (SAND4M)

The focus in this clause is on media resources and transport resources that conform to DASH Segments as specified in ISO/IEC 23009-1 [32].

In particular, the base URL concept and service locations as defined in [32] are leveraged to support multiple service locations. A service location in MPEG-DASH is defined as "a collection of network resources that share commonalities and can be referred to by a common label". The **`BaseURL`** element may be used to specify one or more common locations for DASH Segments and other resources. Service locations are labels for base URLs, indicated in the MPD by an attribute of the **`BaseURL`** element.

Clause 5.6 of [32] specifies how to apply base URLs to each relative URL in the MPD to construct an absolute URL suitable for retrieving transport objects containing media resources, as well as specifying rules for handling multiple alternative base URLs, one of the key aspects to support multiple service locations in the 5GMS System.

For content hosting:

- To provision DASH downlink media streaming in the 5GMSd AF, the 5GMSd Application Provider shall use the Content Hosting Provisioning (M1) API specified in clause 7.6, following the Content Hosting provisioning (M1) procedures specified in clause 4.3.3.

- To configure downlink media streaming in the 5GMSd AS, the 5GMSd AF shall use the Content Hosting Configuration (M3) API specified in clause 9.4, following the Content Hosting configuration (M3) procedures specified in clause 4.5.4.

G.4.2 General Content Preparation requirements to support multiple service locations

G.4.2.1 Introduction

This clause specifies general requirements for preparing DASH content to make it suitable for distribution via different service locations in the 5GMS System, in particular the configuration information required by the 5GMSd AS in order to perform any necessary content preparation.

To provision Content Preparation of DASH transport resources in the 5GMSd AF, the 5GMSd Application Provider shall use the Content Preparation Templates Provisioning (M1) API specified in clause 7.4, following the Content Preparation Template provisioning (M1) procedures specified in clause 4.3.5.

To configure Content Preparation of DASH transport resources in the 5GMSd AS, the 5GMSd AF shall use the Content Preparation Templates Configuration (M3) API specified in clause 9.3, following the Content Preparation Template configuration (M3) procedures specified in clause 4.5.3.

Two different variants are supported:

1. DASH content ingested by the 5GMSd AS through reference point M2d, including a provided DASH MPD, already includes different base URLs with service location tags. The content preparation shall provide a mapping of the service locations in the MPD to the provisioned service locations on the 5GMSd AS. For details refer to clause G.4.2.2.

2. DASH content ingested through reference point M2d conforms to TS 26.511 [35] or at least CMAF per ISO/IEC 23000-19 [40], and the 5GMSd AS creates an MPD that includes multiple service locations based on the provisioning information and the content preparation information. For details refer to clause G.4.2.3.

The following assumptions are common to both scenarios:

1. Multiple service locations have been provisioned at reference point M1d as distinct distribution configurations using the Content Hosting Provisioning (M1) API specified in clause 7.6 and following the Content Hosting provisioning (M1) procedures specified in clause 4.3.3.

G.4.2.2 Pre-configured service locations

In this case, content ingested by the 5GMSd AS through reference point M2d, including a provided DASH MPD, already includes different base URLs with service location tags. The content preparation shall the base URLs of each service location in the provided MPD with a provisioned service location on the 5GMSd AS, shall make the modified MPD available to the 5GMS Client via reference point M4d.

The ingested Media Presentation is described by an MPD that shall include a **`BaseURL`** element including a `@serviceLocation` attribute that fulfils the requirements for a service location as specified in ISO/IEC 23009-1 [32]. If this service location is to be exposed by the 5GMSd AS at reference point M4d, it shall be associated with a Distribution Configuration provisioned in the 5GMSd AF Content Hosting Configuration by matching its `@serviceLocaton` attribute value with the value of a *DistributionConfiguration.distributionId* property (see clause 8.8.3.1 of TS 26.510 [56]).

In addition, the **`BaseURL`** may include the following optional parameters

1. An `@availabilityTimeOffset` attribute adjusting the Segment availability time for this service location.

2. An `@availabilityComplete` attribute indicating whether all Segments of all associated Representations are complete at the adjusted availability start time.

3. A `@timeShiftBufferDepth` attribute indicating, for a DASH Media Presentation of type *dynamic*, the duration of the smallest timeshift buffer for which Segments of any Representation are guaranteed to remain available.

4. A `@rangeAccess` attribute indicating whether or not partially available Segments may be accessed with an HTTP byte range request.

   - In the case where the attribute value indicates that partially available Segments may be accessed with an HTTP byte range request, if a Media Player makes a byte-range request against a partially available Segment, and the *first-byte* position of that range request is non-zero, and the Media Player desires a continuously aggregating/live response, then it should signal this using the convention of RFC 8673 [61]. Specifically, it should use a *last-byte* value of 9007199254740991. When responding to Segment requests through this service location, the 5GMSd AS shall return an HTTP *206 (Partial Content)* aggregating response instead of responding with a *200 (OK)* response and a `Content-Length` response header after waiting for the end of the Segment to become available.

   - Otherwise, the Media Player should not expect a response corresponding to the requested byte range.

Based on this information, the 5GMSd AS offers DASH content associated with its provisioned service locations accordingly to the 5GMS Client via reference point M4d, taking into account the optional parameters provided above, if provided.

G.4.2.3 Configuration of service locations in the 5GMSd AS

In this case, content ingested by the 5GMSd AS through reference point M2d conforms to TS 26.511 [35] or at least CMAF per ISO/IEC 23000-19 [40], and the 5GMSd AS creates an MPD that includes multiple service locations based on the provisioning information and the content preparation information. The ingested content may be described by an MPD conforming to a DASH profile for CMAF content as specified in ISO/IEC 23009-1 [32].

In order to create an MPD for M4d distribution using the provisioning information as well as the CMAF content annotation, the content preparation information shall include a mapping of included CMAF tracks to one or multiple service locations. If this service location is to be exposed by the 5GMSd AS at reference point M4d, it shall be associated with a Distribution Configuration provisioned in the 5GMSd AF Content Hosting Configuration by matching its `@serviceLocaton` attribute value with the value of a *DistributionConfiguration.distributionId* property (see clause 8.8.3.1 of TS 26.510 [56]).Based on this information, the 5GMSd AS offers DASH content associated with its provisioned service locations accordingly to the 5GMS Client via reference point M4d.

G.4.3 Client-side switching between service locations

G.4.3.1 Introduction

For client-side switching, DASH content is offered from multiple service locations, but switching between them, for example based on throughput estimation, error behaviour, etc. is a decision made on the client side.

G.4.3.2 DASH content offering requirements and recommendations

A 5GMSd AS offering a DASH Media Presentation with multiple service locations shall provide a service offering that conforms to a DASH Media Profile for CMAF content as defined in ISO/IEC 23009-1 [32], with the following additional requirements:

1. For each Representation that represents a CMAF track, all **`BaseURL`** elements shall be associated with this Representation whereby the value of the element is the base URL provided in the template and the value of the `@serviceLocation` attribute shall be set to the *distributionBaseURL* of the corresponding service location provisioned in a Distribution Configuration of the Content Hosting Configuration.

2. For each base URL, the associated parameters specified in clause G.4.2.2 may be set as well.

3. The resources shall be made available by the 5GMSd AS at reference point M4d according to the configuration for each base URL.

G.4.3.3 Media Player requirements and recommendations

A Media Player supporting multiple service locations with DASH and client-side switching shall support the general requirements for the media access client as defined in clause 10.3A.1 and the Media Access Client capabilities to support switching between multiple service locations during content distribution as defined in clause 10.3A.2.

In order to support service offerings, for which different Representations and/or Adaptation Sets are offered on different service locations, and the Representations are expected to the be played together, the Media Player should support the Media Access Client capabilities to support concurrent use of multiple service locations for content distribution as defined in clause 10.3A.2.

G.4.3.4 Examples (informative)

Listing G.4.3.4-1 provides an example for which two service locations are specified in the MPD.

Listing G.4.3.4-1 MPD with multiple service locations

```
<?xml version="1.0" encoding="UTF-8"?>
<MPD
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns="urn:mpeg:dash:schema:mpd:2011"
  xsi:schemaLocation="urn:mpeg:dash:schema:mpd:2011 DASH-MPD.xsd"
  type="dynamic"
  minimumUpdatePeriod="PT2S"
  timeShiftBufferDepth="PT30M"
  availabilityStartTime="2014-10-17T17:17:05Z"
  minBufferTime="PT4S"
  profiles="urn:mpeg:dash:profile:isoff-live:2011"
  publishTime="2014-10-17T17:17:05Z">

  <BaseURL serviceLocation="dist1">http://distribution-1.com-provider-
service.ms.as.3gppservices.org</BaseURL>
  <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org</BaseURL>

  <Period id="1">
    <!-- Video -->
    <AdaptationSet
      mimeType="video/mp4"
      codecs="avc1.4D401F"
      frameRate="30000/1001"
      segmentAlignment="true"
      startWithSAP="1">
      <BaseURL>video/</BaseURL>
      <SegmentTemplate timescale="90000" initialization="$Bandwidth$/init.mp4v"
media="$Bandwidth$/$Time$.mp4v">
        <SegmentTimeline>
          <S t="0" d="180180" r="432"/>
        </SegmentTimeline>
      </SegmentTemplate>
      <Representation id="v0" width="320" height="240" bandwidth="250000"/>
      <Representation id="v1" width="640" height="480" bandwidth="500000"/>
      <Representation id="v2" width="960" height="720" bandwidth="1000000"/>
    </AdaptationSet>
    <!-- English Audio -->
    <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40" lang="en" segmentAlignment="0"
startWithSAP="1">
      <SegmentTemplate timescale="48000" initialization="audio/en/init.mp4a"
media="audio/en/$Time$.mp4a">
        <SegmentTimeline>
          <S t="0" d="96000" r="432"/>
        </SegmentTimeline>
      </SegmentTemplate>
      <Representation id="a0" bandwidth="64000"/>
    </AdaptationSet>
    <!-- French Audio -->
    <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40" lang="fr" segmentAlignment="0"
startWithSAP="1">
      <SegmentTemplate timescale="48000" initialization="audio/fr/init.mp4a"
media="audio/fr/$Time$.mp4a">
        <SegmentTimeline>
          <S t="0" d="96000" r="432"/>
        </SegmentTimeline>
```

```
      </SegmentTemplate>
      <Representation id="b0" bandwidth="64000"/>
    </AdaptationSet>
  </Period>
</MPD>
```

G.4.4 Switching between service locations using Content Steering for DASH

G.4.4.1 Introduction

Content steering provides a deterministic capability for a content distributor to switch the content source that a media player uses, either at presentation start-up or at any point during the presentation, by means of a remote steering service. Content Steering for DASH is specified in ETSI TS 103 998 [76]. This clause defines the requirements and recommendations to support content distribution at reference point M4d in DASH-based downlink media streaming in combination with Content Steering as specified in [76].

G.4.4.2 DASH content offering requirements and recommendations

A 5GMSd AS offering a DASH Media Presentation to be used with Content Steering shall offer a Media Presentation according to the requirements in clause 4.3.2.

In addition, the 5GMSd AS shall:

- Add Content Steering signalling in the DASH MPD as specified in clause 5 of ETSI TS 103 998 [76], and
- Support the functionalities of a DASH Steering Server as specified in clause 6 of ETSI TS 103 998 [76].

G.4.4.3 Media Player requirements and recommendations

A Media Player supporting Content Steering for DASH shall support:

- The normative DASH client steering behaviour specified in clause 7 of ETSI TS 103 998 [76].
- The media player requirements and recommendations in clause G.4.3.3.
- The Media Access Client capabilities to support content steering for content distribution as defined in clause 10.3A.4 where the content steering service is the Content Steering server as provided in the MPD according to clause G.4.4.2.

G.4.4.4 Examples (informative)

Listing G.4.4.4-1 provides an example in which three service locations are specified in the MPD. In this example, some content representations are available from service location *dist1*, a subset is available from service location *dist2*, and complementary content is available from service location *dist3*.

Listing G.4.4.4-1 MPD with multiple service locations and Content Steering

```
<MPD
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="urn:mpeg:dash:schema:mpd:2011"
    type="dynamic" minimumUpdatePeriod="PT10s"
    timeShiftBufferDepth="PT600S"
    minBufferTime="PT2S"
    profiles="urn:3GPP:PSS:profile:DASH10"
    publishTime="2014-10-17T17:17:05Z"
    availabilityStartTime="2014-10-17T17:17:05Z">
    <Location>distribution-1.com-provider-service.ms.as.3gppservices.org/MPD2.mpd</Location>
    <Location>distribution-2.com-provider-service.ms.as.3gppservices.org/MPD2.mpd</Location>
    <Location>distribution-3.com-provider-service.ms.as.3gppservices.org/MPD2.mpd</Location>
    <ServiceDescription id="420">
      <ContentSteering defaultServiceLocation="dist1" queryBeforeStart="true">
```

```
        https://com-example.ms.as.3gppservices.org/steering</ContentSteering>
    </ServiceDescription>
    <Period id="1" start="PT0S">
        <SegmentTemplate media="./$RepresentationID$/$Number$.m4s"
initialization="$RepresentationID$-init.mp4"/>
        <!-- Video 720p -->
        <AdaptationSet mimeType="video/mp4" codecs="hvc1.1.2.L93.B0" startWithSAP="1"
maxWidth="1280" maxHeight="720" frameRate="30" profile="urn:3GPP:video:op:h265-720p-HD">
            <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <SegmentTemplate timescale="30" duration="60"/>
            <Representation id="v2048" bandwidth="2048000">
                <BaseURL serviceLocation="dist1"> http://distribution-1.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            </Representation>
            <Representation id="v1024" bandwidth="1024000"/>
            <Representation id="v512" bandwidth="512000"/>
            <Representation id="v128" bandwidth="128000"/>
        </AdaptationSet>
        <!--  Video HDR -->
        <AdaptationSet mimeType="video/mp4" codecs="hvc1.2.4.L113.B0" startWithSAP="1"
maxWidth="1920" maxHeight="1080" frameRate="30" profile="urn:3GPP:video:op:h265-Full-HD-HDR">
            <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <EssentialDescriptor schemeIdUri="urn:mpeg:mpegB:cicp:MatrixCoefficients" value="9"/>
            <EssentialDescriptor schemeIdUri="urn:mpeg:mpegB:cicp:TransferCharacteristics"
value="16"/>
            <EssentialDescriptor schemeIdUri="urn:mpeg:mpegB:cicp:ColourPrimaries" value="9"/>
            <SegmentTemplate timescale="30" duration="60"/>
            <Representation id="8M" bandwidth="8192000">
            <Representation id="6M" bandwidth="6144000"/>
            <Representation id="4M" bandwidth="4096000"/>
            <Representation id="2M" bandwidth="2048000"/>
        </AdaptationSet>
        <!-- Audio English -->
        <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40.2" segmentAlignment="true"
startWithSAP="1" language="en">
            <BaseURL serviceLocation="dist3">http://distribution-3.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <SegmentTemplate timescale="20" duration="40"/>
            <Representation id="a128" bandwidth="128000">
                <BaseURL serviceLocation="dist3">http://distribution-3.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            </Representation>
            <Representation id="a64" bandwidth="64000">
        </AdaptationSet>
        <!-- Audio Spanish -->
        <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40.2" segmentAlignment="true"
startWithSAP="1" language="es">
            <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <SegmentTemplate timescale="20" duration="40"/>
            <Representation id="a128" bandwidth="128000">
            <Representation id="a64" bandwidth="64000">
        </AdaptationSet>
    </Period>
</MPD>
```

The Content Steering message is provided to the Media Player by the DASH Steering Server instantiated in the 5GMSd. The example in listing G.4.4.4-1, it indicates a single service location that hosts one Representation is available; the HDR video and the Spanish language are available as well from service location *dist3*.

Listing G.4.4.4-1: Example DASH Content Steering Message indicating availability of content from 5GMSd AS service locations

```
{
   "VERSION": 1,
   "TTL": 5,
   "RELOAD-URI": "https://com-example.ms.as.3gppservices.org/steering"
   "PATHWAY-PRIORITY": ["dist2","dist3","dist1"]
}
```

In case the 5GMSd AS fails to pre-populate content for service location *dist2*, it recommends switching to *dist1* or *dist3*.

Listing 5.12.2.2.2-3: Example DASH Content Steering Message indicating non-availability of content from 5GMSd AS service locations

```
{
   "VERSION": 1,
   "TTL": 5,
   "RELOAD-URI": "https://com-example.ms.as.3gppservices.org/steering"
   "PATHWAY-PRIORITY": ["dist1","dist3"]
}
```

G.4.5 Switching between service locations using SAND4M

G.4.5.1 Introduction

3GPP DASH as specified in clause 13.10 of 3GPP TS 26.247 [4], defines the *Server-Assisted Network Delivery* (SAND) functionality that enables SAND for Multi-network support (SAND4M), and specifies required and recommended functionalities for both a DASH-Aware Network Element (DANE) and a DASH client. In the context of the 5GMS System, the DANE is mapped to the 5GMSd AS and the DASH client is realised by the 5GMSd Client.

Specifically, the following cases are relevant in the context of multiple service locations:

- Not all resources announced in the presentation manifest (e.g. MPEG-DASH MPD) are always accessible on all access networks, e.g. certain resources are unavailable when the UE is for example in a roaming scenario.
- Even on a particular access network, resources may not be available all the time.
- Networks may have different availability times.
- Networks go down dynamically and may re-appear.
- The DANE may issue preferences for one network.
- Information about the dynamic availability of resources from different service locations may be established via in-band and/or out-of-band channels.

G.4.5.2 DASH content offering requirements and recommendations

A 5GMSd AS offering a DASH Media Presentation to be used with SAND4M shall offer a Media Presentation according to the requirements in clause 4.3.2.

In addition, the 5GMSd AS shall support the functionalities of a DANE for SAND4M specified in clause 13.10.2 of TS 26.247 [4].

G.4.5.3 Media Player requirements and recommendations

A Media Player supporting SAND4M with multiple service locations shall support:

- The DASH Client functionalities of SAND4M specified in clause 13.10.3 of TS 26.247 [4].
- The Media Player requirements and recommendations in clause G.4.3.3.
- The Media Access Client capabilities to support content steering for content distribution as defined in clause 10.3A.4 where the content steering service is the DANE according to clause G.4.5.2.

G.4.5.4 Examples (informative)

Listing G.4.5.4-1 provides an example for which three service locations are specified in the MPD. In this example, some content representations are available on service location *dist1*, a subset is available from service location *dist2*, and complementary content is available from service location *dist3*.

Listing G.4.5.4-1 MPD with multiple service locations

```
<MPD
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="urn:mpeg:dash:schema:mpd:2011"
    type="dynamic" minimumUpdatePeriod="PT10s"
    timeShiftBufferDepth="PT600S"
    minBufferTime="PT2S"
    profiles="urn:3GPP:PSS:profile:DASH10"
    publishTime="2014-10-17T17:17:05Z"
    availabilityStartTime="2014-10-17T17:17:05Z">
    <Location>distribution-1.com-provider-service.ms.as.3gppservices.org/MPD2.mpd</Location>
    <Location>distribution-2.com-provider-service.ms.as.3gppservices.org/MPD2.mpd</Location>
    <Location>distribution-3.com-provider-service.ms.as.3gppservices.org/MPD2.mpd</Location>
    <Period id="1" start="PT0S">
        <SegmentTemplate media="./$RepresentationID$/$Number$.m4s"
initialization="$RepresentationID$-init.mp4"/>
        <!-- Video 720p -->
        <AdaptationSet mimeType="video/mp4" codecs="hvc1.1.2.L93.B0" startWithSAP="1"
maxWidth="1280" maxHeight="720" frameRate="30" profile="urn:3GPP:video:op:h265-720p-HD">
            <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <SegmentTemplate timescale="30" duration="60"/>
            <Representation id="v2048" bandwidth="2048000">
                <BaseURL serviceLocation="dist1"> http://distribution-1.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            </Representation>
            <Representation id="v1024" bandwidth="1024000"/>
            <Representation id="v512" bandwidth="512000"/>
            <Representation id="v128" bandwidth="128000"/>
        </AdaptationSet>
        <!--  Video HDR -->
        <AdaptationSet mimeType="video/mp4" codecs="hvc1.2.4.L113.B0" startWithSAP="1"
maxWidth="1920" maxHeight="1080" frameRate="30" profile="urn:3GPP:video:op:h265-Full-HD-HDR">
            <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <EssentialDescriptor schemeIdUri="urn:mpeg:mpegB:cicp:MatrixCoefficients"
value="9"/>
            <EssentialDescriptor schemeIdUri="urn:mpeg:mpegB:cicp:TransferCharacteristics"
value="16"/>
            <EssentialDescriptor schemeIdUri="urn:mpeg:mpegB:cicp:ColourPrimaries" value="9"/>
            <SegmentTemplate timescale="30" duration="60"/>
            <Representation id="8M" bandwidth="8192000">
            <Representation id="6M" bandwidth="6144000"/>
            <Representation id="4M" bandwidth="4096000"/>
            <Representation id="2M" bandwidth="2048000"/>
        </AdaptationSet>
        <!-- Audio English -->
        <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40.2" segmentAlignment="true"
startWithSAP="1" language="en">
            <BaseURL serviceLocation="dist3">http://distribution-3.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <SegmentTemplate timescale="20" duration="40"/>
            <Representation id="a128" bandwidth="128000">
                <BaseURL serviceLocation="dist3">http://distribution-3.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            </Representation>
            <Representation id="a64" bandwidth="64000">
        </AdaptationSet>
        <!-- Audio Spanish -->
        <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40.2" segmentAlignment="true"
startWithSAP="1" language="es">
            <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org</BaseURL>
            <SegmentTemplate timescale="20" duration="40"/>
            <Representation id="a128" bandwidth="128000">
            <Representation id="a64" bandwidth="64000">
        </AdaptationSet>
    </Period>
</MPD>
```

The SAND message is provided to the Media Player by the DANE instantiated in the 5GMSd AS. The example in listing G.4.5.4-1, indicates that only a service location that hosts one Representation is available; the HDR video and the Spanish language are available as well from service location *dist3*. The SAND message indicates that service location *dist2* is available in the 5GMSd AS.

Listing G.4.5.4-1: Example SAND message indicating availability of content in 5GMSd AS

```
<SAND>
    <Status baseURL="http://distribution-2.com-provider-service.ms.as.3gppservices.org">
        <ResourceStatus status="cached"/>
    </Status>
    <Status baseURL="http://distribution-1.com-provider-service.ms.as.3gppservices.org">
        <ResourceStatus status="unavailable"/>
    </Status>
    <Status baseURL="http://distribution-3.com-provider-service.ms.as.3gppservices.org">
        <ResourceStatus status="available"/>
    </Status>
</SAND>
```

In case the 5GMSd AS fails to pre-populate content for service location *dist2*, it recommends switching to service location *dist1* or *dist3*.

Listing G.4.5.4-2: Example SAND message indicating non-availability of content in 5GMSd AS

```
<SAND>
    <Status baseURL="http://distribution-2.com-provider-service.ms.as.3gppservices.org">
        <ResourceStatus status="unavailable"/>
    </Status>
    <Status baseURL="http://distribution-1.com-provider-service.ms.as.3gppservices.org">
        <ResourceStatus status="available"/>
    </Status>
    <Status baseURL="http://distribution-3.com-provider-service.ms.as.3gppservices.org">
        <ResourceStatus status="available"/>
    </Status>
</SAND>
```

G.5 In-band client data reporting in DASH

G.5.1 Overview

This clause provides details on how to instantiate in-band client data reporting as specified in clause 10.5.1 in DASH. In particular, this clause provides requirements and recommendations for the 5GMSd AS and 5GMSd Client to support the following features:

- CMCD-based client data reporting

G.5.2 CMCD-based client data reporting

G.5.2.1 Introduction

CMCD-based client data reporting is specified in clause 10.5.1 based on CTA-5004 [70].

G.5.2.2 DASH content offering requirements and recommendations

A 5GMSd AS offering CMCD-based client reporting shall support:

- The requirements and recommendations for the 5GMS AS for CMCD-based client data reporting as specified in clause 10.5.1.

- The configuration of this feature by means of the client data reporting configuration for CMCD defined in clause K.3.7 of ISO/IEC 23009-1 [32], using the in-band Service Description defined in clause K.4.3.7.2 of ISO/IEC 23009-1 [32]. This includes for example:
  - The ability to restrict the requests that include CMCD information to specific Service Locations, Adaptations Sets and/or media object types .
  - The ability to declare a content identifier for the purpose of client data reporting by populating the CMCD *cid* key based on the details in clauses K.3.7 and K.4.2.7 of ISO/IEC 23009-1 [32].
  - The ability to explicitly specify a set of CMCD keys to be reported based on the details in clauses K.3.7 and K.4.2.7 of ISO/IEC 23009-1 [32].

G.5.2.3 Media Player requirements and recommendations

A Media Player supporting CMCD-based client data reporting shall support:

- The requirements and recommendations for the Media Player for CMCD-based client data reporting specified in clause 10.5.1.
- The derivation and reporting of all metrics defined in CTA-5004 [70].
- The response to the content offering for CMCD-based client reporting as defined in clause G.5.2.2. If values are configured via the Media Player configuration and settings API at reference point M7d/M11d (see clause 13.2.4) and in the MPD, then the value provided on the M7d/M11d API shall take precedence as specified in clause 10.5.1.

G.5.2.4 Examples (informative)

Listing G.5.2.4-1 provides an example for CMCD in-band configuration with the following properties:

- Reporting is restricted to a single service location for either *dist1* or *dist2*.
- Reporting is restricted to requests for Segments assigned to video Adaptations Sets only.
- Reporting is restricted to Media Segments.
- Content identifier reporting is requested because the CMCD *cid* key is included and the respective content identifier is specified.
- A set of CMCD keys to be reported is specified as `keys="br,bl,cid,dl,mtp,nor,ot,sf,v"`.

Listing G.5.2.4-1 MPD with CMCD inband configuration

```
<?xml version="1.0" encoding="UTF-8"?>
<MPD
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns="urn:mpeg:dash:schema:mpd:2011"
  xsi:schemaLocation="urn:mpeg:dash:schema:mpd:2011 DASH-MPD.xsd"
  type="dynamic"
  minimumUpdatePeriod="PT2S"
  timeShiftBufferDepth="PT30M"
  availabilityStartTime="2014-10-17T17:17:05Z"
  minBufferTime="PT4S"
  profiles="urn:mpeg:dash:profile:isoff-live:2011"
  publishTime="2014-10-17T17:17:05Z">

  <BaseURL serviceLocation="dist1">http://distribution-1.com-provider-
service.ms.as.3gppservices.org/</BaseURL>
  <BaseURL serviceLocation="dist2">http://distribution-2.com-provider-
service.ms.as.3gppservices.org/</BaseURL>

  <ServiceDescription id="1250">
    <Scope schemeIdUri="urn:mpeg:dash:event:service-description:2024"/>
    <Latency min="750" max="4200" target="1250" referenceId="7"/>
    <PlaybackRate min="0.96" max="1.04"/>
    <ClientDataReporting scheme="urn:mpeg:dash:cta-5004:2023"
                         serviceLocations="dist1"
```

```
                          adaptationSets="video">
      <CMCDParameters mode="header"
                      contentID="md:cid:EIDR:10.5240%2f0EFB-02CD-126E-8092-1E49-W"
                      includeInRequests="segment"
                      keys="br,bl,cid,dl,mtp,nor,ot,sf,v"/>
    </ClientDataReporting>
  </ServiceDescription>

  <ServiceDescription id="2500">
    <Scope schemeIdUri="urn:mpeg:dash:event:service-description:2024"/>
    <Latency min="750" max="4200" target="2500" referenceId="7"/>
    <PlaybackRate min="0.96" max="1.04"/>
    <ClientDataReporting scheme="urn:mpeg:dash:cta-5004:2023"
                         serviceLocations="dist2"
                         adaptationSets="video">
      <CMCDParameters mode="header"
                      contentID="md:cid:EIDR:10.5240%2f0EFB-02CD-126E-8092-1E49-W"
                      includeInRequests="segment"
                      keys="br,bl,cid,dl,mtp,nor"/>
    </ClientDataReporting>
  </ServiceDescription>

  <Period id="1">
    <!-- Video -->
    <AdaptationSet
      mimeType="video/mp4"
      codecs="avc1.4D401F"
      frameRate="30000/1001"
      segmentAlignment="true"
      startWithSAP="1">
      <BaseURL>video/</BaseURL>
      <SegmentTemplate timescale="90000" initialization="$Bandwidth$/init.mp4v"
media="$Bandwidth$/$Time$.mp4v">
        <SegmentTimeline>
          <S t="0" d="180180" r="432"/>
        </SegmentTimeline>
      </SegmentTemplate>
      <Representation id="v0" width="320" height="240" bandwidth="250000"/>
      <Representation id="v1" width="640" height="480" bandwidth="500000"/>
      <Representation id="v2" width="960" height="720" bandwidth="1000000"/>
    </AdaptationSet>
    <!-- English Audio -->
    <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40" lang="en" segmentAlignment="0"
startWithSAP="1">
      <SegmentTemplate timescale="48000" initialization="audio/en/init.mp4a"
media="audio/en/$Time$.mp4a">
        <SegmentTimeline>
          <S t="0" d="96000" r="432"/>
        </SegmentTimeline>
      </SegmentTemplate>
      <Representation id="a0" bandwidth="64000"/>
    </AdaptationSet>
    <!-- French Audio -->
    <AdaptationSet mimeType="audio/mp4" codecs="mp4a.40" lang="fr" segmentAlignment="0"
startWithSAP="1">
      <SegmentTemplate timescale="48000" initialization="audio/fr/init.mp4a"
media="audio/fr/$Time$.mp4a">
        <SegmentTimeline>
          <S t="0" d="96000" r="432"/>
        </SegmentTimeline>
      </SegmentTemplate>
      <Representation id="b0" bandwidth="64000"/>
    </AdaptationSet>
  </Period>
</MPD>
```

Annex H (normative): Media streaming in the 5GMS System using CMMF

H.1 Introduction

Coded Multisource Media Format (CMMF) as specified in ETSI TS 103 973 [77] is an extensible container format designed to facilitate the management and interchange of audio−visual media and metadata in one or more coded representations (e.g., encoded with Application Layer Forward Error Correction, linear, network, or channel codes). The coded media representations supported by CMMF enable the efficient use of multi-source, multipath, and multi-access connectivity for network-delivered applications.

This annex specifies the use of multi-source object coding using CMMF within the 5GMS System.

H.2 Downlink media streaming from multiple service locations using CMMF

H.2.1 General

CMMF as specified in ETSI TS 103 973 [77] may be employed to enable efficient downlink media streaming when concurrently using multiple service locations exposed by the 5GMSd AS at reference point M4d and/or by the 5GMSd Application Provider at reference point M2d or M13d. Media resources (e.g., CMAF segments) shall first be encoded into one or more CMMF transport resources using the profile specified in clause H.2.3.1, either by the 5GMSd Application Provider or by using Content Preparation in the 5GMSd AS. These CMMF transport resources shall be made available to 5GMSd Clients from one or more reference point M4d and/or M13d service locations. A CMMF Media Access Client of the Media Player in the 5GMSd Client shall then request and download these CMMF transport resources from the service locations in parallel. The CMMF Media Access Client decodes the received CMMF transport resources, recovering the media resource required for playback and presents this to its Media Playback and Content Decryption Platform.

The remainder of this annex specifies requirements necessary to enable downlink media streaming from multiple reference point M4d service locations using CMMF within the 5GMS System. Further discussion of reference points M8d and M13d for use in streaming media with CMMF are omitted since they are out of scope.

H.2.2 Procedures for downlink media streaming from multiple service locations using CMMF

H.2.2.1 Provisioning (M1d) and configuration (M3d) interfaces for downlink media delivery using CMMF

H.2.2.1.1 Overview

This clause defines all procedures and APIs used to provision and configure the 5GMS System to use CMMF for the purposes of downlink media delivery from multiple service locations.

H.2.2.1.2 Content Hosting

To provision CMMF downlink media streaming in the 5GMSd AF, the 5GMSd Application Provider shall use the Content Hosting Provisioning (M1) API specified in clause 7.6, following the Content Hosting provisioning (M1) procedures specified in clause 4.3.3.

To configure CMMF downlink media streaming in the 5GMSd AS, the 5GMSd AF shall use the Content Hosting Configuration (M3) API specified in clause 9.4, following the Content Hosting configuration (M3) procedures specified in clause 4.5.4.

When Content Hosting is provisioned, a service location exposed by the 5GMSd AS at reference point M4d shall host CMMF transport resources, each of which should comprise a unique CMMF representation or variant of the media resources to be streamed. Each CMMF transport resource shall be uniquely addressable as specified in clause 10.1.

H.2.2.1.3 Content Preparation

To provision Content Preparation of CMMF transport resources in the 5GMSd AF, the 5GMSd Application Provider shall use the Content Preparation Templates Provisioning (M1) API specified in clause 7.4, following the Content Preparation Template provisioning (M1) procedures specified in clause 4.3.5.

To configure Content Preparation of CMMF transport resources in the 5GMSd AS, the 5GMSd AF shall use the Content Preparation Templates Configuration (M3) API specified in clause 9.3, following the Content Preparation Template configuration (M3) procedures specified in clause 4.5.3.

When Content Preparation is provisioned to encode media resources into CMMF transport resources, a CMMF Encoder (as specified in clauses 4.2.1 and 4.3.3 of ETSI TS 103 973 [77]) is instantiated and configured in the 5GMSd AS by means of a Content Preparation Template. The following Content Preparation Template requirements apply:

- A Content Preparation Template used to encode media resources (e.g., audio segments, video segments, etc.) shall create CMMF transport resources whose bitstreams conform to the profile specified in clause H.2.3.1.
- Depending on the Content Hosting Configuration, the CMMF Encoder may be a single input, single output or a single input, multiple output process. A media resource (e.g., audio segment, video segment, etc.) made available to the 5GMSd AS at reference point M2d or M10d shall be ingested into the CMMF Encoder where it is encoded and packaged as one or more CMMF transport resources (each a unique CMMF representation or variant of the media resource), and those transport resources shall be made available at service locations exposed by the 5GMSd AS at reference point M4d.
- CMMF transport resources shall be accessible via URLs that allow for the appropriate mapping between CMMF resource URLs used at service locations exposed by the 5GMSd AS at reference point M4d and their corresponding media resource URLs used for media ingest at reference points M2d or M10d. This mapping shall be determined using a member of the *DistributionConfiguration.pathRewriteRules* array specified in table 8.8.3.1-1 of TS 26.510 [56].

NOTE: Differences between the URLs of media resources and their associated CMMF transport resources are restricted to differences in the path part of the URL between and including the leading "/" and the final "/" owing to restrictions imposed by the definition of the *PathRewriteRule* data type.

The Content Preparation Template for CMMF-enabled downlink media delivery is specified in clause H.2.3.2.

H.2.2.2 Media Streaming (M4d) interface for downlink media delivery using CMMF

To serve CMMF transport resources to a CMMF Media Access Client at reference point M4d, the 5GMSd AS shall use the media streaming (M4) interface specified in clause 10.3A.3, following the media streaming (M4d) procedures specified in clause 4.6.3.

A media resource required by the Media Playback and Decryption Platform for immediate or delayed consumption shall be obtained by a CMMF Access Client downloading one or more CMMF transport resources (either partially or in full) where each CMMF transport resource is typically downloaded from a different service location exposed at reference point M4d.

If recovery of a media resource from a set of downloaded CMMF transport resources fails, the CMMF Media Access Client may attempt to acquire the media resource using an alternate method (for example, by retrieving the original media resource directly from a single service location exposed by the 5GMSd AS at reference point M4d).

H.2.2.3 Media Session Handling (M5d) interface for downlink media streaming using CMMF

To perform media session handling for a media streaming session using CMMF, the Media Session Handler shall use the APIs specified in clause 11, following the Media Session Handling (M5) procedures specified in clause 4.7.

In cases where the Service Access Information for media streaming includes one or more Media Player Entries, at least one of these Media Player Entries shall be a URL to a CMMF Media Entry Point as specified in clause H.2.3.3.

H.2.2.4 UE Media Player interface (M7d/M11d) for downlink media delivery using CMMF

To control a Media Player during a media streaming session using CMMF, the 5GMSd-Aware Application or Media Session Handler shall use the Media Stream Handler (M7/M11) APIs specified in clause 13, following the UE Media Player (M7d) procedures specified in clause 4.9.

H.2.2.5 Media Player requirements and recommendations

A Media Player shall configure and use a CMMF Media Access Client upon receiving a CMMF Media Entry Point as specified in clause H.2.3.3. At a minimum, the CMMF Media Access Client shall support the following:

- Translation of media resource (e.g., CMAF segment) URLs and the corresponding URLs pointing to CMMF transport resources containing encoded representations or variants of those media resources that are available from service locations exposed by the 5GMSd AS at reference point M4d. This mapping is provided by the CMMF Media Entry Point.
- Decoding a received CMMF transport resource or jointly decoding multiple partially received CMMF transport resources that conform to the profile specified in clause H.2.3.1.
- Recovery of media resources (e.g., CMAF segments) and delivery of these media resources to the Media Playback and Encryption Platform.

H.2.3 Protocols and formats for downlink media streaming using CMMF

H.2.3.1 HTTP-based CMMF delivery conformance profile

CMMF transport resources communicated within the 5GMS System shall conform to ETSI TS 103 973 [77].

If the 5GMSd Client supports CMMF, then it shall support a CMMF Receiver conforming to CMMF bitstream profile A (CMMF profile identifier *org.etsi.cmmf.a*) of ETSI TS 103 973 [77] for at least one of the CMMF code types as specified in clause 6.1.4.5 of ETSI TS 103 973 [77]. Additionally, the 5GMSd Client that supports CMMF should support a CMMF Receiver conforming to CMMF bitstream profile A for CMMF code type 1 of ETSI TS 103 973 [77].

NOTE: The CMMF code type 1 (i.e., "Raptor") is specified in IETF RFC 5053 [78] and is the 3GPP MBMS FEC code as defined in TS 26.346 [51].

H.2.3.2 Content Preparation Template for downlink media streaming using CMMF

When Content Preparation is provisioned to create CMMF transport resources within the 5GMS System, the CMMF Encoder Configuration document with MIME media type *application/vnd.cmmf-encoder-configuration+json* as specified in clause E.1 of ETSI TS 103 973 [77] should be used as the Content Preparation Template.

H.2.3.3 CMMF Media Player Entry

H.2.3.3.1 General

A CMMF Media Player Entry shall be either a CMMF Extended File Delivery Table (EFDT) with MIME media type *application/vnd.cmmf-efd+xml* as specified in clause D.2.3 of ETSI TS 103 973 [77] and further elaborated in clause H.2.3.3.2 below or a CMMF Configuration Information document with MIME media type *application/vnd.cmmf-configuration-information+json* as specified in clause D.3.3.3 of ETSI TS 103 973 [77] and further elaborated in clause H.2.3.3.3 below.

The CMMF Media Player Entry shall comprise:

- A media presentation or a pointer to a media presentation (e.g., URL to a MPD for DASH, URL to a video file, etc.) and
- Any necessary CMMF configuration information required by the CMMF Media Access Client to download CMMF transport resources, decode them, and recover the associated media resource(s) (e.g., CMAF segments).

When Content Preparation is provisioned within the 5GMS System to encode media resources (e.g., CMAF segments) into CMMF transport resources, URLs of the CMMF transport resources provided in the CMMF Media Player Entry shall be formatted in such a way that they can be mapped back to the URLs of their corresponding media resources (e.g., CMAF segment) by the 5GMSd AS so that the media resource may be ingested at reference point M2d. As specified in clause H.2.2.1.3, the URL format of these CMMF transport resources is determined by the provisioned Content Hosting Configuration (see the definition of *DistributionConfiguration.PathRewriteRules* specified in table 8.8.3.1-1 of TS 26.510 [56]).

Example CMMF Media Player Entries referencing a single video file and a DASH MPD can be found in clause H.3.2.

H.2.3.3.2 CMMF Media Player Entry based on Extended FDT

In the case where the media presentation (e.g., MPD for DASH, HLS playlist, etc.) referenced by a CMMF Media Entry Point is a document containing pointers to media resources (e.g., URLs to CMAF segments), the reference (e.g., URL) to the media presentation description shall be listed within the EFDT as *TOI="0"*.

The EFDT shall provide a mapping between each of the URLs contained within the media presentation description and the URLs of the CMMF transport resources containing encoded representations or variants of those media resources that are available from service locations exposed by the 5GMSd AS at reference point M4d. For the EFDT, these mappings are provided using TOI values greater than 0.

The EFDT may also include media resource and CMMF transport resource availability information, including availability information relating to media resources and CMMF transport resources that may be available after the EFDT is made available to the CMMF Media Access Client. The CMMF Media Access Client may use this information to download relevant CMMF transport resources at appropriate times. Details are left for implementation.

H.2.3.3.3 CMMF Media Player Entry based on CMMF Configuration Information document

URLs to media presentations (e.g., MPD for DASH, HLS playlist, etc.) or media resources (e.g., video file) intended for playback shall be referenced within the CMMF Configuration Information document using the *applicationResource Locators.locator* property as specified in clause D.3.3.3.2 of ETSI TS 103 973 [77].

The CMMF Configuration Information document shall provide the necessary information within the *application ResourceConfigurations.serviceLocations.requestPathMaps* array as specified in clause D.3.3.3.2 of ETSI TS 103 973 [77] to map URLs contained within a media presentation description to the URLs of the CMMF transport resources containing encoded representations or variants of those media resources that are available from service locations exposed by the 5GMSd AS at reference point M4d.

H.3 Examples of downlink media streaming using CMMF (informative)

H.3.1 General

This annex provides examples of CMMF Media Player Entries and Content Hosting Configurations that support the delivery of CMMF-encoded media within the 5GMS System.

H.3.2 CMMF Media Player Entry examples

H.3.2.1 General

This clause provides examples showing how CMMF configuration information required by a 5GMS Client may be communicated within a CMMF Media Player Entry as specified in clause H.2.3.3. When applicable, a general example of a media asset is used, described by the DASH MPD shown in listing H.3.2.1-1, and notionally hosted by a hypothetical 5GMSd Application Provider at the URL `https://example.com/manifest.mpd`.

Listing H.3.2.1-1: Example MPD

```
<?xml version="1.0"?>
<MPD
    profiles="urn:3GPP:PSS:profile:DASH10"
    type="static"
    minBufferTime="PT10S"
    mediaPresentationDuration="PT2H"
    availabilityStartTime="2010-04-01T09:30:47Z"
    availabilityEndTime="2010-04-07T09:30:47Z"
    xsi:schemaLocation="urn:mpeg:dash:schema:mpd:2011 3GPP-Rel10-MPD.xsd"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="urn:mpeg:dash:schema:mpd:2011">
    <ProgramInformation moreInformationURL="http://www.example.com">
        <Title>Example</Title>
    </ProgramInformation>
    <Period start="PT0S">
        <AdaptationSet mimeType="video/3gpp">
            <ContentComponent contentType="video"/>
            <ContentComponent contentType="audio" lang="en"/>
            <Representation codecs="s263, samr" bandwidth="256000" id="256">
                <BaseURL>"rep1"</BaseURL>
                <SegmentList duration="1000" timescale="100">
                    <Initialization sourceURL="seg-init.3gp"/>
                    <SegmentURL media="seg-1.3gp"/>
                    <SegmentURL media="seg-2.3gp"/>
                    <SegmentURL media="seg-3.3gp"/>
                </SegmentList>
            </Representation>
            <Representation codecs="mp4v.20.9, mp4a.E1" bandwidth="128000" id="128">
                <BaseURL>"rep2"</BaseURL>
                <SegmentList duration="10">
                    <Initialization sourceURL="seg-init.3gp"/>
                    <SegmentURL media="seg-1.3gp"/>
                    <SegmentURL media="seg-2.3gp"/>
                    <SegmentURL media="seg-3.3gp"/>
                </SegmentList>
            </Representation>
        </AdaptationSet>
    </Period>
    <Period start="PT30S">
        <SegmentTemplate
            duration="10"
            initialization="seg-init-$RepresentationId$.3gp"
            media="http://example.com/$RepresentationId$/$Number$.3gp"/>
        <AdaptationSet mimeType="video/3gpp" codecs="mp4v.20.9, mp4a.E1">
            <ContentComponent contentType="video"/>
            <ContentComponent contentType="audio" lang="en"/>
            <Representation bandwidth="256000" id="1"/>
            <Representation bandwidth="128000" id="2"/>
```

```
        </AdaptationSet>
    </Period>
</MPD>
```

H.3.2.2 Single file EFDT example

The following example shows an EFDT where a single MP4 file is delivered using CMMF from three service locations exposed by the 5GMSd AS at reference point M4d.

Listing H.3.2.2-1: Single MP4 CMMF EFDT example

```
<?xml version="1.0" encoding="UTF-8"?>
<FDTInstance xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
             xmlns="urn:ETSI:CMMF:2023:FDT"
             xsi:schemaLocation="urn:ETSI:CMMF:2023:FDT extendedFDT.xsd"
             Expires="2010-04-07T09:30:47Z"
             Complete="true"
             ContentType="video/mp4 codecs="mp4v.20.9, mp4a.E1"
             FEC-OTI-FEC-Encoding-ID="1">
    <File TOI="0" Content-Location="https://example.com/video.mp4" Content-Length="64000">
        <EncodedObjects type="source" complete="true">
            https://example.com/video.mp4
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/cmmf-a/video.mp4
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/cmmf-b/video.mp4
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/cmmf-c/video.mp4
        </EncodedObjects>
    </File>
</FDTInstance>
```

H.3.2.3 EFDT referencing a DASH MPD

The following example shows an EFDT where the contents of the presentation described by the MPD shown in listing H.3.2.1-1 are delivered using CMMF. It is assumed that CMMF transport resources for the media resources referenced within the MPD are accessible from three service locations exposed by the 5GMSd AS at reference point M4d.

In this example, a CMMF Media Access Client uses the EFDT shown in listing H.3.2.3-1 combined with the MPD shown in listing H.3.2.1-1 as the Media Player Entry. The EFDT includes a reference to the MPD (**`File`** element with TOI 0) and references to all segments referenced within the MPD are mapped to CMMF transport resources assigned TOI values 1 through 16.

Listing H.3.2.3-1: CMMF EFDT example referencing MPD

```
<?xml version="1.0" encoding="UTF-8"?>
<FDTInstance xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
             xmlns="urn:ETSI:CMMF:2023:FDT"
             xsi:schemaLocation="urn:ETSI:CMMF:2023:FDT extendedFDT.xsd"
             Expires="2010-04-07T09:30:47Z"
             Complete="true"
             FEC-OTI-FEC-Encoding-ID="1">
    <File TOI="0" Content-Location="https://example.com/manifest.mpd">
        <EncodedObjects type="source" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/manifest.mpd
        </EncodedObjects>
        <EncodedObjects type="source" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/manifest.mpd
        </EncodedObjects>
        <EncodedObjects type="source" complete="true">
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/manifest.mpd
        </EncodedObjects>
    </File>
    <File TOI="1" Content-Location="https://example.com/rep1/seg-init.3gp">
        <EncodedObjects type="source" complete="true">
```

```
            https://example.com/rep1/seg-init.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-a
            /seg-init.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-b
            /seg-init.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-c
            /seg-init.3gp
        </EncodedObjects>
    </File>
    <File TOI="2" Content-Location="https://example.com/rep1/seg-1.3gp">
        <EncodedObjects type="source" complete="true">
            https://example.com/rep1/seg-1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-a
            /seg-1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-b
            /seg-1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-c
            /seg-1.3gp
        </EncodedObjects>
    </File>
    ...
    <File TOI="8" Content-Location="https://example.com/rep2/seg-3.3gp">
        <EncodedObjects type="source" complete="true">
            https://example.com/rep2/seg-3.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/rep2/cmmf-a
            /seg-3.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/rep2/cmmf-b
            /seg-3.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/rep2/cmmf-c
            /seg-3.3gp
        </EncodedObjects>
    </File>
    <File TOI="9" Content-Location="https://example.com/seg-init-1.3gp">
        <EncodedObjects type="source" complete="true">
            https://example.com/seg-init-1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/cmmf-a
            /seg-init-1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/cmmf-b
            /seg-init-1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/cmmf-c
            /seg-init-1.3gp
        </EncodedObjects>
    </File>
    <File TOI="10" Content-Location="https://example.com/1/1.3gp">
        <EncodedObjects type="source" complete="true">
            https://example.com/1/1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/1/cmmf-a/1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/1/cmmf-b/1.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
```

```
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/1/cmmf-c/1.3gp
        </EncodedObjects>
    </File>
    ...
    <File TOI="16" Content-Location="https://example.com/2/3.3gp">
        <EncodedObjects type="source" complete="true">
            https://example.com/2/3.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-a.com-provider-service.ms.as.3gppservices.org/2/cmmf-a/3.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-b.com-provider-service.ms.as.3gppservices.org/2/cmmf-b/3.3gp
        </EncodedObjects>
        <EncodedObjects type="cmmf" complete="true">
            https://distribution-c.com-provider-service.ms.as.3gppservices.org/2/cmmf-c/3.3gp
        </EncodedObjects>
    </File>
</FDTInstance>
```

H.3.2.4 Single file CMMF Configuration Information document example

The following example shows a CMMF Configuration Information document used as a Media Player Entry in which a single MP4 file is delivered using CMMF from three service locations exposed by the 5GMSd AS at reference point M4d.

Listing H.3.2.4-1: Single MP4 CMMF Configuration Information document example

```
{
  "version": 0,
  "applicationResourceLocators": [
    {
      "locator": "https://example.com/video.mp4",
      "contentType": "video/3gpp",
      "isManifest": false,
    },
  ],
  "applicationResourceConfigurations": [
    {
      "serviceLocations": [
        {
          "variantId": "cmmf-a",
          "baseUrl": "https://distribution-a.com-provider-service.ms.as.3gppservices.org",
          "requestPathMaps": [
            {
              "requestPathPattern": "^",
              "mappedPath": "/cmmf-a",
            },
          ],
        },
        {
          "variantId": "cmmf-b",
          "baseUrl": "https://distribution-b.com-provider-service.ms.as.3gppservices.org",
          "requestPathMaps": [
            {
              "requestPathPattern": "^",
              "mappedPath": "/cmmf-b",
            },
          ],
        },
        {
          "variantId": "cmmf-c",
          "baseUrl": "https://distribution-c.com-provider-service.ms.as.3gppservices.org",
          "requestPathMaps": [
            {
              "requestPathPattern": "^",
              "mappedPath": "/cmmf-c",
            },
          ],
        },
      ],
      "cmmfCodingConfiguration": {
        "cmmfVersion": 0,
        "cmmfCodeType": 0,
        "cmmfProfile": "org.etsi.cmmf.a",
      },
    },
  ],
}
```

H.3.2.5 CMMF Configuration Information document referencing a DASH MPD

The following example shows a CMMF Configuration Information document used as a Media Player Entry in which the contents of the presentation described by the MPD shown in listing H.3.2.1-1 are delivered using CMMF. It is assumed that CMMF transport resources for the media resources referenced within the MPD are accessible from three service locations exposed by the 5GMSd AS at reference point M4d.

In this example, a CMMF Media Access Client uses the CMMF Configuration Information document shown in listing H.3.2.5-1 combined with the MPD shown in listing H.3.2.1-1 as the Media Player Entry. The URLs to media

resources referenced within the MPD are mapped to CMMF transport resource URLs using information contained within the CMMF Configuration Information document.

Listing H.3.2.5-1: CMMF Configuration Information document example referencing MPD

```
{
  "version": 0,
  "applicationResourceLocators": [
    {
      "locator": "https://example.com/manifest.mpd",
      "contentType": "application/dash+xml",
      "isManifest": true,
    },
  ],
  "applicationResourceConfigurations": [
    {
      "serviceLocations": [
        {
          "variantId": "cmmf-a",
          "baseUrl": "https://distribution-a.com-provider-service.ms.as.3gppservices.org",
          "requestPathMaps": [
            {
              "requestPathPattern": "\/( ?=[^\/*$)",
              "mappedPath": "/cmmf-a/",
            },
          ],
        },
        {
          "variantId": "cmmf-b",
          "baseUrl": "https://distribution-b.com-provider-service.ms.as.3gppservices.org",
          "requestPathMaps": [
            {
              "requestPathPattern": "\/( ?=[^\/*$)",
              "mappedPath": "/cmmf-b/",
            },
          ],
        },
        {
          "variantId": "cmmf-c",
          "baseUrl": "https://distribution-c.com-provider-service.ms.as.3gppservices.org",
          "requestPathMaps": [
            {
              "requestPathPattern": "\/( ?=[^\/*$)",
              "mappedPath": "/cmmf-c/",
            },
          ],
        },
      ],
      "cmmfCodingConfiguration": {
        "cmmfVersion": 0,
        "cmmfCodeType": 0,
        "cmmfProfile": "org.etsi.cmmf.a",
      },
    },
  ],
}
```

H.3.3 Provisioning Session and Content Hosting Configuration examples

H.3.3.1 General

This clause provides examples of provisioning the 5GMS System to deliver media resources using CMMF from multiple service locations exposed by the 5GMSd AS at reference point M4d. It is assumed within these examples that the CMMF Media Access Client extends functionality of the DASH Access Client (in particular the Downloader sub-function) specified in clause 13.2.1 to enable download and decoding of media resources encoded within CMMF transport resources.

H.3.3.2 Example of media delivery from multiple service locations using CMMF

H.3.3.2.1 Overview

This example shows how CMMF can be used in a basic deployment where a single Provisioning Session is configured by the 5GMSd Application Provider to expose multiple service locations. This example assumes the 5GMSd Application Provider provisions the system in the following manner:

1. A 5GMSd AS is provisioned to ingest media resources at reference point M2d.
2. Media resources ingested at reference point M2d are encoded into CMMF transport resources by the 5GMSd AS using an appropriate CMMF Content Preparation Template as specified in clause H.2.3.2.
3. As part of the Service Access Information, a CMMF Media Player Entry is provided as specified in clause H.2.3.3.

This implementation example is illustrated in figure H.3.3.2.1-1.

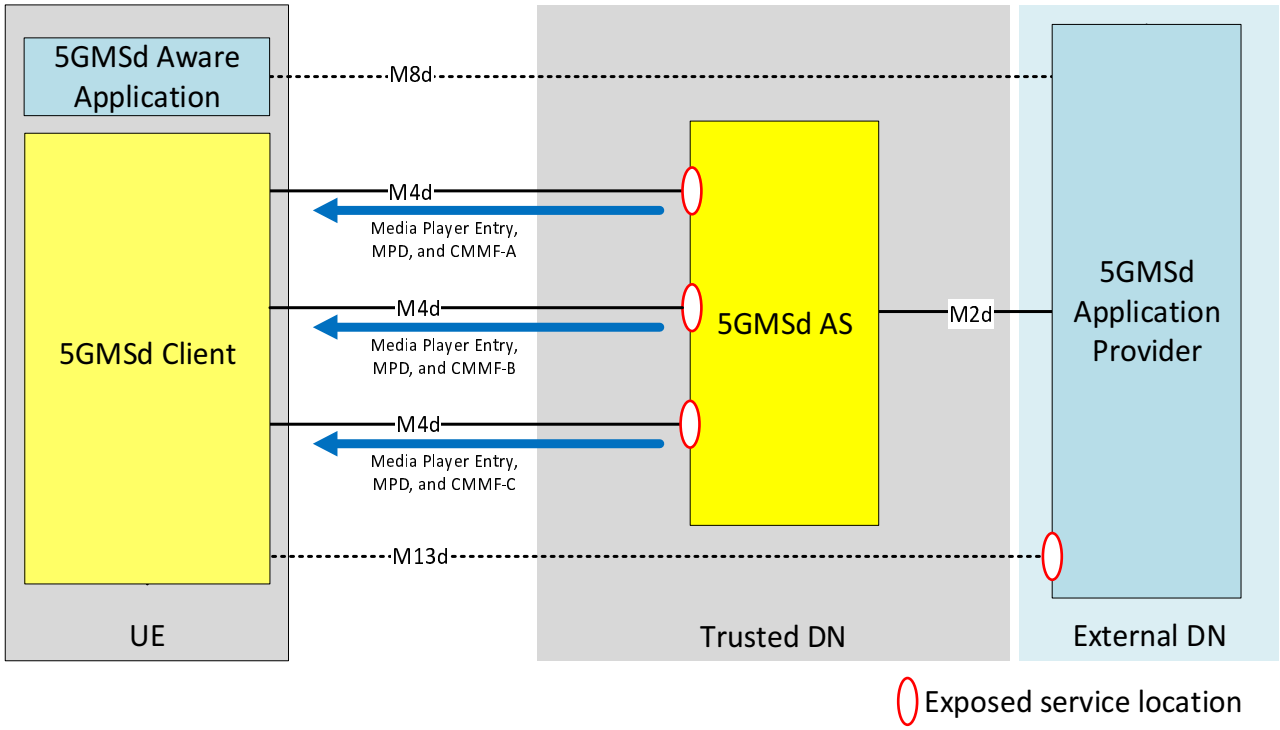


Figure H.3.3.2.1-1: Basic deployment example for 5GMSd AS ingest, content preparation, and delivery using CMMF

The following clauses describe how the 5GMS APIs and protocols are used to realize this example.

H.3.3.2.2 Provisioning Session provisioning

A Provisioning Session is created by the 5GMSd Application Provider using the operation specified in clause 5.2.2.3 of TS 26.510 [56] and the API specified in clause 8.2 of [56]. Example values of the provisioning session API parameters used are shown in table H.3.3.2.2-1.

Table H.3.3.2.2-1: Example ProvisioningSession resource parameters

Property name	Property value	Assigned by
provisioningSessionId	provisioning.session.a	5GMSd AF
provisioningSessionType	*MS_DOWNLINK*	Media Application Provider
externalServiceId	com.provider.service	
appId	dash.downlink.streaming	
NOTE: Values for *provisioningSessionId, externalServiceId,* and *appId* are notional and may not reflect actual values.		

H.3.3.2.3 Content Preparation Templates provisioning

The 5GMSd Application Provider provisions one CMMF encoder Content Preparation Template within the *provisioning.session.a* Provisioning Session using the operation specified in clause 5.2.5.2 of TS 26.510 [56] and the API defined in clause 8.5 of [56]. The following is assumed:

- A sub-path URL added by the 5GMSd Client in the M4d request URL is used to determine which CMMF transport resource representation of the media resource is generated. For the purposes of this example, the Content Preparation Template generates representation "CMMF-A" if a M4d request URL containing the sub-path `cmmf-a` is received; it generates representation "CMMF-B" if the URL contains the sub-path `cmmf-b` is received; and it generates representation "CMMF-C" if the URL contains the sub-path `cmmf-c` is received.

- Content Preparation is triggered upon the receipt of a pull-based request to the 5GMSd AS at a service location exposed at reference point M4d.

- The path of the requested resource is available to the Content Preparation Template associated with the service location that received the request.

- Media resources (e.g., video segments, audio segments, etc.) are ingested by the 5GMSd AS at reference point M2d and encoded into CMMF transport resources according to the Content Preparation Template.

- Upon completion of the CMMF encoding operation, the resulting CMMF transport resource is made available for caching by the 5GMSd AS and/or delivery to the requesting 5GMSd Client at the service location where the request was received.

Upon successful provisioning of the Content Preparation Template, the 5GMSd AF returns the Content Preparation Template ID *cmmf.content.preparation.template* to the Media Application Provider. An example CMMF Content Preparation Template is shown in clause H.3.4.

H.3.3.2.4 Server Certificates provisioning

The 5GMSd Application Provider provisions the Server Certificates for the Provisioning Session using the operation specified in clause 5.2.4.2 of TS 26.510 [56] and the API defined in clause 8.4 of [56]. The Server Certificates assume a canonical domain name as specified in clause 6.0.2.2. Example values for the certificates generated by the 5GMSd AF are provided in table H.3.3.2.4-1.

Table H.3.3.2.4-1: Example Server Certificate parameters

CN	subjectAltName	certificateId
*.com-provider-service.ms.as.3gppservices.org	*.com-provider-service.ms.as.3gppservices.org	server.certificate.a

H.3.3.2.5 Content Hosting provisioning and configuration

The 5GMSd Application Provider provisions the Content Hosting Configuration for each Provisioning Session using the operation specified in clause 5.2.8.2 of TS 26.510 [56] and the API defined in clause 8.8 of [56].

Configuring the Content Hosting Configuration in the 5GMSd AS is performed according to:

- The example base URL of the 5GMSd Application Provider's origin server is `https://origin.media-application-provider.com`.

The 5GMSd Application Provider provides the Media Player Entry URL to the 5GMSd-Aware Application via reference point M8d. Based on this, the 5GMSd Client can download the Media Player Entry from a service location exposed by the 5GMSd AS at reference point M4d. Example Media Player Entry documents are provided in clause H.3.2.

Table H.3.3.2.5-1 provides example values for the Content Hosting Configuration parameters.

Table H.3.3.2.5-1: Content Hosting Configuration resource parameters

Property name	Property value	Assigned by
name	content-hosting-configuration-a	5GMSd Application Provider
ingestConfiguration		
mode	*PULL*	5GMSd Application Provider
protocol	*urn:3gpp:5gms:content-protocol:http-pull*	
baseURL	`https://origin.media-application-provider.com`	
distributionConfiguration		
affinityGroup	affinity.group.a	5GMSd Application Provider
contentPreparationTemplateId	cmmf.content.preparation.template	
certificateId	server.certificate.a	
canonicalDomainName	distribution-a.com-provider-service.ms.as.3gppservices.org	5GMSd AF
baseURL	`https://distribution-a.com-provider-service.ms.as.3gppservices.org`	
pathRewriteRule		
requestPathPattern	`cmmf-a/`	5GMSd Application Provider
mappedPath		
distributionConfiguration		
affinityGroup	affinity.group.b	5GMSd Application Provider
contentPreparationTemplateId	cmmf.content.preparation.template	
certificateId	server.certificate.a	
canonicalDomainName	distribution-b.com-provider-service.ms.as.3gppservices.org	5GMSd AF
baseURL	`https://distribution-b.com-provider-service.ms.as.3gppservices.org`	
pathRewriteRule		
requestPathPattern	`cmmf-b/`	5GMSd Application Provider
mappedPath		
distributionConfiguration		
affinityGroup	affinity.group.c	5GMSd Application Provider
contentPreparationTemplateId	cmmf.content.preparation.template	
certificateId	server.certificate.a	
canonicalDomainName	distribution-c.com-provider-service.ms.as.3gppservices.org	5GMSd AF
baseURL	`https://distribution-c.com-provider-service.ms.as.3gppservices.org`	
pathRewriteRule		
requestPathPattern	`cmmf-c/`	5GMSd Application Provider
mappedPath		

H.3.3.2.6 End-to-end URL mapping

Table H.3.3.2.6-1 provides an example of the end-to-end mapping for requests initiated by the Media Player for a subset of the URLs provided in the example MPD shown in listing H.3.2.1-1 where the CMMF Media Access Client uses the corresponding URL mapping provided in the EFDT shown in listing H.3.2.3-1 or in the CMMF Configuration Information document shown in listing H.3.2.5-1.

Table H.3.3.2.6-1: End-to-End URL mapping example

MPD URL	M4d Request URLs	M2d Request URL
rep1/seg-1.3gp	https://distribution-a.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-a/seg-1.3gp	https://origin.media-application-provider.com/rep1/seg-1.3gp
	https://distribution-b.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-b/seg-1.3gp	
	https://distribution-c.com-provider-service.ms.as.3gppservices.org/rep1/cmmf-c/seg-1.3gp	
http://example.com/1/1.3gp	https://distribution-a.com-provider-service.ms.as.3gppservices.org/1/cmmf-a/1.3gp	https://origin.media-application-provider.com/1/1.3gp
	https://distribution-b.com-provider-service.ms.as.3gppservices.org/1/cmmf-b/1.3gp	
	https://distribution-c.com-provider-service.ms.as.3gppservices.org/1/cmmf-c/1.3gp	

H.3.3.3 Example of media delivery from multiple service locations using CMMF and 5GMSd AS service chaining at reference point M10d

H.3.3.3.1 Overview

This example shows how CMMF can be used in a deployment where 5GMSd AS service chaining at reference point M10d is provisioned. This example assumes the 5GMSd Application Provider provisions the system in the following manner:

1. A 5GMSd AS is provisioned to serve as an origin shield intended to reduce requests for content at reference point M2d. Furthermore, the creation of CMMF transport resources from ingested content at reference point M2d using Content Preparation is performed within this 5GMSd AS.

2. Two 5GMSd AS instances are provisioned to serve 5GMSd Clients from exposed service locations at reference point M4d. Both 5GMSd AS instances are configured to ingest CMMF transport resources (unique to their exposed service locations) from the 5GMSd AS serving as the origin shield at reference M10d.

3. Media resources ingested at reference point M2d are encoded into CMMF transport resources using an appropriate CMMF Content Preparation Template as specified in clause H.2.3.2.

4. A CMMF Media Player Entry as specified in clause H.2.3.3.

This implementation example is illustrated in figure H.3.3.3.1-1.

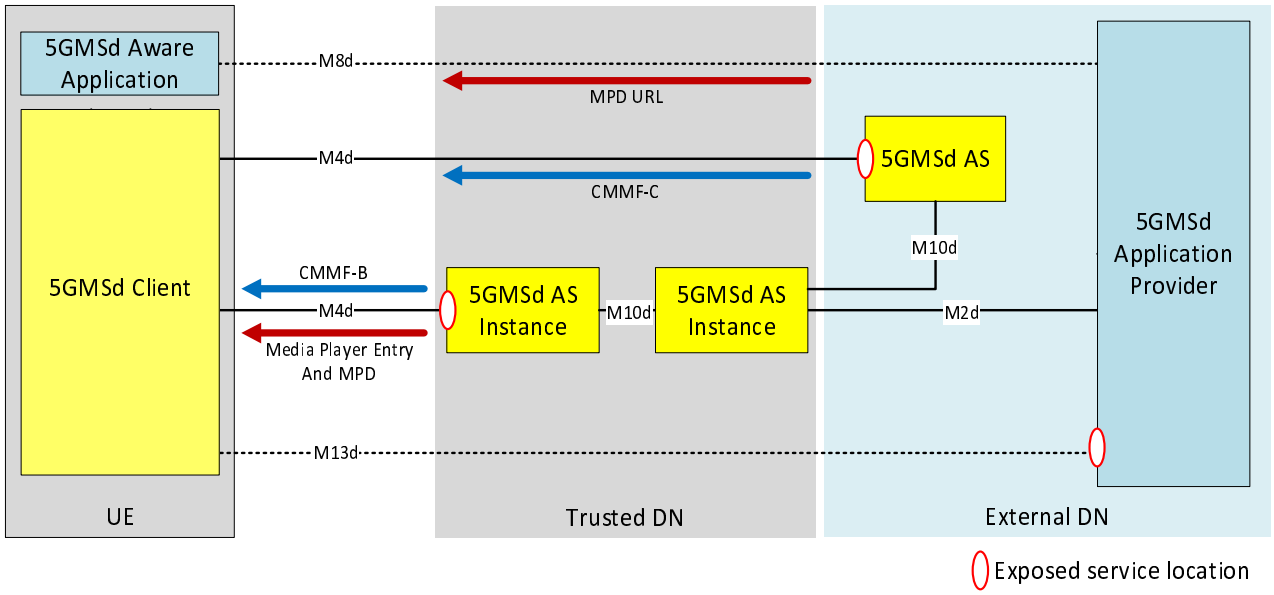


Figure H.3.3.3.1-1: Centralized 5GMSd AS content preparation and ingest deployment example

The following clauses describe how the 5GMS APIs and protocols are used to realize this example.

H.3.3.3.2 Provisioning Session provisioning

Three provisioning sessions are created by the 5GMSd Application Provider using the operation specified in clause 5.2.2.3 of TS 26.510 [56] and the API specified in clause 8.2 of [56]. Example values for the three Provisioning Session parameters used are shown in table H.3.3.3.2-1.

Table H.3.3.3.2-1: ProvisioningSession resource parameters

Property name	Provisioning Session A (Trusted DN)	Provisioning Session B (Trusted DN)	Provisioning Session C (External DN)	Assigned by
provisioningSessionId	provisioning. session.a	provisioning. session.b	provisioning. session.c	Media AF
provisioningSession Type	*MS_DOWNLINK*	*MS_DOWNLINK*	*MS_DOWNLINK*	Media Application Provider
externalServiceId	com.provider.service.a	com.provider.service.b	com.provider.service.c	
appId	dash.downlink. streaming	dash.downlink. streaming	dash.downlink. streaming	
NOTE: Values for *provisioningSessionId*, *externalServiceId*, and *appId* are notional and may not reflect actual values.				

H.3.3.3.3 Content Preparation Templates provisioning

The 5GMSd Application Provider provisions one CMMF encoder Content Preparation Template within the *provisioning.session.a* Provisioning Session using the operation specified in clause 5.2.5.2 of TS 26.510 [56] and the API defined in clause 8.5 of [56]. The following is assumed:

- Content Preparation is triggered upon the receipt of a pull-based request at reference point M10d.
- The path of the requested resource is available to the Content Preparation Template.
- Media resources (e.g., video segments, audio segments, etc.) are ingested by the 5GMSd AS at reference point M2d and encoded into CMMF transport resources by the Content Preparation Template.
- A sub-path URL added by the 5GMSd Client in the M4d request URL is used to determine which CMMF transport resource representation of the media resource is generated. For the purposes of this example, the

Content Preparation Template generates representation "CMMF-B" if the URL contains the sub-path `cmmf-b` is received, and it generates representation "CMMF-C" if the URL contains the sub-path `cmmf-c` is received.

- Upon completion of the CMMF encoding operation, the CMMF transport resource is made available for caching by the 5GMSd AS and/or delivery to the requesting entity.

Upon successful provisioning of the Content Preparation Template, the 5GMSd AF returns the Content Preparation Template ID *cmmf.content.preparation.template* to the Media Application Provider. An example CMMF Content Preparation Template is shown in clause H.3.4.

H.3.3.3.4 Server Certificates provisioning

The 5GMSd Application Provider provisions the Server Certificates for each Provisioning Session using the operation specified in clause 5.2.4.2 of TS 26.510 [56] and the API defined in clause 8.4 of [56]. The Server Certificates assume a canonical domain name as specified in clause 6.0.2.2. Example values for the certificates generated by the 5GMSd AF are provided in table H.3.3.3.4-1.

Table H.3.3.3.4-1: Server Certificates

provisioningSessionId	CN	subjectAltName	certificateId
provisioning.session.a	*.com-provider-service-a.ms.as.3gppservices.org	*.com-provider-service-a.ms.as.3gppservices.org	server.certificate.a
provisioning.session.b	*.com-provider-service-b.ms.as.3gppservices.org	*.com-provider-service-b.ms.as.3gppservices.org	server.certificate.b
provisioning.session.c	*.com-provider-service-c.ms.as.3gppservices.org	*.com-provider-service-c.ms.as.3gppservices.org	server.certificate.c

H.3.3.3.5 Content Hosting provisioning and configuration

The 5GMSd Application Provider provisions the Content Hosting Configuration for each Provisioning Session using the operation specified in clause 5.2.8.2 of TS 26.510 [56] and the API defined in clause 8.8 of [56].

Configuring the Content Hosting Configuration in each 5GMSd AS is performed according to:

- The example base URL of the Media Application Provider's origin server is `https://origin.media-application-provider.com`.

- The creation of the Content Hosting Configuration for Provisioning Session *provisioning.session.a* is completed prior to the creation of the Content Hosting Configuration for Provisioning Sessions *provisioning.session.b* and *provisioning.session.c*.

The 5GMSd Application Provider provides the Media Player Entry URL to the 5GMSd-Aware Application via reference point M8d. Based on this, the 5GMSd Client can download the Media Player Entry from a service location exposed by the 5GMSd AS at reference point M4d. Example Media Player Entry documents are provided in clause H.3.2.

Table H.3.3.3.5-1 provides example values for the Content Hosting Configuration API parameters for all three Provisioning Sessions.

Table H.3.3.3.5-1: ContentHostingConfiguration resource parameters

Property name	ProvisioningSession A	ProvisioningSession B	Provisioning Session C	Assigned by
name	content-hosting-configuration-a	content-hosting-configuration-b	content-hosting-configuration-c	5GMSd Application Provider
ingestConfiguration				
mode	*PULL*	*PULL*	*PULL*	5GMSd Application Provider
protocol	*urn:3gpp:5gms:content-protocol:http-pull*	*urn:3gpp:5gms:content-protocol:http-pull*	*urn:3gpp:5gms:content-protocol:http-pull*	
baseURL	`https://origin.media-application-provider.com`	`https://distribution-a.com-provider-service-a.ms.as.3gppservices.org`	`https://distribution-a.com-provider-service-a.ms.as.3gppservices.org`	
distributionConfiguration				
contentPreparationTemplateId	*cmmf.content.preparation.template*			5GMSd Application Provider
certificateId	*server.certificate.a*	*server.certificate.b*	*server.certificate.c*	
canonicalDomainName	*distribution-a.com-provider-service-a.ms.as.3gppservices.org*	*distribution-a.com-provider-service-b.ms.as.3gppservices.org*	*distribution-a.com-provider-service-c.ms.as.3gppservices.org*	5GMSd AF
baseURL	`https://distribution-a.com-provider-service-a.ms.as.3gppservices.org`	`https://distribution-a.com-provider-service-b.ms.as.3gppservices.org`	`https://distribution-a.com-provider-service-c.ms.as.3gppservices.org`	
pathRewriteRule				
requestPathPattern	`(cmmf-[b-c]/)$`			5GMSd Application Provider
mappedPath				

H.3.3.3.6 End-to-end URL mapping

Table H.3.3.3.6-1 provides an example of the end-to-end mapping for requests initiated by the Media Player for a subset of the URLs provided in the example MPD shown in listing H.3.2.1-1 where the CMMF Media Access Client uses the corresponding URL mapping provided in the EFDT shown in listing H.3.2.3-1 or in the CMMF Configuration Information document shown in listing H.3.2.5-1.

Table H.3.3.3.6-1: End-to-End URL mapping example

MPD URL	M4d Request URL	M10d Request URL	M2d Request URL
`rep1/seg-1.3gp`	`https://distribution-a.com-provider-service-b.ms.as.3gppservices.org/rep1/cmmf-b/seg-1.3gp`	`https://distribution-a.com-provider-service-b.ms.as.3gppservices.org/rep1/cmmf-b/seg-1.3gp`	`https://origin.media-application-provider.com/rep1/seg-1.3gp`
	`https://distribution-a.com-provider-service-c.ms.as.3gppservices.org/rep1/cmmf-c/seg-1.3gp`	`https://distribution-a.com-provider-service-c.ms.as.3gppservices.org/rep1/cmmf-c/seg-1.3gp`	
`http://example.com/1/1.3gp`	`https://distribution-a.com-provider-service-b.ms.as.3gppservices.org/1/cmmf-b/1.3gp`	`https://distribution-a.com-provider-service-b.ms.as.3gppservices.org/1/cmmf-b/1.3gp`	`https://origin.media-application-provider.com/1/1.3gp`
	`https://distribution-a.com-provider-service-c.ms.as.3gppservices.org/1/cmmf-c/1.3gp`	`https://distribution-a.com-provider-service-c.ms.as.3gppservices.org/1/cmmf-c/1.3gp`	

H.3.4 Example CMMF Content Preparation Template

This clause provides an example showing a CMMF Content Preparation Template as specified in clause H.2.3.2. The Content Preparation Template, which is a CMMF Encoder Configuration document, is shown in listing H.3.4-1. This CMMF Encoder Configuration provides information necessary for a CMMF Encoder provisioned within the 5GMS System to encode different CMMF transport resources (identified as variants `cmmf-a`, `cmmf-b`, and `cmmf-c`) that may be requested by a 5GMSd Client. It is assumed in the example that the CMMF Media Player Entry shown in listing H.3.2.5-1 is available at `https://distribution-a.com-provider-service.ms.as.3gppservices.org/cmmf-configuration-information.json`.

Listing H.3.4-1: CMMF Content Preparation Template example

```
{
  "cmmfConfigurationInformationLocator": "https://distribution-a.com-provider-service.ms.as.
3gppservices.org/cmmf-configuration-information.json",
  "bitstreamVersion": 0,
  "contentSourceType": "001b",
  "codeType": "0",
  "profile": "org.etsi.cmmf.a",
  "blockSizeToBlockNumSymbolsMaps" : [
    {
      "blockSizeMin": 0,
      "blockSizeMax": 128,
      "blockNumSymbols": 1
    },
    {
      "blockSizeMin": 129,
      "blockSizeMax": 1048576,
      "blockNumSymbols": 4
    },
    {
      "blockSizeMin": 1048577,
      "blockSizeMax": 2097152,
      "blockNumSymbols": 8
    },

    {
      "blockSizeMin": 2097153,
      "blockSizeMax": 4194304,
      "blockNumSymbols": 16
    }
  ],
  "xcdInformation": {
    "coefficientType": "vector",
    "signalSystematicSymbols": true,
    "coefficientInformation": {
      "coefficientVectorMaps": [
        {
          "variantId": "cmmf-a",
          "numSymbols": 4,
          "fieldSizeExp": 1,
          "coefficientVectors": ["1000b", "0001b", "0110b", "1111b"],
        },
        {
          "variantId": "cmmf-b",
          "numSymbols": 4,
          "fieldSizeExp": 1,
          "coefficientVectors": ["0100b", "1010b", "0011b", "1111b"],
        },

        {
          "variantId": "cmmf-c",
          "numSymbols": 4,
          "fieldSizeExp": 1,
          "coefficientVectors": ["0010b", "1100b", "1001b", "1111b"],
        },
        {
          "variantId": "cmmf-a",
          "numSymbols": 8,
          "fieldSizeExp": 1,
          "coefficientVectors": […],
        },
        {
          "variantId": "cmmf-b",
```

```
          "numSymbols": 8,
          "fieldSizeExp": 1,
          "coefficientVectors": […],
        },

        {
          "variantId": "cmmf-c",
          "numSymbols": 8,
          "fieldSizeExp": 1,
          "coefficientVectors": […],
        },
        {
          "variantId": "cmmf-a",
          "numSymbols": 16,
          "fieldSizeExp": 1,
          "coefficientVectors": […],
        },
        {
          "variantId": "cmmf-b",
          "numSymbols": 16,
          "fieldSizeExp": 1,
          "coefficientVectors": […],
        },

        {
          "variantId": "cmmf-c",
          "numSymbols": 16,
          "fieldSizeExp": 1,
          "coefficientVectors": […],
        }
      ]
    }
  }
}
```

Annex I (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
25.6.2019	SA4#104	S4-190649				Initial Version	0.0.1
23.1.2020	SA4#107	S4-200077, S4-200238, S4-200239, S4-200318				Updates during SA4#107	0.3.0
07.02.2020	ConfCall	S4-AHI931, S4-AHI932				Scope, editorial improvements and online edits from Conf Call (6th Feb 2020)	0.3.1
11.02.2020	offline					Editorial updates according to offline email discussions	0.3.2
2020-02	ConfCall	S4-AHI950				Editorial updates from Conf Call (Online, 13th Feb 2020)	0.4.0
2020-03	-	SP-200237				Specification to TSG: 5G Media Streaming (5GMS); Protocols TS 26.512, Version 1.0.0	1.0.0
		S4-AHI953					1.0.1
2020-04	SA4#108e	S4-200513, S4-200514, S4-200633				Renaming entities in the 5GMS Provisioning API, Additional clauses to specify procedures for manipulating Ingest Protocols, Content Preparation Templates and Server Certificates, Consumption Reporting Procedure API- M1d and M5d	1.0.2
2020-05	Conf Call	S4-AHI989				New Structure	1.1.0
2020-06	SA4#109e	S4-200920, S4-200886, S4-200889, S4-200883				920: Consumption reporting in M7d interface, 886: RAN Signaling-based Network Assistance, 889: API for Service Access information acquisition, 883: APIs for Server Certificates, Content Preparation Templates and Ingest Protocols	1.2.0
2020-08	SA4#110e	S4-AHI996 S4-AHI998 S4-AHIA33				996: Completion of Content Preparation Template procedures, 998: Completion of content distribution geofencing feature, A33: Completion of Server Certificates Provisioning API	1.3.0
2020-08	SA4#110e	Cor of S4-AHI998				Correction of S4-AHI998 implementation, Editorial Correction in Clause 11.2.4	1.3.1
2020-08	SA4#110e	S4-201092, S4-201114, S4-201210, S4-201208, S4-201213, S4-201230, S4-201004, S4-201229, S4-201221, S4-201231, S4-201225, S4-201271, S4-201266, S4-201282, S4-201281				1092: Editorial Improvements, 1114: Specification structure – interfaces and APIs, 1210: Completion of Ingest Protocols API, 1208: Informative Annex on Parameter Population, 1213: Addition of General Sections, 1230: M6d APIs for 5GMS, 1004: Informative annex on Content Hosting Configuration examples, 1229: Correction of the Policy Template resource state transitions, 1221: DASH/CMAF in 5GMSd, 1231: M7d APIs, 1225: Update on consumption reporting, 1271: Update on Metrics Reporting, 1266: Updated on M5 Dynamic Policy activation API and M1 Policy Template Provisioning API, 1282: 5GMS3: AF-based Network Assistance, 1281: Provisioning Sessions API	1.4.0
2020-09	SA#89-e	SP-200666				5G Media Streaming (5GMS); Protocols (This was the presentation of Specification to TSG: 5G Media Streaming (5GMS); Protocols TS 26.512, Version 2.0.0 to bring UCC)	16.0.0

2020-12	SA#90-e	SP-200935	0004	3	F	Cumulative corrections of 5GMS3 APIs [CRs implemented: S4-201432: Cumulative corrections of 5GMS3 APIs, Ericsson S4-201305: Editorial corrections, BBC S4-201363: Additions and Modifications to M1 API on Metrics Reporting Configuration, Qualcomm S4-201622: Text on Procedures for Uplink Streaming, Qualcomm, Ericsson S4-201580: Correction of the missing SdfMethod type definition, Ericsson S4-201593: Correction of the missing CRUD operation notation, Ericsson S4-201594: Correction of the MediaPlayerEntry and ClientMetricsReportingConfiguration cardinality in the Service Access Information resource, Ericsson S4-201596: Correction of the Service Access Information subresource (URL), Ericsson S4-201597: Annex for OpenAPI Implementation, Ericsson S4-201595, Update Consumption reporting, Enensys Technology, BBC S4-201590: Bug Fixes on Metrics Reporting Functionality, Ericsson LM, Qualcomm Incorporated S4-201486: AF-based Network Assistance, Sony Europe B.V., Ericsson LM S4-201608: CR on AT Commands for RAN-based Assistance, Qualcomm Inc.]	16.1.0
2021-05	SA#91-e	SP-210039	0007	2	F	OpenAPI implementation and aggregated essential corrections	16.2.1
2021-06	SA#92-e	SP-210566	0010	1	F	ClientId for Consumption and Metrics Reporting	16.3.0
2021-06	SA#92-e	SP-210566	0011	1	F	Corrections to 5GMS stage 3 specification	16.3.0
2021-06	SA#92-e	SP-210566	0012	1	F	Correction on Dynamic Policy parameters	16.3.0
2021-09	SA#93-e	SP-210826	0013	1	F	Corrections to TS 26.512	16.4.0
2022-03	SA#95-e	SP-220055	0014	0	F	Correction to QoE metrics reporting client configuration	16.5.0
2022-03	SA#95-e	SP-220238	0019	0	F	Add support for VR QoE metrics	16.5.0
2022-03	SA#95-e						17.0.0
2022-06	SA#96	SP-220595	0020	2	B	CR on Adding Edge Support	17.1.0
2022-06	SA#96	SP-220595	0021	1	F	Fixing api-version in 5GMS RESTful APIs	17.1.0
2022-06	SA#96	SP-220598	0023	1	B	Support for Data Collection and Reporting for 5G Media Streaming	17.1.0
2022-06	SA#96	SP-220597	0024	1	B	[5MBP3] 5GMS Protocol Extensions for 5GMS via eMBMS	17.1.0
2022-07	SA#96					Editorial Corrections	17.1.1
2022-08	SA#96					Editorial Corrections	17.1.2
2022-09	SA#97-e	SP-220757	0025	1	F	[5GMS_EDGE_3] [EVEX] Rel-17 API corrections	17.2.0
2022-12	SA#98-e	SP-221043	0028	2	F	[5GMS3] Rel-17 clarifications and corrections	17.3.0
2023-03	SA#99	SP-230253	0032	1	F	[5GMS3] Rel-17 corrections	17.4.0
2023-06	SA#100	SP-230546	0033	5	F	[5GMS3, TEI17] Rel-17 corrections	17.5.0
2023-09	SA#101	SP-231053	0037	3	F	[5GMS3, TEI17] Essential maintenance	17.6.0
2023-09	SA#101	SP-230913	0053	3	F	[5GMS, TEI17] Content hosting essential correction	17.6.0
2023-09	SA#101	SP-230918	0039	5	F	[5GMS_EDGE_3] Correction of EAS Discovery	17.6.0
2023-12	SA#102	SP-231368	0054	1	F	[5GMS3, TEI17] Essential maintenance	17.7.0
2023-12	SA#102	SP-231761	0060	5	F	[5GMS3, TEI17] Correction of Server Certificate handling	17.7.0
2023-12	SA#102	SP-231347	0040	4	B	[5GMS_Pro_Ph2] ANBR-based network assistance data reporting	18.0.0
2023-12	SA#102	SP-231229	0041	4	B	[5GMS_Pro_Ph2] Event exposure APIs	18.0.0
2024-03	SA#103	SP-240040	0036	5	B	[5GMS_Pro_Ph2] Default 5GMS AF address	18.1.0
2024-06	SA#104	SP-240568	0070	2	F	[EVEX, TEI18] Add missing explicit data reporting and event exposure parameters	18.2.0
2024-06	SA#104	SP-240859	0046	4	B	[5GMS_Pro_Ph2] Service URL Handling	18.2.0
2024-06	SA#104	SP-240925	0057	8	D	[5GMS_Pro_Ph2] Remove M1, M5 and M6 procedures and APIs	18.2.0
2024-06	SA#104	SP-240689	0055	4	B	[5GMS_Pro_Ph2] 5GMS AS configuration procedures and APIs at M3	18.2.0
2024-06	SA#104	SP-240689	0048	9	B	[5GMS_Pro_Ph2] 5GMS over MBS and 5GMS hybrid services	18.2.0
2024-06	SA#104	SP-240689	0047	12	B	[5GMS_Pro_Ph2] Consolidated media plane enhancements	18.2.0
2024-06	SA#104	SP-240689	0066	7	B	[5GMS_Pro_Ph2] Media delivery session identifier at M4+M7+M11	18.2.0
2024-06	SA#104	SP-240697	0067	-	F	[EVEX, TEI18] Data packaging strategies for 5GMS	18.2.0

2024-07						Fixing implementation of CR0066r7 and CR0070r2	18.2.1
2024-09	SA#105	SP-241108	0075	1		[5GMS_Pro_Ph2] Alignment with renamed data types	18.3.0
2024-09	SA#105	SP-241108	0077			[5GMS_Pro_Ph2] Essential corrections	18.3.0
2024-09	SA#105	SP-241108	0078			[5GMS_Pro_Ph2] Pre-selection of Service Operation Point in Service URL	18.3.0
2024-09	SA#105	SP-241108	0079	1		[5GMS_Pro_Ph2]: OAuth 2.0 security at Reference Point M3	18.3.0
2024-09	SA#105	SP-241118	0076	1		[EVEX, TEI18] Add context identifiers to BaseRecord	18.3.0
2025-01	SA#106	SP-241742	0081	3	F	[5GMS_Pro_Ph2] Alignment of 3GPP Service URL parameter	18.4.0
2025-01	SA#106	SP-241742	0083	3		[5GMS_Pro_Ph2] Completion of media stream handling client API for Background Data	18.4.0
2025-01	SA#106	SP-241587	0084	2		[5GMS_Pro_Ph2] Canonical 5GMS AS host name for universal Media Entry Point URL embedded in portable Service URL	18.4.0
2025-03	SA#107	SP-250131	0085			[5GMS3, TEI18] Align OpenAPI YAML with normative description	18.5.0
2025-06	SA#108	SP-250695	0094	1	F	[5MBP3, TEI18] References, reference points, and other corrections	18.6.0
2025-06	SA#108	SP-250567	0095		F	Update of info and externalDocs fields	18.6.0
2025-09	SA#109	SP-250922	0096	2	F	[5GMS_Pro_Ph2] Uplink procedures alignment	18.7.0
2025-09	SA#109	SP-250931	0097	2	B	[AMD_PRO-MED] Consolidated additions	19.0.0
2025-12	SA#110	SP-251443	0100	1	B	[AMD_PRO-MED] Security procedures for metrics reporting at M3	19.1.0
2025-12	SA#110	SP-251443	0101	1	F	[AMD_PRO-MED] Correction to M3 data models	19.1.0
2025-12	SA#110	SP-251443	0102	1	F	[AMD_PRO-MED] Updates to normative CMMF Content Preparation Template and Media Player Entry references.	19.1.0
2025-12	SA#110	SP-251313	0103	-	F	Update of info and externalDocs fields	19.1.0
2026-03	SA#111	SP-260124	0104	1	F	[AMD_PRO-MED] Corrections to CMCD phase 1	19.2.0
2026-06	SA#112	SP-260399	0105	1	F	[AMD_PRO-MED] Alignment to update ETSI TS 103 973 reference with latest published version	19.3.0
2026-06	SA#112	SP-260608	0106		F	Update of info and externalDocs fields	19.3.0

History

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V19.0.0	October 2025	Publication
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V19.3.0	August 2026	Publication